



US 20250287484A1

(19) **United States**

(12) **Patent Application Publication**
ZHANG et al.

(10) **Pub. No.: US 2025/0287484 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **COLOR-TEMPERATURE-TUNABLE LIGHT
EMITTING DEVICES**

Publication Classification

(51) **Int. Cl.**
H05B 45/20 (2020.01)
(52) **U.S. Cl.**
CPC **H05B 45/20** (2020.01)

(71) Applicant: **BRIDGELUX, INC.**, Fremont, CA
(US)
(72) Inventors: **Jingqiong ZHANG**, XIAMEN CITY
(CN); **Yi-Qun LI LI**, DANVILLE, CA
(US); **Jungang ZHAO**, SUZHOU (CN)

(57) **ABSTRACT**

(21) Appl. No.: **19/219,601**

(22) Filed: **May 27, 2025**

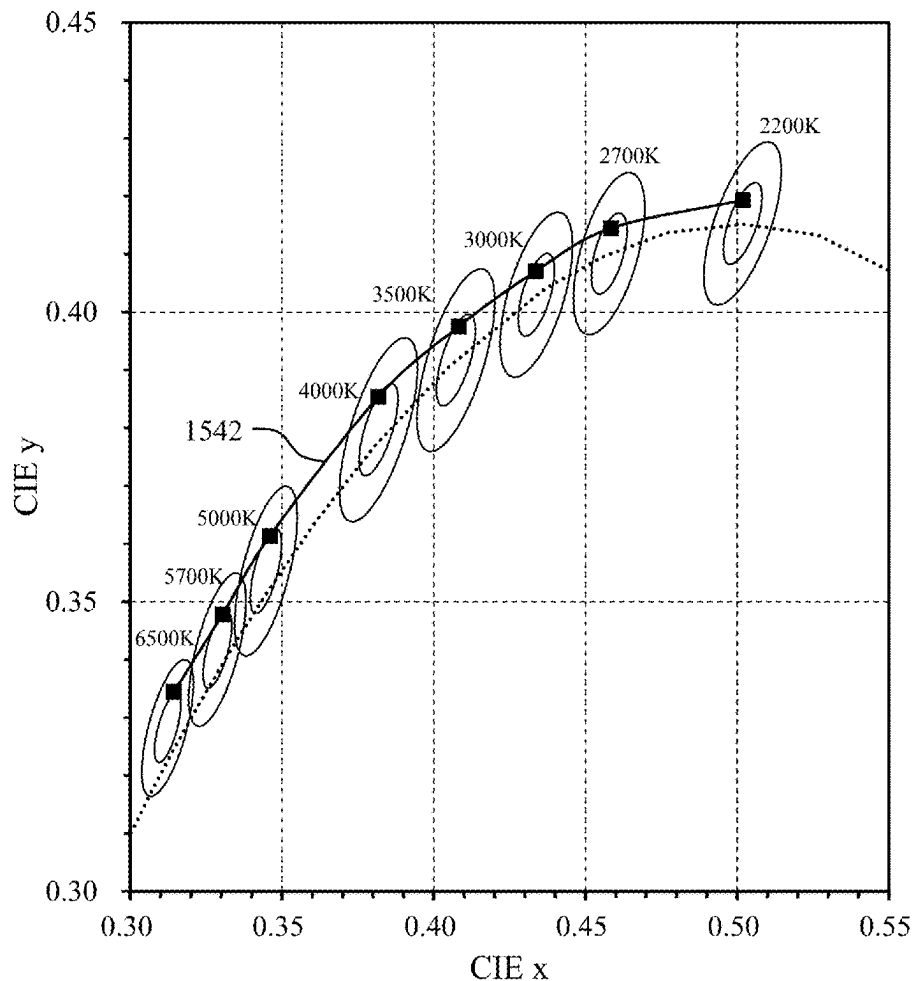
Related U.S. Application Data

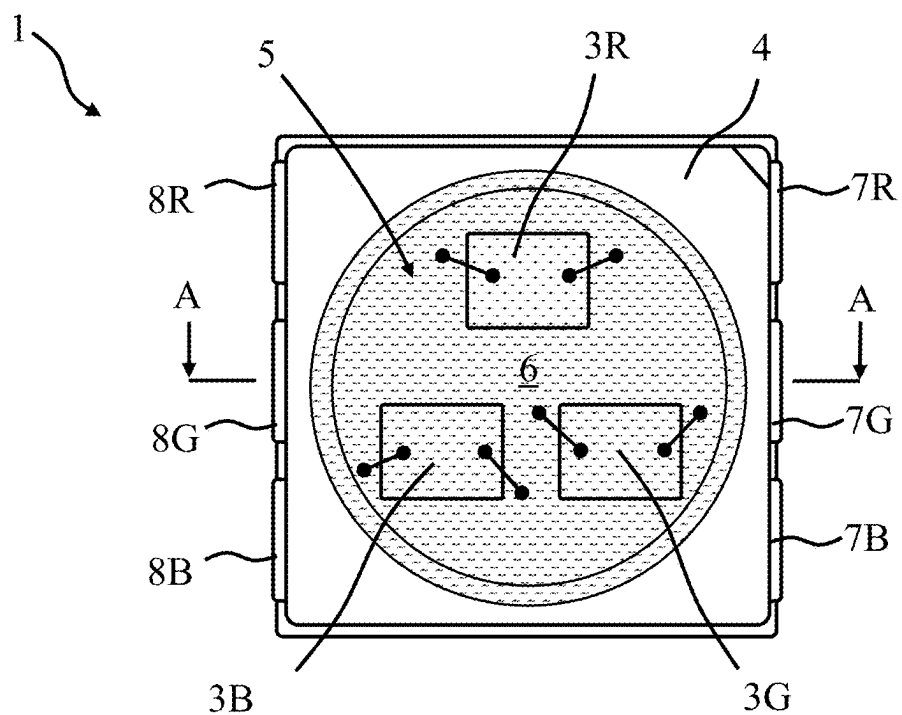
(63) Continuation of application No. PCT/US2023/
081478, filed on Nov. 29, 2023, which is a continu-
ation-in-part of application No. 18/361,814, filed on
Jul. 28, 2023.
(60) Provisional application No. 63/510,341, filed on Jun.
26, 2023.

Foreign Application Priority Data

Nov. 29, 2022 (WO) PCT/CN2022/135022

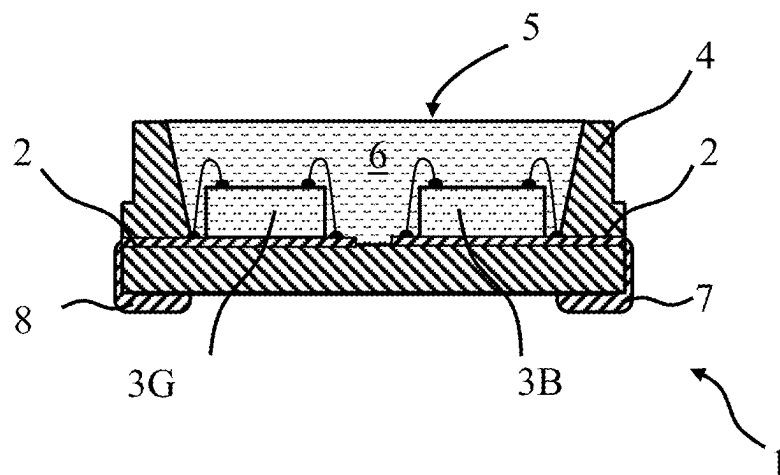
A light emitting device comprising: a first LED for gener-
ating light of a first chromaticity in a first chromaticity
region; a second LED for generating light of a second
chromaticity in a second chromaticity region; and a third
LED for generating light of a third chromaticity in a third
chromaticity region, wherein the device is for generating
light with a CCT that is tunable within a range of CCTs from
1800K to 6500K and has a chromaticity that is within 5
SDCM of the black body locus; wherein the device has a
luminous efficacy of at least 95 lm/W; and wherein the light
generated by the device comprises a combination of light
generated by the first, second, and third LEDs and wherein
the CCT is tunable by controlling power to the first, second
and third LEDs.





PRIOR ART

FIG. 1A



SECTION A-A

FIG. 1B

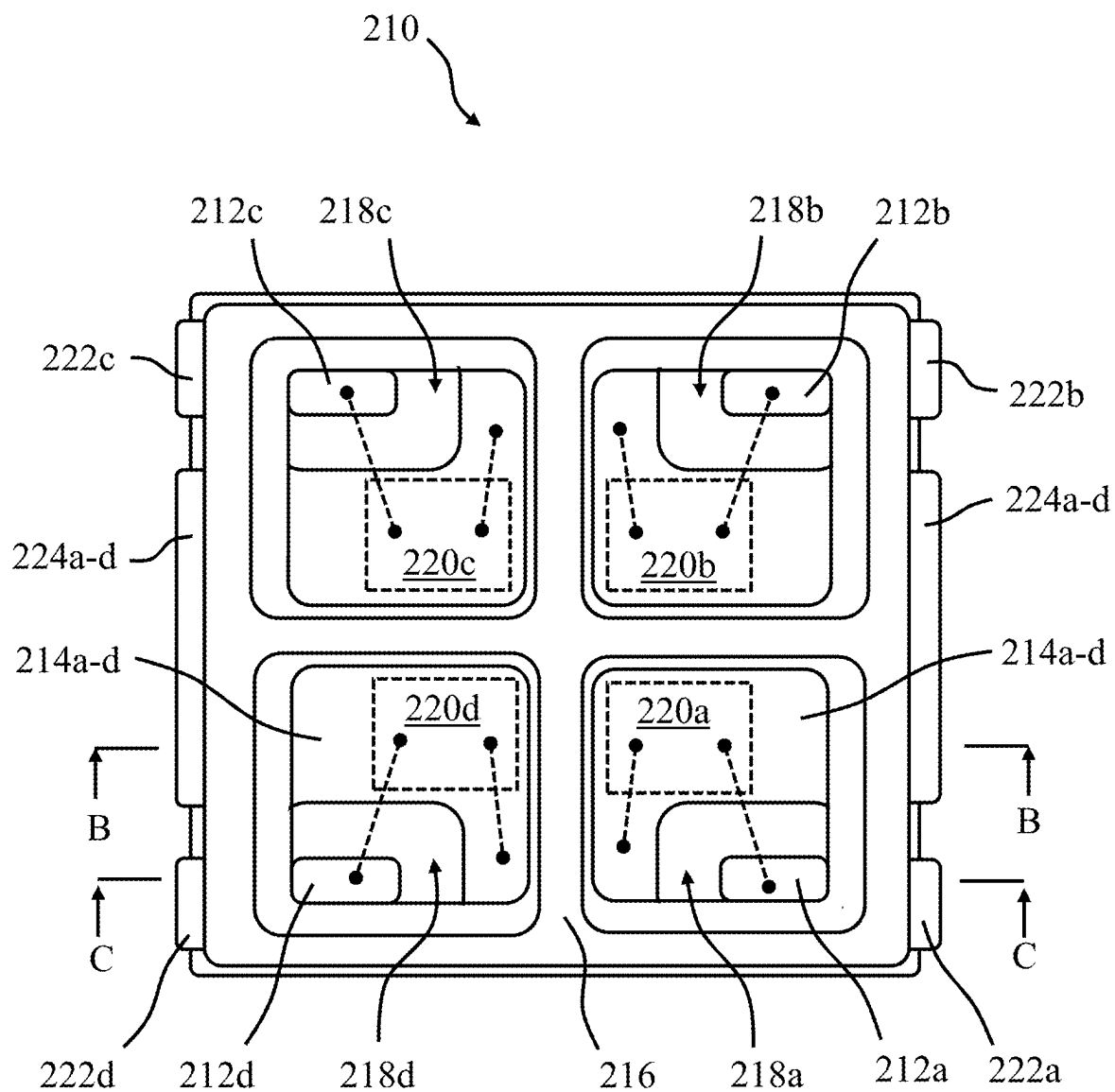
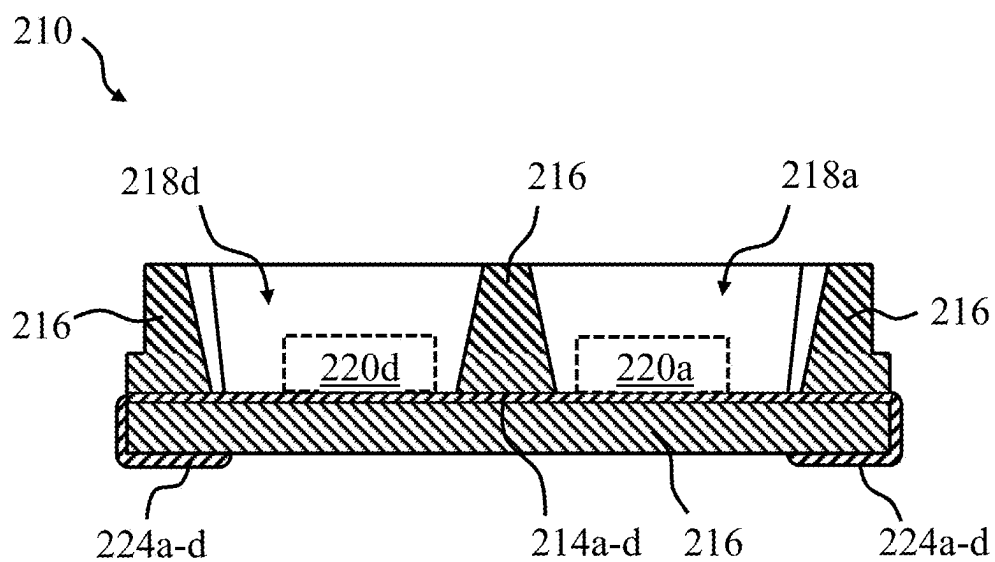
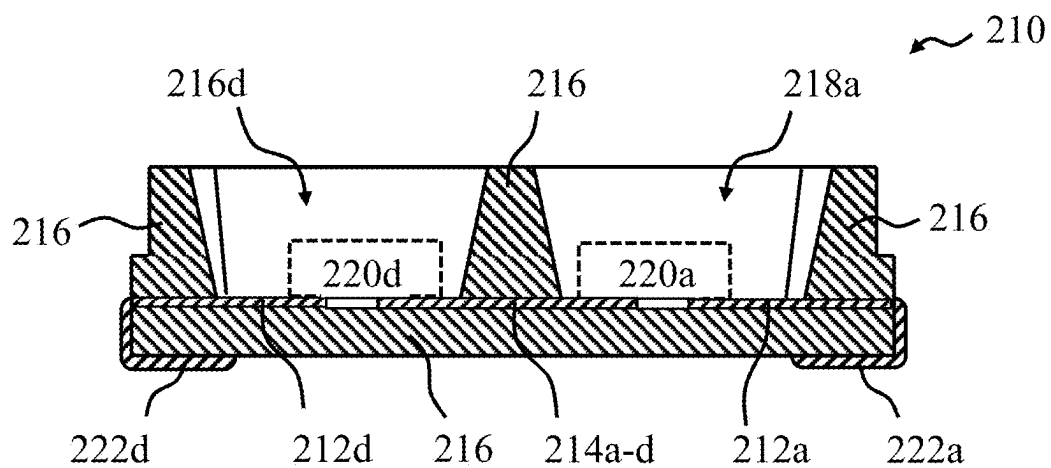


FIG. 2A



SECTION B-B

FIG. 2B



SECTION C-C

FIG. 2C

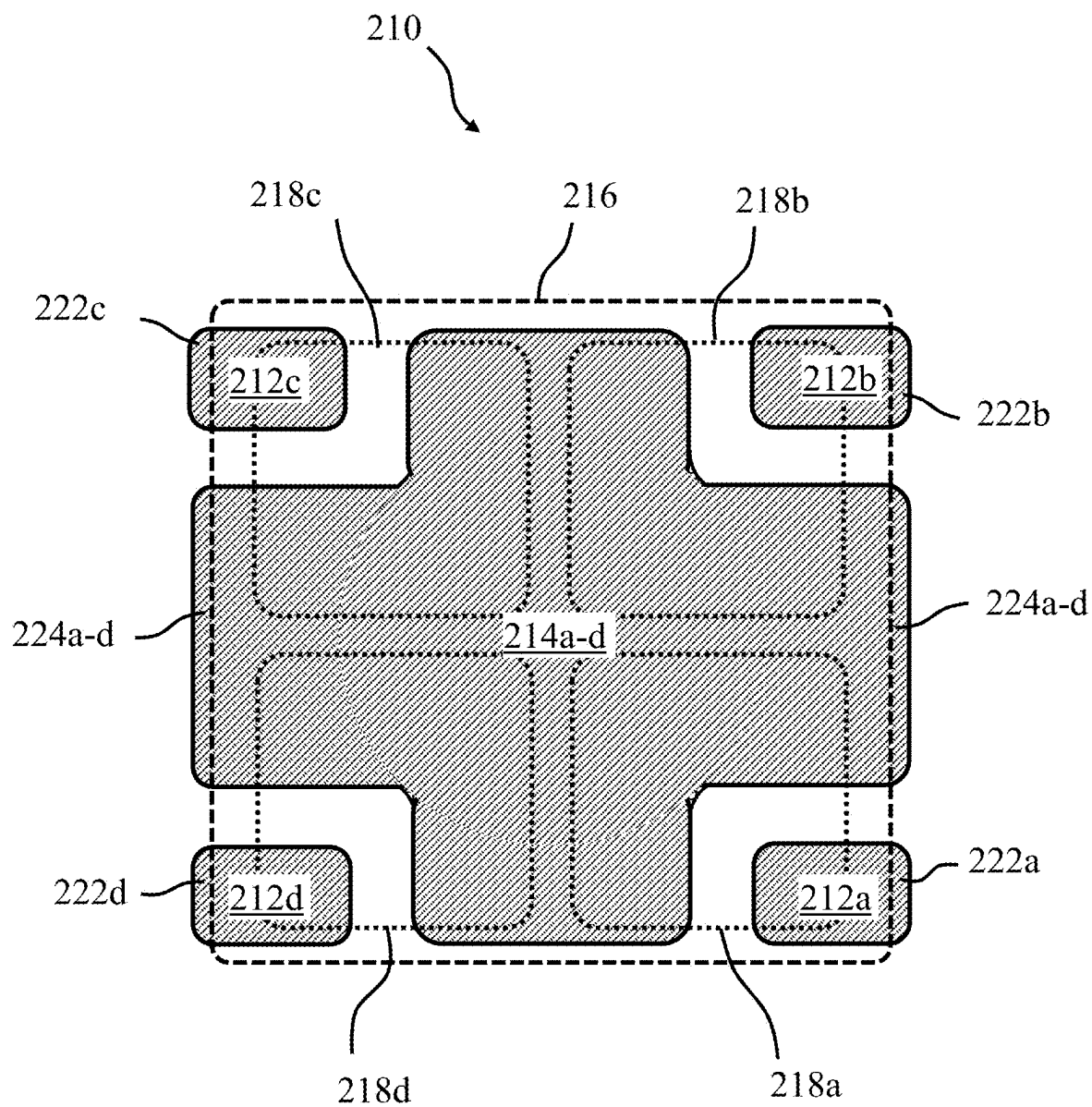


FIG. 2D

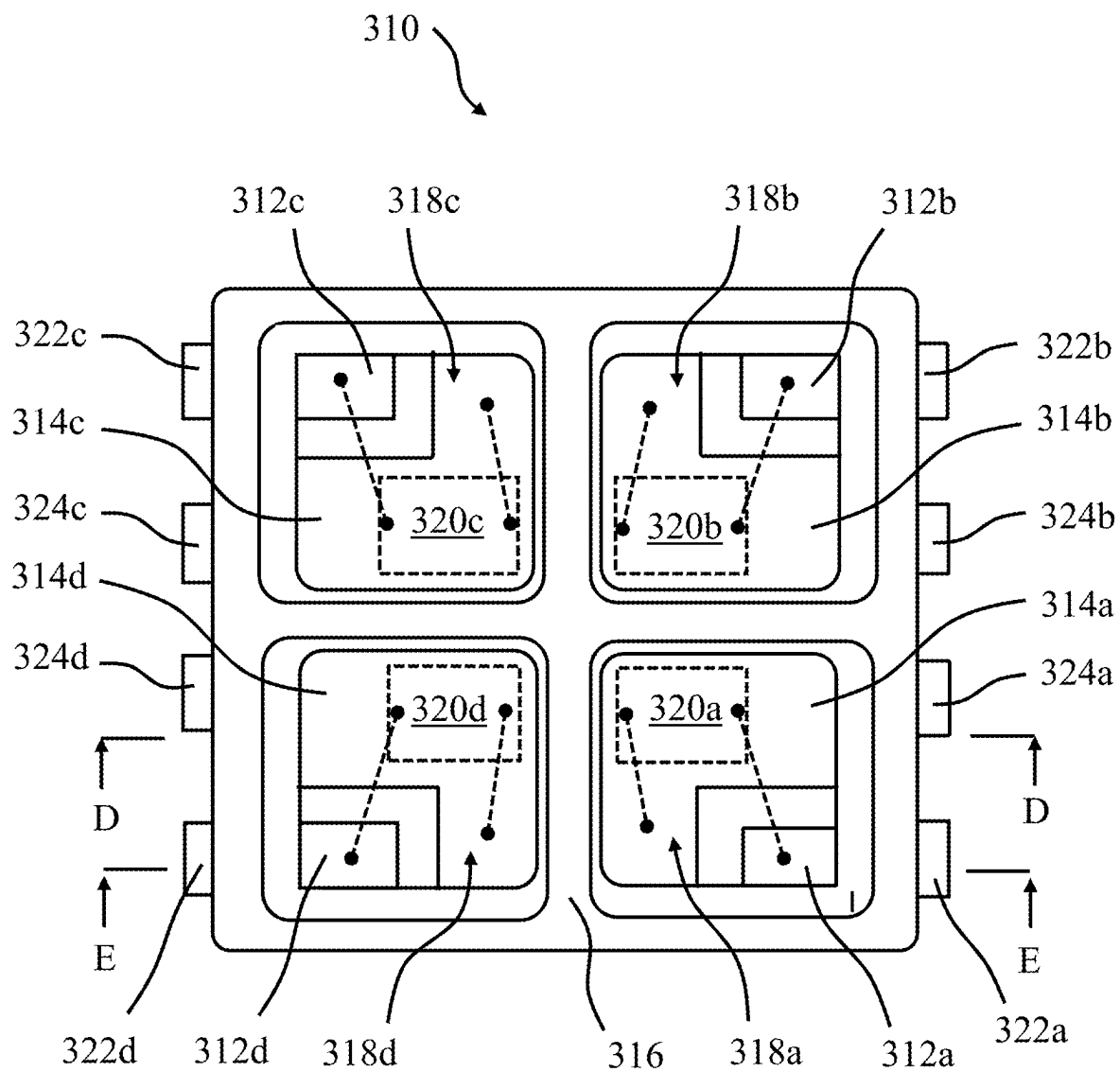
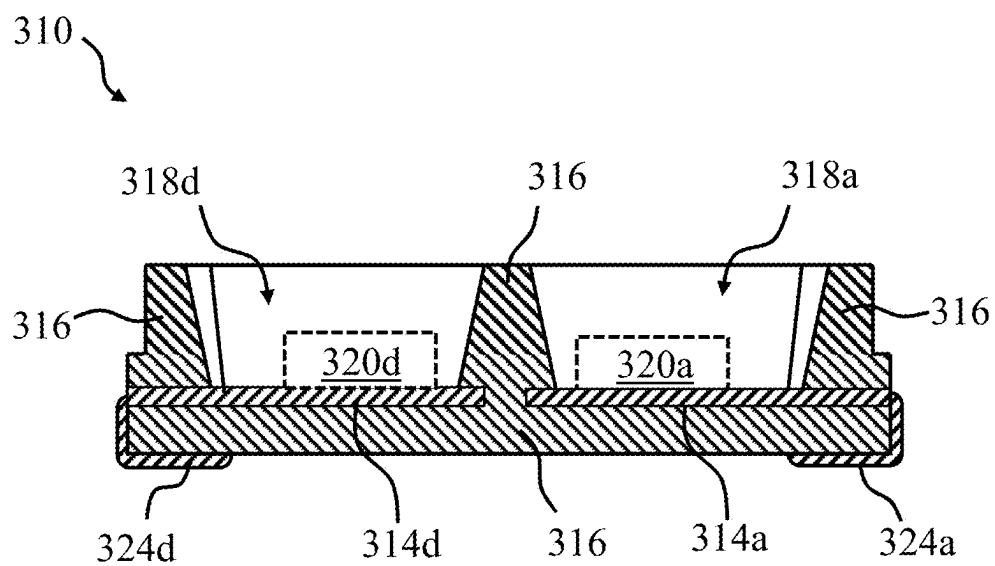
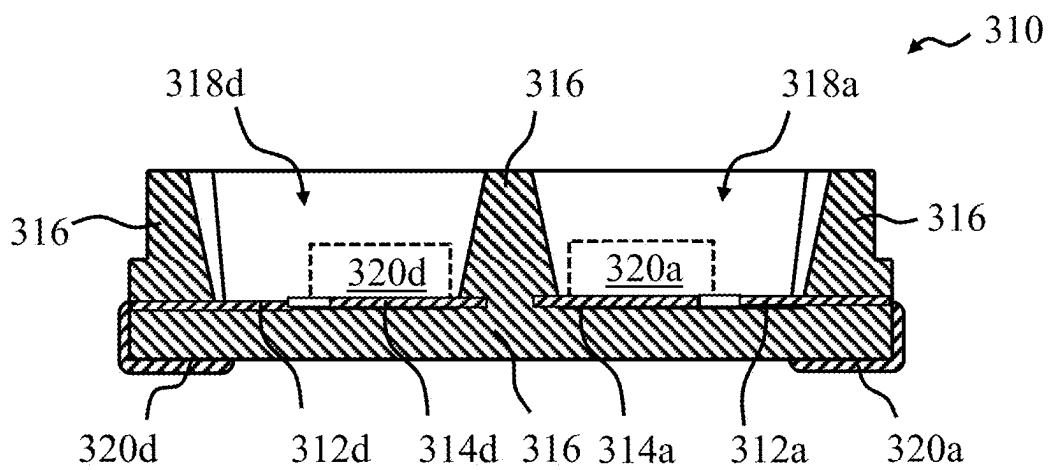


FIG. 3A



SECTION D-D

FIG. 3B



SECTION E-E

FIG. 3C

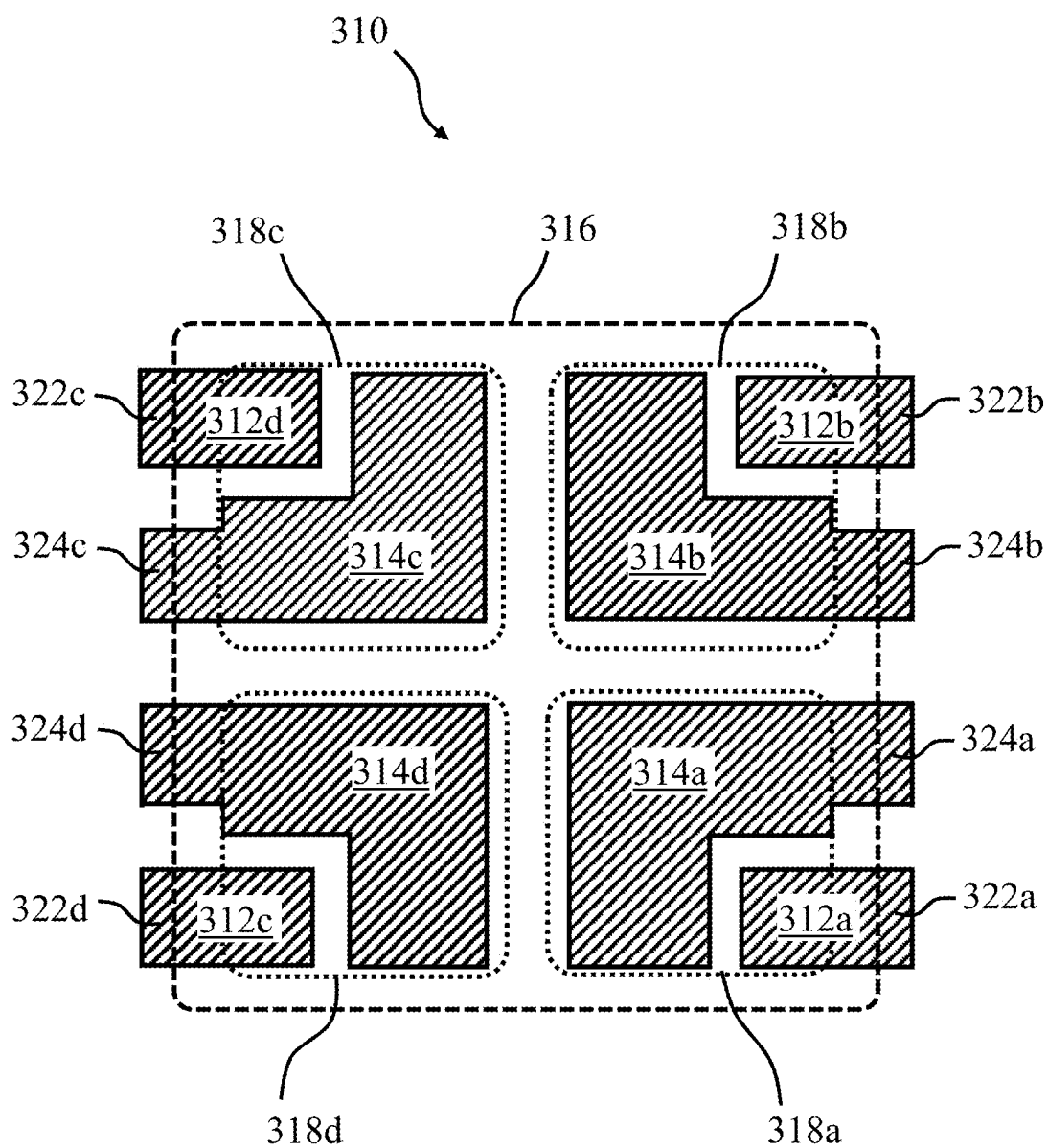


FIG. 3D

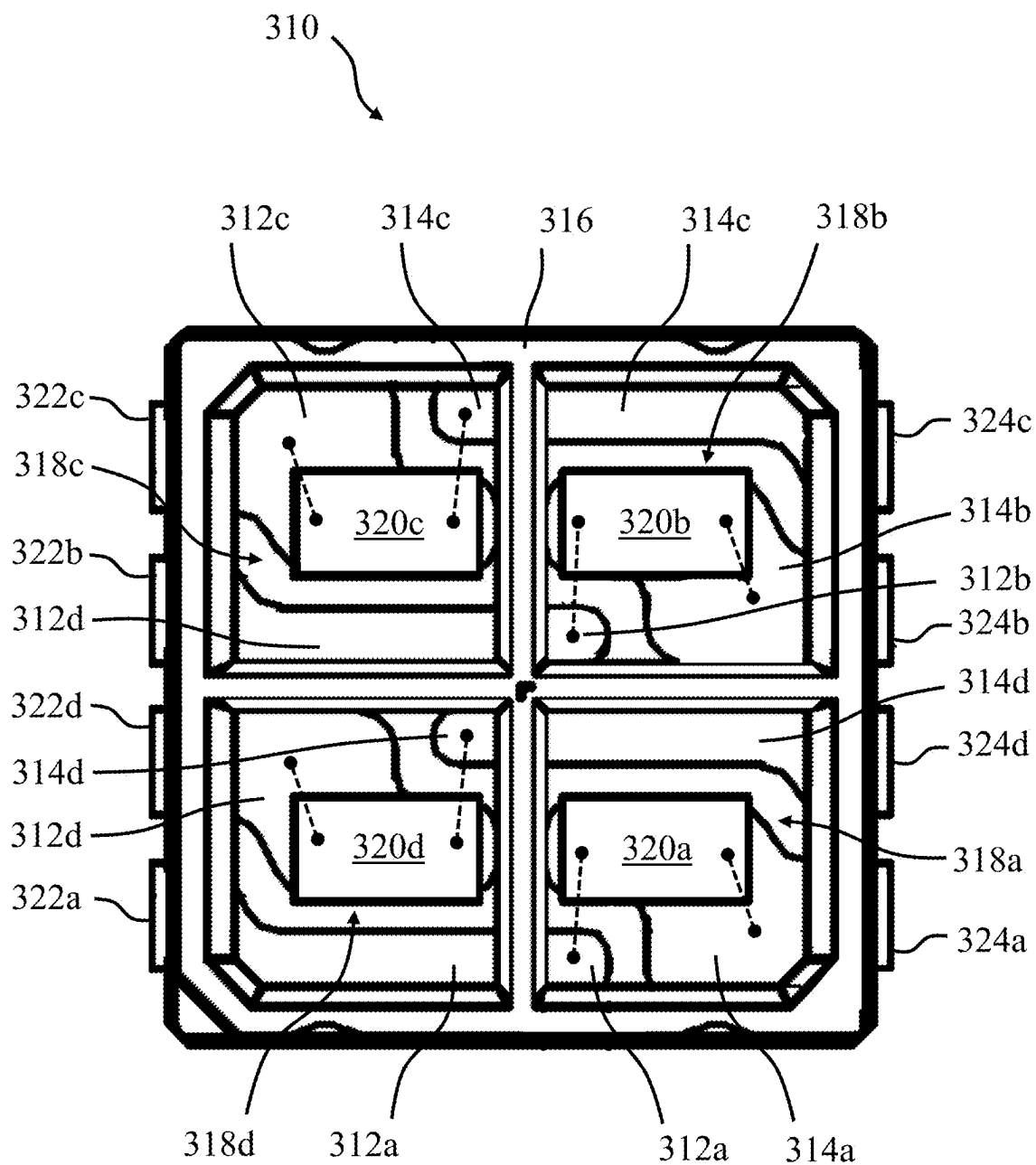


FIG. 3E

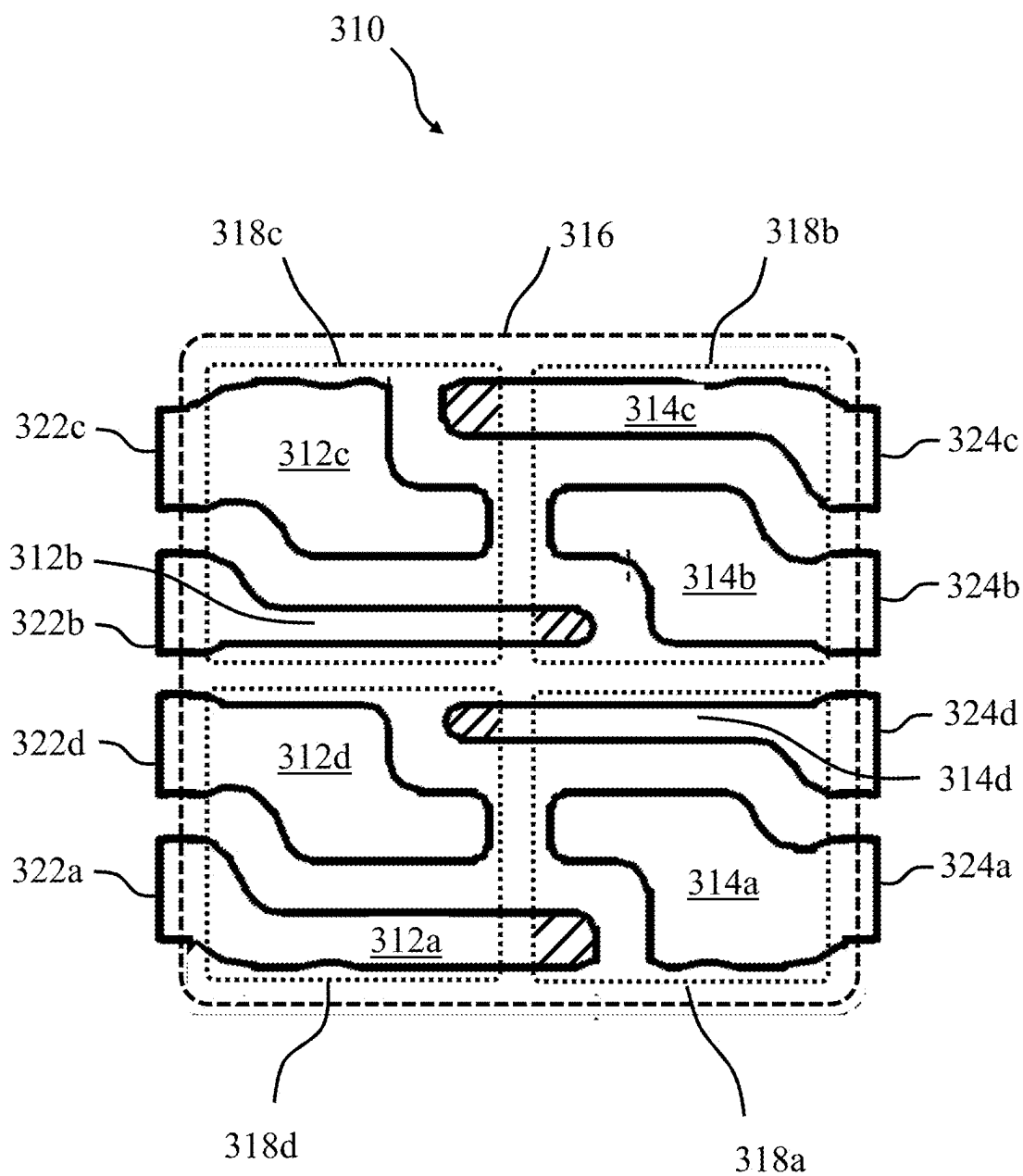


FIG. 3F

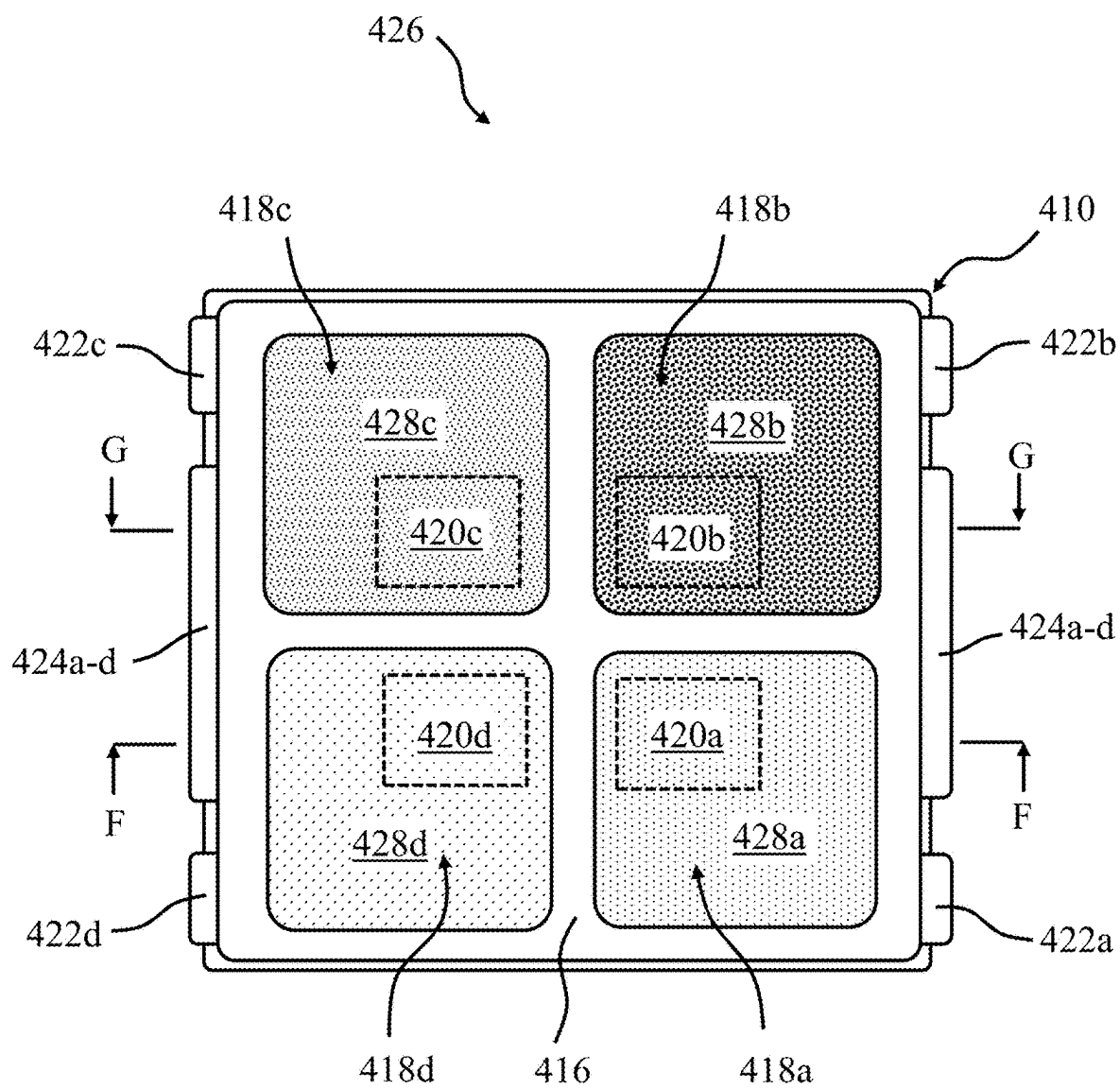
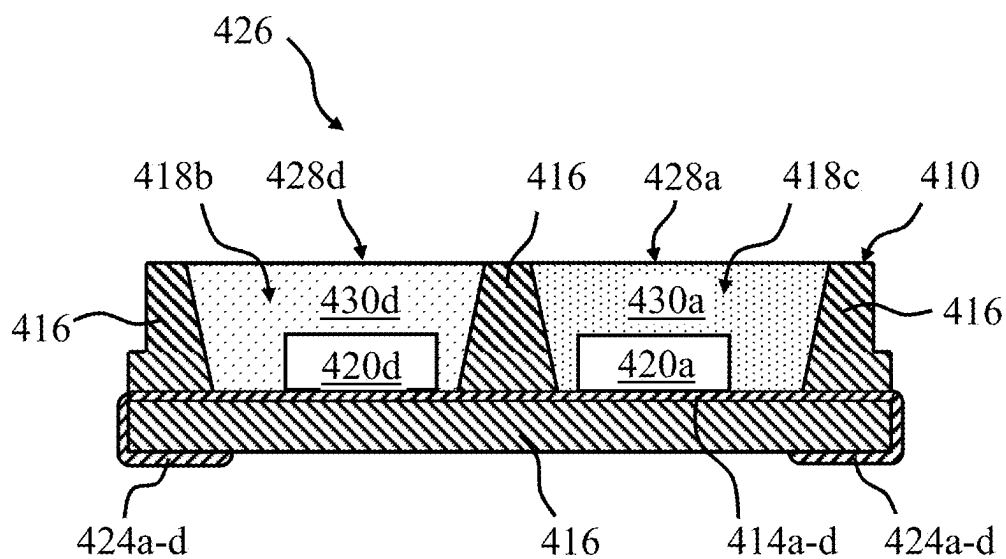
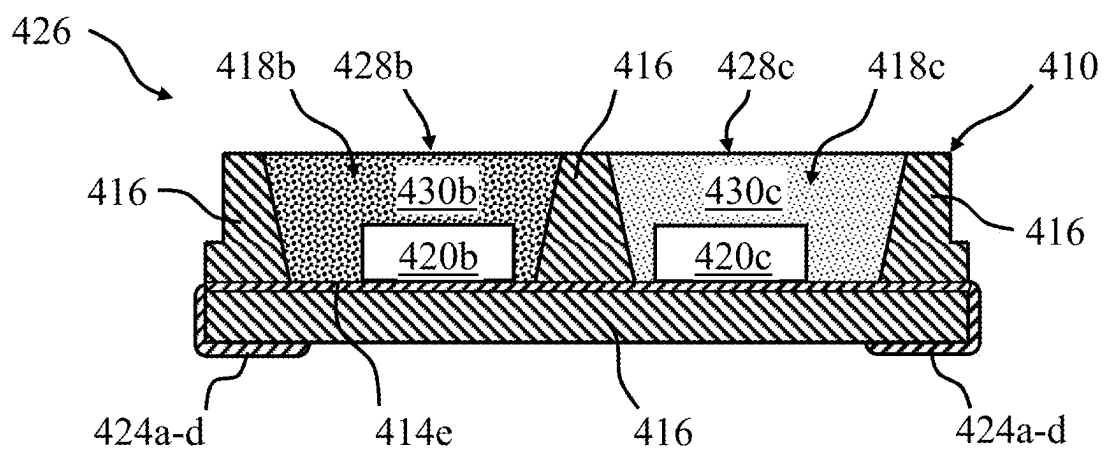


FIG. 4A



SECTION F-F

FIG. 4B



SECTION G-G

FIG. 4C

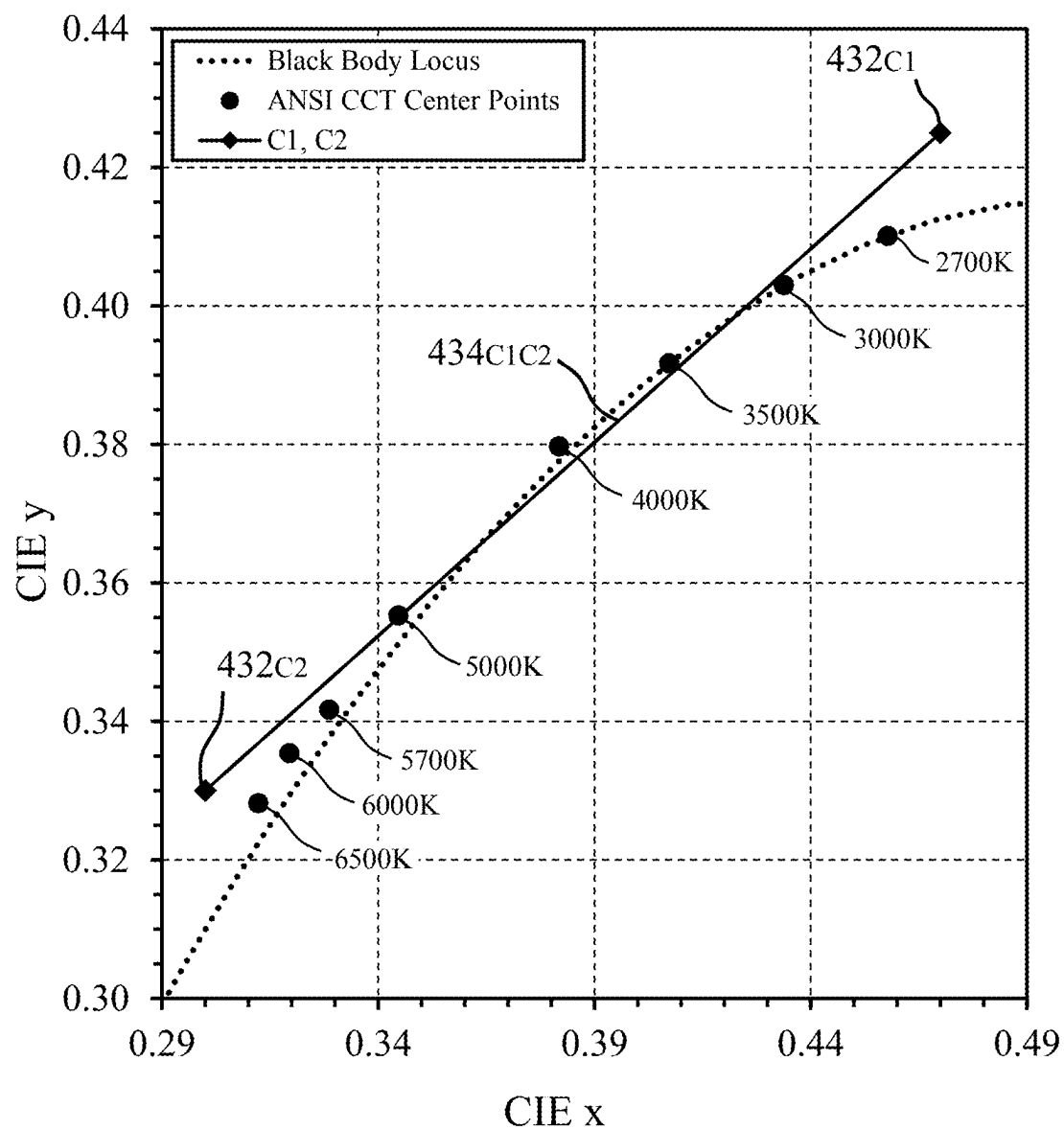


FIG. 4D

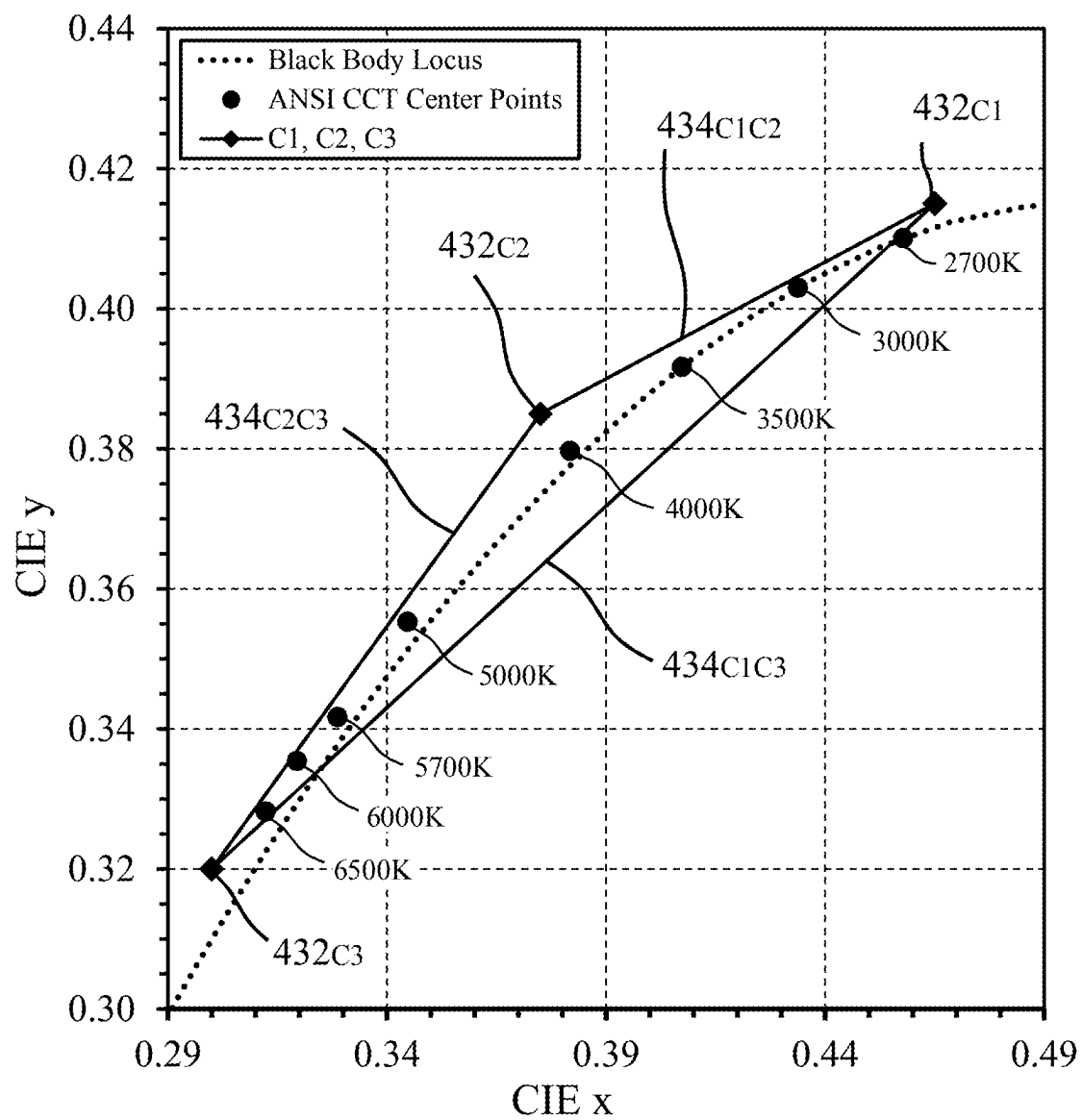


FIG. 4E

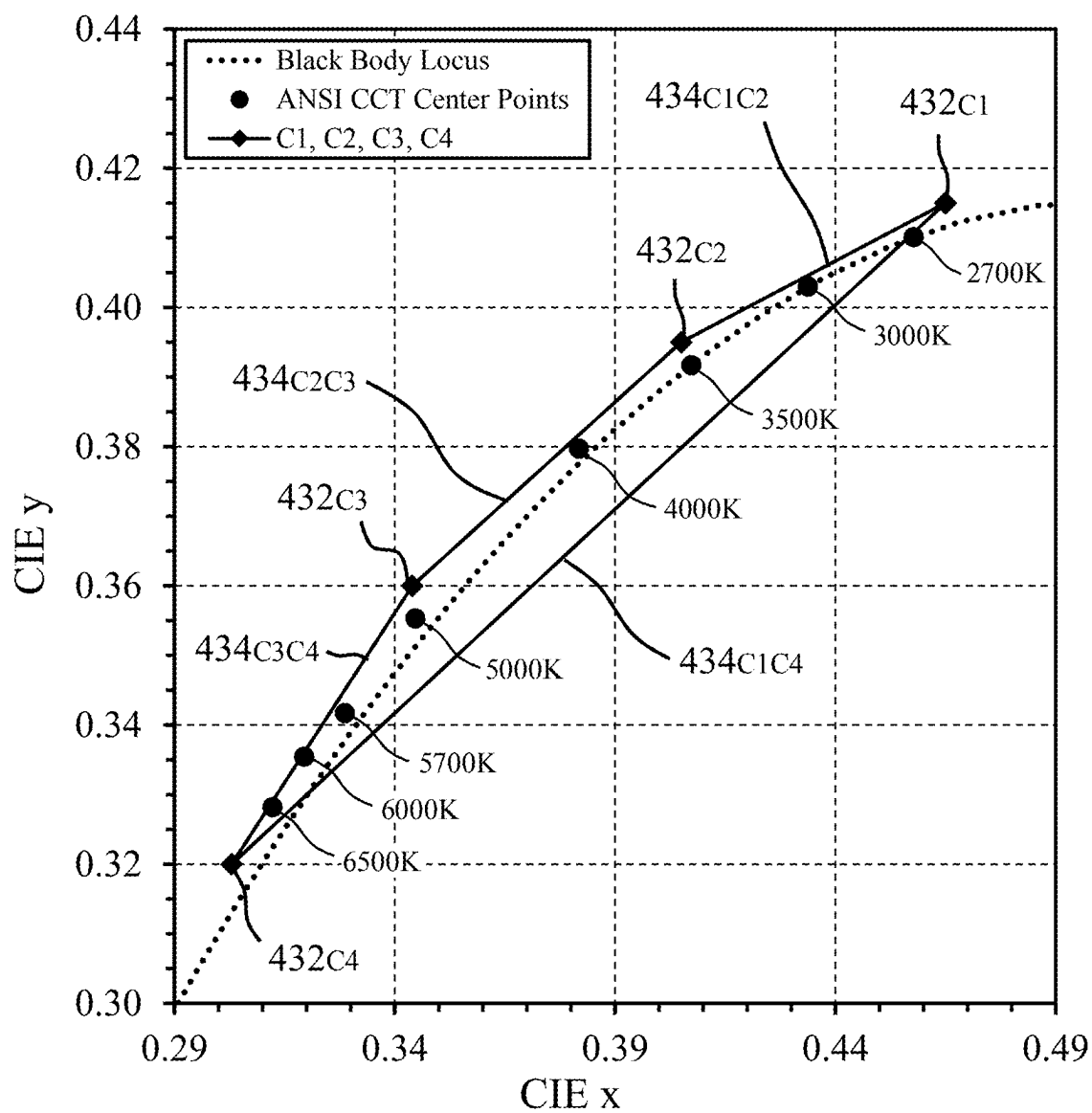


FIG. 4F

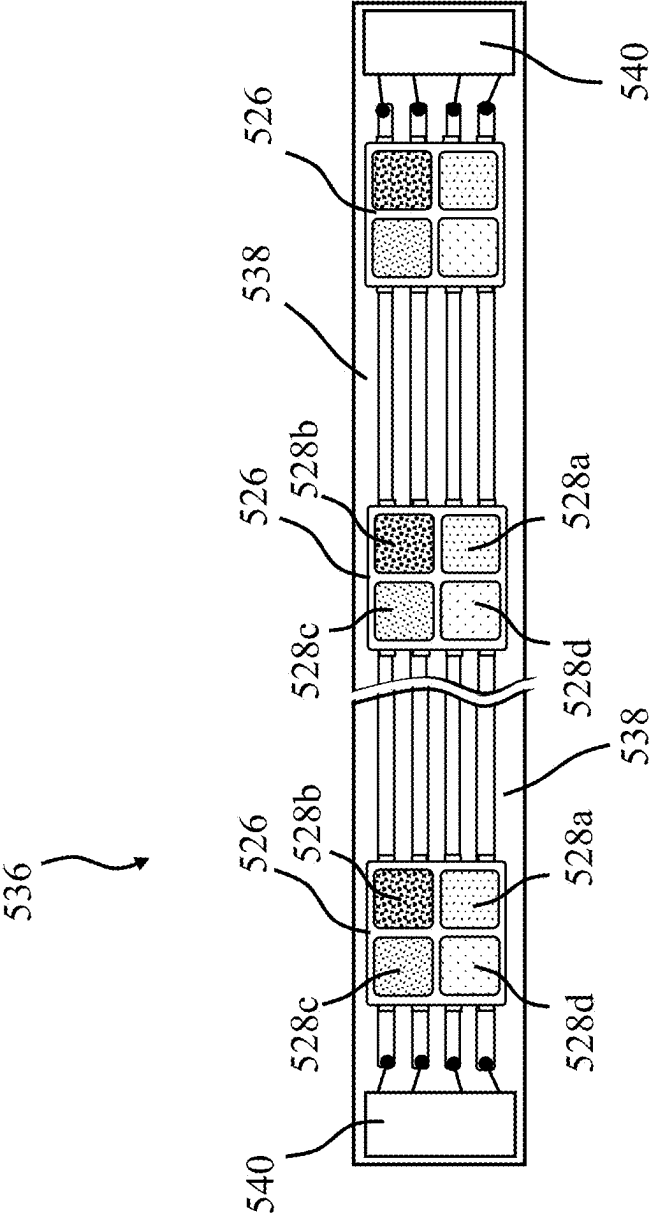


FIG. 5

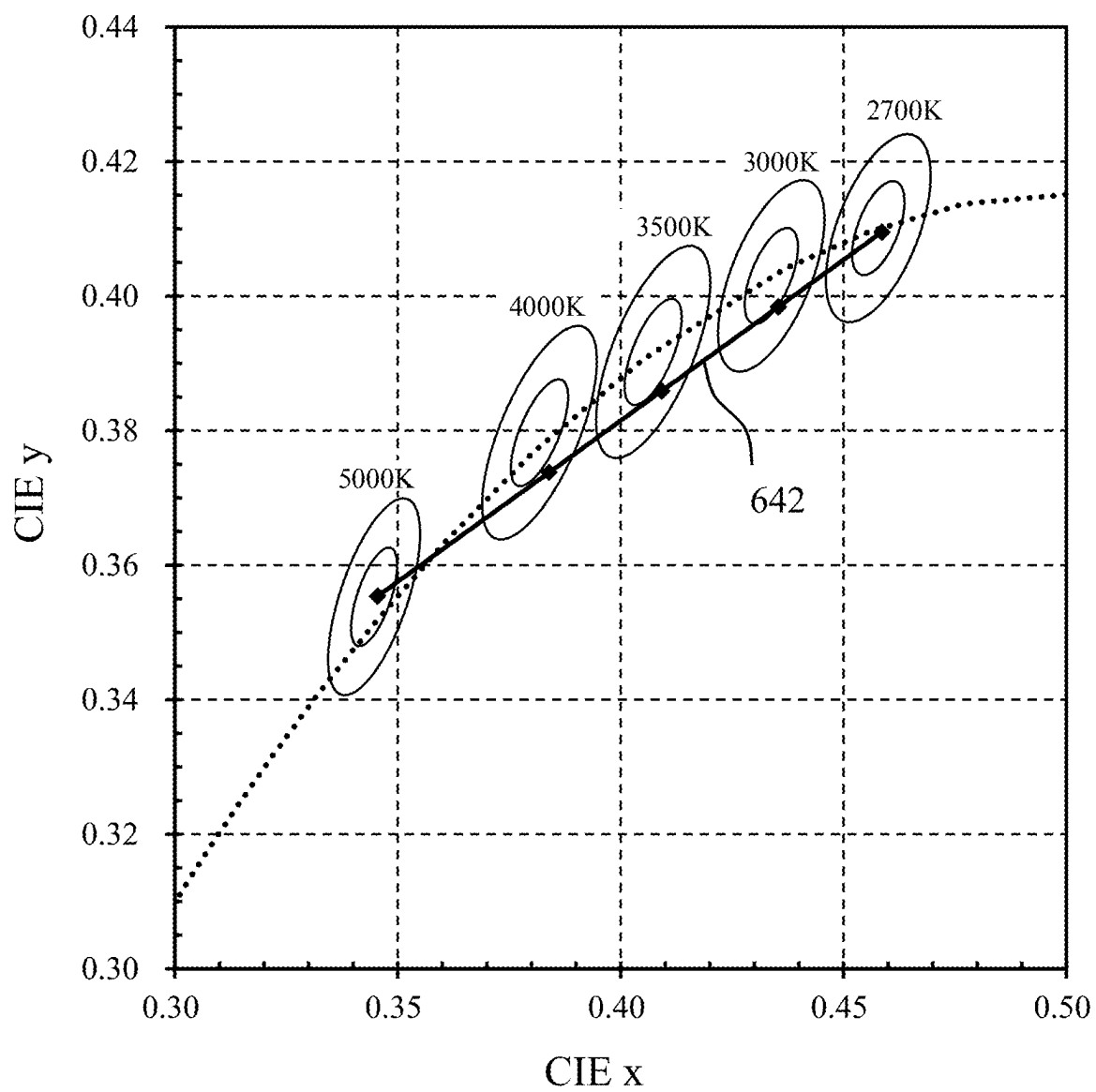


FIG. 6

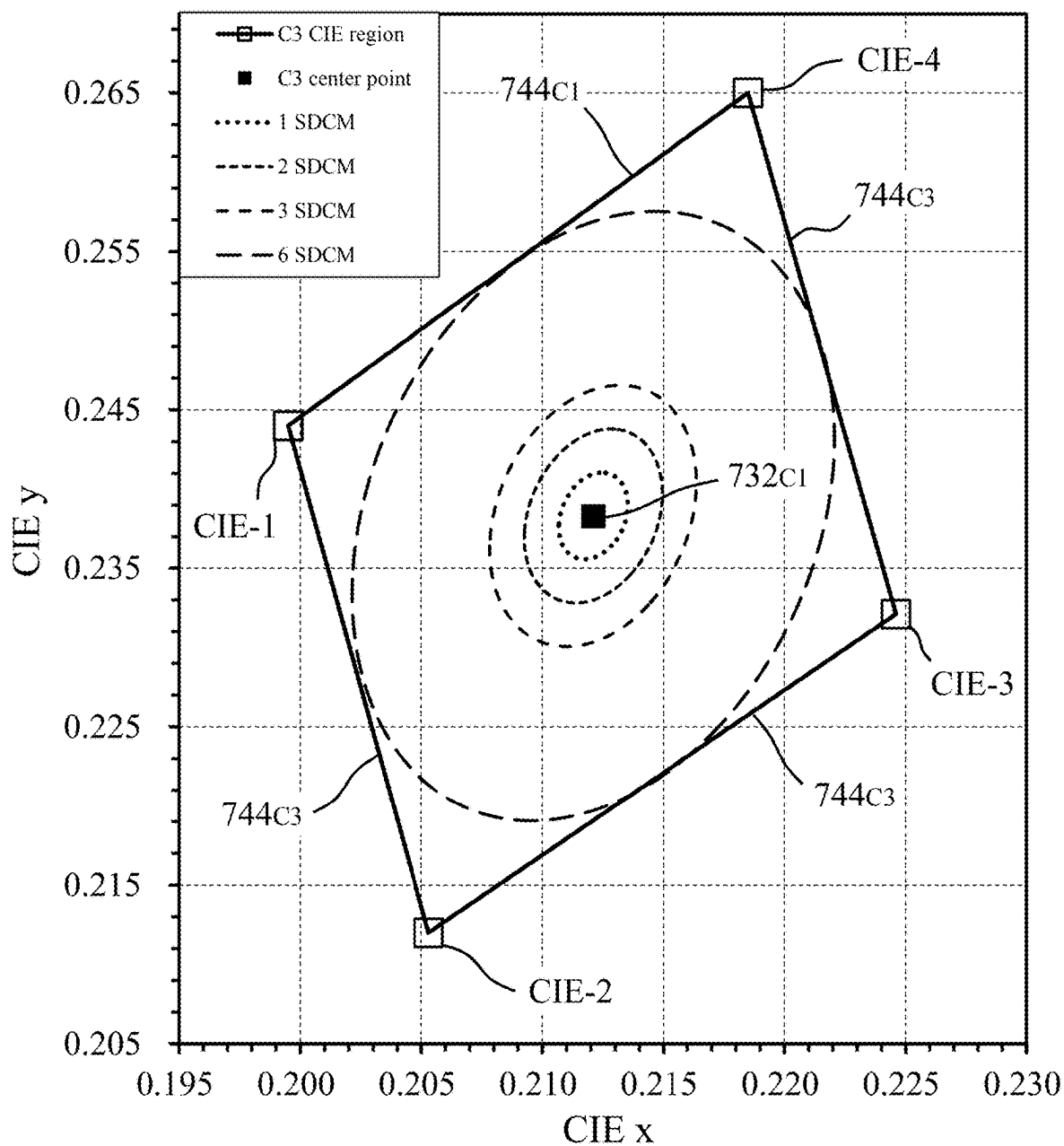


FIG. 7A

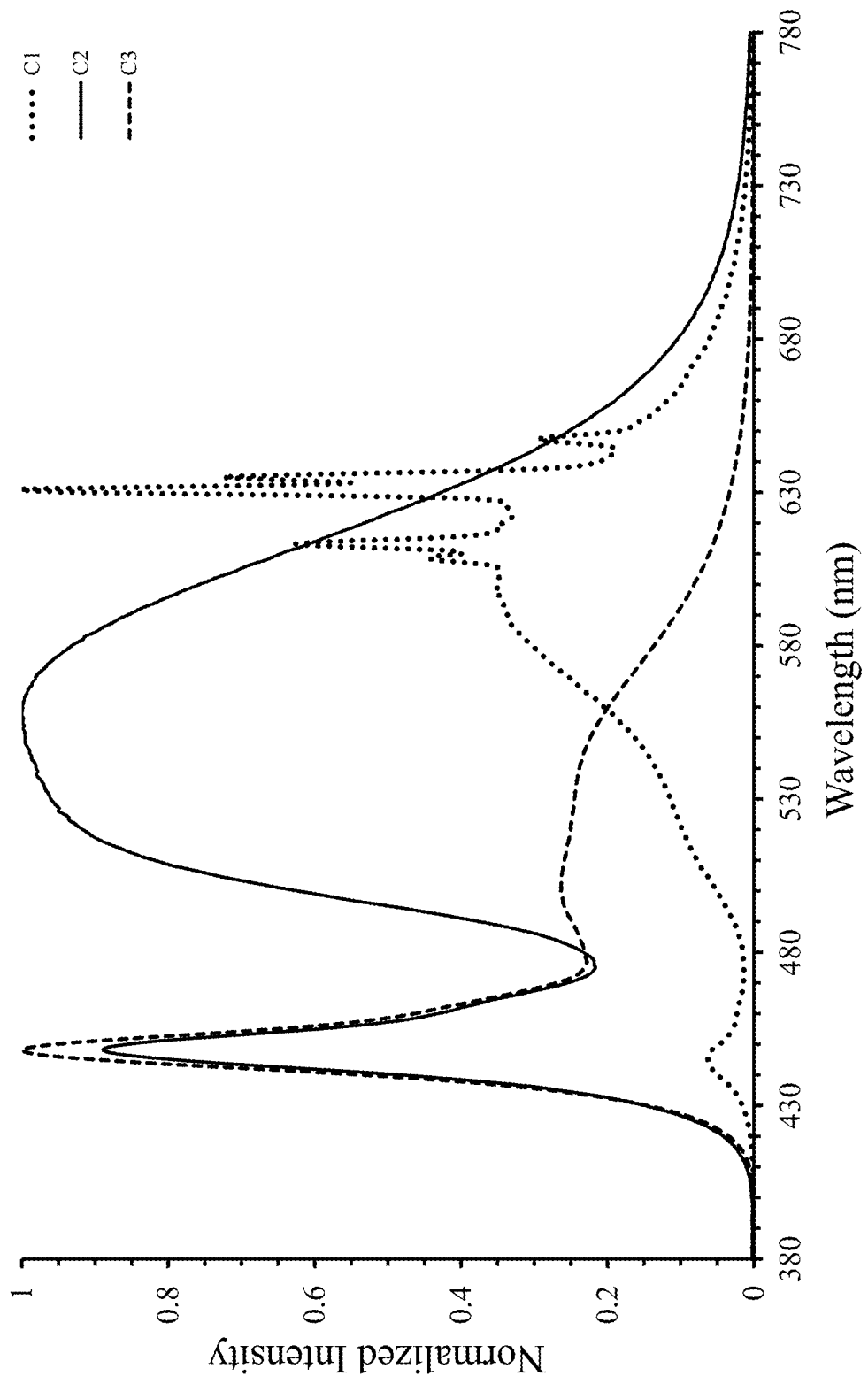


FIG. 7B

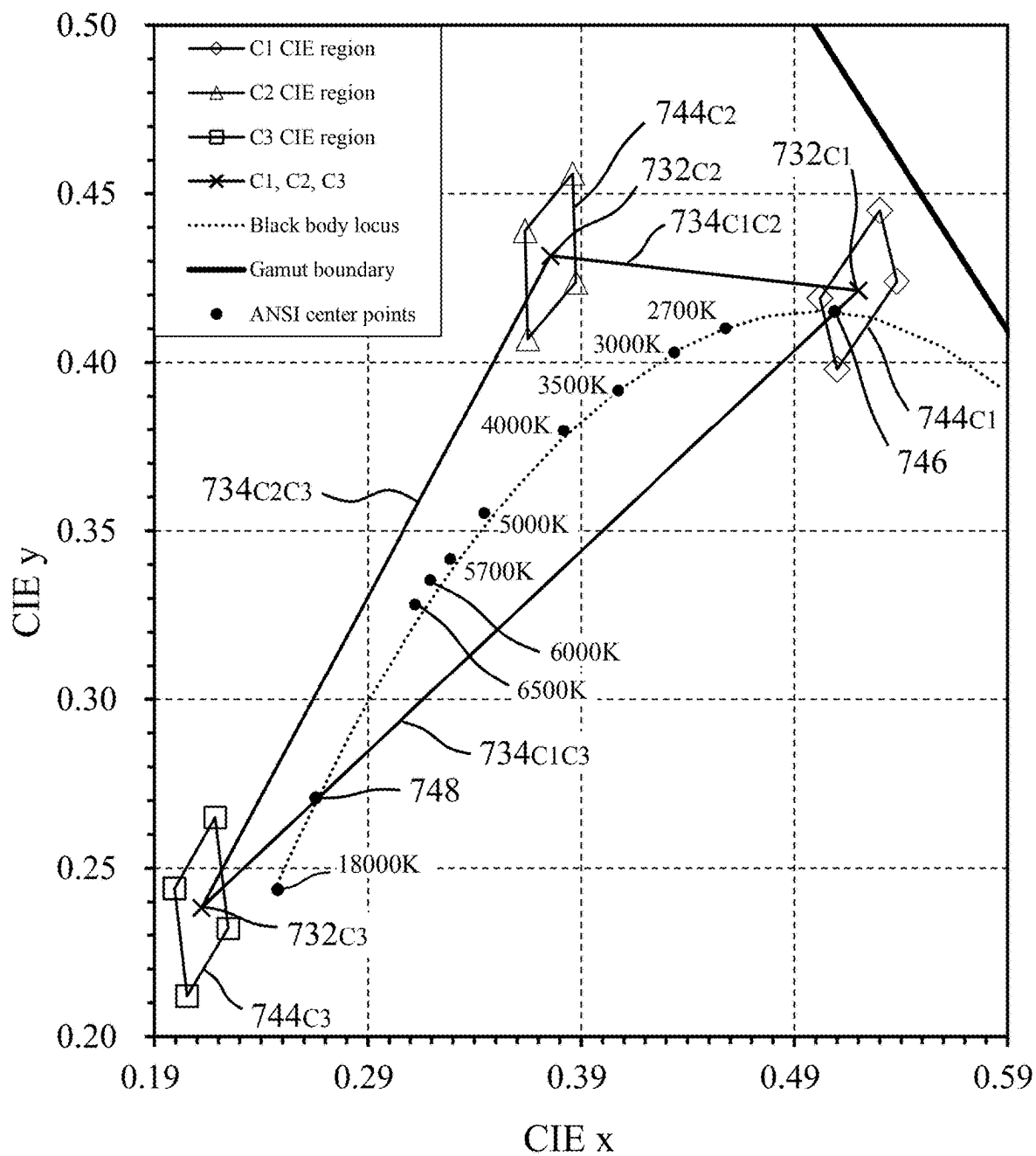


FIG. 7C

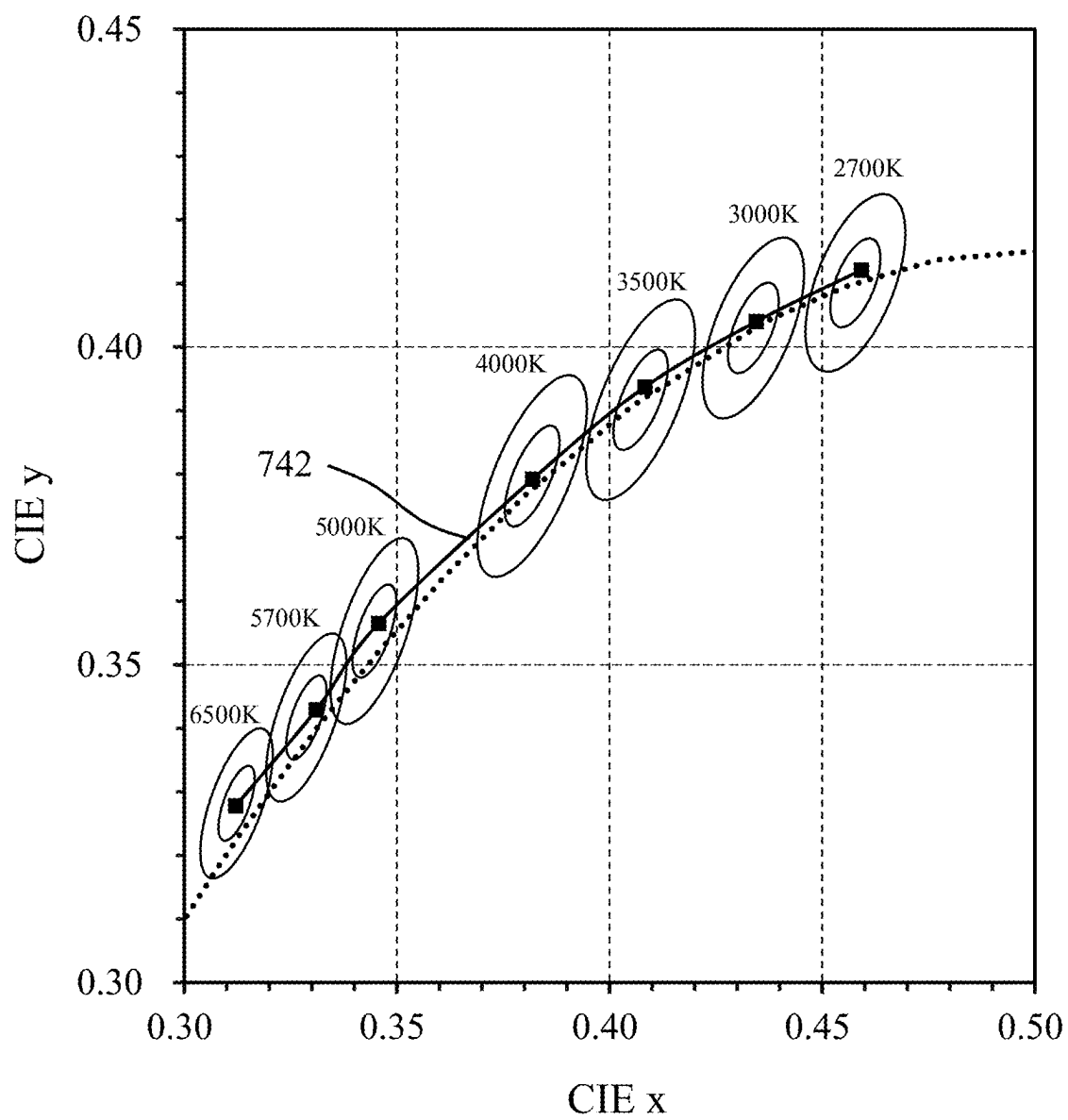


FIG. 7D

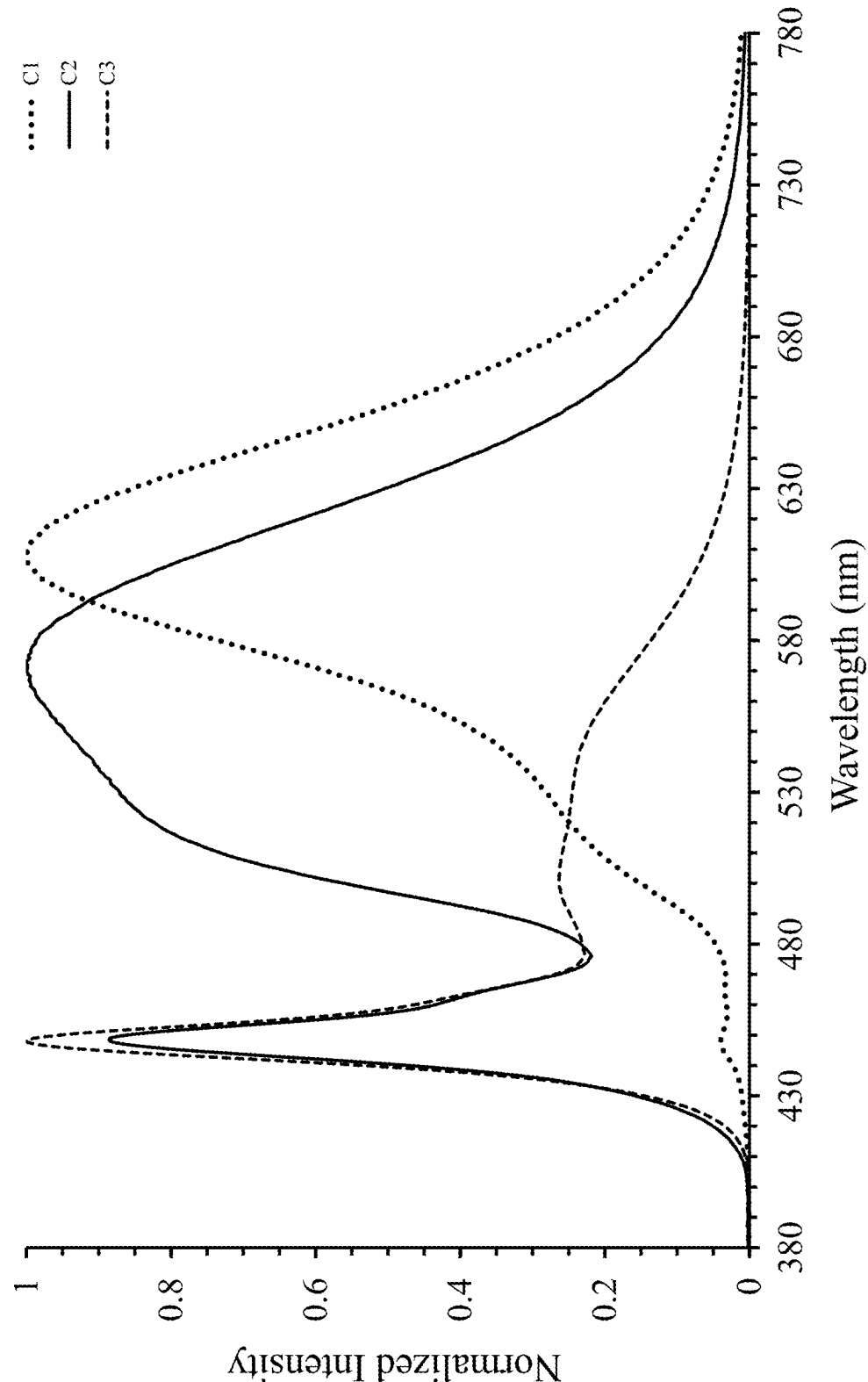


FIG. 8A

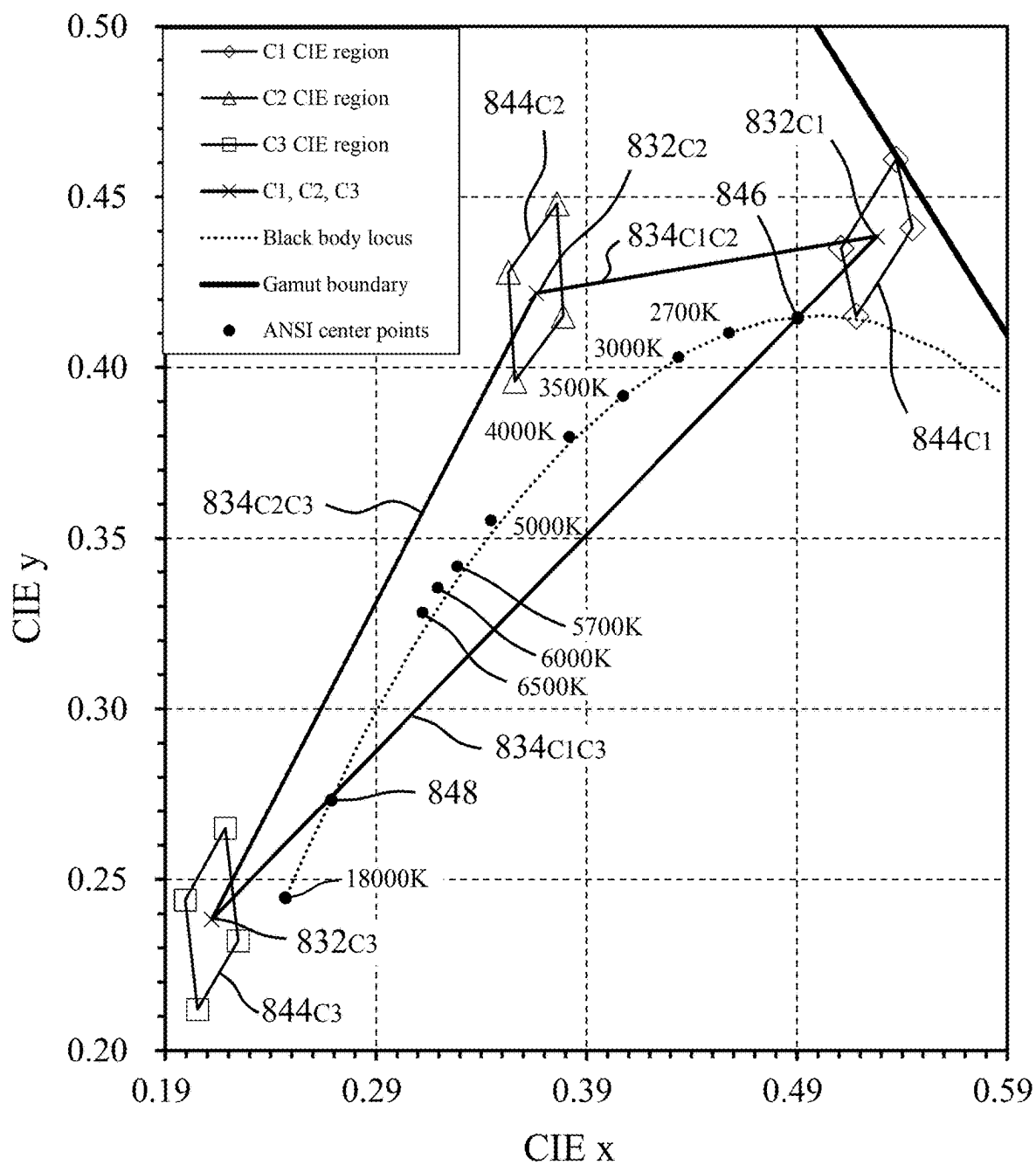


FIG. 8B

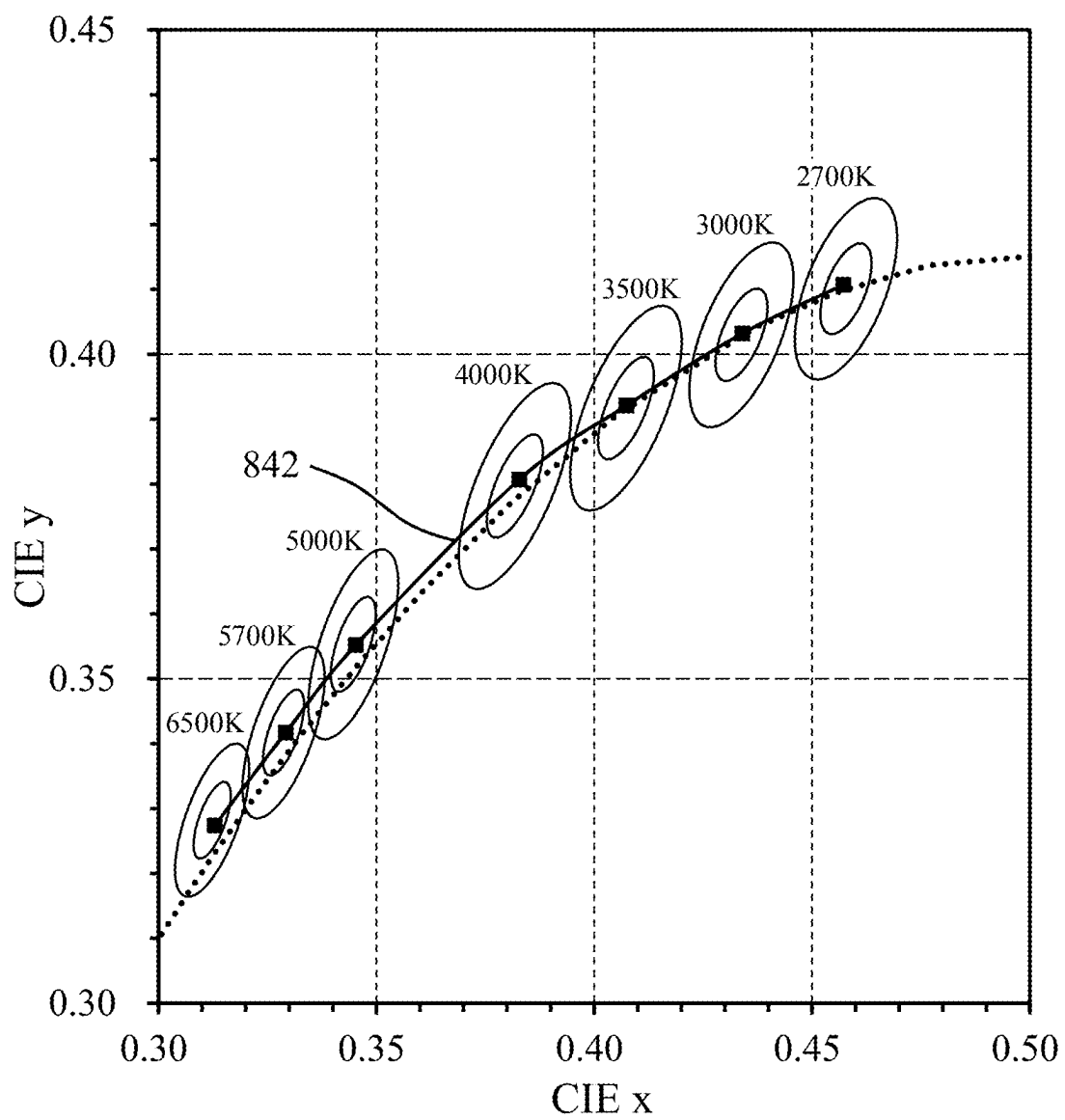


FIG. 8C

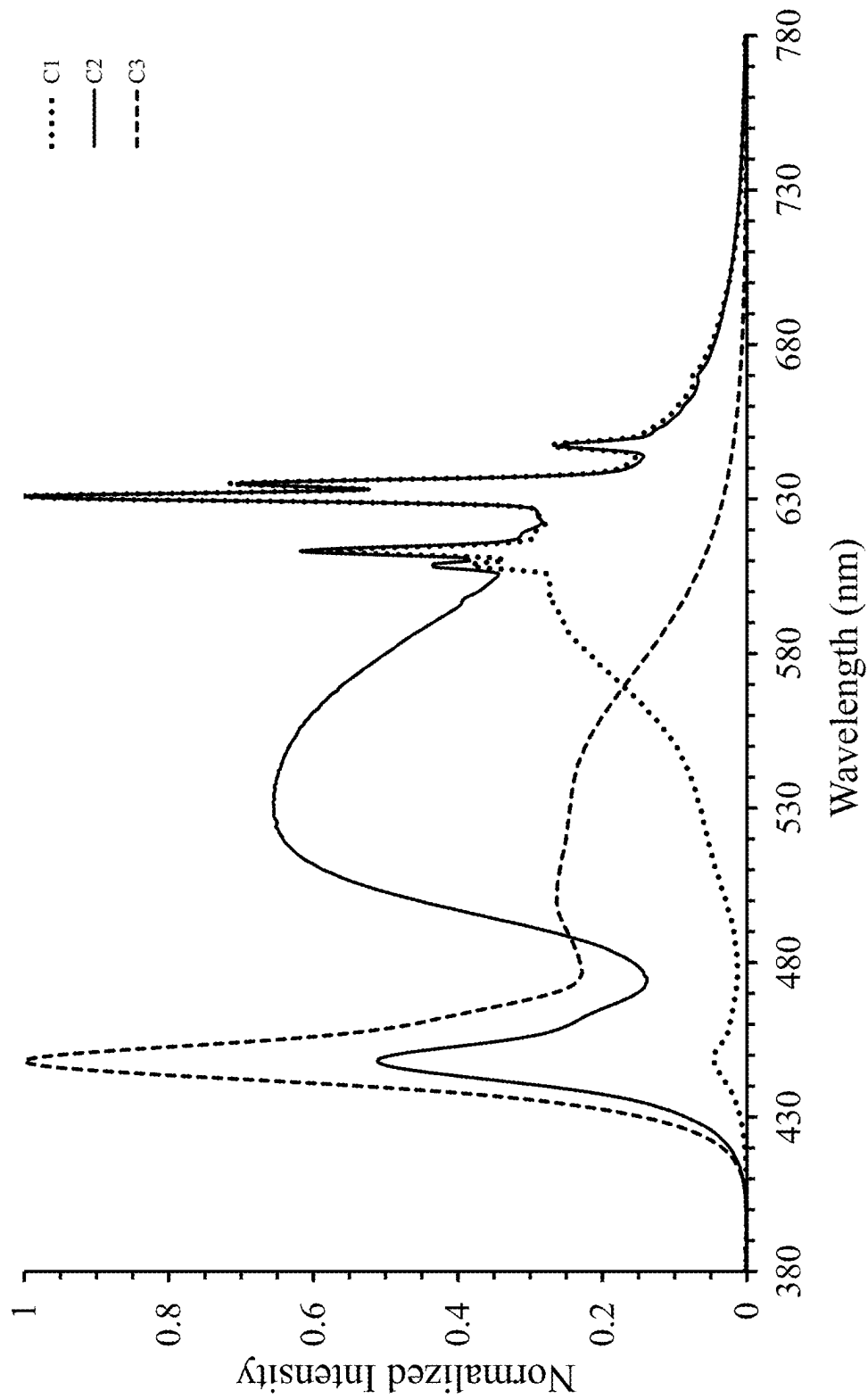


FIG. 9A

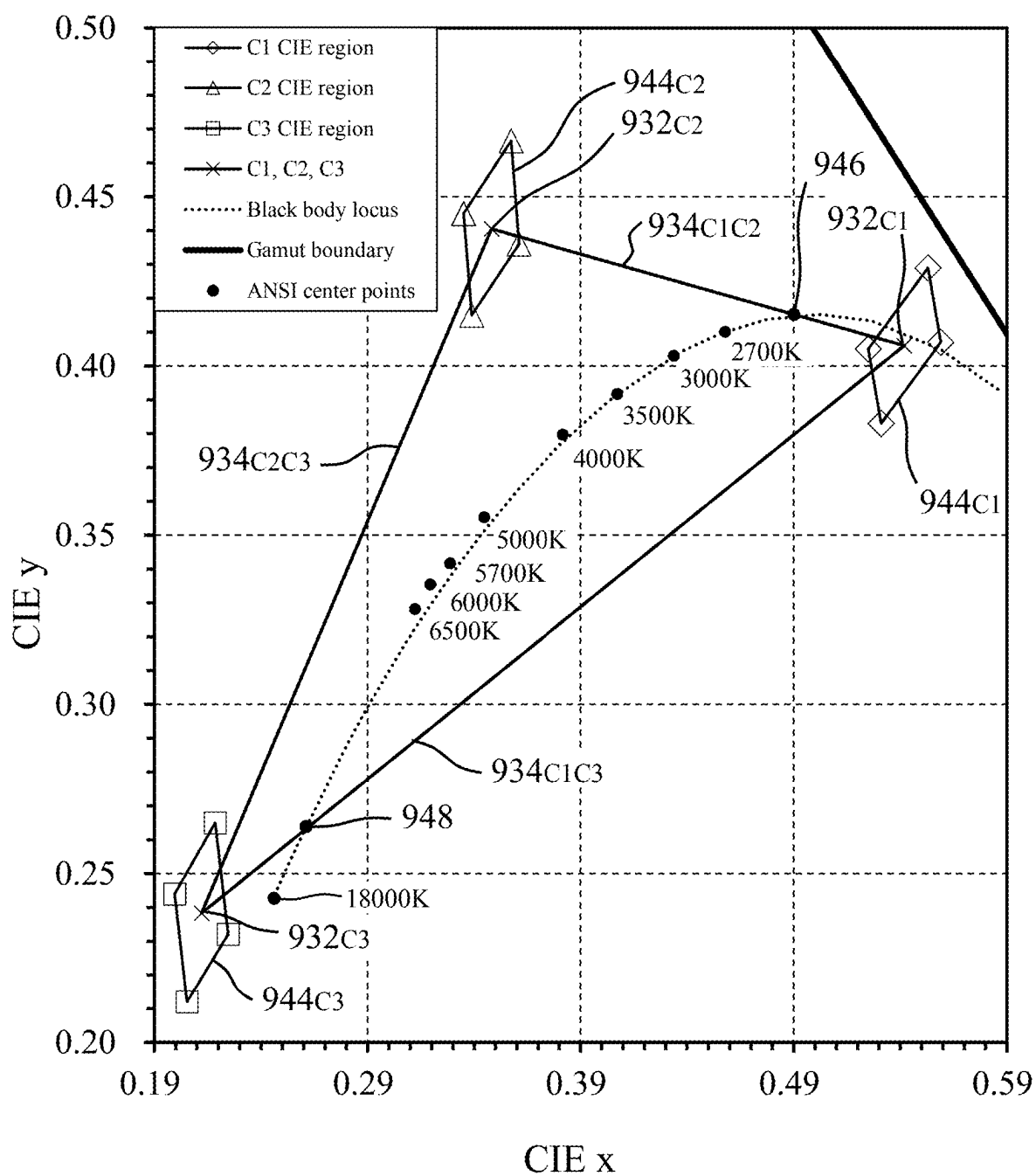


FIG. 9B

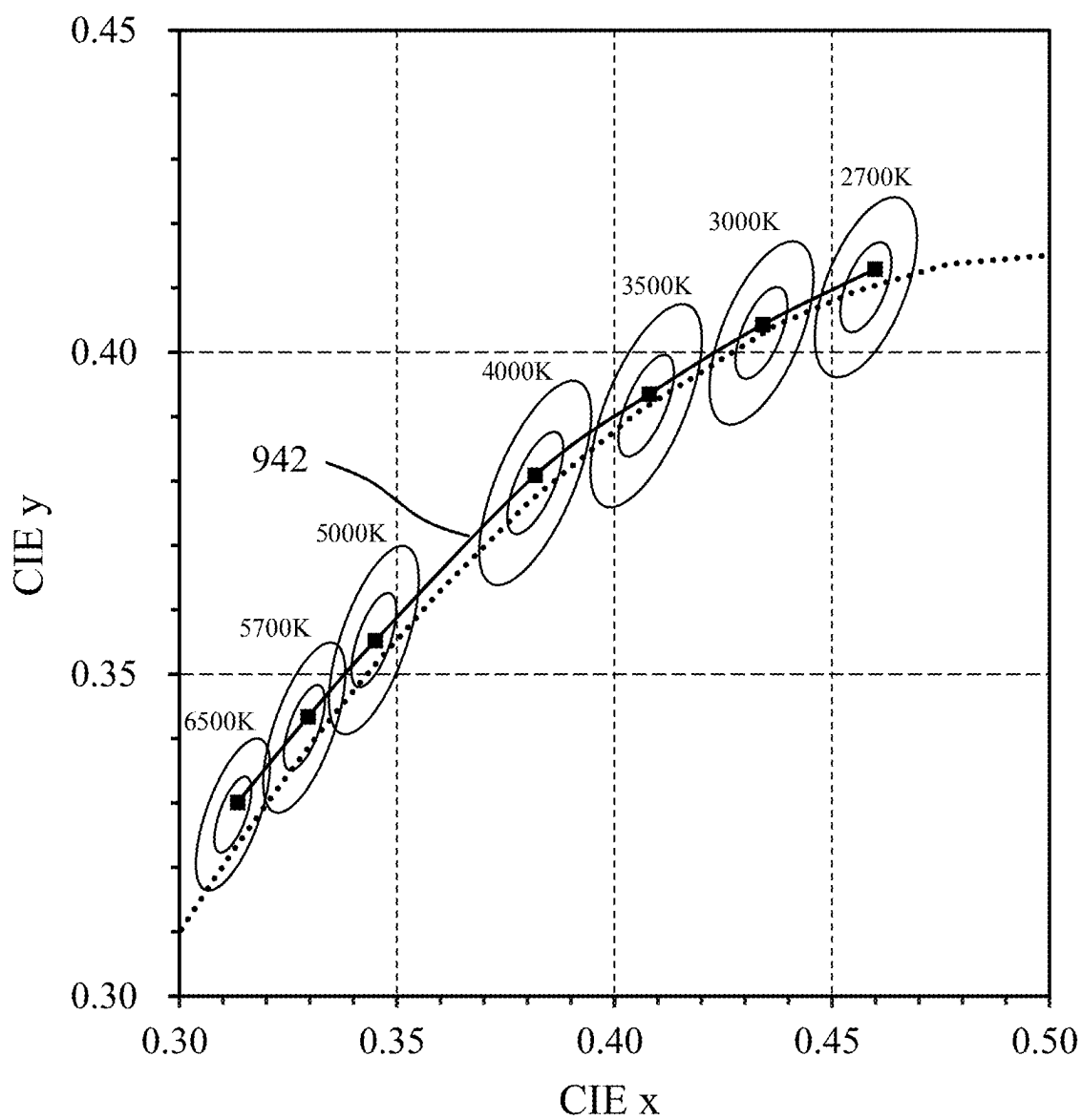


FIG. 9C

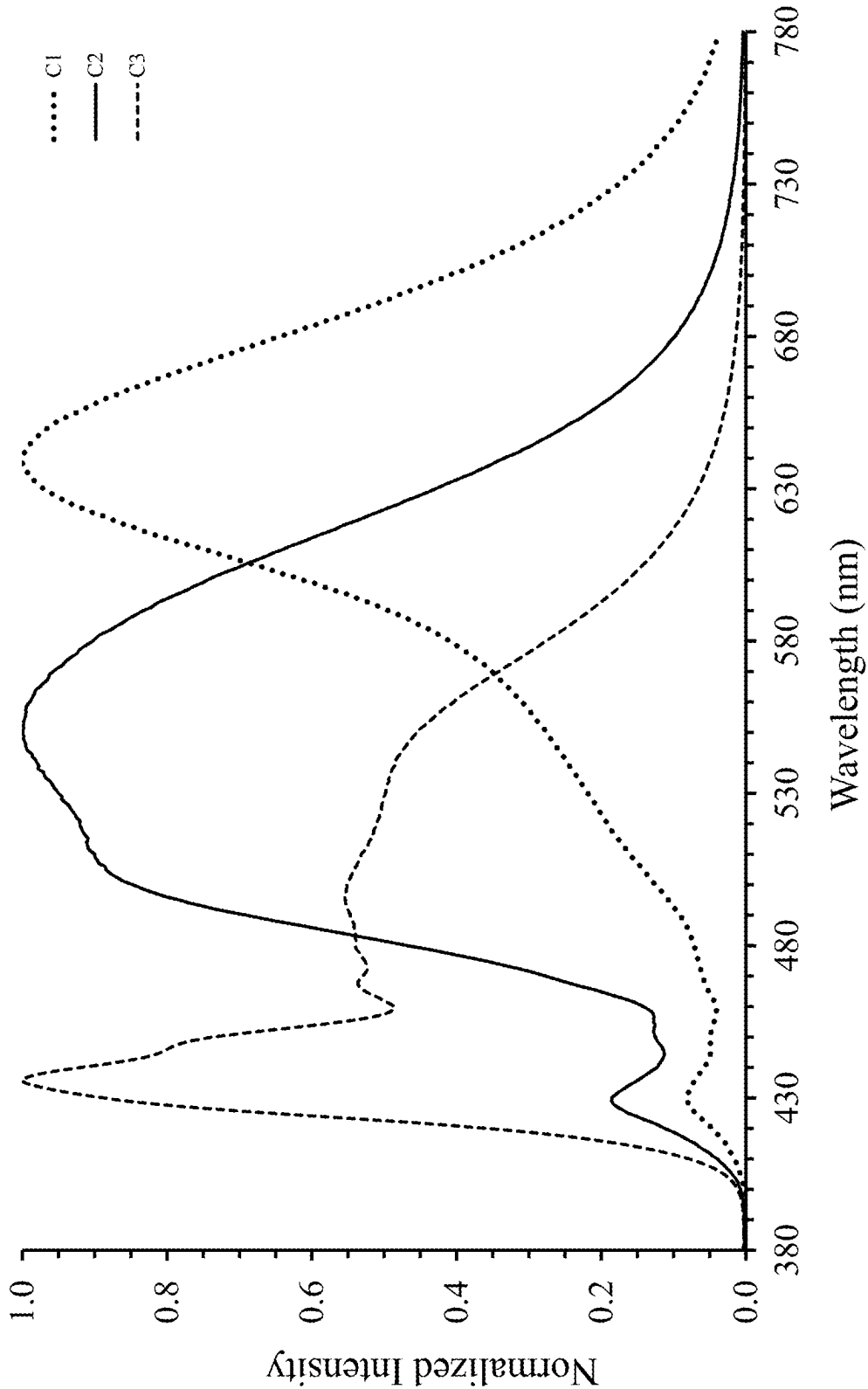


FIG. 10A

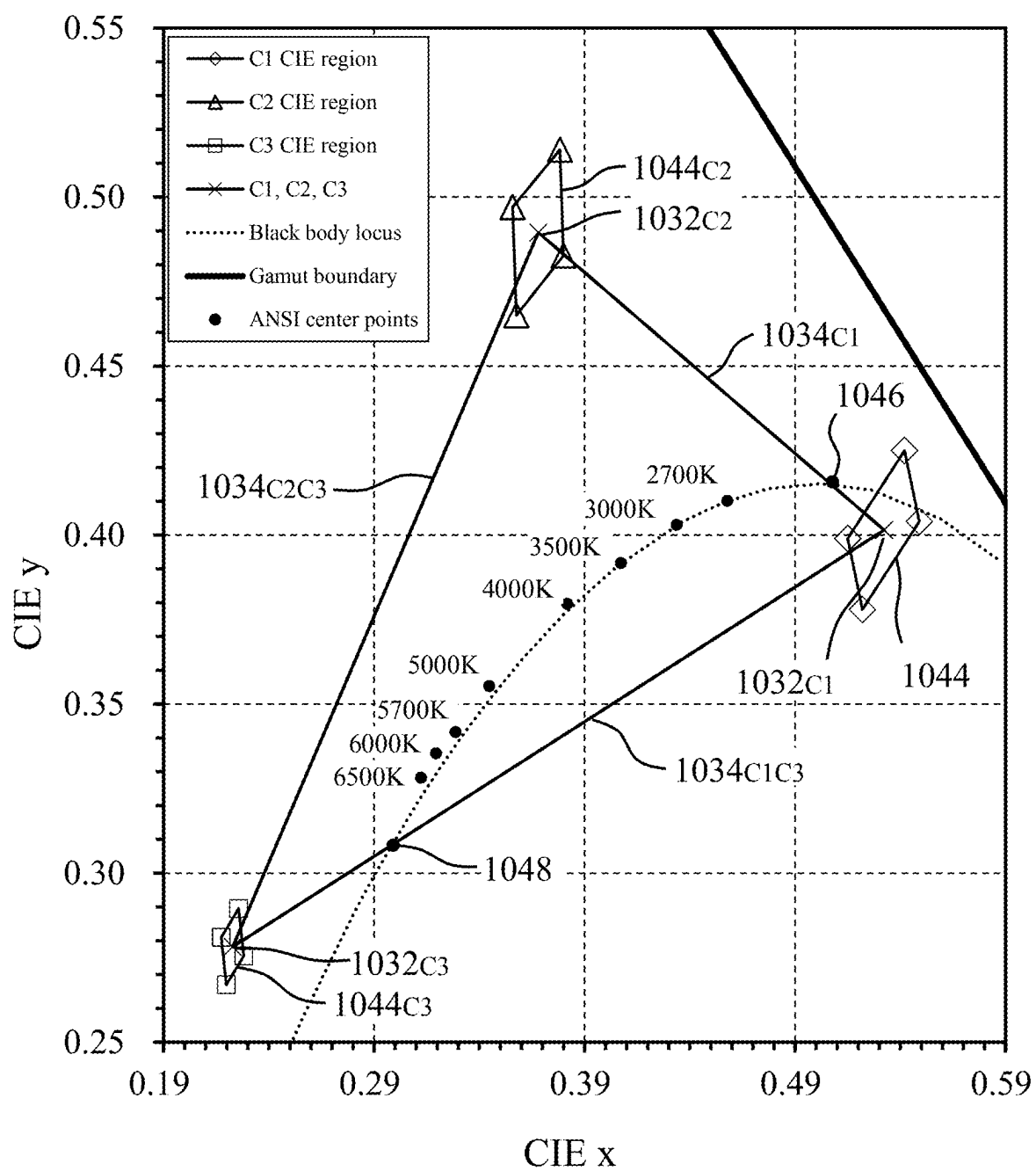


FIG. 10B

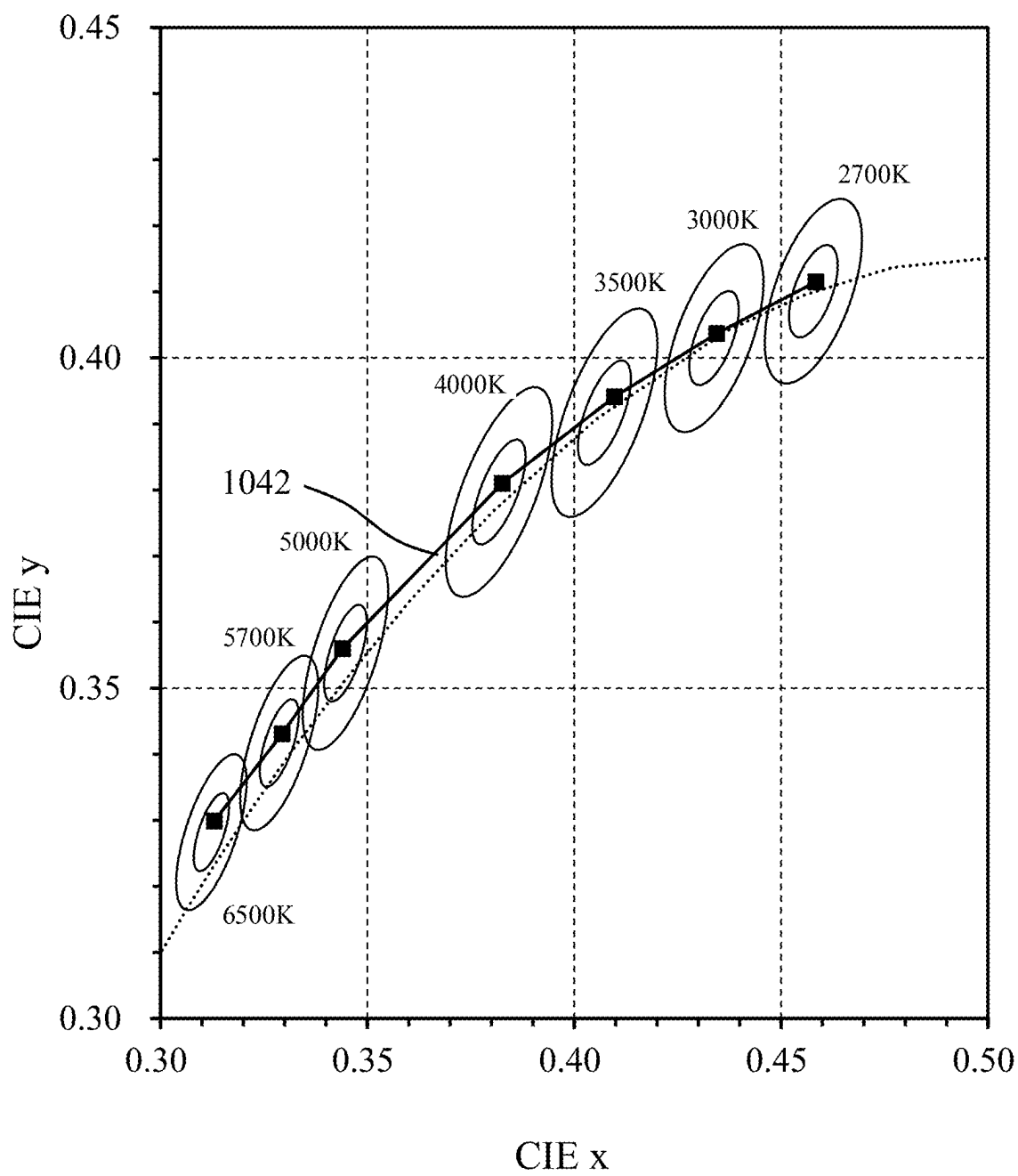


FIG. 10C

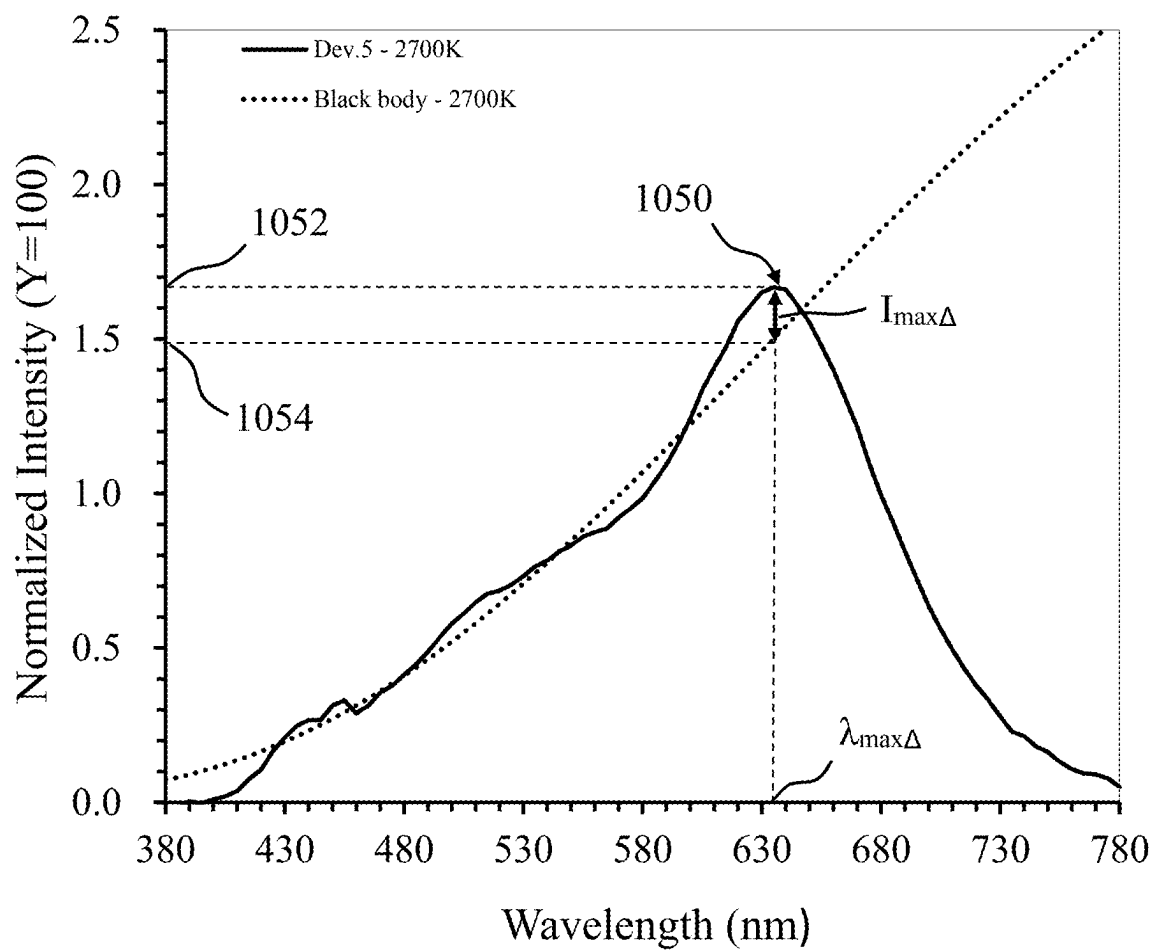


FIG. 10D

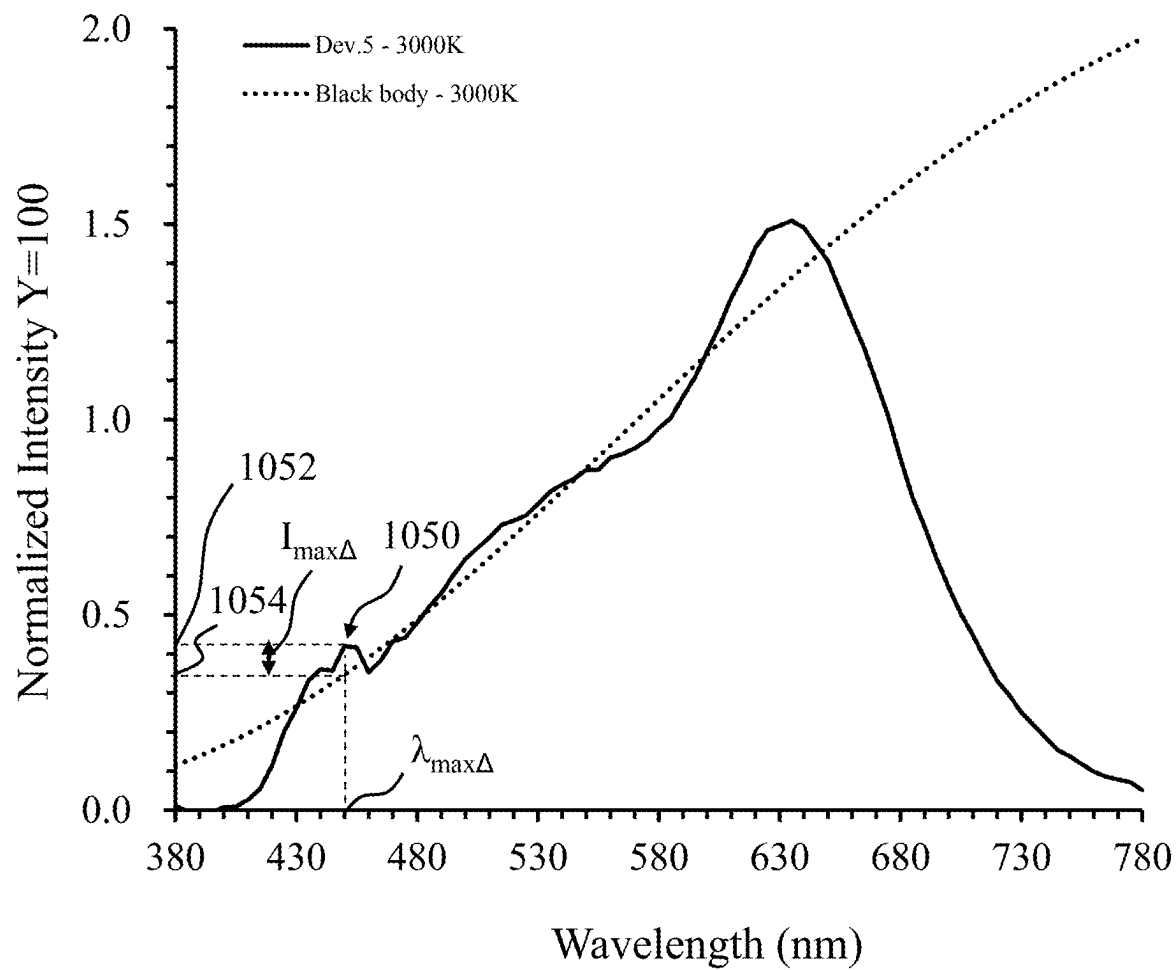


FIG. 10E

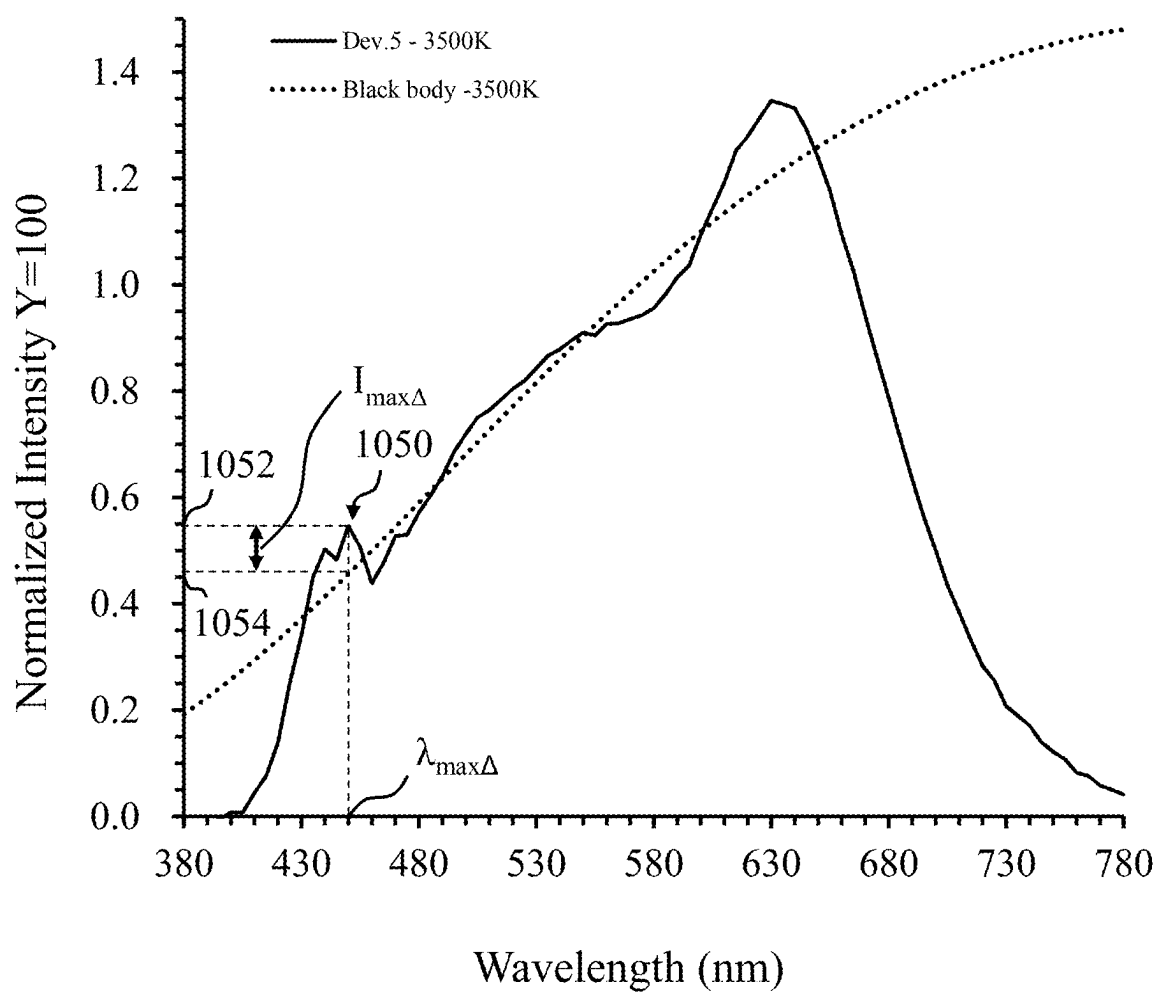


FIG. 10F

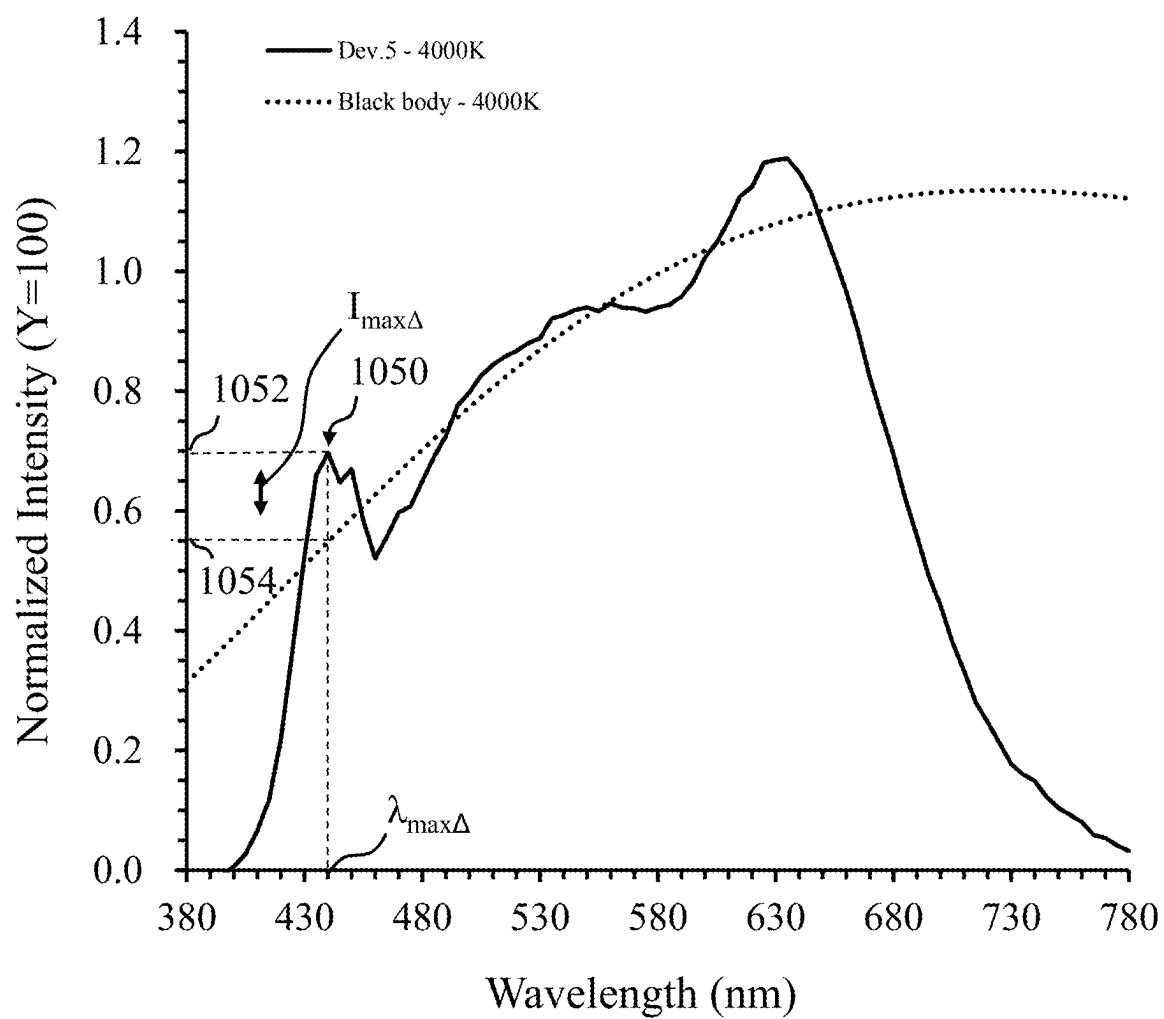


FIG. 10G

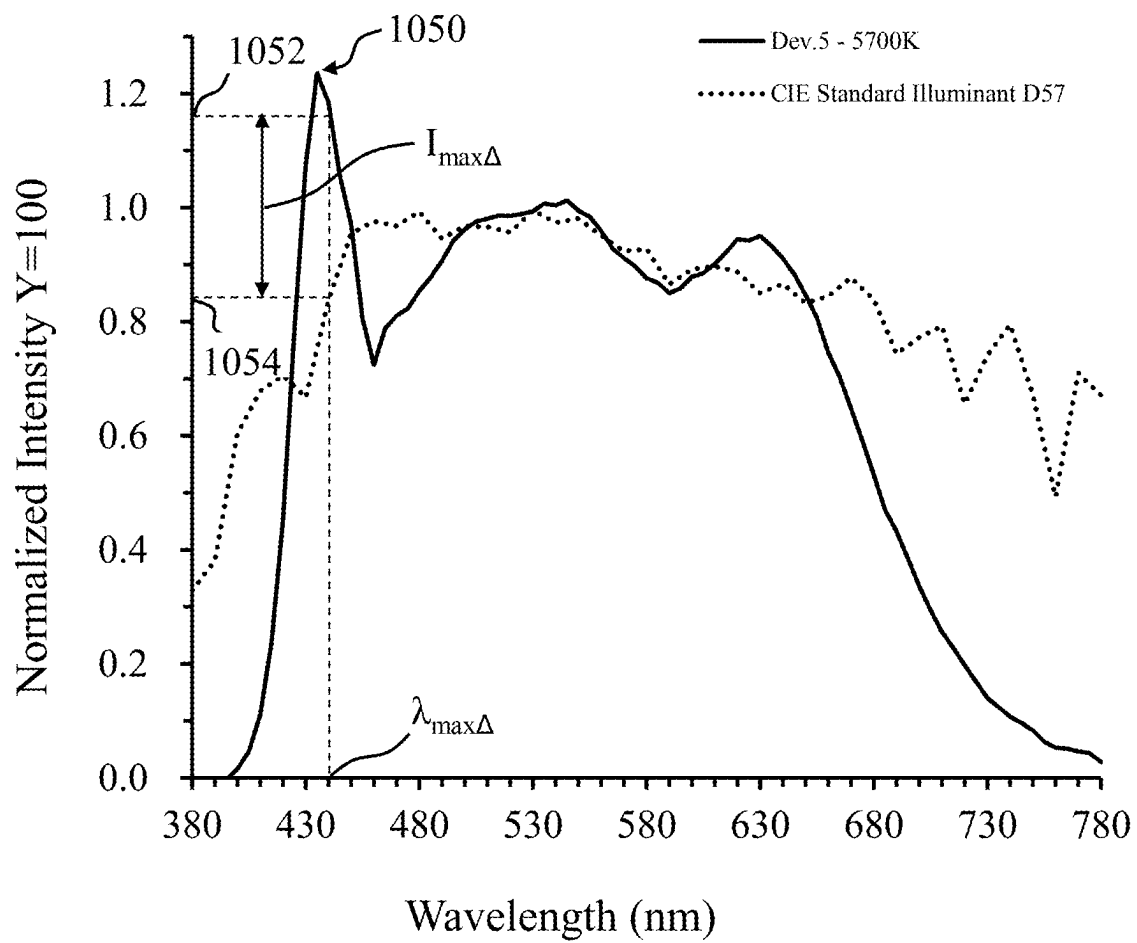


FIG. 10H

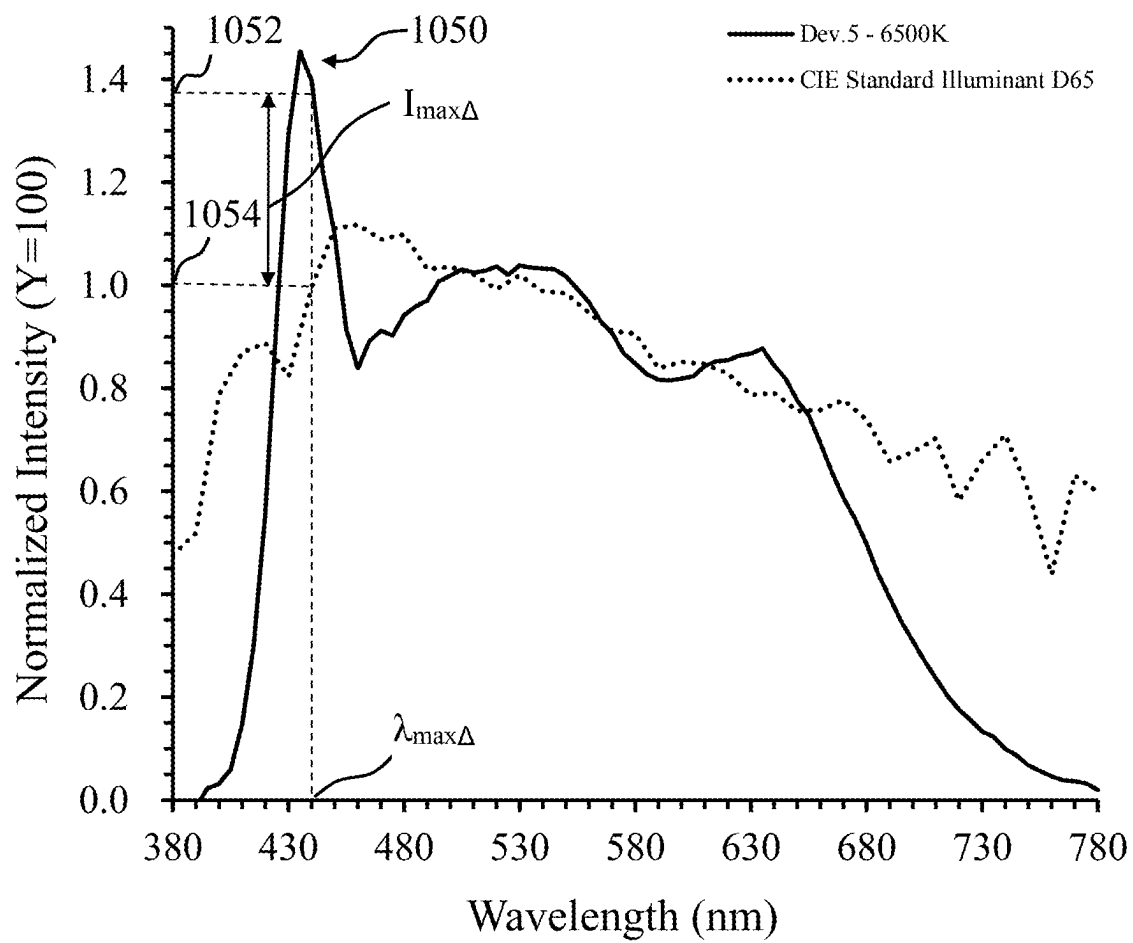


FIG. 10I

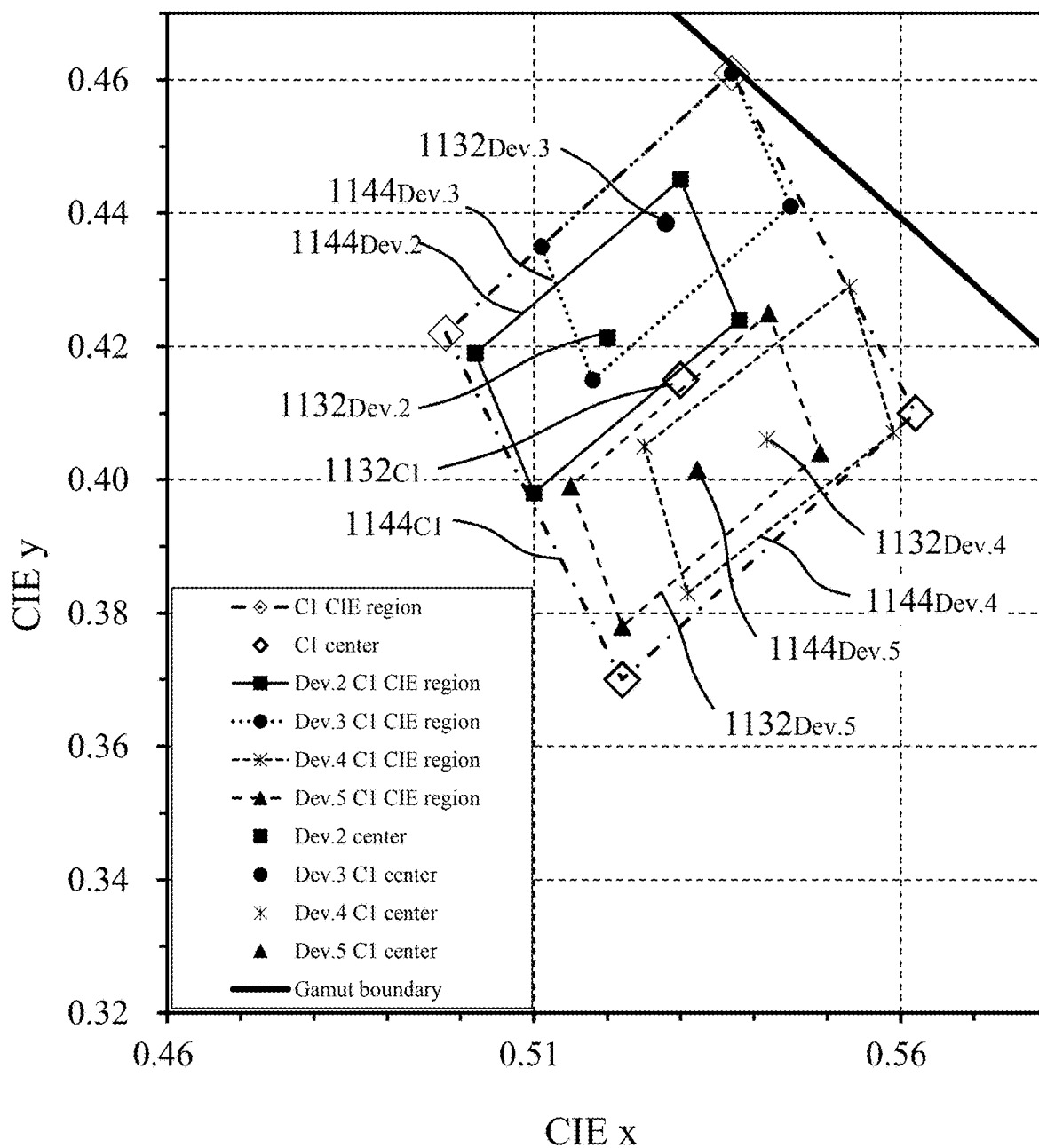


FIG. 11A

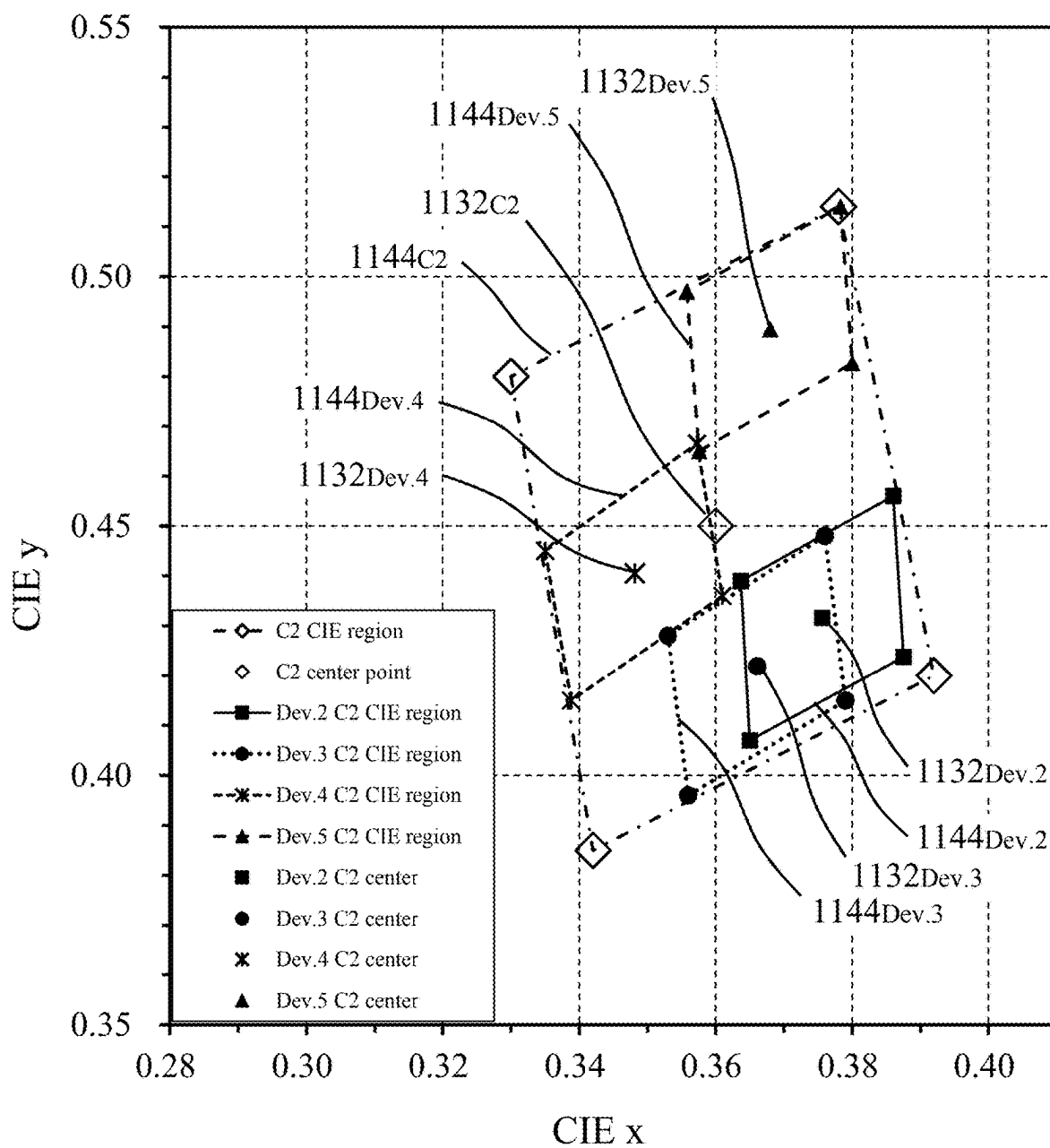


FIG. 11B

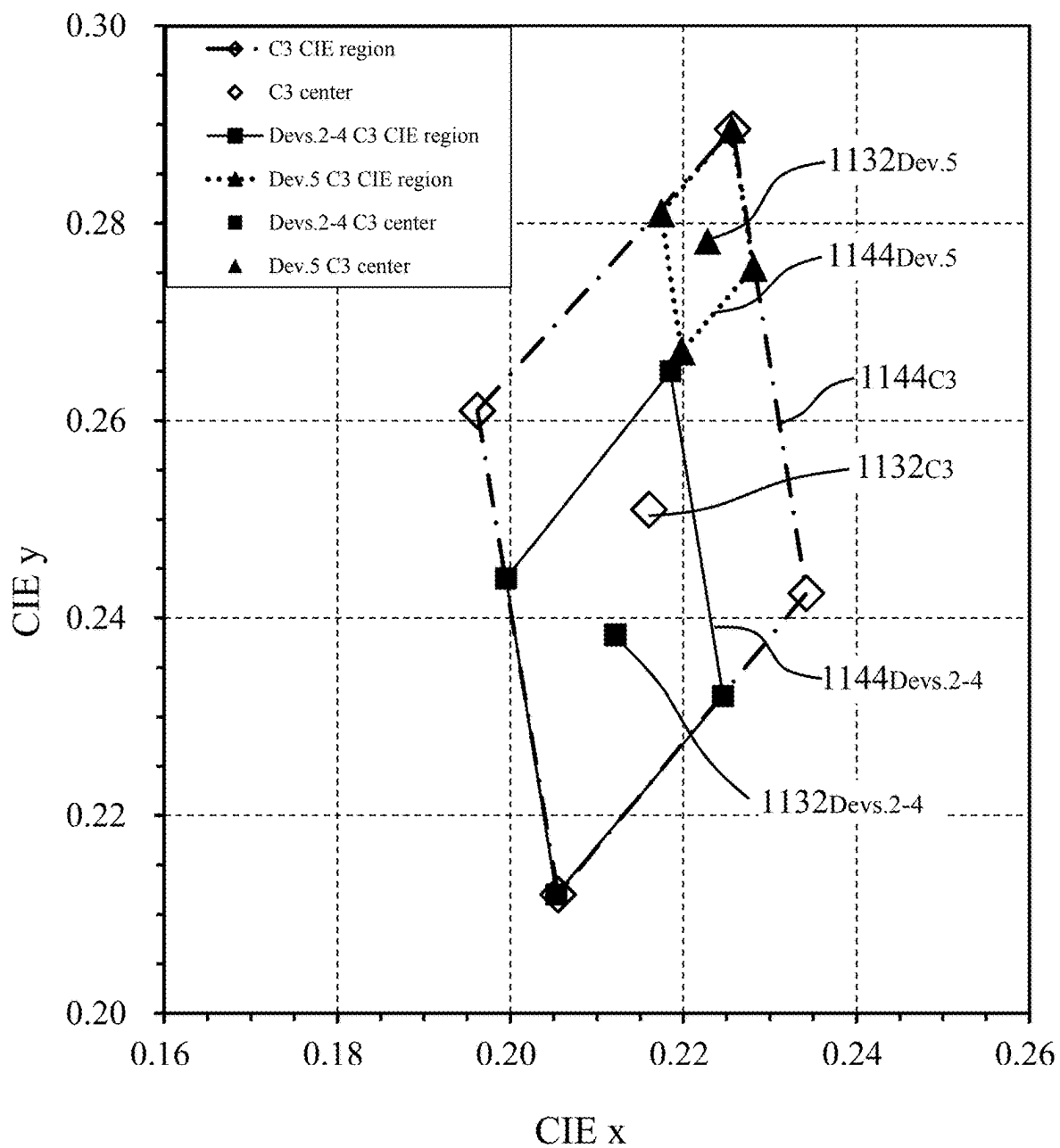


FIG. 11C

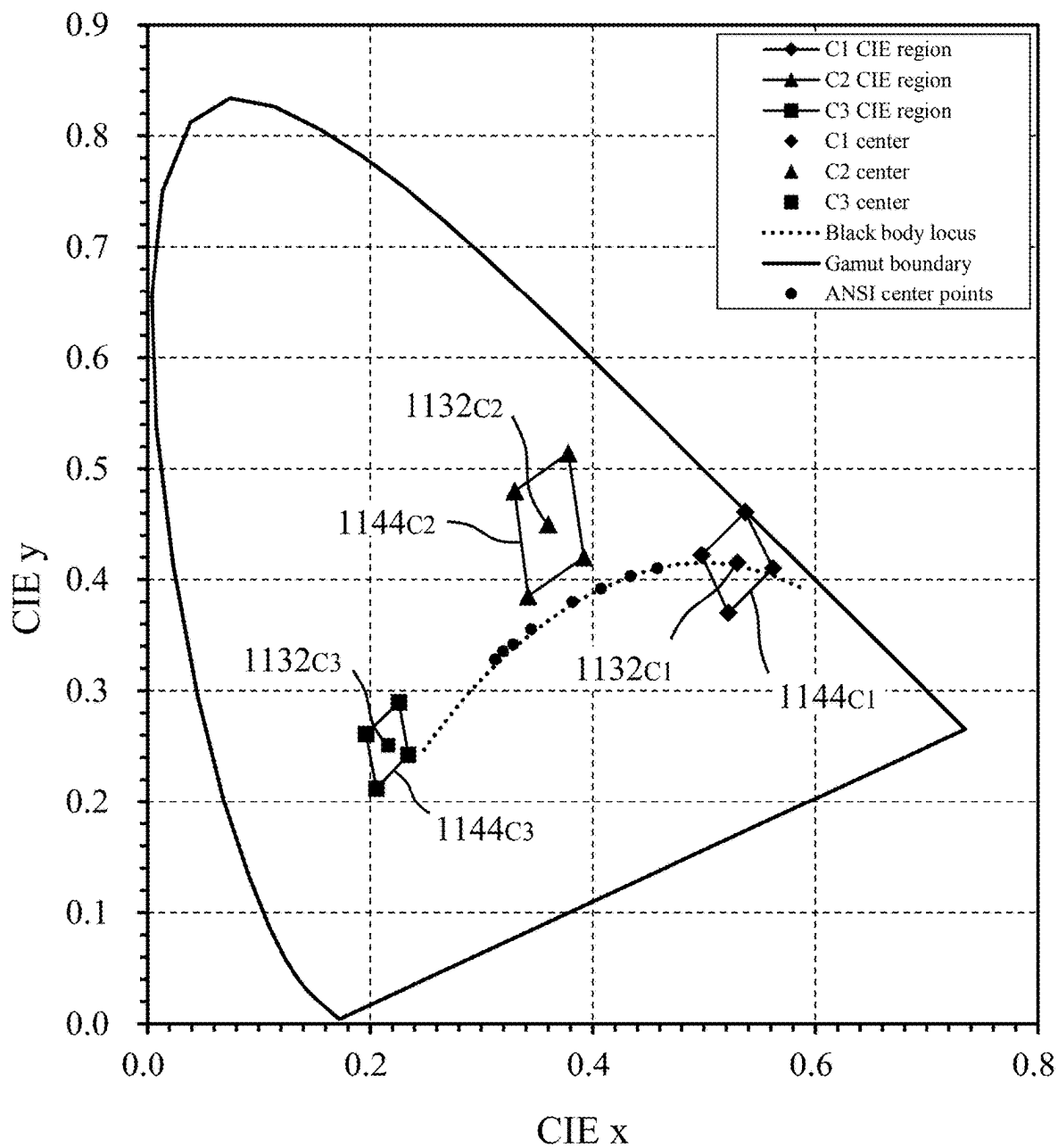


FIG. 11D

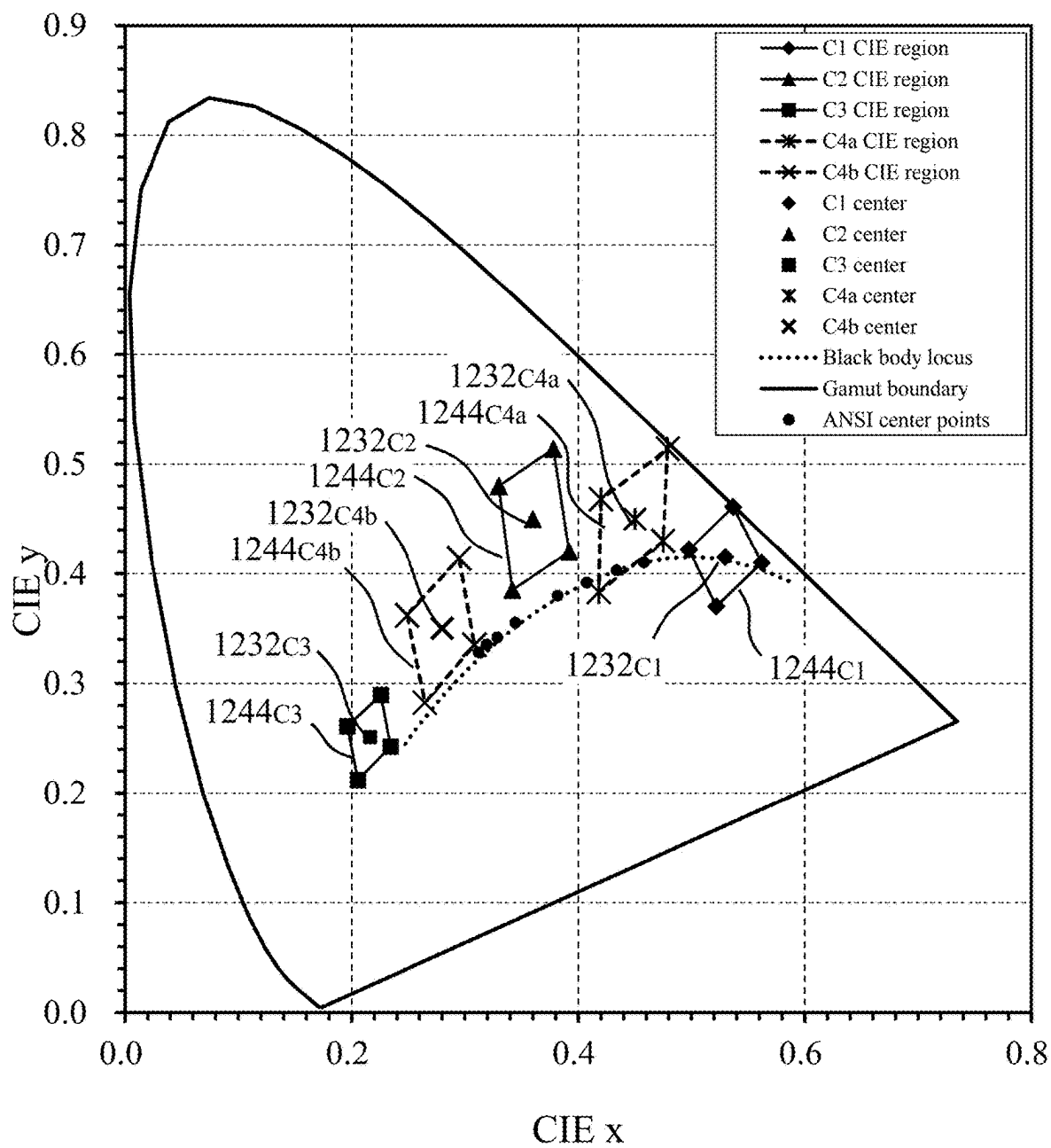


FIG. 12

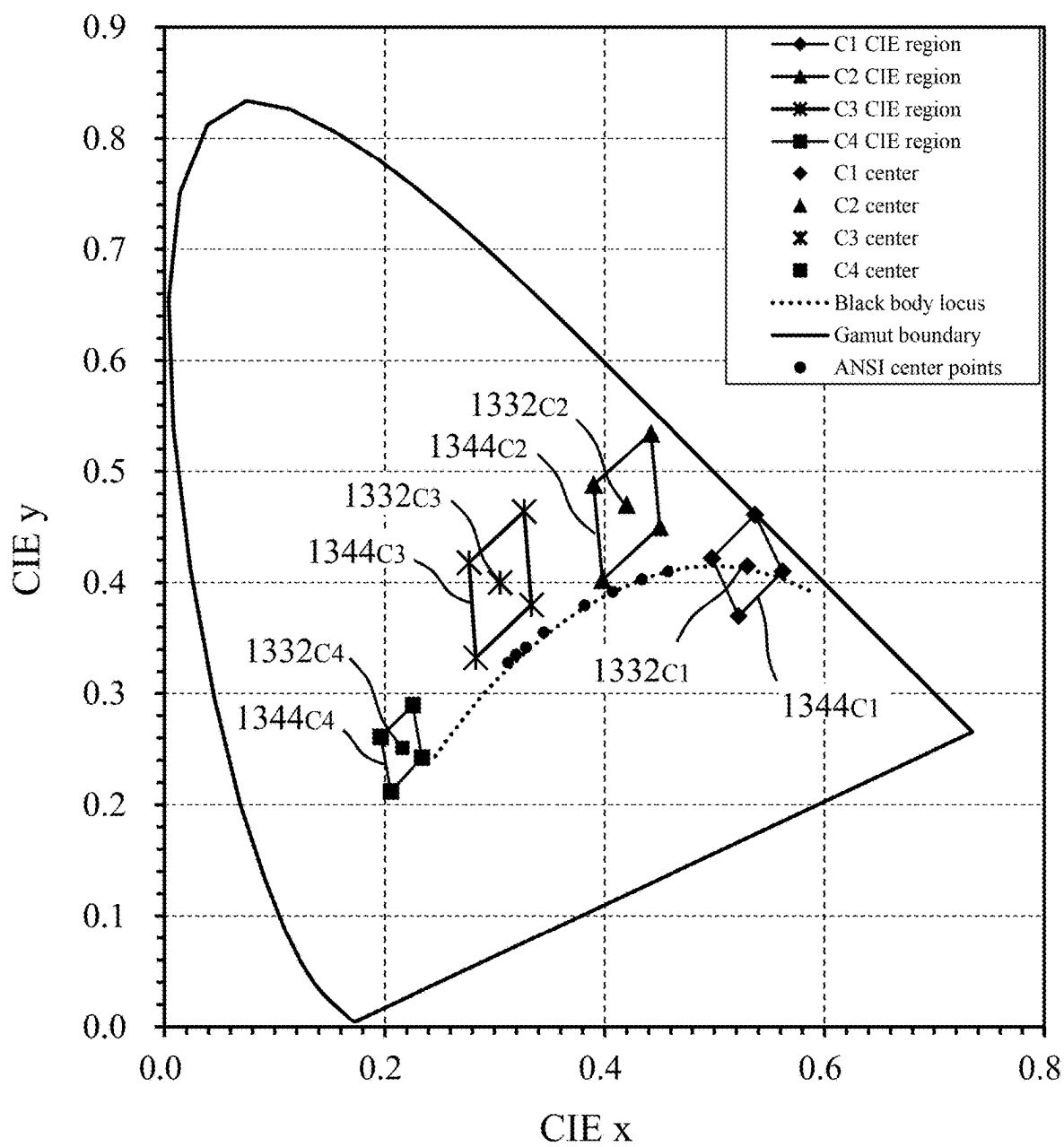


FIG. 13

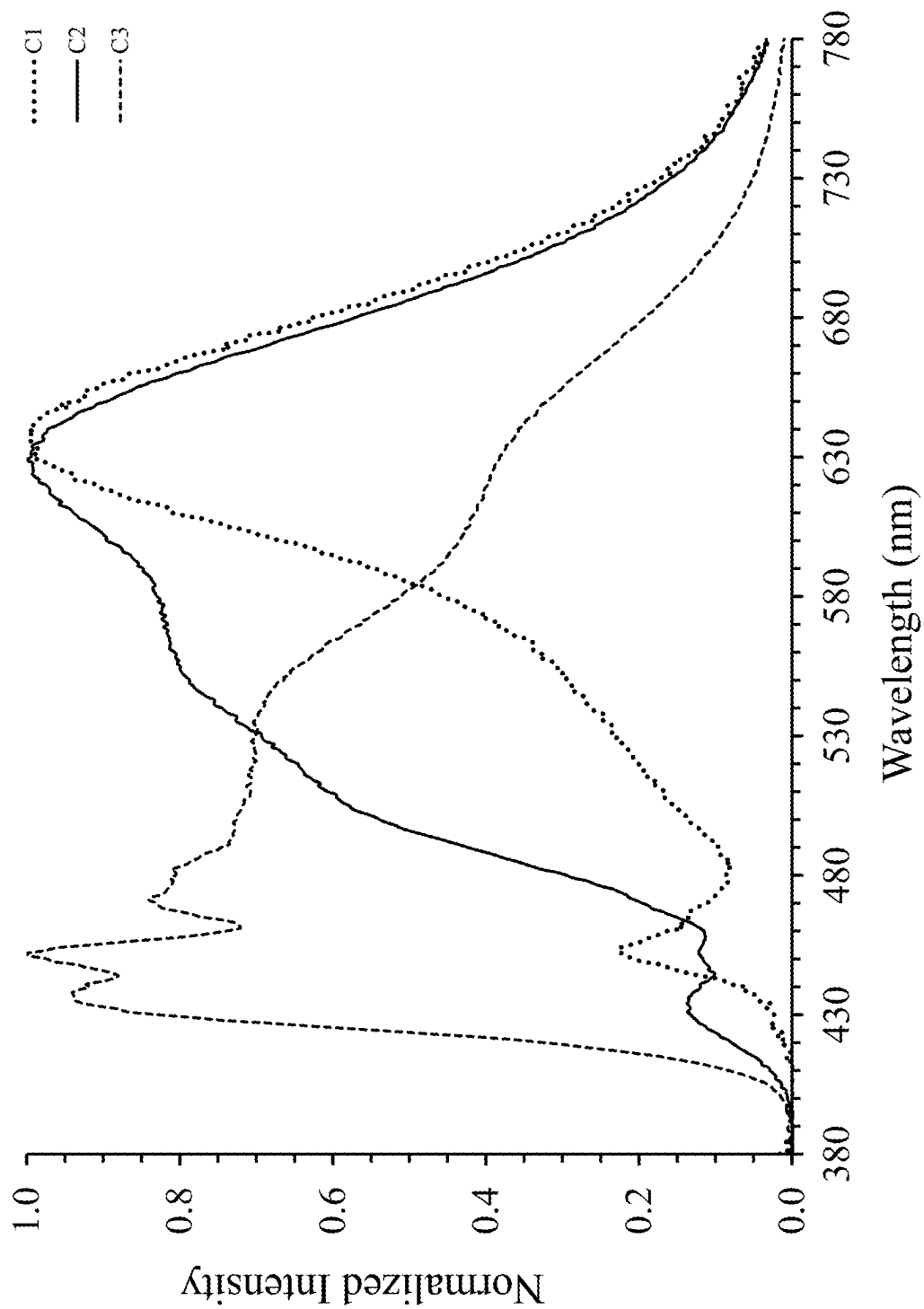


FIG. 14A

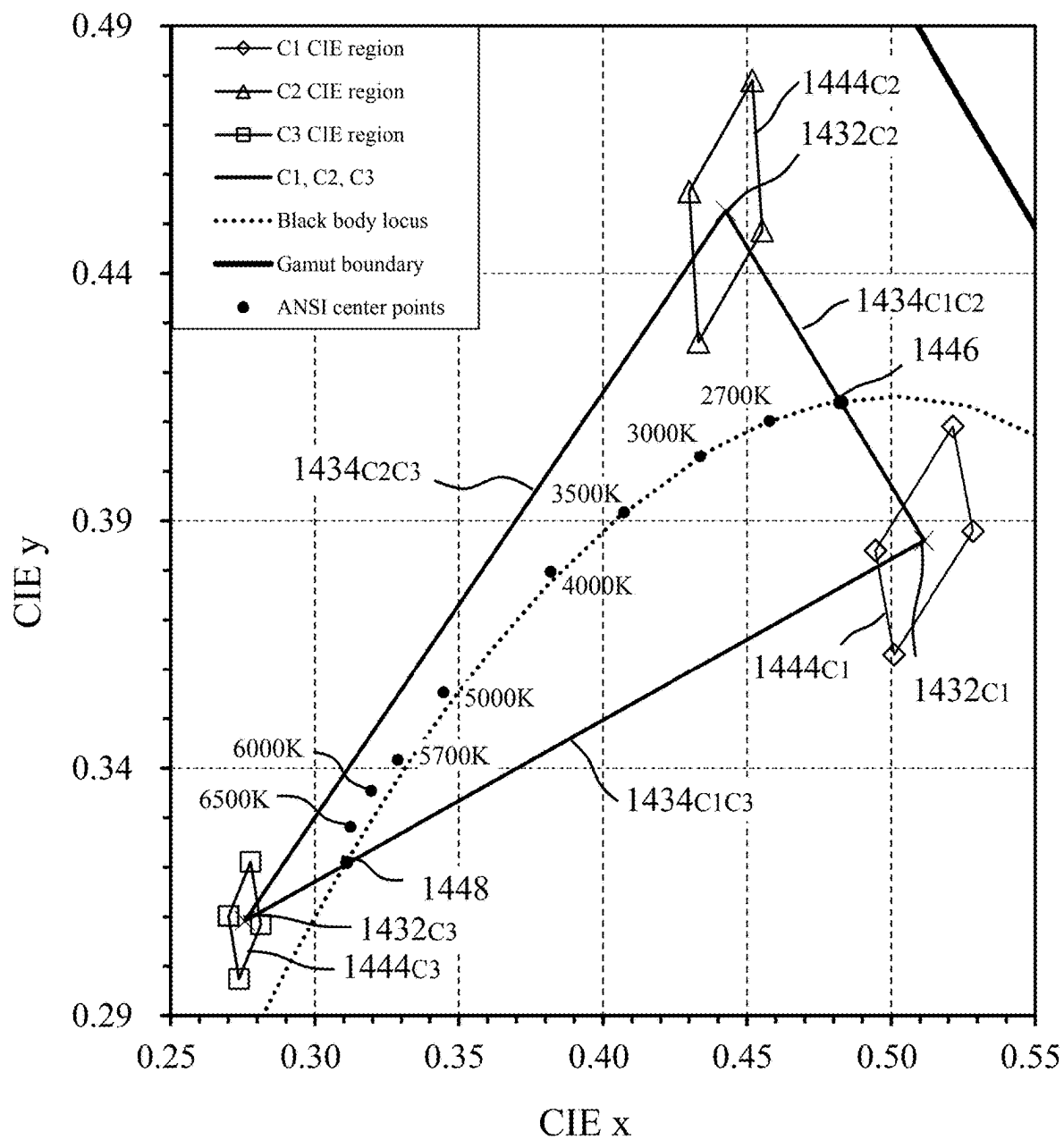


FIG. 14B

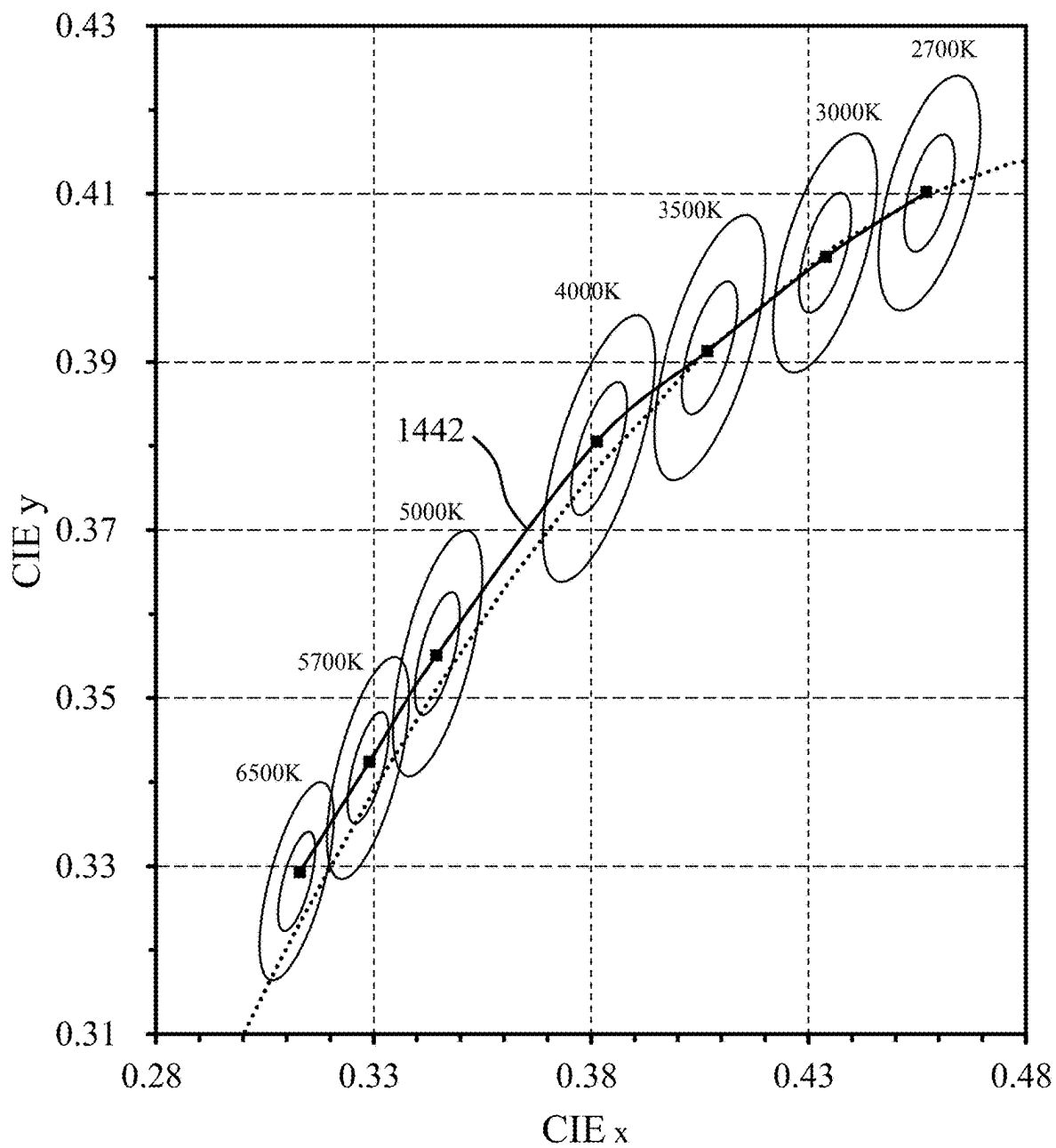


FIG. 14C

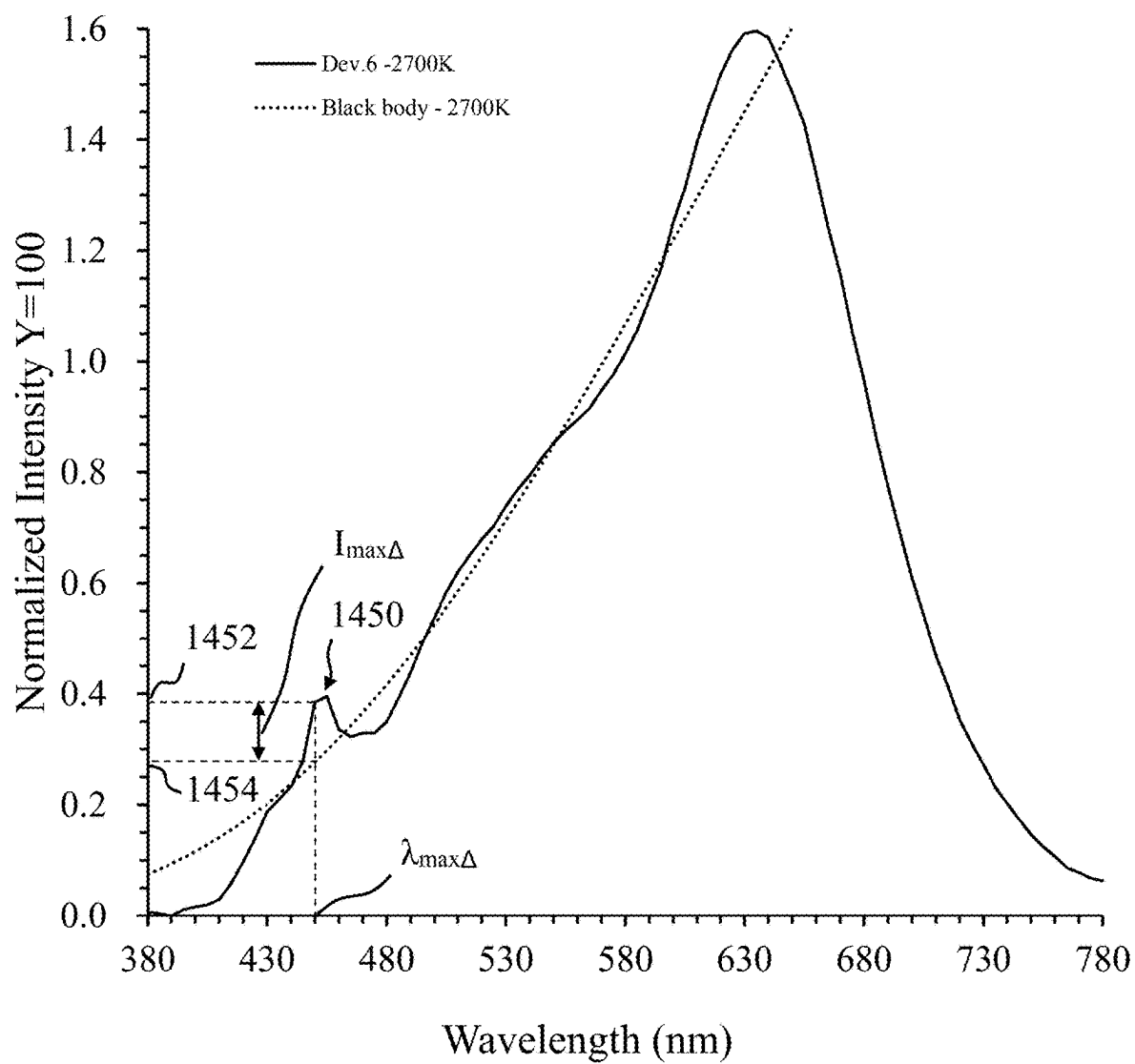


FIG. 14D

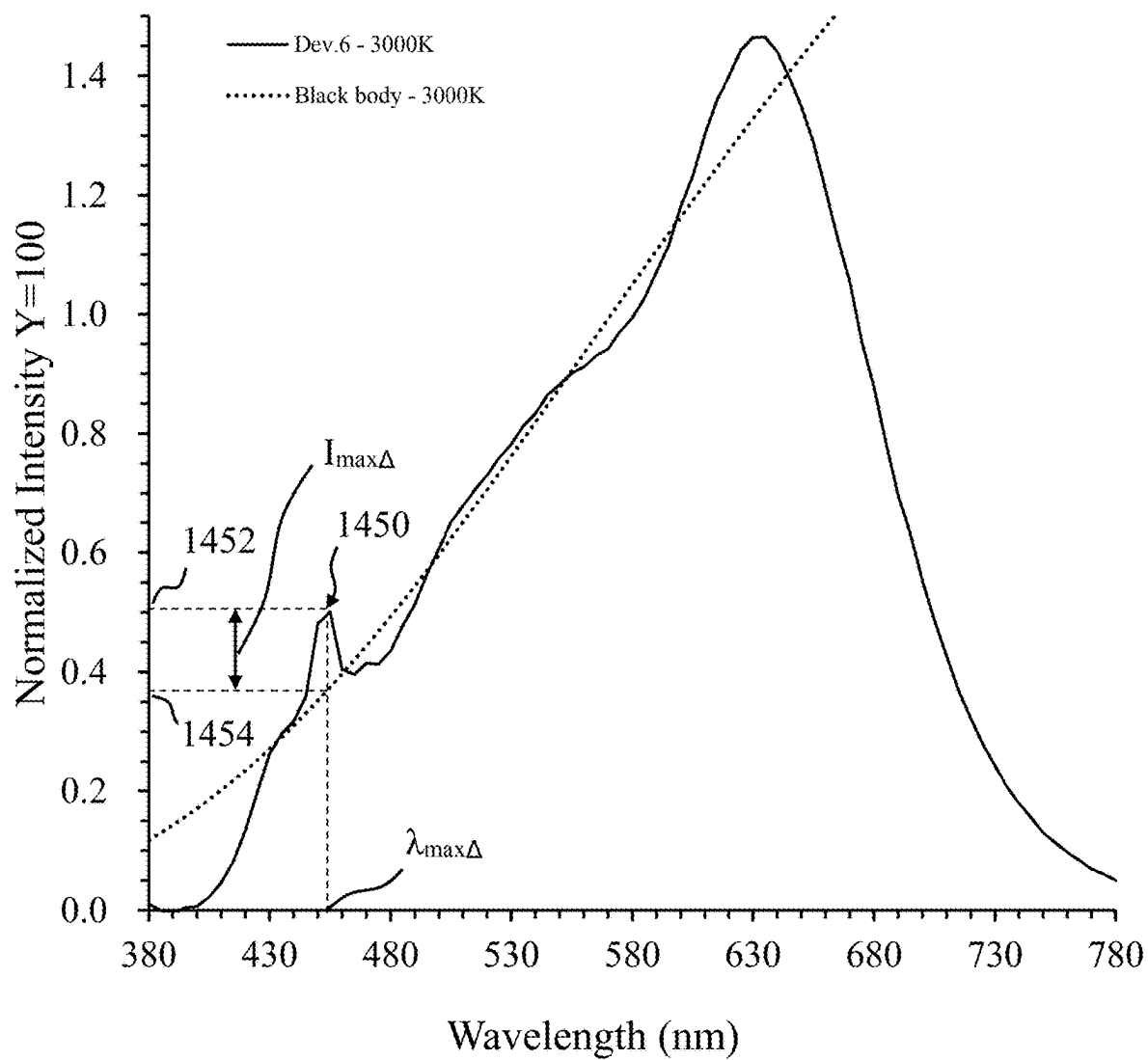


FIG. 14E

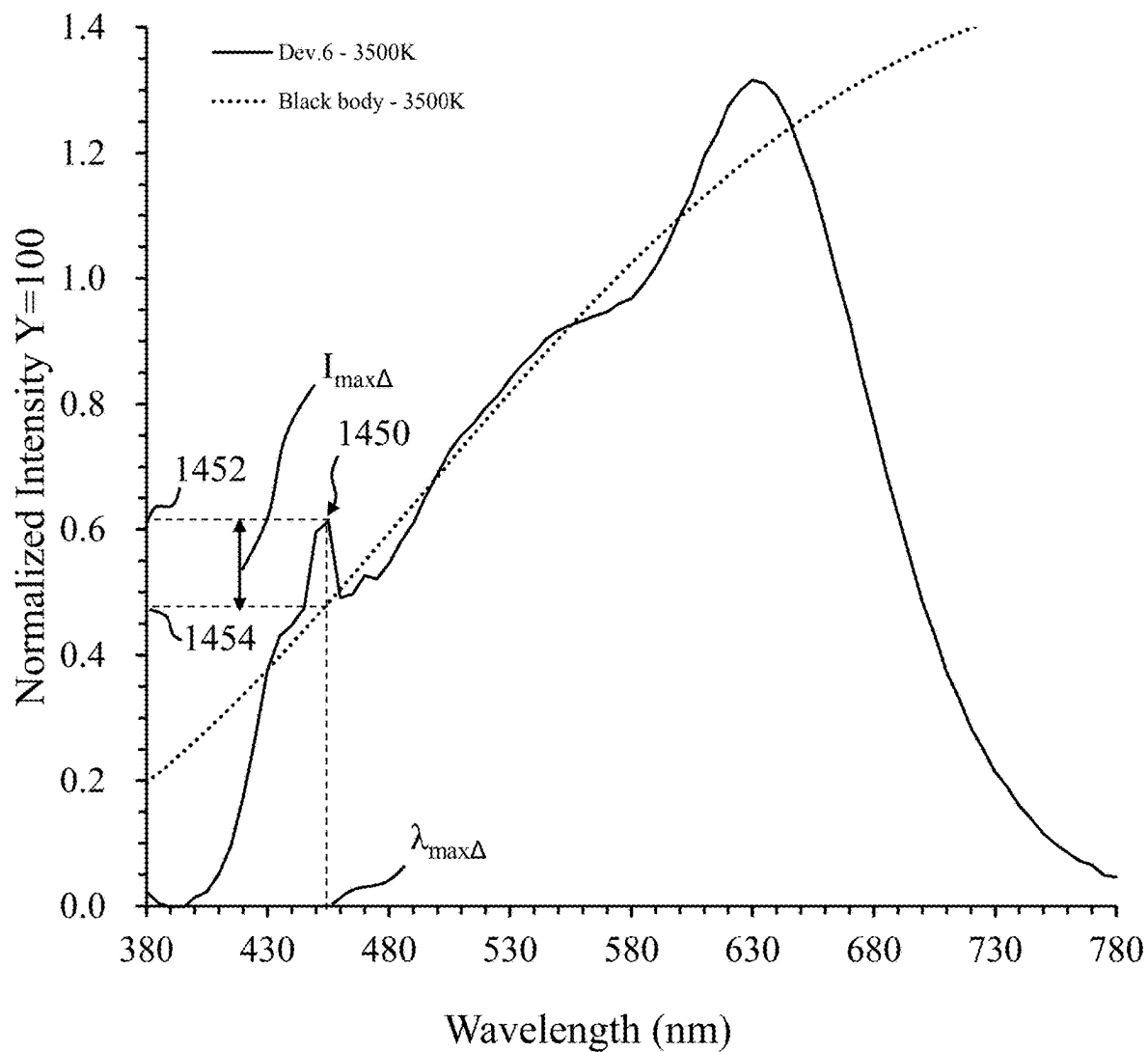


FIG. 14F

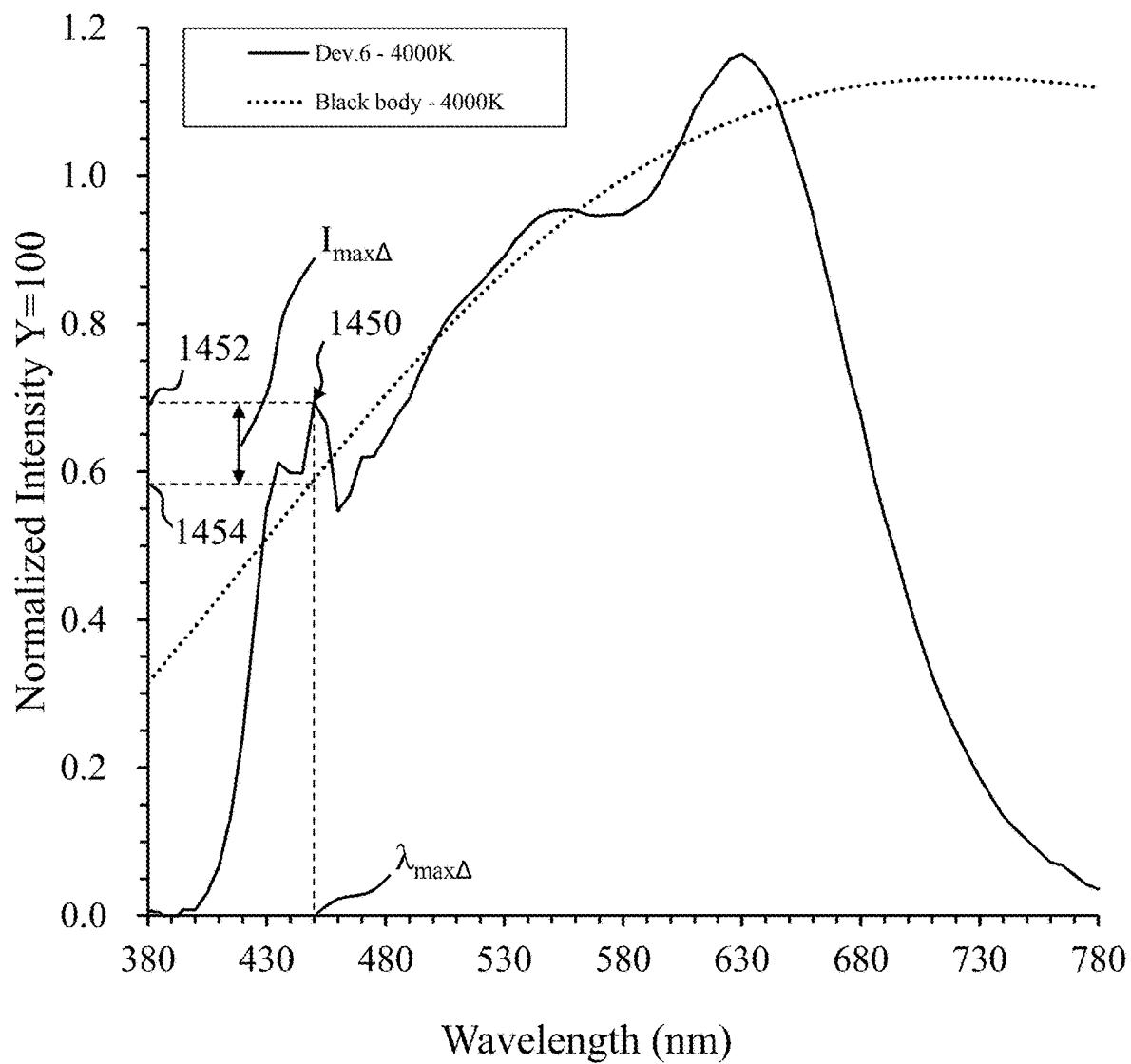


FIG. 14G

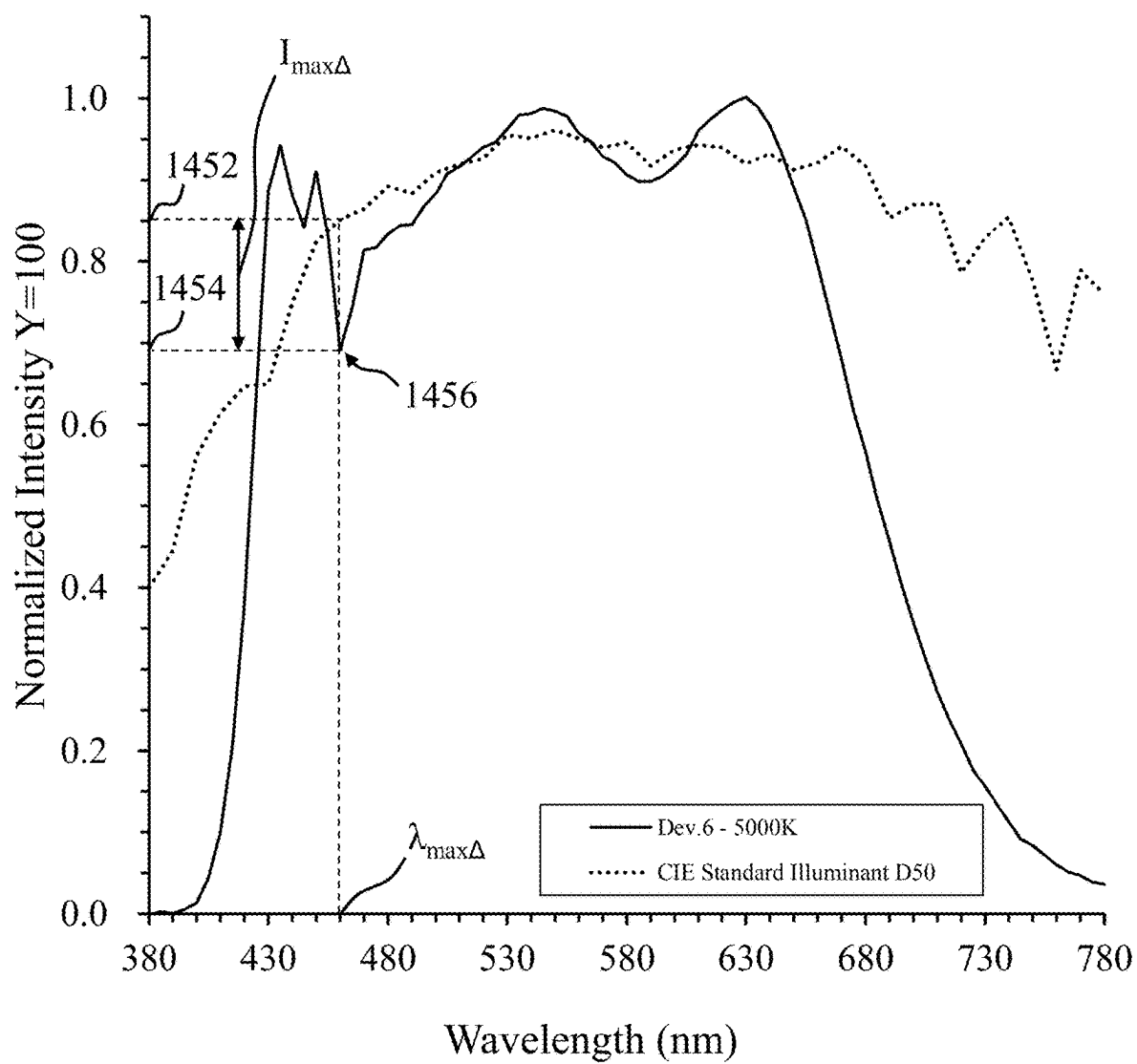


FIG. 14H

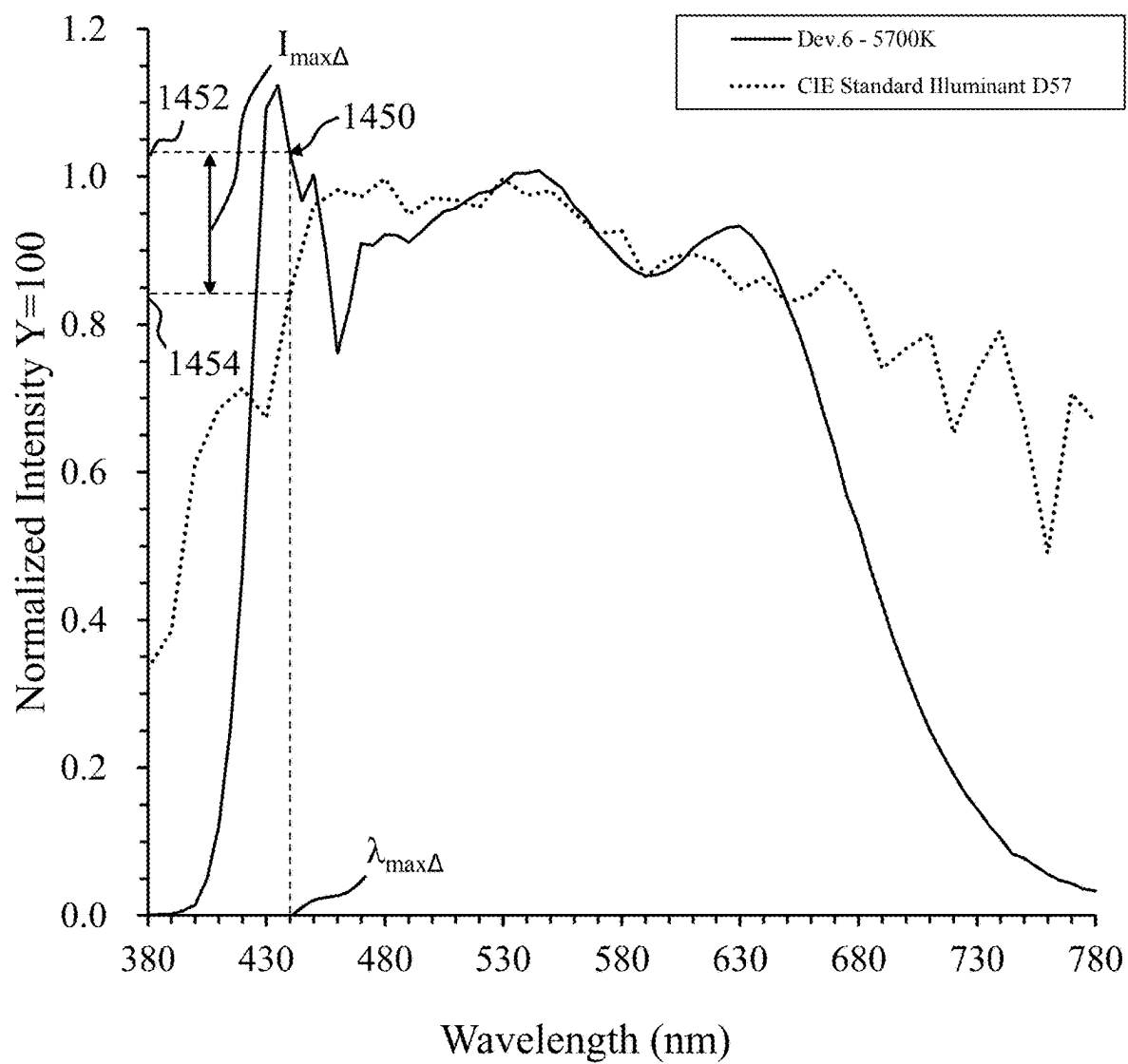


FIG. 14I

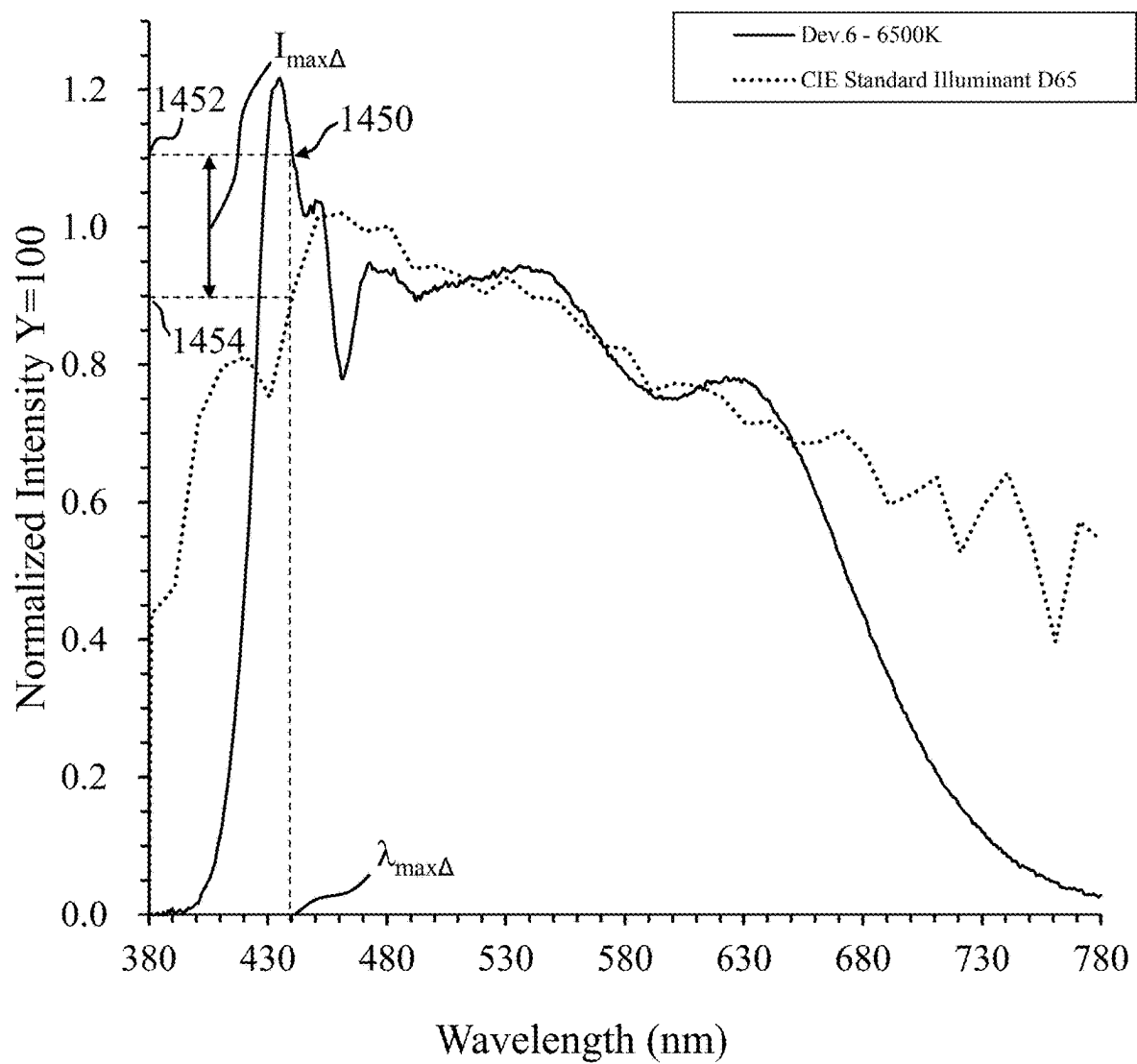


FIG. 14J

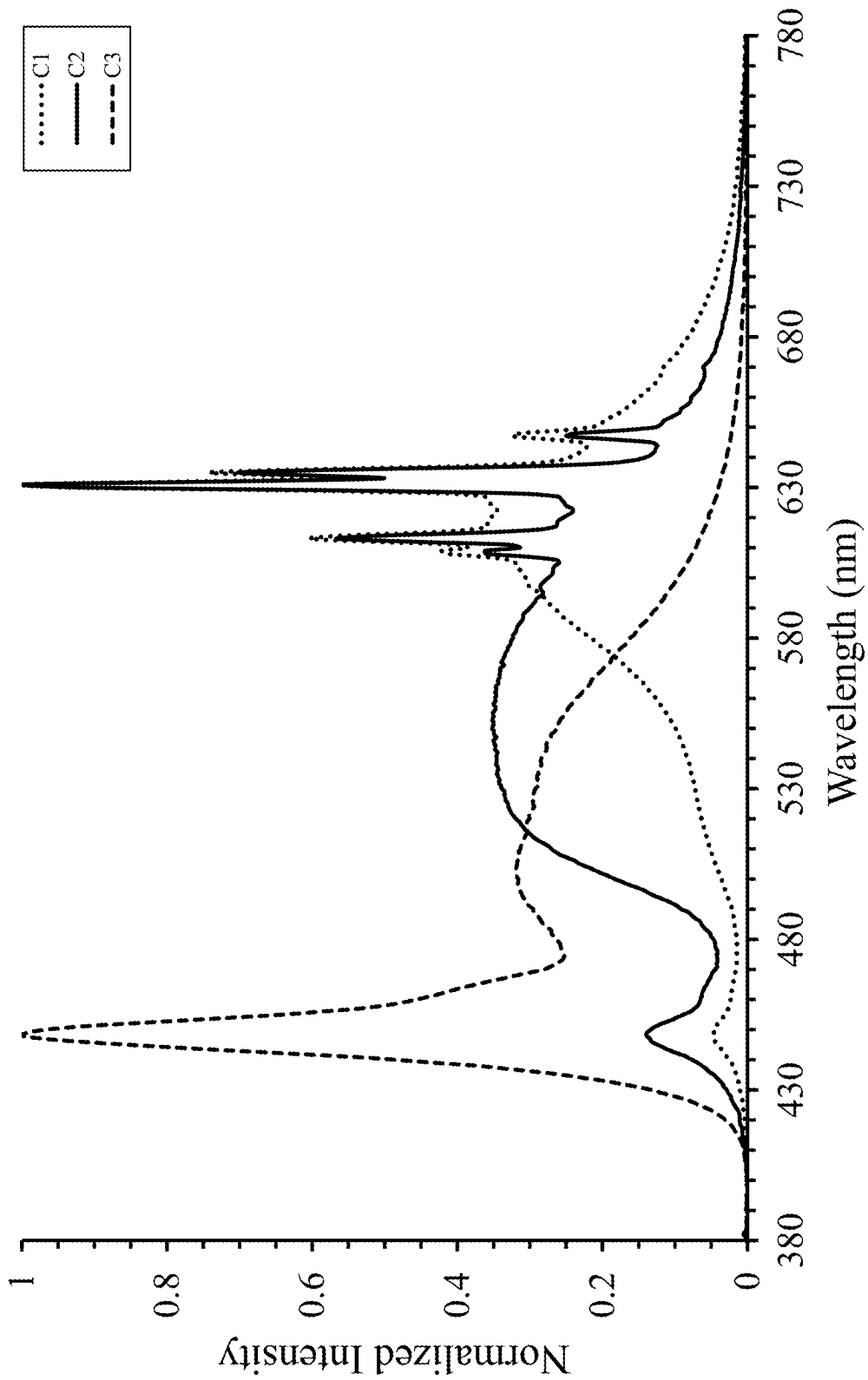


FIG. 15A

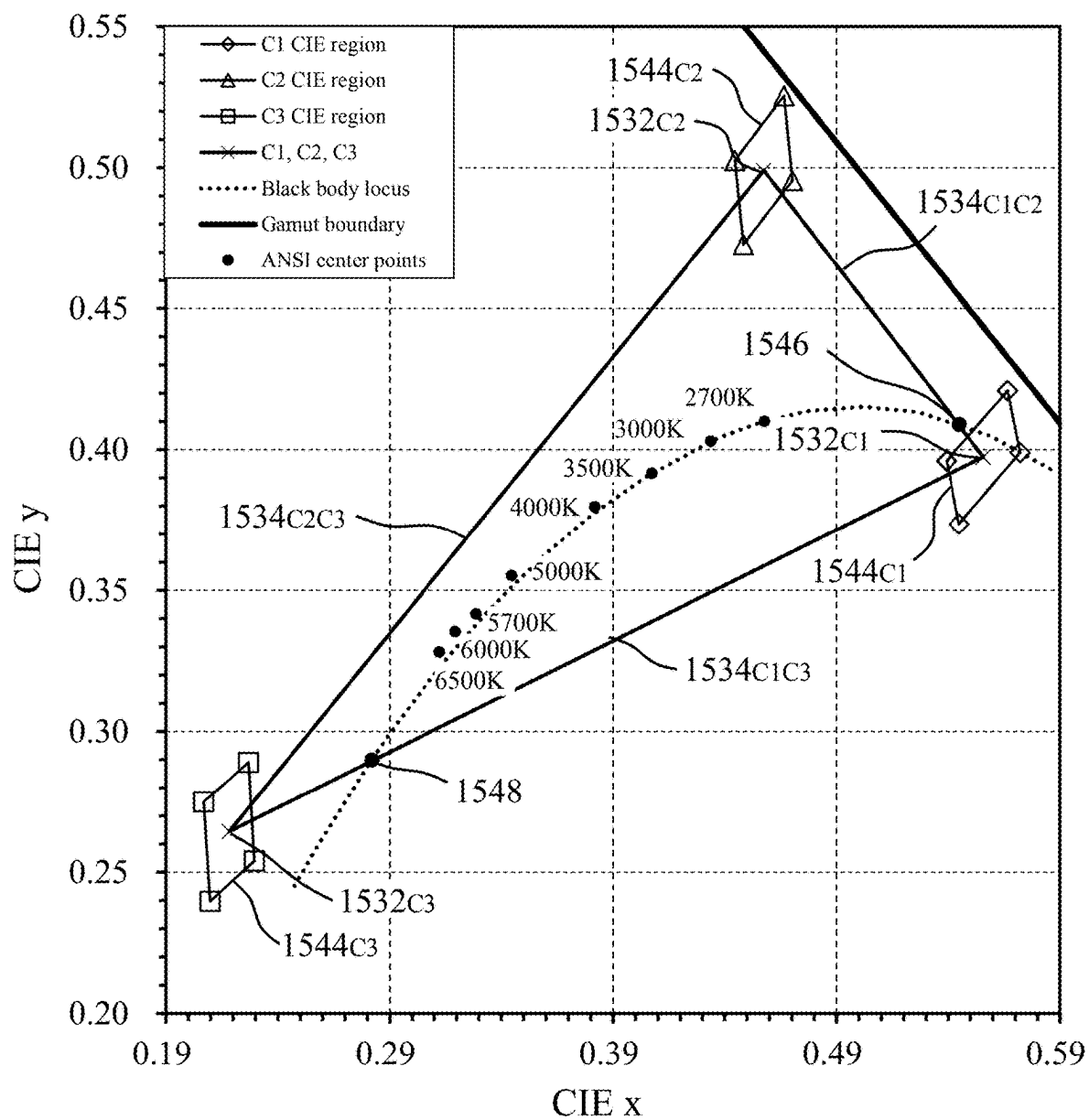


FIG. 15B

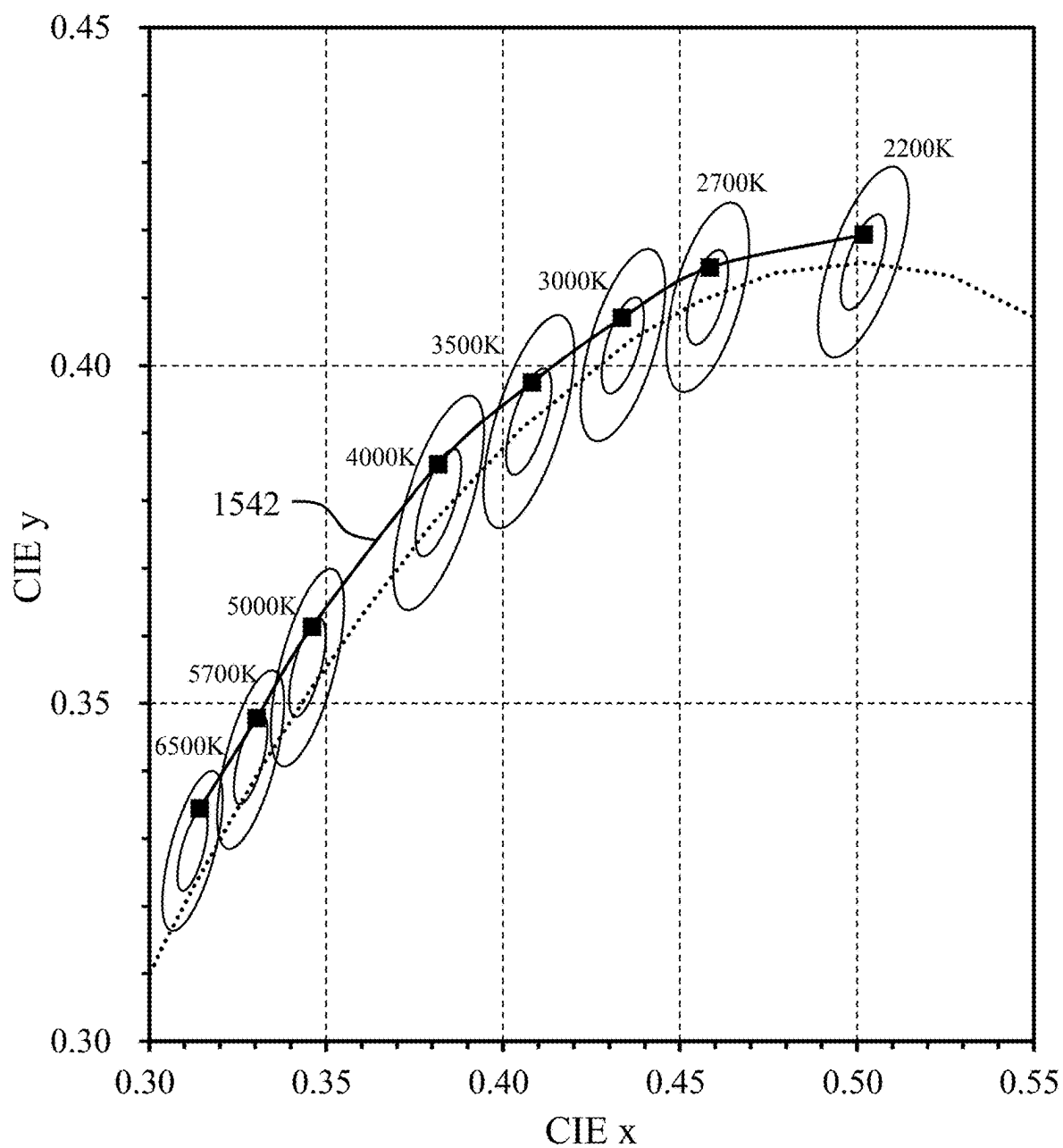


FIG. 15C

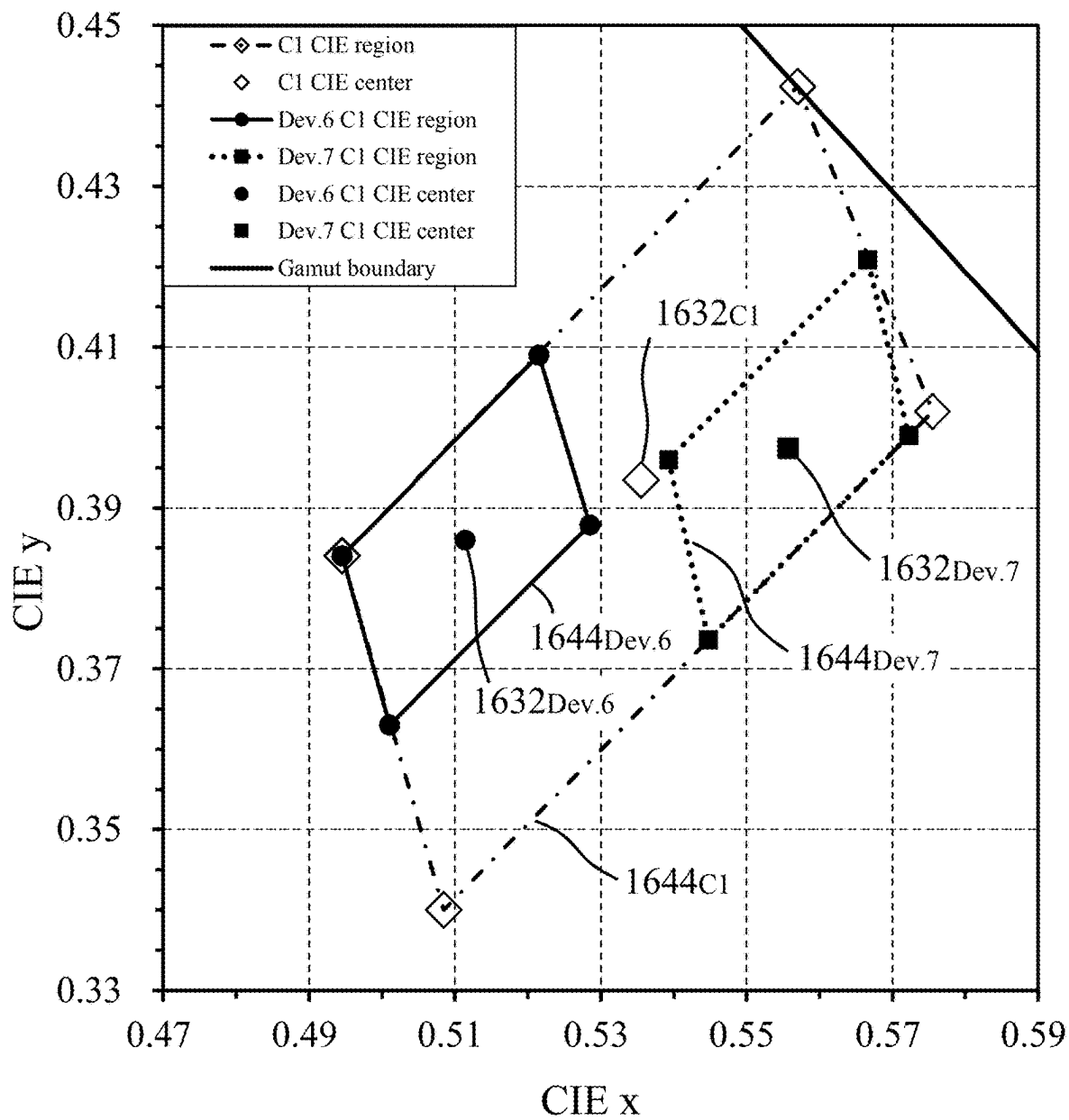


FIG. 16A

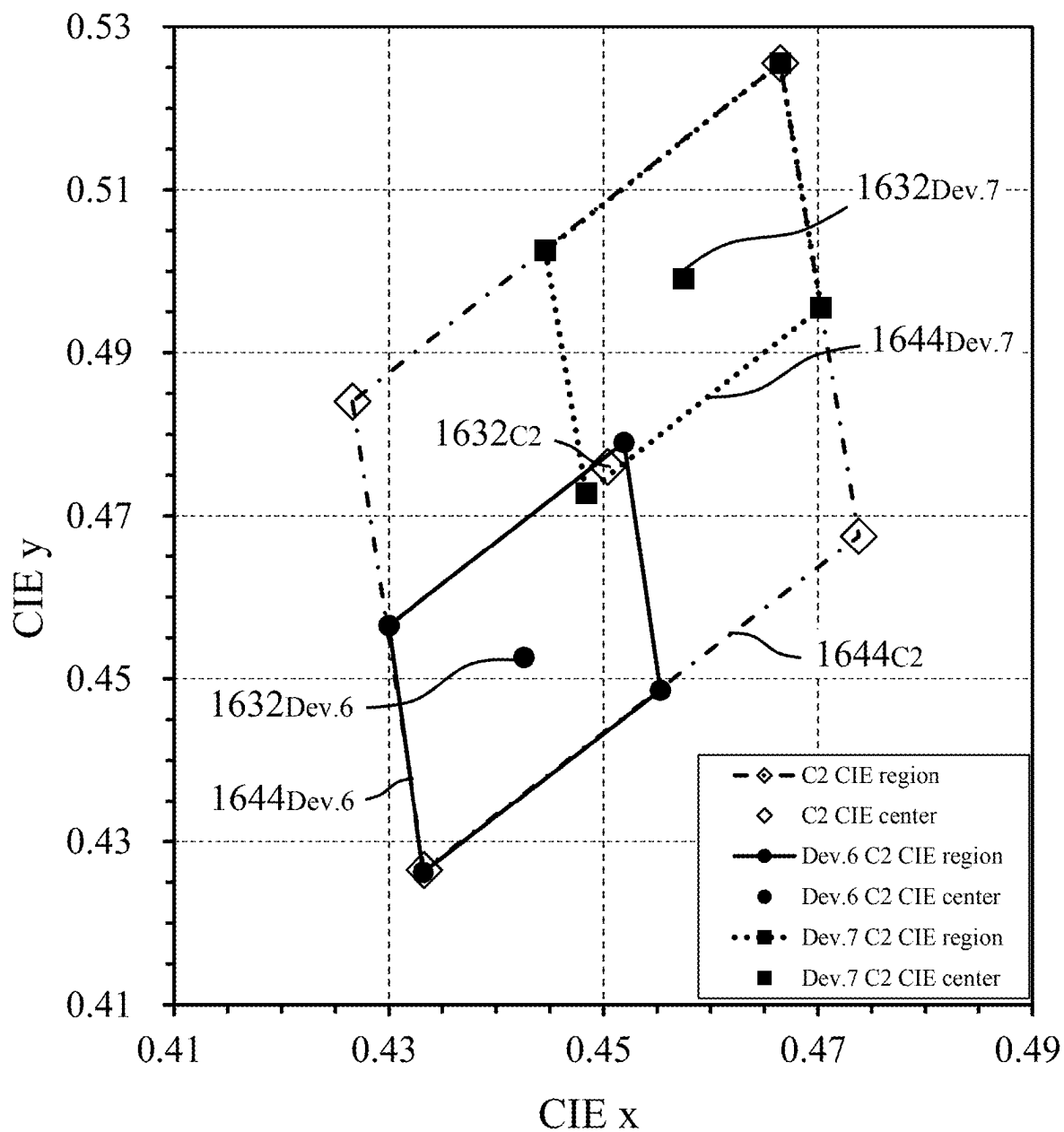


FIG. 16B

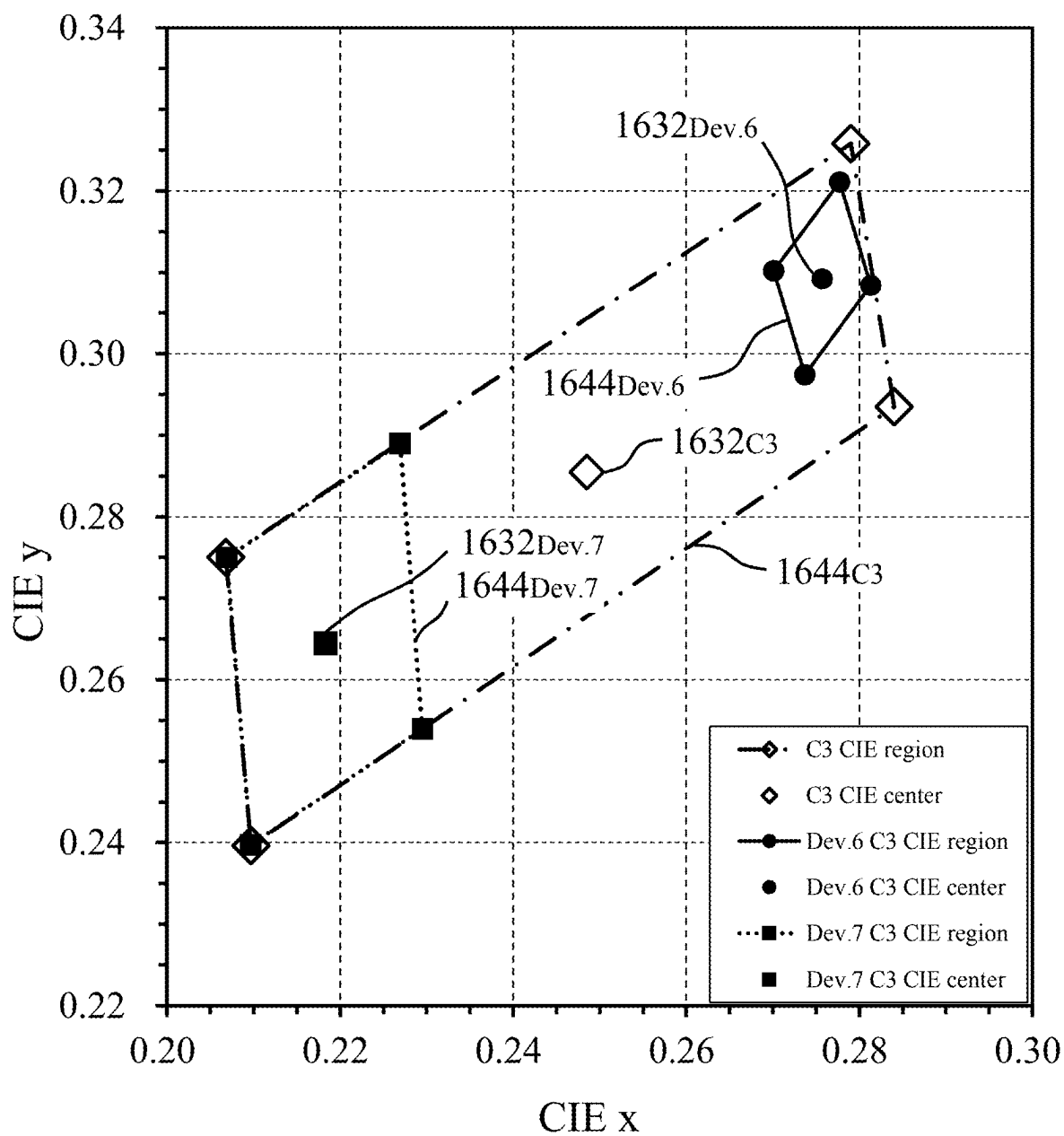


FIG. 16C

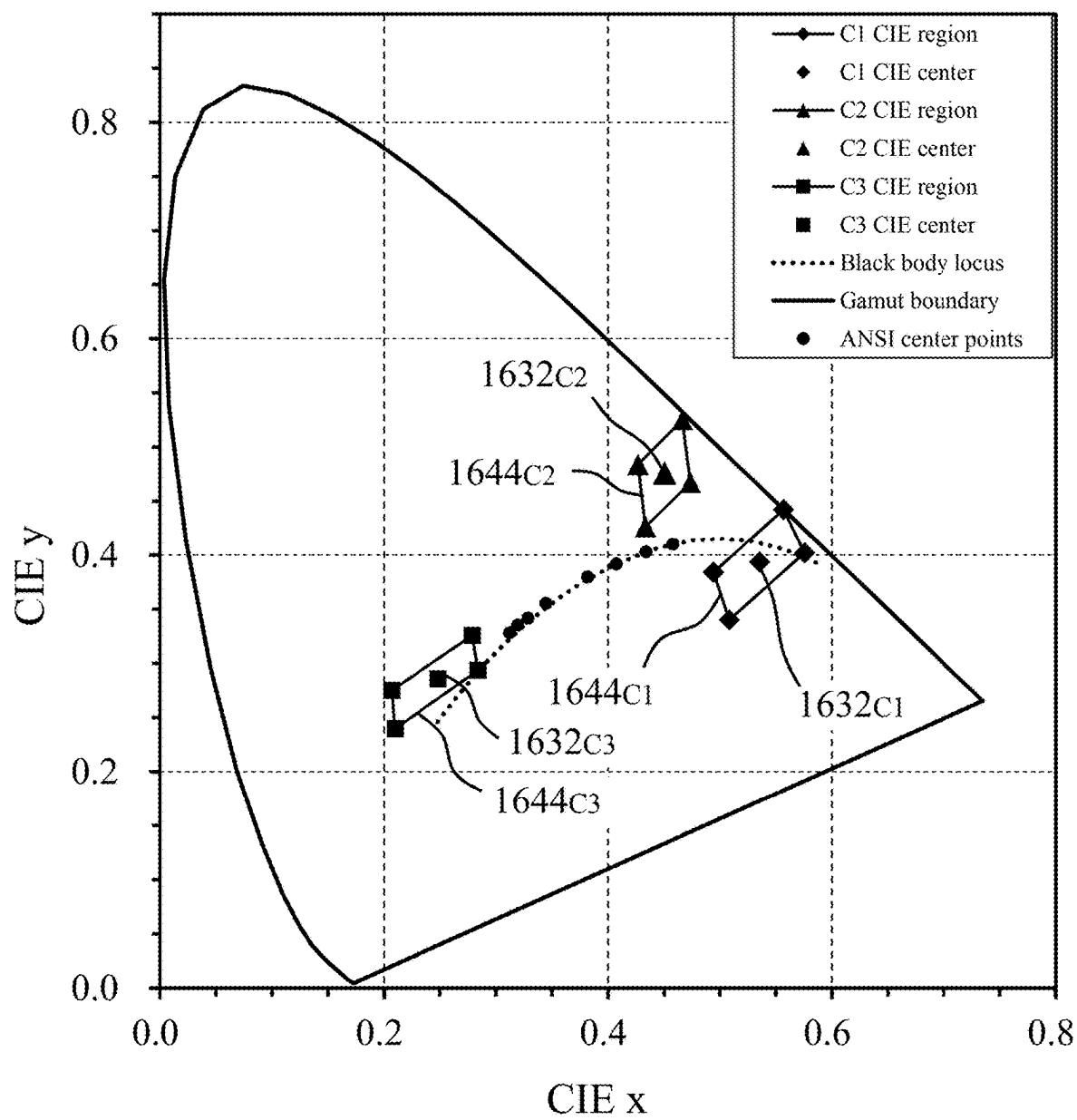


FIG. 16D

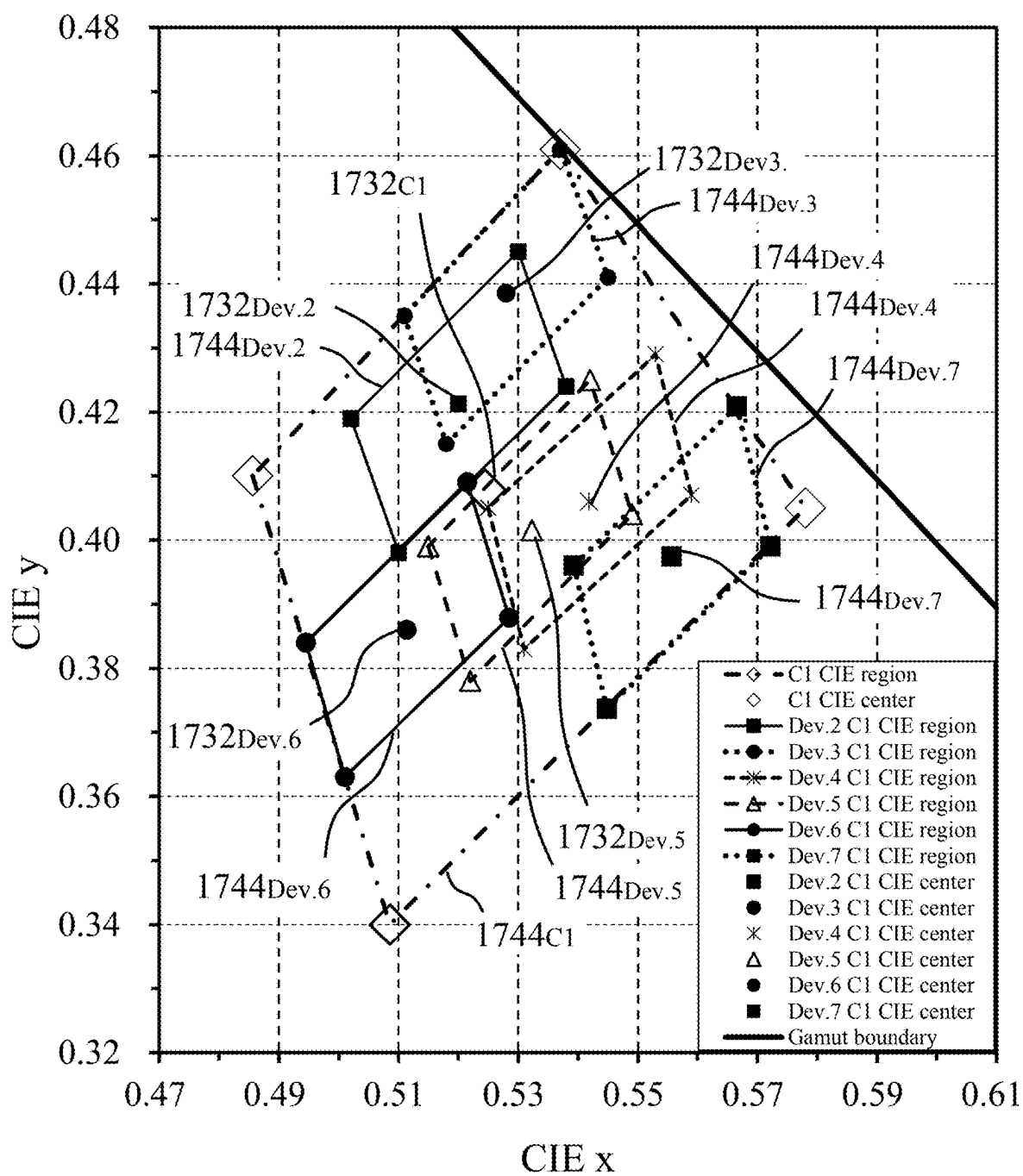


FIG. 17A

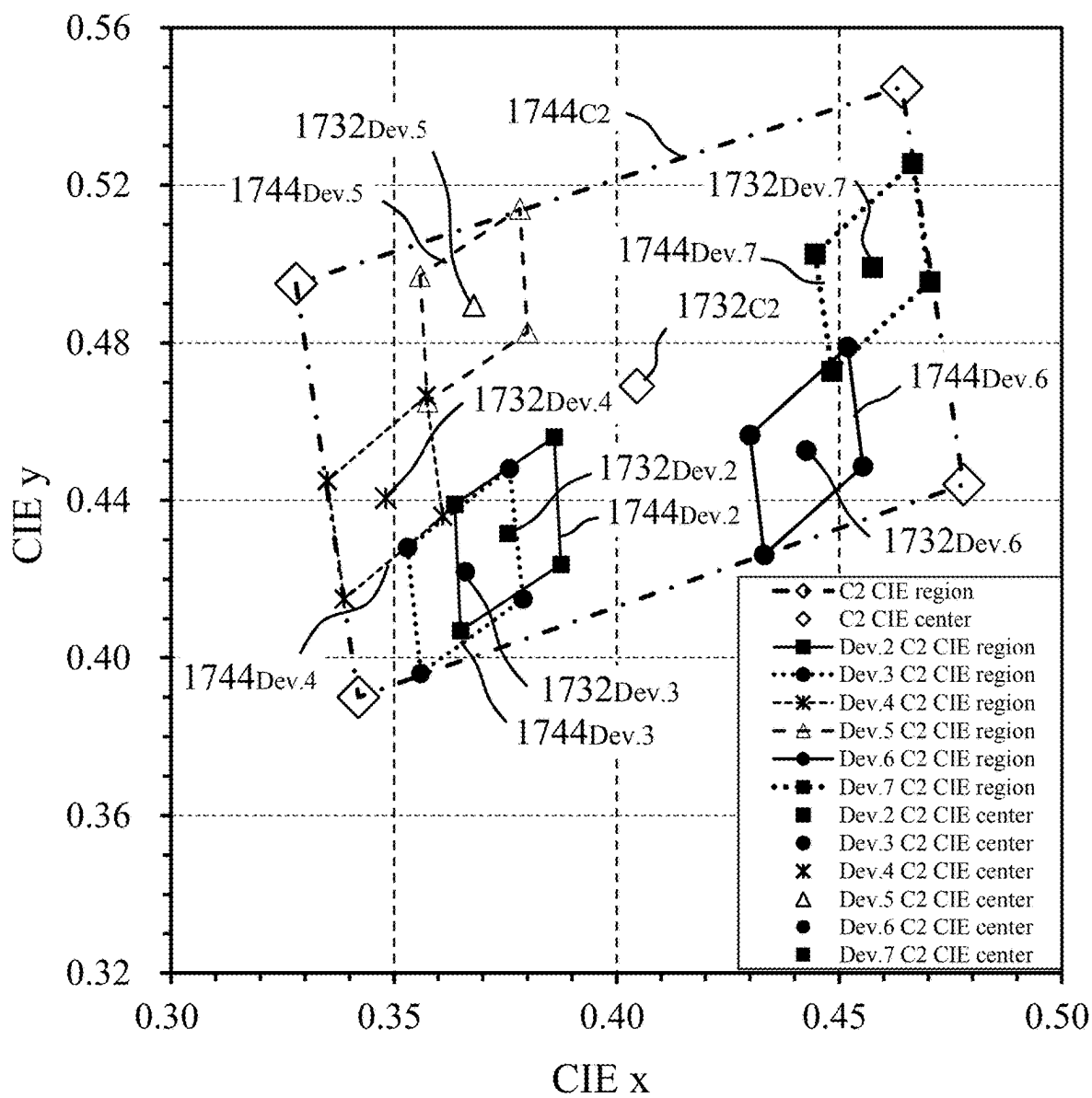


FIG. 17B

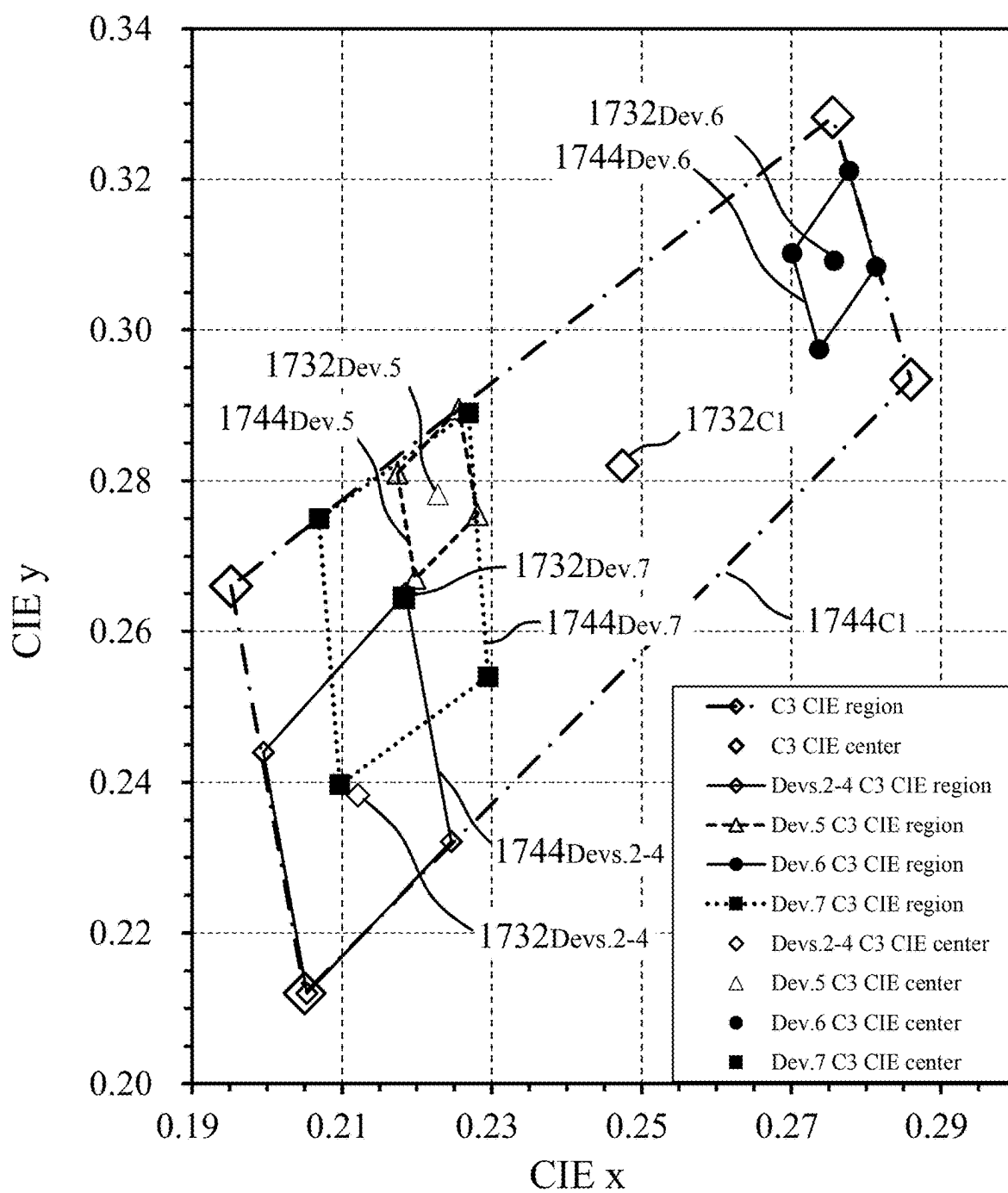


FIG. 17C

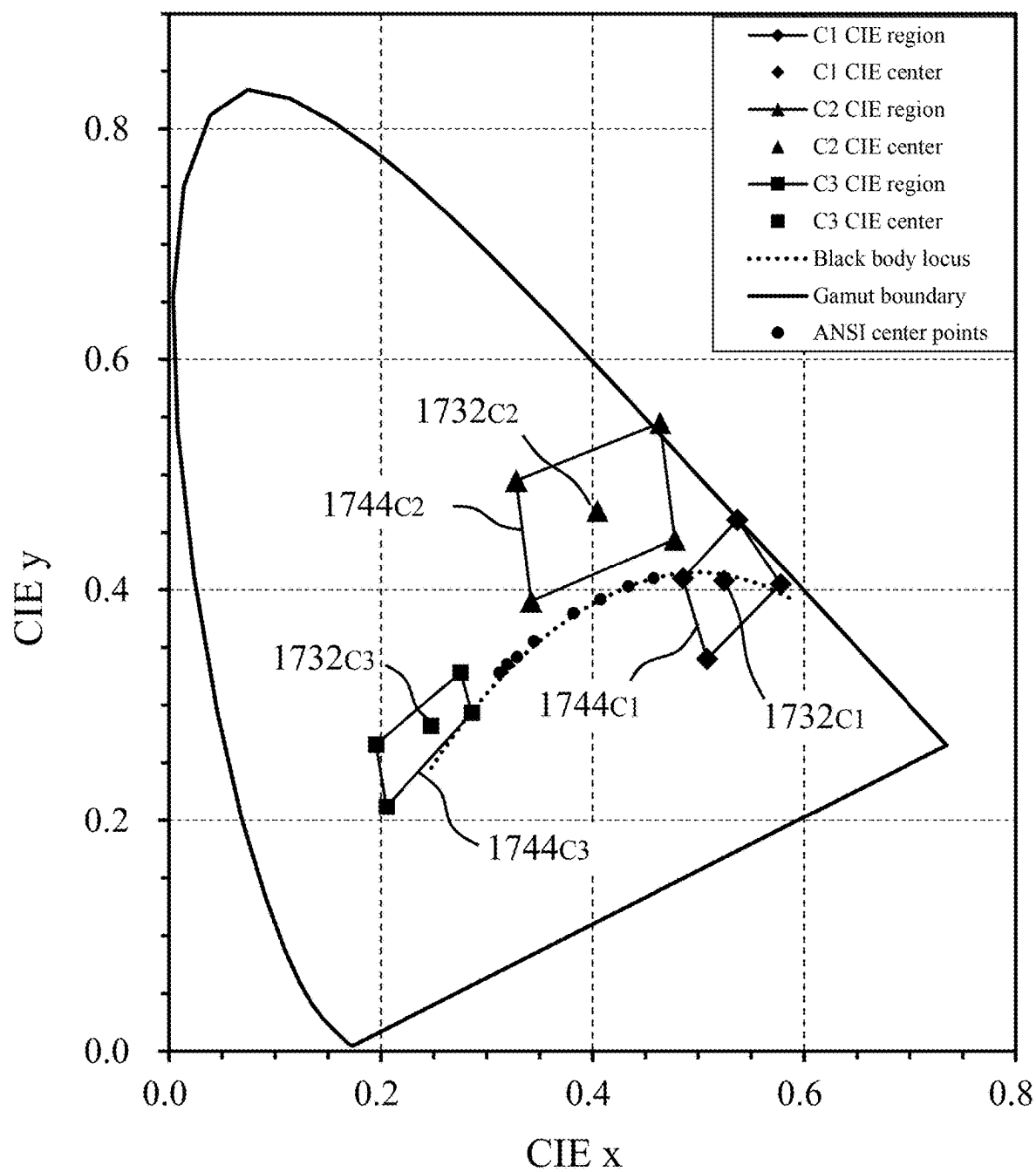


FIG. 17D

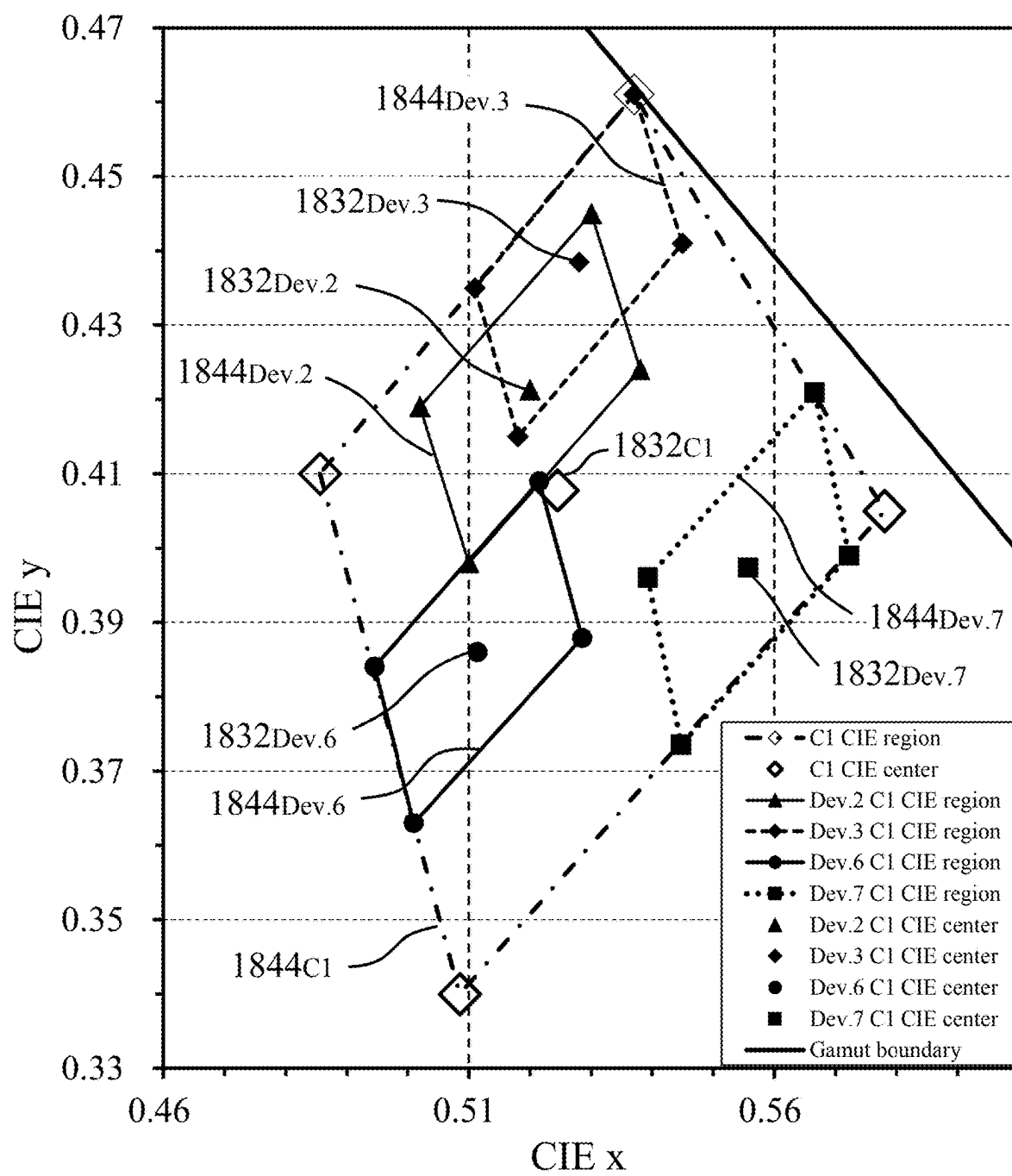


FIG. 18A

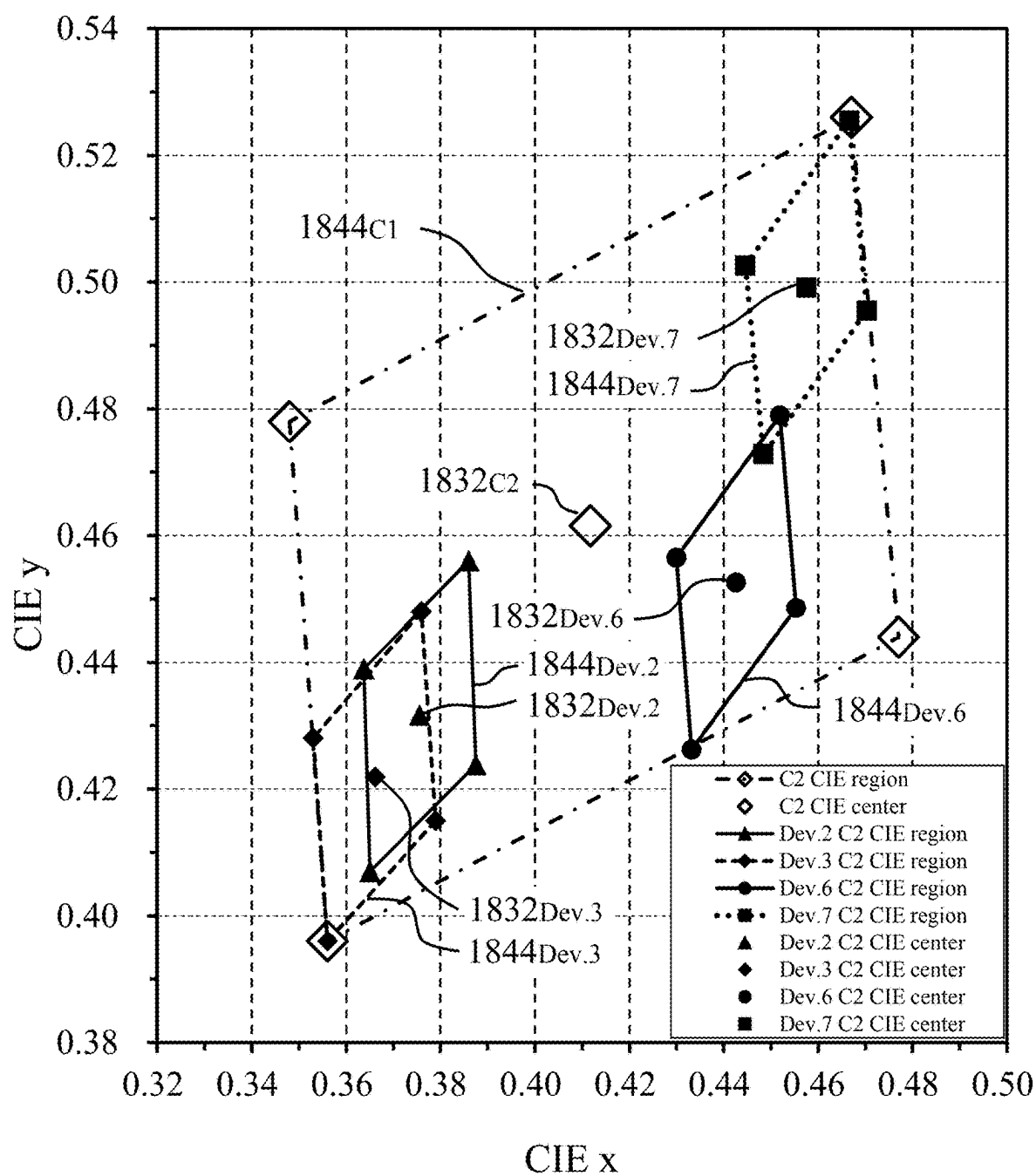


FIG. 18B

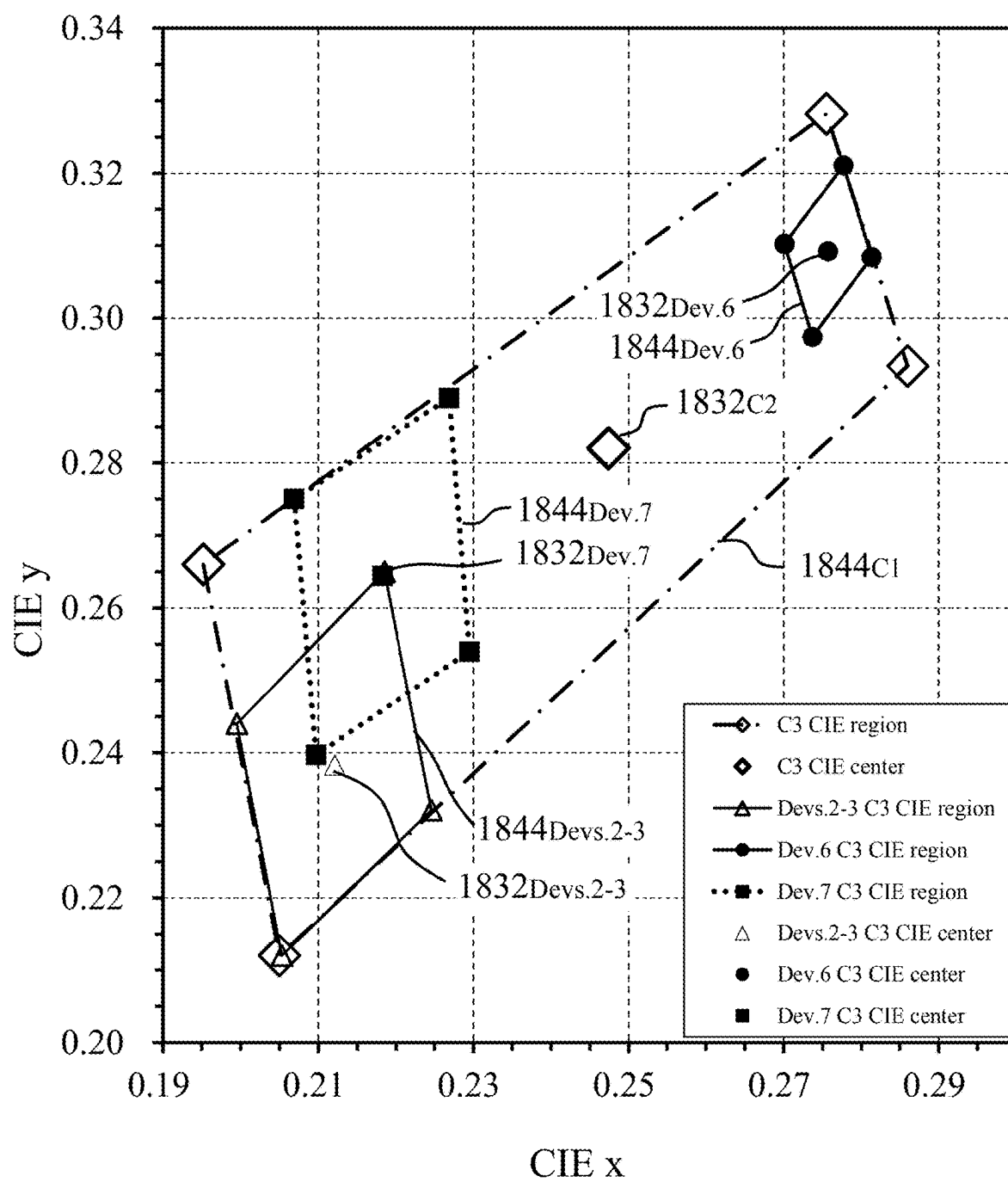


FIG. 18C

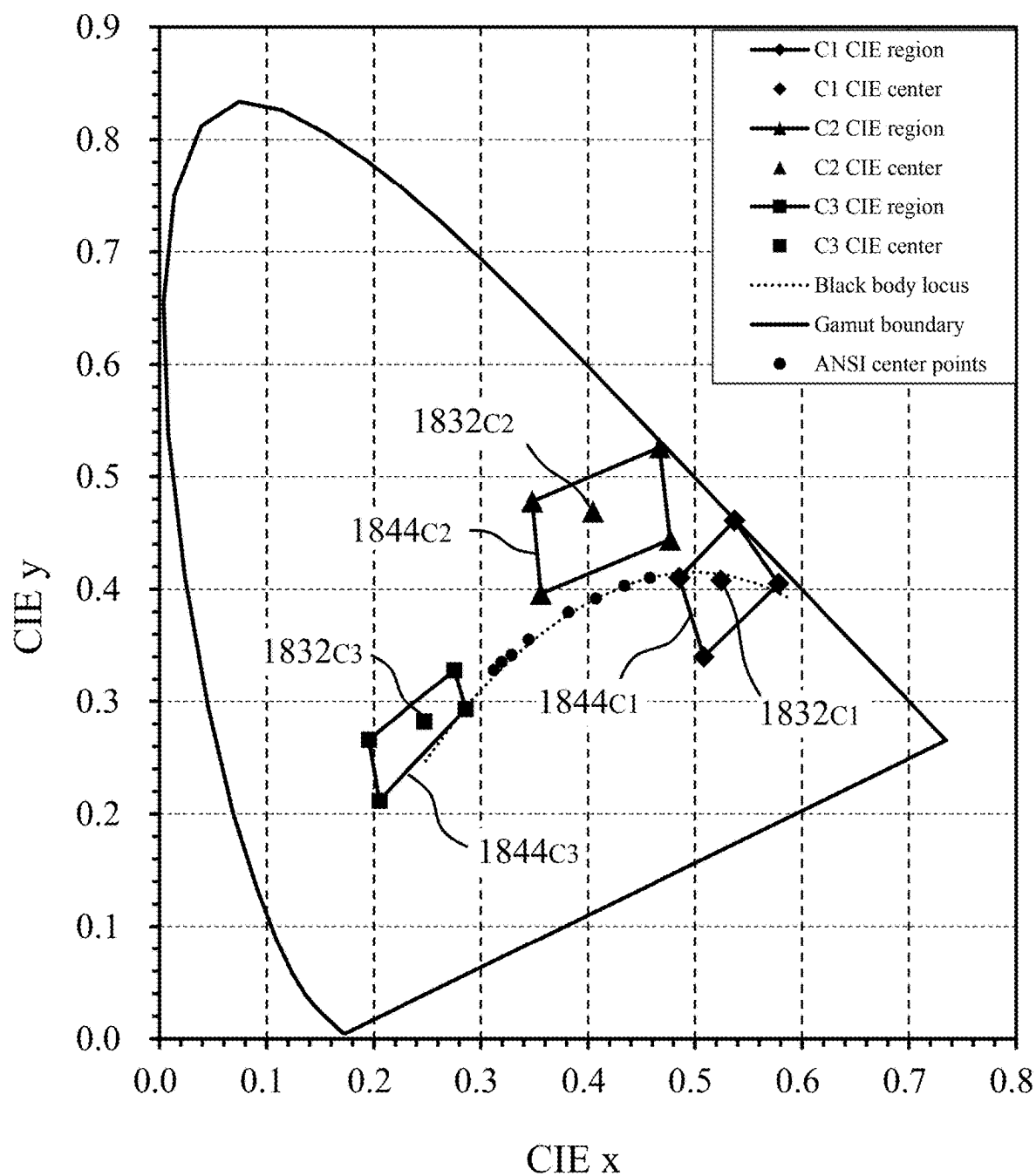


FIG. 18D

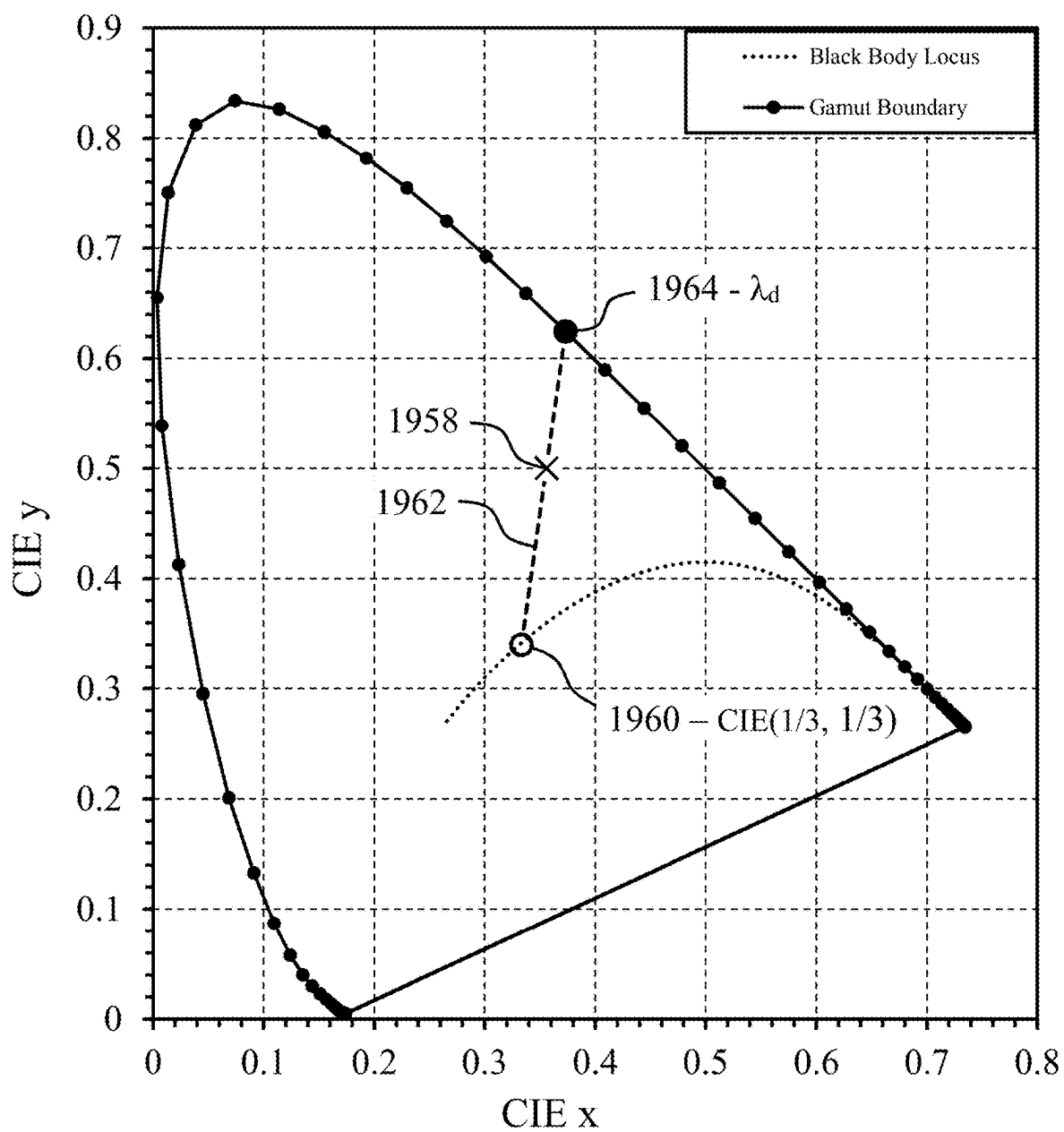


FIG. 19

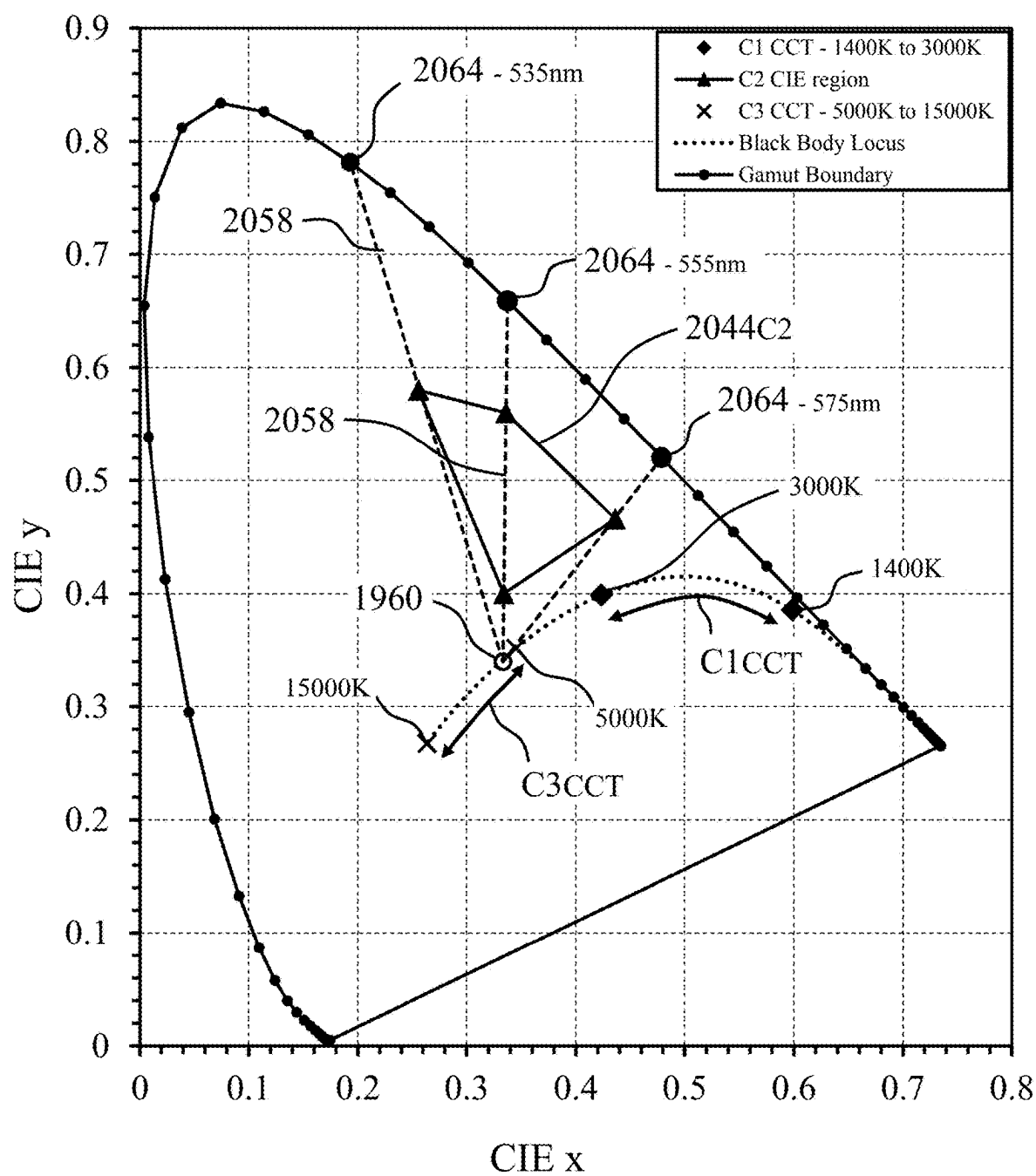


FIG. 20

COLOR-TEMPERATURE-TUNABLE LIGHT EMITTING DEVICES

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is a bypass continuation application of International PCT Application Number PCT/US2023/081478 filed Nov. 29, 2023, entitled “Color-Temperature-Tunable Light Emitting Devices”, which claims the benefit of priority to: i) International Patent Application Number PCT/CN2022/135022, filed Nov. 29, 2022, entitled “Color-Temperature-Tunable Lighting Devices”, ii) U.S. Provisional Patent Application No. 63/510,341 filed Jun. 26, 2023, entitled “Color-Temperature-Tunable Lighting Devices”, and iii) U.S. Utility patent application Ser. No. 18/361,814 filed Jul. 28, 2023, entitled “Color-Temperature-Tunable Lighting Devices”, all of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

[0002] Embodiments of the invention relate to color-temperature-tunable lighting devices (light emitting devices) that for generating light with a Correlated Color Temperature (CCT) in a range from 1500K to 7500K. More particularly, though not exclusively, embodiments concern multi-color LED (Light Emitting Diode) packaged devices and multi-LED packaging arrangements.

BACKGROUND OF THE INVENTION

[0003] An example of a known color-tunable multi-LED packaged lighting device is shown in FIGS. 1A and 1B, in which FIG. 1A shows a top view and FIG. 1B shows a sectional side view through A-A of the multi-LED packaged lighting device. The known color-tunable lighting device 1 comprises a lead frame 2 for providing power to the red, green, and blue direct-emitting LED chips 3R, 3G, 3B. A housing 4 is molded onto the lead frame and comprises a single cavity (recess) 5 (e.g., circular in shape). The red, green, and blue LED direct-emitting LED chips 3R, 3G, 3B are mounted on the floor of cavity 5 and electrically connected to the lead frame 2. To protect the LED chips 3R, 3G, 3B from the external environment, the cavity 5 is typically filled with a light-transmissive encapsulant 6 such as a silicone material. Portions of the lead frame 2 extend laterally to the outside edges of the housing 4 and form respective electrical terminals 7, 8 along opposing edges and base of the package allowing electrical power to be independently (individually) applied to the anode and cathode of each of the red, green, and blue direct-emitting LED chips 3R, 3G, 3B.

[0004] Phosphor Converted (PC) LEDs are to be contrasted with Direct-Emitting Color LED chips, wherein PC LEDs comprise a Direct-Emitting blue LED chip and a photoluminescence material, typically a phosphor material, that converts a portion blue excitation light generated by the LED chip, with the remainder of the blue light contributing to the final emission product. The phosphor material may be incorporated in the light-transmissive encapsulant in the LED package.

[0005] A disadvantage of color-tunable multi-LED packaged lighting devices based on Direct-Emitting Color LEDs, however, is that since they are based on different semiconductor material systems each Color LED has different char-

acteristics such as thermal stability, ageing characteristics, drive requirements etc. As a result of these different characteristics, the light output of Red, Green and Blue LEDs will change differently from one another with temperature and time. The color composition of light generated by an RGB system based on Color LEDs will consequently change with temperature and time and such RGB systems may employ complex drive circuitry to compensate for these differing characteristics which can lead to additional cost during manufacture and maintenance.

[0006] Prior art color-temperature-tunable lighting devices typically comprise two color temperature LEDs, one warm white (e.g., CCTs from 1800K to 3000K) and one cool white (e.g., CCT from 4000K to 6500K). While such devices can generate warm light and cool light that corresponds to the black body locus, for color temperatures in between, in which light generated by the devices is a combination of warm white and cool white, said light deviates from the black body locus. A further limitation of the known color-temperature-tunable lighting devices is that their emission spectrum, intensity versus wavelength, does not resemble the spectrum of a black body radiator.

[0007] The present invention intends to address and/or overcome the limitations discussed above by presenting new designs and methods not hitherto contemplated nor possible by known constructions. More particularly, although not exclusively, embodiments of the invention concern improvements relating to increasing the luminous efficacy of color-tunable multi-LED packaged lighting devices.

SUMMARY OF THE INVENTION

[0008] The invention relates generally to color-temperature-tunable lighting devices comprising multi-color LEDs that can generate light of a color temperature that is tunable in a range of CCTs from 1800K to 6500K with a chromaticity/color temperature lying on or near the black body locus. Advantageously, lighting devices according to the invention generate light with a spectrum, intensity versus wavelength, that closely resembles (matches) the spectrum of a black body radiator. The characteristic of being able to generate light with a spectrum that closely matches a black body radiator over a full range of color temperatures is desirable for circadian lighting in which color tuning is used to mimic the color temperature cycle of sunlight from sunrise to sunset to match the human circadian rhythm.

[0009] Aspects of the invention relate to color-temperature-tunable light emitting (lighting) devices comprising a plurality of LEDs that generate light of two, three, or four different chromaticity/CCTs. To improve the luminous efficacy of the lighting device, at least one of the plurality of LEDs, for example, the LED that generates light with the lowest color temperature, may comprise a narrowband red phosphor such as a narrowband red fluoride phosphor; for example, $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and/or $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0010] In this specification “chromaticity” of light, “color of light”, and “color point” of light may be used interchangeably and refer to the chromaticity/color of light as represented by chromaticity coordinates CIE x, y on a CIE 1931 chromaticity diagram.

[0011] According to an aspect of the present invention, there is provided a lighting (i.e. light emitting) device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED; a second LED;

a third LED, wherein the first LED is for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.498, 0.422), (0.522, 0.370), (0.562, 0.410), and (0.537, 0.461); the second LED is for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.330, 0.480), (0.342, 0.385), (0.392, 0.420), and (0.378, 0.514); and the third LED is for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.196, 0.261), (0.206, 0.212), (0.234, 0.243), and (0.226, 0.290). In this specification “a chromaticity region”—CIE region—is as an area on the CIE 1931 chromaticity diagram and may be defined as a quadrilateral area composed of four chromaticity coordinates CIE x, y connected by four straight lines. The chromaticity of light for the chromaticity region (CIE region) can comprise any chromaticity value (CIE x, y) within the quadrilateral or lying on the straight line sides of the quadrilateral. Light generated by the device comprises a combination of light generated by the first, second, and third LEDs and wherein a chromaticity of light generatable by the device is tunable by controlling power to the first, second and third LEDs. In terms of CCT, light of the first chromaticity may have a CCT from 1720K to 2350K and light of the second chromaticity may have a CCT from 4000K to 5580K. Such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, from 80 to 97 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT.

[0012] It may be that a chromaticity of light generated by the device are within 0.003 Δuv of the black body locus for CCTs from 1800K to 6500K. Δuv (Delta uv) is a metric that quantifies how close light of a given color temperature is to the black body locus. As is known, Δuv is the Euclidean difference of chromaticity coordinate uv between a test light source to the closest point on the black body locus and is defined in ANSI_NEMA_ANSI C78.377-2008: American National Standard for electric lamps—Specifications for the Chromaticity of Solid State Lighting Products. Δuv is on the 1976 CIE u, v chromaticity diagram, a measure of the distance of the color point of light of a given CCT (Correlated Color Temperature) from the black body locus (Planckian locus of black body radiation) along the iso-CCT line (Lines of Constant Color Temperature). A positive Δuv value indicates that the color point is above the black body locus (i.e., on a 1931 CIE x, y chromaticity diagram CIE y is greater than the CIE y value of the black body locus) with a yellowish/greenish color shift from the black body locus. A negative value the color point is below the black body locus (i.e., on a 1931 CIE x, y chromaticity diagram CIE y is less than the CIE y value of the black body locus) with a pinkish color shift from the black body locus.

[0013] It may be that the first chromaticity region is 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses) centered on a chromaticity (0.530, 0.415); the second chromaticity region is 6 SDCM centered on a chromaticity (0.360, 0.450); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.216, 0.251). Again, such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, from 80 to 97 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003).

[0014] In an embodiment, it may be that the first chromaticity region is defined by chromaticity coordinates (0.502, 0.419), (0.510, 0.398), (0.538, 0.424), and (0.530, 0.445); the second chromaticity region is defined by chromaticity coordinates (0.364, 0.439), (0.365, 0.407), (0.388, 0.424), and (0.386, 0.456); and the third chromaticity region is defined by chromaticity coordinates (0.200, 0.244), (0.205, 0.212), (0.225, 0.232), and (0.219, 0.265). Such a lighting device may include a narrowband red fluoride phosphor; for example, $K_2SiF_6:Mn^{4+}$, $K_2GeF_6:Mn^{4+}$, and/or $K_2TiF_6:Mn^{4+}$ and can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 80 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically, Δuv is less than 0.003). In terms of CCT, light of the first chromaticity may have a CCT from 1970K to 2270K, and light of the second chromaticity may have a CCT from 4100K to 4750K.

[0015] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.520, 0.421); the second chromaticity region is 6 SDCM centered on a chromaticity (0.376, 0.432); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.212, 0.238). Again, such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 80 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003) and may include a narrowband red fluoride phosphor.

[0016] In another embodiment, it may be that the first chromaticity region is defined by chromaticity coordinates (0.511, 0.435), (0.518, 0.415), (0.545, 0.441), and (0.537, 0.461); the second chromaticity region is defined by chromaticity coordinates (0.353, 0.428), (0.356, 0.396), (0.379, 0.415), and (0.376, 0.448); and the third chromaticity region is defined by chromaticity coordinates (0.200, 0.244), (0.205, 0.212), (0.225, 0.232), and (0.219, 0.265). Such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 80 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003). In terms of CCT, light of the first chromaticity may have a CCT from 2200K to 2280K, and light of the second chromaticity may have a CCT from 4250K to 4980K.

[0017] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.528, 0.439); the second chromaticity region is 6 SDCM centered on a chromaticity (0.366, 0.422); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.212, 0.238). Again, such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 80 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003).

[0018] In a further embodiment, it may be that the first chromaticity region is defined by chromaticity coordinates (0.525, 0.405), (0.531, 0.383), (0.559, 0.407), and (0.553, 0.429); the second chromaticity region is defined by chromaticity coordinates (0.335, 0.445), (0.339, 0.415), (0.361, 0.436), and (0.357, 0.467); and the third chromaticity region is defined by chromaticity coordinates (0.200, 0.244), (0.205, 0.212), (0.225, 0.232), and (0.219, 0.265). Such a lighting device may include a narrowband red fluoride

phosphor; for example, $K_2SiF_6:Mn^{4+}$, $K_2GeF_6:Mn^{4+}$, and/or $K_2TiF_6:Mn^{4+}$ and can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 90 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003). In terms of CCT, light of the first chromaticity may have a CCT from 1700K to 1970K and light of the second chromaticity may have a CCT from 4800K to 5460K.

[0019] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.5412, 0.406); the second chromaticity region is 6 SDCM centered on a chromaticity (0.348, 0.441); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.212, 0.238). Again, such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 90 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003) and may include a narrowband red fluoride phosphor.

[0020] In a yet further embodiment, it may be that the first chromaticity region is defined by chromaticity coordinates (0.515, 0.399), (0.522, 0.378), (0.549, 0.404), and (0.542, 0.425); the second chromaticity region is defined by chromaticity coordinates (0.356, 0.497), (0.358, 0.465), (0.380, 0.483), and (0.378, 0.514); and the third chromaticity region is defined by chromaticity coordinates (0.217, 0.281), (0.220, 0.267), (0.228, 0.276), and (0.226, 0.290). Such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 97 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003). In terms of CCT, light of the first chromaticity may have a CCT from 1750K to 2020K and light of the second chromaticity may have a CCT from 4500K to 5050K.

[0021] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.532, 0.4012); the second chromaticity region is 6 SDCM centered on a chromaticity (0.368, 0.490); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.223, 0.278). Again, such a lighting device can generate light with a CCT from 1800K to 6500K with a General Color Rendering Index, CRI Ra, of about 97 and have a chromaticity corresponding to, or close to, the black body locus or Standard Illuminant of the same CCT (typically Δuv is less than 0.003).

[0022] The light emitting (lighting) device, according to various embodiments, may comprise at least two first LEDs. It may be beneficial to utilize at least two first LEDs that generate light of the first chromaticity rather than using a single first LED to reduce a need to overdrive the first LED or underdrive the second and third LEDs; that is it can equalize the maximum drive current to the first, second, and third LEDs. This is because light of the first chromaticity corresponds to light with the lowest color temperature and, as a result, the luminous efficacy of the first LED(s) is lower than that of the second and third LEDs.

[0023] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0024] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED that generates light with a fourth chromaticity in a fourth chromaticity region.

[0025] The fourth chromaticity region may be between the first and second chromaticity regions. For example, the fourth chromaticity region may be defined by chromaticity coordinates (0.420, 0.468), (0.418, 0.383), (0.475, 0.430), and (0.479, 0.514). In terms of CCT, light of the fourth chromaticity may have a CCT from 2620K to 3750K.

[0026] Alternatively, the fourth chromaticity region may be between the second and third chromaticity regions. For example, the fourth chromaticity region may be defined by chromaticity coordinates (0.249, 0.362), (0.265, 0.283), (0.309, 0.335), and (0.295, 0.414). In terms of CCT, light of the fourth chromaticity may have a CCT from 6650K to 12500K.

[0027] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0028] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0029] It may be that the array is a two dimensional (2D) array.

[0030] It may be that the array is a 2x2 (two-by-two) array.

[0031] The array of cavities may be a square array.

[0032] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0033] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0034] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0035] Lighting devices according to various embodiments can generate light of a chromaticity that may be within 0.003 Δuv of the black body locus for CCTs from 1800K to 6500K.

[0036] The at least one of the first, second, or third LEDs may comprise a broadband LED chip that generates light with a dominant wavelength from 420 nm to 480 nm with a FWHM of at about 30 nm to about 80 nm. A broadband LED can be beneficial for generating light having a spectrum that resembles sunlight.

[0037] The light generated by the device may have a selected Color Temperature and an intensity versus wavelength spectrum, which over a wavelength range 460 nm to 650 nm, a maximum percentage deviation between the normalized intensity of light generated by the lighting

device and the normalized intensity of light of the spectrum of a black body radiator or standard illuminant of the same Color Temperature is at least one of less than 30%, less than 20%, and/or less than 10%, said intensity versus wavelength spectra of the light and black body radiator being normalized to the same luminance. The characteristic of generating light with a spectrum that matches a black body radiator is desirable for circadian lighting in which color tuning is used to mimic the color temperature cycle of sunlight from sunrise to sunset to match the human circadian rhythm.

[0038] The at least one of the first, second, or third LEDs may comprise a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0039] Light emitting (lighting) devices according to embodiments may further comprise a package comprising a lead frame; and a housing comprising a first recess having the first LED, a second recess having the second LED, and a third recess having the third LED; and wherein the lead frame may comprise a first (e.g., cathode) electrode common to each recess and a respective second (e.g., anode) electrode to each recess.

[0040] It may be that each recess comprises a first terminal connected to the first electrode and a second terminal connected to the second electrode and wherein the first and second terminals for each recess are located on opposing edges of the housing across from one another. For instance, it could be said that the first and second terminals for each recess are located on opposing edges of the housing and face one another (or positioned facing one another).

[0041] It may be that the first (e.g., cathode) terminals are common to a recess.

[0042] The lighting device may comprise respective first and second terminals on opposite edges of housing.

[0043] In another aspect, the present invention contemplates a lighting device comprising: a first LED for generating light of a first CCT from about 1700K to about 3500K; and a second LED for generating light of a second CCT from about 4000K to about 5600K; wherein the first LED comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0044] The light emitting (lighting) device may comprise a third LED for generating light with a chromaticity in a chromaticity region defined by chromaticity coordinates (0.196, 0.261), (0.206, 0.212), (0.234, 0.243), and (0.226, 0.290).

[0045] In another aspect, the present invention envisages a light emitting (lighting) device comprising: a first, second, and third LED for generating light with different CIE color points, wherein the device can generate white light of different CCTs from 1800K to 8000K by controlling the light output of the three LEDs, wherein the chromaticity of white light generated by the device is along the black body locus, and wherein at least one of the LEDs comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0046] The chromaticity of light generated by the device may be within 0.003 $\Delta u v$ of the black body locus or Standard Illuminant for CCTs from 1800K to 6500K.

[0047] The light emitting (lighting) device, according to various embodiments, may comprise at least two first LEDs. It may be beneficial to utilize at least two first LEDs for generating light of the first chromaticity rather than using a

single first LED to reduce a need to overdrive the first LED or underdrive the second and third LEDs; that is this can equalize the maximum drive current to the first, second, and third LEDs. This is because light of the first chromaticity corresponds to light with the lowest color temperature and as a result the luminous efficacy of the first LED(s) is lower than that of the second and third LEDs.

[0048] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0049] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0050] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0051] It may be that the array is a two dimensional (2D) array.

[0052] It may be that the array is a 2x2 (two-by-two) array.

[0053] The array of cavities may be a square array.

[0054] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0055] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0056] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0057] In another aspect, the present invention comprehends a lighting (i.e. light emitting) device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED; a second LED; a third LED, and a fourth LED, wherein the first LED for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.498, 0.422), (0.522, 0.370), (0.562, 0.410), and (0.537, 0.461); the second LED for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.390, 0.488), (0.398, 0.403), (0.450, 0.450), and (0.442, 0.534); the third LED for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.277, 0.418), (0.283, 0.333), (0.334, 0.380), and (0.327, 0.464); and the fourth LED for generating light of a fourth chromaticity in a fourth chromaticity region defined by chromaticity coordinates (0.196, 0.261), (0.206, 0.212), (0.234, 0.243), and (0.226, 0.290). Light generated by the device comprises a combination of light generated by the first, second, third, and fourth LEDs and wherein a chromaticity of light generated by the device is

tunable by controlling power to the first, second, third, and fourth LEDs. In terms of CCT, light of the first chromaticity may have a CCT from 1700K to 1970K, light of the second chromaticity may have a CCT from 4800K to 5460K, and light of the third chromaticity may have a CCT from 4800K to 5460K.

[0058] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.530, 0.415); the second chromaticity region is 6 SDCM centered on a chromaticity (0.420, 0.470); the third chromaticity region is 6 SDCM centered on a chromaticity (0.305, 0.400); and the fourth chromaticity region is 6 SDCM centered on a chromaticity (0.216, 0.251).

[0059] In another aspect, the present invention encompasses a light emitting (lighting) device comprising: a circuit board and a plurality of lighting devices as defined herein.

[0060] The circuit board may comprise a flexible circuit board.

[0061] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0062] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0063] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0064] It may be that the array is a two dimensional (2D) array.

[0065] It may be that the array is a 2x2 (two-by-two) array.

[0066] The array of cavities may be a square array.

[0067] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0068] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0069] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0070] In another aspect, the present invention comprehends a light emitting (lighting) device comprising: at least two first LEDs for generating light of a first chromaticity; a second LED for generating light of a second chromaticity; and a third LED for generating light of a third chromaticity; wherein the first chromaticity has the lowest color temperature. Light generated by the device comprises a combination of light generated by the first, second, and third LEDs and wherein a chromaticity of light generated by the device is tunable by controlling power to the first, second and third

LEDs. It may be beneficial to utilize at least two first LEDs for generating light of the lowest color temperature rather than using a single first LED to reduce a need to overdrive the first LED or underdrive the second and third LEDs; that is, it can equalize the maximum drive current to the first, second, and third LEDs. This is because light of the first chromaticity corresponds to light with the lowest color temperature and as a result the luminous efficacy of the first LED(s) is lower than that of the second and third LEDs.

[0071] It will be understood that any of the first, second, or third chromaticity described herein may be defined by any of the chromaticity regions defined herein.

[0072] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0073] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0074] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0075] It may be that the array is a two dimensional (2D) array.

[0076] It may be that the array is a 2x2 (two-by-two) array.

[0077] The array of cavities may be a square array.

[0078] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0079] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0080] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0081] In another aspect, the present invention contemplates an LED package comprising: a lead frame; and a housing comprising a first recess for receiving a first LED, a second recess for receiving a second LED, a third recess for receiving a third LED, and a fourth recess for receiving a fourth LED; wherein the lead frame comprises a respective first (e.g., cathode) electrode to each recess and a respective second (e.g., anode) electrode to each recess.

[0082] It may be that each recess comprises a first terminal connected to the first electrode and a second terminal connected to the second electrode and wherein the first and second terminals for each recess are located on opposing edges of the housing across from one another. For instance, it could be said that the first and second terminals for each recess are located on opposing edges of the housing and face one another (or positioned facing one another).

[0083] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0084] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0085] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0086] It may be that the array is a two dimensional (2D) array.

[0087] It may be that the array is a 2x2 (two-by-two) array.

[0088] The array of cavities may be a square array.

[0089] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0090] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0091] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0092] According to further aspect of the invention, there is contemplated a light emitting (lighting) device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED; a second LED; a third LED, wherein the first LED is for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.486, 0.410), (0.509, 0.340), (0.578, 0.405), and (0.537, 0.461); the second LED is for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.328, 0.495), (0.342, 0.390), (0.478, 0.444), and (0.464, 0.545); and the third LED is for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.195, 0.266), (0.205, 0.212), (0.286, 0.293), and (0.276, 0.328). It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.525, 0.408); the second chromaticity region is 6 SDCM centered on a chromaticity (0.405, 0.469); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.247, 0.282).

[0093] It may be that the second chromaticity region is defined by chromaticity coordinates (0.348, 0.478), (0.356, 0.396), (0.477, 0.444), and (0.467, 0.526).

[0094] In embodiments, the first chromaticity region may be defined by chromaticity coordinates (0.495, 0.384), (0.509, 0.340), (0.576, 0.402), and (0.557, 0.442); the second chromaticity region may be defined by chromaticity coordinates (0.427, 0.484), (0.433, 0.427), (0.474, 0.468), and (0.467, 0.526); and the third chromaticity region may be

defined by chromaticity coordinates (0.207, 0.275), (0.210, 0.240), (0.284, 0.294), and (0.279, 0.326). It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.536, 0.394); the second chromaticity region is 6 SDCM centered on a chromaticity (0.450, 0.476); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.249, 0.286).

[0095] It may be that the first chromaticity region is defined by chromaticity coordinates (0.495, 0.384), (0.501, 0.363), (0.529, 0.388), and (0.522, 0.409); the second chromaticity region is defined by chromaticity coordinates (0.430, 0.457), (0.433, 0.426), (0.455, 0.449), and (0.452, 0.479); and the third chromaticity region is defined by chromaticity coordinates (0.270, 0.310), (0.274, 0.297), (0.281, 0.308), and (0.278, 0.321). It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.511, 0.386); the second chromaticity region is 6 SDCM centered on a chromaticity (0.443, 0.453); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.276, 0.309).

[0096] In an embodiment the first chromaticity region be defined by chromaticity coordinates (0.539, 0.396), (0.545, 0.374), (0.572, 0.399), and (0.567, 0.421); the second chromaticity region be defined by chromaticity coordinates (0.445, 0.503), (0.448, 0.473), (0.470, 0.496), and (0.467, 0.526); and the third chromaticity region is defined by chromaticity coordinates (0.207, 0.275), (0.210, 0.240), (0.230, 0.254), and (0.227, 0.289). It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.556, 0.397); the second chromaticity region is 6 SDCM centered on a chromaticity (0.458, 0.499); and the third chromaticity region is 6 SDCM centered on a chromaticity (0.218, 0.264).

[0097] In various embodiments, the chromaticity of light generatable by the device may be within 0.004 Δuv or 0.003 Δuv of the black body locus for a CCT in a range from 1800K to 6500K.

[0098] It may be that the device is for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a chromaticity that is within 0.004 Δuv of the black body locus.

[0099] In embodiments, at least one of the first, second, or third LEDs may comprise a LED chip for generating blue light with a FWHM of at about 30 nm to about 80 nm, for example a broadband LED chip.

[0100] In various embodiments, light generated light by the device can have a selected Color Temperature and an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, a maximum percentage deviation between the intensity of light generated by the lighting device and the intensity of light of the spectrum of a black body radiator or standard illuminant (for instance of the same Color Temperature) is at least one of less than 30%, less than 20%, or less than 10%. The intensity versus wavelength spectra of the light and black body radiator may be normalized to the same relative luminance.

[0101] In embodiments, the light emitting (lighting) device is for generating light with a CCT that is tunable within a range from 1800K to 6500K and has a chromaticity that is within 3 SDCM of the black body locus.

[0102] In embodiments, at least one of the first, second, or third LEDs comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0103] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0104] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0105] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0106] It may be that the array is a two dimension (2D) array.

[0107] It may be that the array is a 2x2 (two-by-two) array.

[0108] The array of cavities may be a square array.

[0109] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0110] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0111] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

[0112] The light emitting (lighting) device may further comprise a package comprising a lead frame; and a housing comprising a first recess having the first LED, a second recess having the second LED, and a third recess having the third LED; and wherein the lead frame comprises a first (e.g., cathode) electrode that is common to each recess and a respective second (e.g., anode) electrode to each recess.

[0113] Alternatively, the light emitting (lighting) device may further comprise a package comprising a lead frame; and a housing comprising a first recess having the first LED, a second recess having the second LED, and a third recess having the third LED; and wherein the lead frame comprises a respective first (e.g., cathode) electrode to each recess and a respective second (e.g., anode) electrode to each recess.

[0114] According to a further aspect, there is contemplated a light emitting (lighting) device comprises: a first LED, a second LED, and a third LED that for generating light with different CIE color points, and wherein at least one of the first, second, or third LEDs comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0115] According to another aspect, there is provided a light emitting (lighting) device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED; a second LED; a third LED, wherein the first LED is for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.495, 0.384), (0.501, 0.363), (0.529, 0.388), and

(0.522, 0.409); the second LED is for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.430, 0.457), (0.433, 0.426), (0.455, 0.449), and (0.452, 0.479); and the third LED is for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.270, 0.310), (0.274, 0.297), (0.281, 0.308), and (0.278, 0.321); and wherein at least one of the first, second, or third LEDs comprises an LED chip for generating blue light having a FWHM of about 30 nm to about 80 nm.

[0116] Advantageously, the light emitting (lighting) device is for generating light of a CCT within a range 1800 k to 6500K and has an intensity versus wavelength spectrum that closely resembles sunlight/daylight. It may be that the device is for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K. In embodiments, the device is for generating light with an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, a maximum percentage deviation between the intensity of light generated by the lighting device and the intensity of light of the spectrum of a black body radiator or standard illuminant is at least one of less than 30%, less than 20%, or less than 10%. Preferably, the lighting device is for generating light with a CRI of at least 95.

[0117] The light emitting (lighting) device may be for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a chromaticity that is within 5 SDCM of the black body locus.

[0118] The light emitting (lighting) device of Claim 15, wherein the device has a luminous efficacy of at least 95 lm/W.

[0119] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0120] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0121] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0122] It may be that the array is a two dimension (2D) array.

[0123] It may be that the array is a 2x2 (two-by-two) array.

[0124] The array of cavities may be a square array.

[0125] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0126] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0127] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDS of the same chromaticity in series, for instance.

[0128] According to further aspect, there is provided a light emitting (lighting) device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED; a second LED; a third LED, wherein the first LED is for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.539, 0.396), (0.545, 0.374), (0.572, 0.399), and (0.567, 0.421); the second LED is for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.445, 0.503), (0.448, 0.473), (0.470, 0.496), and (0.467, 0.526); and the third LED is for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.207, 0.275), (0.210, 0.240), (0.230, 0.254), and (0.227, 0.289); and wherein at least one of the first, second, or third LEDs comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$. Inclusion of $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$ can increase luminous efficacy of the lighting device and the device may have a luminous efficacy of at least 135 lm/W.

[0129] The light emitting (lighting) device may be for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a chromaticity that is within 5 SDCM of the black body locus.

[0130] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0131] In at least some embodiments disclosed herein, the multi-cavity lead frame may comprise a fourth independent cavity for a/the fourth LED.

[0132] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0133] It may be that the array is a two dimension (2D) array.

[0134] It may be that the array is a 2x2 (two-by-two) array.

[0135] The array of cavities may be a square array.

[0136] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0137] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0138] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear

arrangements for connecting the LEDS of the same chromaticity in series, for instance.

[0139] In at least some embodiments, the chromaticity regions can be defined by a range of dominant wavelengths or a range of color temperatures. Accordingly, the present invention envisages a light emitting device comprising: a multi-cavity lead frame package comprising at least three independent cavities; a first LED for generating “warm white” light; a second LED for generating light with a dominant wavelength from 535 nm to 575 nm; and a third LED for generating “cool white” light. Use of an LED with a dominant wavelength from 535 nm to 575 nm has the benefit of increasing luminous efficacy of the light emitting device since such a dominant wavelength is close to, or at, the maxima of the photopic response curve at 555 nm.

[0140] It may be that the first LED is for generating “warm white” light with a CCT from 1400K to 3000K, optionally 1400K to 2500K, optionally 1400K to 2000K.

[0141] It may be that the third LED is for generating “cool white” light with a CCT from 5000K to 15000K, optionally 10000K to 15000K.

[0142] It may be that the second LED may be for generating light with a dominant wavelength from 545 nm to 565 nm, optionally from 550 nm to 560 nm, optionally about 555 nm.

[0143] The light generated by the first LED and third LED may have a chromaticity that is within 0.003 Δuv of the black body locus.

[0144] The second LED may be for generating light of a chromaticity in a chromaticity region defined by chromaticity coordinates (0.256, 0.580), (0.334, 0.400), (0.436, 0.466), and (0.336, 0.560);

[0145] The light generated by the second LED may be at least 0.007 away from the black body locus.

[0146] It may be that the device is for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a chromaticity that is within 0.004 Δuv or 0.003 Δuv of the black body locus.

[0147] In embodiments, at least one of the first, second, or third LEDs may comprise a LED chip for generating blue light with a FWHM of at about 30 nm to about 80 nm, for example a broadband LED chip.

[0148] In various embodiments, light generated light by the device can have a selected Color Temperature and an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, a maximum percentage deviation between the intensity of light generated by the lighting device and the intensity of light of the spectrum of a black body radiator or standard illuminant (for instance of the same Color Temperature) is at least one of less than 30%, less than 20%, or less than 10%. The intensity versus wavelength spectra of the light and black body radiator may be normalized to the same relative luminance.

[0149] In embodiments, the light emitting (lighting) device is for generating light with a CCT that is tunable within a range from 1800K to 6500K and has a chromaticity that is within 3SDCM of the black body locus.

[0150] In embodiments, at least one of the first, second, or third LEDs comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6\text{:Mn}^{4+}$, $\text{K}_2\text{GeF}_6\text{:Mn}^{4+}$, and $\text{K}_2\text{TiF}_6\text{:Mn}^{4+}$.

[0151] According to further aspects of the invention, there are contemplated lighting devices comprising four LEDs for generating light of four different chromaticity. In an embodi-

ment, a lighting device comprises: a first LED is for generating light of a first chromaticity in a first chromaticity region defined by chromaticity coordinates (0.498, 0.422), (0.522, 0.370), (0.562, 0.410), and (0.537, 0.461); a second LED is for generating light of a second chromaticity in a second chromaticity region defined by chromaticity coordinates (0.390, 0.488), (0.398, 0.403), (0.450, 0.450), and (0.442, 0.534); a third LED is for generating light of a third chromaticity in a third chromaticity region defined by chromaticity coordinates (0.277, 0.418), (0.283, 0.333), (0.334, 0.380), and (0.327, 0.464); and a fourth LED is for generating light of a fourth chromaticity in a fourth chromaticity region defined by chromaticity coordinates (0.196, 0.261), (0.206, 0.212), (0.234, 0.243), and (0.226, 0.290).

[0152] It may be that the first chromaticity region is 6 SDCM centered on a chromaticity (0.530, 0.415); the second chromaticity region is 6 SDCM centered on a chromaticity (0.420, 0.470); the third chromaticity region is 6 SDCM centered on a chromaticity (0.305, 0.400); and the fourth chromaticity region is 6 SDCM centered on a chromaticity (0.216, 0.251).

[0153] Lighting devices (light emitting devices), according to various embodiments, may further comprise a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

[0154] It may be that multi-cavity lead frame comprises: a housing comprising an array of cavities comprising a first cavity for the first LED, a second cavity for the second LED, a third cavity for the third LED, and a fourth cavity for the fourth LED; and a lead frame comprising respective anode and cathode regions for each cavity, wherein each anode region comprises an anode electrode on a floor of the cavity and an anode terminal for providing power to the anode electrode, and each cathode region comprises a cathode electrode on the floor of the cavity and a cathode terminal for providing power to the cathode electrode, and wherein the respective anode and cathode terminals for each cavity are located along opposing edges of the housing and are aligned with each other.

[0155] It may be that the array is a two dimensional (2D) array.

[0156] It may be that the array is a 2x2 (two-by-two) array.

[0157] The array of cavities may be a square array.

[0158] It may be that the anode or the cathode region for cavities in a row passes through the adjacent cavity of said row.

[0159] Each of the anode terminals may be located on a first edge of the housing and each of the anode terminals may be located on a second opposing edge of the housing.

[0160] It may be that each of the anode and cathode terminals are located along a line across from one another. This configuration may be particularly beneficial in linear arrangements for connecting the LEDs of the same chromaticity in series, for instance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0161] These and other aspects and features of the present invention will become apparent to those ordinarily skilled in the art upon review of the following description of specific embodiments of the invention in conjunction with the accompanying figures, in which:

[0162] FIGS. 1A and 1B are schematic representations of a known color-tunable multi-LED package in which FIG. 1A shows a top view and FIG. 1B shows a sectional side view through A-A;

[0163] FIGS. 2A-2D are schematic representations of a multi-LED (Four-LED) package in accordance with an embodiment of the invention comprising a common cathode arrangement in which FIG. 2A shows a top view, FIG. 2B shows a sectional side view through B-B, FIG. 2C shows a sectional side view through C-C, and FIG. 2D is top view of the lead frame of the multi-LED package;

[0164] FIGS. 3A-3D are schematic representations of a multi-LED (Four-LED) package in accordance with a further embodiment of the invention in which FIG. 3A shows a top view, FIG. 3B shows a sectional side view through D-D, FIG. 3C shows a sectional side view through E-E, and FIG. 3D is top view of the lead frame of the multi-LED package;

[0165] FIGS. 3E and 3F are schematic representations of a multi-LED (Four-LED) package in accordance with another embodiment of the invention in which FIG. 3E shows a top view, and FIG. 3F is top view of the lead frame of the multi-LED package;

[0166] FIGS. 4A-4C show a schematic top view, a sectional side view through F-F, and a sectional side view through G-G of a color-temperature-tunable lighting device in accordance with an embodiment of the invention, respectively;

[0167] FIG. 4D is a CIE 1931 chromaticity diagram illustrating the gamut of light (solid line) that the color-temperature-tunable lighting device of FIGS. 4A-4C can generate when it comprises LEDs of two different chromaticity/color temperatures C1 and C2 (solid diamond);

[0168] FIG. 4E is a CIE 1931 chromaticity diagram illustrating the gamut of light (solid line) that the color-temperature-tunable lighting device of FIGS. 4A-4C can generate when it comprises LEDs of three different chromaticity/color temperatures C1, C2 and C3 (solid diamond);

[0169] FIG. 4F is a CIE 1931 chromaticity diagram illustrating the gamut of light (solid line) that the color-temperature-tunable lighting device of FIGS. 4A-4C can generate when it comprises LEDs of four different chromaticity/color temperatures C1, C2, C3, and C4 (solid diamond);

[0170] FIG. 5 is a schematic representation of a color-temperature-tunable linear lighting device in accordance with an embodiment of the invention;

[0171] FIG. 6 is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light generated by a color-temperature-tunable lighting device (Dev.1) comprising LEDs of two chromaticity/color temperatures: C1—Warm White (WW) and C2—Cool White (CW) for CCTs of 2700K, 3000K, 3500K, 4000K, and 5000K, light emission locus (solid line) for CCTs from 2700K to 5000K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0172] FIG. 7A-7D are measured characteristics for color-temperature-tunable lighting device (Dev.2) comprising LEDs of three chromaticity/color temperatures: C1, C2, C3 in which: FIG. 7A is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—solid square) of light C3, 1 SDCM, 2 SDCM, 3 SDCM and 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses—dotted and dashed lines) and CIE region (solid line—square) for light C3, FIG. 7B shows spectra, normal-

ized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 7C is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.2 can generate, and black body locus (dotted line), and FIG. 7D is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.2 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses—thin solid line);

[0173] FIG. 8A-8C are measured characteristics for a color-temperature-tunable lighting device (Dev.3) in which FIG. 8A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 8B is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that the lighting device (Dev.3) can generate, and black body locus (dotted line), and FIG. 8C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.3 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0174] FIG. 9A-9C are measured characteristics for a color-temperature-tunable lighting device (Dev.4) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. 9A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 9B is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.4 can generate, and black body locus (dotted line), and FIG. 9C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.4 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0175] FIG. 10A-10C are measured characteristics for a color-temperature-tunable lighting device (Dev.5) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. 10A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 10B is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.5 can generate, and black body locus (dotted line), and FIG. 10C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.5 for nominal

CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0176] FIGS. 10D-10I are measured emission spectra (solid line), normalized intensity versus wavelength (nm), for lighting device Dev.5 in which: FIG. 10D is the emission spectrum for operation for a nominal CCT of 2700K, FIG. 10E is the emission spectrum for operation for a nominal CCT of 3000K, FIG. 10F is the emission spectrum for operation for a nominal CCT of 3500K, FIG. 10G is the emission spectrum for operation for a nominal CCT of 4000K, FIG. 10H is the emission spectrum for operation for a nominal CCT of 5700K, and FIG. 10I is the emission spectrum for operation for a nominal CCT of 6500K;

[0177] FIGS. 11A-11D are CIE 1931 chromaticity diagrams illustrating various CIE regions and center points for light of chromaticity C1, C2, and C3;

[0178] FIG. 12 is a CIE 1931 chromaticity diagram illustrating CIE regions and center points for a lighting device comprising LEDs that generate light of four chromaticity C1, C2, C3, and C4;

[0179] FIG. 13 is a CIE 1931 chromaticity diagram illustrating CIE regions and center points for a lighting device comprising LEDs that generate light of four chromaticity C1, C2, C3, and C4;

[0180] FIG. 14A-14C are measured characteristics for a color-temperature-tunable lighting device (Dev.6) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. 14A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 14B is a CIE 1931 chromaticity diagram illustrating the chromaticity (CIE center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.6 can generate, and black body locus (dotted line), and FIG. 14C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.6 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dashed line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0181] FIGS. 14D-14J are measured emission spectra (solid line), normalized intensity versus wavelength (nm), for lighting device Dev.6 in which: FIG. 14D is the emission spectrum for operation for a nominal CCT of 2700K, FIG. 14E is the emission spectrum for operation for a nominal CCT of 3000K, FIG. 14F is the emission spectrum for operation for a nominal CCT of 3500K, FIG. 14G is the emission spectrum for operation for a nominal CCT of 4000K, FIG. 14H is the emission spectrum for operation for a nominal CCT of 5000K, FIG. 14I is the emission spectrum for operation for a nominal CCT of 5700K, and FIG. 14J is the emission spectrum for operation for a nominal CCT of 6500K;

[0182] FIG. 15A-15C are measured characteristics for a color-temperature-tunable lighting device (Dev.7) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. 15A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 15B is

a CIE 1931 chromaticity diagram illustrating the chromaticity (CIE center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.7 can generate, and black body locus (dotted line), and FIG. 15C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.7 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line); [0183] FIGS. 16A-16D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light of chromaticity C1, C2, and C3 for lighting devices Dev.6 and Dev.7;

[0184] FIGS. 17A-16D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light of chromaticity C1, C2, and C3 for lighting devices Dev.2-7;

[0185] FIGS. 18A-18D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light of chromaticity C1, C2, and C3 for lighting devices Dev.2-3 and Dev.6-7;

[0186] FIG. 19 is a CIE 1931 chromaticity diagram illustrating calculation of dominant wavelength λ_d of light of a given chromaticity (color) point; and

[0187] FIG. 20 is a CIE 1931 chromaticity diagram illustrating CIE regions for light of chromaticity C1, C2, and C3 for lighting device Dev.8.

DETAILED DESCRIPTION OF THE INVENTION

[0188] Embodiments of the invention relate to color-tunable lighting devices that can generate light of a color temperature that is tunable from 1800K to 6500K.

[0189] Aspects of the invention relate to color-temperature-tunable lighting devices comprising a single package comprising a plurality of LEDs that generate light of two, three, or four different chromaticity/CCTs. To improve the luminous efficacy of the lighting device, at least one of the plurality of LEDs, for example, the LED that generates light with the lowest color temperature, may comprise a narrowband red phosphor such as a narrowband red fluoride phosphor; for example, $\text{K}_2\text{SiF}_6:\text{Mn}^{4+}$, $\text{K}_2\text{GeF}_6:\text{Mn}^{4+}$, and/or $\text{K}_2\text{TiF}_6:\text{Mn}^{4+}$.

[0190] Throughout this specification like reference numerals are used to denote like parts preceded by the figure number #. For example, an LED chip #20 is denoted 220 in FIG. 2 and is denoted 320 in FIG. 3 and so forth.

Multi-LED Packages

[0191] FIGS. 2A-2D are schematic representation of a multi-LED, four-LED, package (Multi-cavity lead frame package) in accordance with an embodiment of the invention comprising a common cathode terminal arrangement in which FIG. 2A shows a top view, FIG. 2B shows a sectional side view through B-B, FIG. 2C shows a sectional side view through C-C, and FIG. 2D is a top view of the lead frame of the multi-LED package.

[0192] As shown in FIGS. 2A-2D, the multi-LED package 210 comprises a lead frame 212a-212d, 214a-d and a housing 216 that can be molded onto the lead frame. The

housing 216 comprises multiple (e.g., four) independent cavities comprising a first cavity (cup) 218a for receiving a respective first LED chip 220a, a second cavity (cup) 218b for receiving a respective second LED chip 220b, a third cavity (cup) 218c for receiving a respective third LED chip 220c, and a fourth cavity 218d for receiving a respective fourth LED chip 220d. The LED chips 220a-220d are indicated in FIG. 2A to 2C by a dashed rectangle and bond wires connecting the LED chips to the lead frame are indicated by dashed lines.

[0193] Referring to FIG. 2D, various regions of the lead frame 212a-212d, 214a-d are indicated by cross-hatching and the relative position of the housing 216 and the cavities 218a-218d are respectively indicated by dashed and dotted lines. The lead frame comprises a central cross-shaped cathode region 214a-d and four rectangular anode regions 212a-212d located at the empty corners of the cross-shaped cathode region 214a-d. As can be seen from FIG. 2D, each cavity 218a-218d comprises on its floor, a respective L-shaped region of the cross-shaped cathode region 214a-d which constitutes a common cathode electrode (connection) to each cavity. The L-shaped cathode region on the floor of each cavity, in addition to providing a common (shared) cathode electrode (connection), may provide a thermally conductive mounting pad for the LED chip, thereby improving thermal dissipation from the LED chip. As can be seen from FIG. 2D, each cavity 218a-218d comprises on its floor, a respective rectangular anode region 212a-212d which constitutes an anode electrode (connection) to the cavity. As illustrated in FIGS. 2A-2D, each respective anode region of the lead frame 212a-212d extends beyond an outer side (edge) of the housing 216 and provides a respective anode electrical terminal 222a-222d for each cavity 218a-218d. Similarly, the cathode region of the lead frame 214a-d extends beyond opposing sides (edges) of the housing and provides a common cathode electrical terminal 224a-d on opposite sides (edges) of the housing.

[0194] As described herein, in embodiments, the multi-LED (e.g., four-LED) package may comprise a single cathode electrical terminal 224a-d that is common to each cavity (LED chip) and a respective anode electrical terminal 222a-222d for each cavity (LED chip). In other embodiments of the invention, the multi-LED package may comprise a respective pair of anode and cathode electrical terminals for each cavity (LED chip). Such an arrangement can be beneficial when using multiple multi-LED packages as it allows the LED chips of a given cavity to be connected in series.

[0195] FIGS. 3A-3D are schematic representations of a multi-LED, four-LED, package (Multi-cavity lead frame package) in accordance with an embodiment of the invention comprising a respective pair of anode and cathode electrical terminals for each cavity (LED chip) in which FIG. 3A shows a top view, FIG. 3B shows a sectional side view through D-D, FIG. 3C shows a sectional side view through E-E, and FIG. 3D is a plan view of the lead frame of the multi-LED package.

[0196] As shown in FIGS. 3A-3D, the multi-LED package 310 comprises a lead frame 312a-312d, 314a-314d and a housing 316 that can be molded onto the lead frame. The housing 316 comprises multiple (e.g., four) independent cavities comprising a first cavity 318a for receiving a respective first LED chip 320a, a second cavity 318b for receiving a respective second LED chip 320b, a third cavity 318c for receiving a respective third LED chip 320c, and a

fourth cavity **318d** for receiving a respective fourth LED chip **320d**. The LED chips **320a-320d** are indicated in FIG. 3A to 3C by a dashed rectangle and bond wires connecting the LED chips to the lead frame are indicated by dashed lines.

[0197] Referring to FIG. 3D, the various regions of the lead frame **312a-312d**, **314a-314d** are indicated by cross-hatching and the relative position of the housing **316** and the cavities **318a-318d** are respectively indicated by dashed and dotted lines. The lead frame comprises four L-shaped cathode region **314a-314d** arranged as a cross and four rectangular anode regions **312a-312d** with a respective anode region located at the empty corners of a respective L-shaped region. As can be seen from FIG. 3D, each cavity **318a-318d** comprises on its floor, a respective L-shaped cathode region **314a-314d** and a respective rectangular shaped anode region **312a-312d**. The L-shaped cathode region on the floor of each cavity, in addition to providing a cathode electrode (connection), may provide a thermally conductive mounting pad for the LED chip, thereby improving thermal dissipation from the LED chips. As can be seen from FIG. 3D, each cavity **318a-318d** comprises on its floor, a respective rectangular anode region **312a-312d** which constitutes an anode electrode (connection) to the cavity. As illustrated in FIGS. 3A-3D, each anode region of the lead frame **312a-312d** extends beyond an outer side (edge) of the housing **316** and provides a respective anode electrical terminal **322a-322d** for each cavity **318a-318d**. Similarly, each cathode region of the lead frame **314a-314d** extends beyond the outer side (edge) of the housing and provides a respective cathode electrical terminal **324a-324d** on the same side (edge) of the housing as the anode electrical terminal.

[0198] FIGS. 3E and 3F are schematic representations of a multi-LED (Four-LED) package (Multi-cavity lead frame package) in accordance with another embodiment of the invention in which FIG. 3E shows a top view, and FIG. 3F is top view of the lead frame of the multi-LED package **310**. This embodiment is similar to the multi-LED package of FIGS. 3A-3D, except that the lead frame regions **312a-312d** and **314a-314d** are configured such that each anode terminal **322a-322d** is aligned with and located on an opposing side (edge) of the housing to its respective cathode terminal **324a-324d**. Such a packaging arrangement can be advantageous in linear lighting arrangements utilizing a plurality of multi-LED packages in which it is preferred to serially connect LEDs of each cavity.

[0199] As shown in FIGS. 3E and 3F the multi-LED package **310** comprises a lead frame **312a-312d**, **314a-314d** and a housing **316** that is typically molded onto the lead frame. The housing **316** comprises multiple (e.g., four) independent cavities comprising a first cavity **318a** for receiving a respective first LED chip **320a**, a second cavity **318b** for receiving a respective second LED chip **320b**, a third cavity **318c** for receiving a respective third LED chip **320c**, and a fourth cavity **318d** for receiving a respective fourth LED chip **320d**. Bond wires connecting the LED chips **320a-320d** to the lead frame are indicated by dashed lines.

[0200] Referring to FIG. 3F, the various regions of the lead frame **312a-312d**, **314a-314d** are shown and the relative position of the housing **316** and the cavities **318a-318d** are respectively indicated by dashed and dotted lines. The lead frame comprises four cathode regions **314a-314d** and four anode regions **312a-312d**.

[0201] For the first and second cavities **318a** and **318b**, the cathode regions **314a** and **314b** are zigzag shaped and the anode regions **312a** and **312b** are elongate in form. The first and second cavities **318a** and **318b** comprise on their floor, a respective zigzag shaped cathode region **314a** and **314b** which constitutes a cathode electrode (connection) to the cavity and a respective square shaped portion (indicated by cross hatching) of the elongate anode region **312a** and **312b** which constitutes an anode electrode (connection) to the cavity. As illustrated in FIG. 3F, each of the zigzag shaped cathode regions **314a**, **314b** extends beyond an outer side (edge) of the housing **316** and provides a respective cathode electrical terminal **324a**, **324b** for the first and second cavities **318a**, **318b**. In this embodiment, the elongate anode region **312a** extends from the first cavity **318a** and passes through the fourth cavity **318d** and then extends from an outer side (edge) of the housing **316** to provide a respective anode electrical terminal **322a** for the first cavity **318a**. Similarly, the elongate anode region **312b** extends from the second cavity **318b** and passes through the third cavity **318c** and then extends from an outer side (edge) of the housing **316** to provide a respective anode electrical terminal **322b** for the second cavity **318a**. In this way, an anode region (for instance, elongate) extends from one cavity and passes through an adjacent cavity in the same row.

[0202] For the third and fourth cavities **318c** and **318d**, the cathode regions **314c** and **314d** are elongate in form and the anode regions **312c** and **312d** are zigzag shaped. The third and fourth cavities **318c** and **318d** comprise on their floor, a respective square shaped cathode portion (indicated by cross hatching) of the elongate cathode region **314c** and **314d** which constitutes a cathode electrode (connection) to the cavity. As illustrated in FIG. 3F, each of the zigzag shaped anode regions **312c**, **312d** extends beyond an outer side (edge) of the housing **316** and provides a respective anode electrical terminal **322c**, **322d** for the third and fourth cavities **318c**, **318d**. In this embodiment, the elongate cathode region **314c** extends from the third cavity **318c** and passes through the second cavity **318b** and then extends from an outer side (edge) of the housing **316** to provide a respective cathode electrical terminal **324c** for the third cavity **318c**. Similarly, in this embodiment, the elongate cathode region **314d** extends from the fourth cavity **318d** and passes through the first cavity **318a** and then extends from an outer side (edge) of the housing **316** to provide a respective cathode electrical terminal **324d** for the fourth cavity **318d**. In this way, a cathode region (for instance, elongate) is able to extend from one cavity to an adjacent cavity. The zigzag shaped region on the floor of each cavity, in addition to providing an electrical connection, may provide a thermally conductive mounting pad for the LED chip, thereby improving thermal dissipation from the LED chips. As illustrated in FIGS. 3E and 3F, each anode region of the lead frame extends beyond an outer edge of the housing **316** and provides a respective anode electrical terminal **322a-322d** for each cavity **318a-318d**. Similarly, each cathode region of the lead frame extends beyond the outer edge of the housing and provides a respective cathode electrical terminal **324a-324d** on the opposite edge of the housing to that of the anode electrical terminal. In at least this embodiment, for example, the anode and cathode terminals are aligned (or in the same linear path, for instance).

Color-Temperature-Tunable Lighting Devices

[0203] FIGS. 4A-4C show a schematic top view, a sectional side view through F-F, and a sectional side view through G-G of a color-temperature-tunable lighting device 426 that utilizes the multi-LED package 210 of FIGS. 2A-2D and comprises four LEDs 428a-428d.

[0204] The lighting device 426 comprises a package 410 comprising a lead frame and a housing 416 molded onto the lead frame. The housing 416 comprises a first cavity 418a, a second cavity 418b, a third cavity 418c, and a fourth cavity 418d that respectively contain a first LED 428a, a second LED 428b, a third LED 428c, and a fourth LED 428d. As illustrated, the first, second, third and fourth LEDs 428a-428d may comprise packaged devices in which each of the first, second, third, and fourth cavities 418a-418d contain a respective violet to blue LED chip 420a-420d and is filled with a phosphor photoluminescence layer 430a-430d that covers the violet to blue LED chip 420a-420d. The chromaticity (color)/CCT (Correlated Color Temperature) of light generated by each LED 428a-428d is dependent on the phosphors within the photoluminescence layer 430a-430d. For example, when an LED is to generate Cool White (CW) light with a CCT from, for example, about 4100K to 8000K the phosphor photoluminescence layer may comprise green to yellow phosphor. When an LED is to generate Warm White (WW) light with a CCT from, for example, about 1800K to 4000K the phosphor photoluminescence layer may comprise green to red phosphors and a narrowband red phosphor such as $\text{K}_2\text{SiF}_6:\text{Mn}^{4+}$, $\text{K}_2\text{GeF}_6:\text{Mn}^{4+}$, and $\text{K}_2\text{TiF}_6:\text{Mn}^{4+}$. As illustrated, the package 410 comprises common cathode terminals 424a-d and respective anode terminals 422a-422d for each cavity 418a-418d allowing electrical power to be independently applied to each of the LED chip 418a-418d.

[0205] The LEDs 430a-430d may be configured generate light of two, three or four different chromaticity (colors)/color temperatures.

[0206] In a first embodiment, the LEDs 430a-430d can be configured to generate light of two chromaticity/color temperatures C1, and C2. The first chromaticity/color temperature of light C1 can, for example, comprise Warm White (WW)—at least 1800K to 4000K and the second chromaticity/color temperature of light C2 can, for example, comprise Cool White (CW)—at least 4100K and up to 25000K; more typically up to about 8000K. In such an arrangement, the light emitting device 426 can generate light of different color temperatures from C1 to C2. FIG. 4D is a CIE 1931 chromaticity diagram illustrating the gamut of light that the light emitting device 426 can generate when it comprises white LEDs of two different color temperatures C1 (2670K) and C2 (7250K). The CIE chromaticity diagram shows the chromaticity (color point: chromaticity coordinates CIE 0.4700, 0.4250-Δuv 0.0044) 432_{C1} of light of a first color C1 generated by the C1 LED(s) and chromaticity (color point: chromaticity coordinates CIE 0.3000, 0.3300) 432_{C2} of light of a second color C2 generated by the C2 LED(s). It is to be noted that the color points 432_{C1} and 432_{C2} lie off of (away from or not directly on) the black body locus (i.e., the points do not lie on the black body locus) and, as illustrated, they both lie above the black body locus. A straight line 434_{C1C2} connecting the points 432_{C1} and 432_{C2} represents the gamut of color temperatures (chromaticity) of light that light emitting device 426 can generate—i.e., the device can generate color temperatures of light lying on the

straight line 434_{C1C2} from 432_{C1} (2670K) to 432_{C2} 7205K. It is to be noted the since the black body locus (dotted line) is a curved line the device cannot generate CCTs of light lying on the black body locus for all CCTs from 432_{C1} to 432_{C2}.

[0207] In a second embodiment, the LEDs 428a-428d can be configured to generate light of three different chromaticity/CCTs: C1, C2, and C3. FIG. 4E is a CIE 1931 chromaticity diagram illustrating the gamut of light that the light emitting device 426 can generate when it comprises white LEDs of three different color temperatures C1 (2665K), C2 (4205K) and C3 (7355K). The CIE chromaticity diagram shows the chromaticity (color point: chromaticity coordinates CIE 0.4650, 0.4150-Δuv 0.0012) 432_{C1} for light of the first CCT (C1), the chromaticity (color point: chromaticity coordinates CIE 0.3750, 0.3850-Δuv 0.0055) 432_{C2} for light of the second CCT (C2), and the chromaticity (color point: chromaticity coordinates CIE 0.3000, 0.3200-Δuv 0.0052) 432_{C3} for light of the third CCT (C3). It is to be noted that the color points 432_{C1}, 432_{C2}, and 432_{C3} lie off of the black body locus (i.e., the points do not lie on the black body locus) and, as illustrated, all three points lie above the black body locus. Straight lines 434_{C1C2}, 434_{C2C3}, and 434_{C1C3} connecting the points 432_{C1}, 432_{C2}, and 432_{C3} define a triangle that represents the gamut of color temperatures (chromaticity) of light that light emitting device 426 can generate—i.e. the device can generate any chromaticity/color temperature of light lying within the triangle or lying on the boundary of the triangle. As will be noted from the chromaticity diagram, the device 426 can generate color temperatures of light from 2700K to 6500K lying on the black body locus (dotted line) or corresponding to the ANSI CCT center points (indicated by dots).

[0208] In a third embodiment, the four LEDs 428a-428d can be configured to generate light of four different chromaticity/CCTs: C1, C2, C3, C4. FIG. 4F is a CIE 1931 chromaticity diagram illustrating the gamut of light that the light device 426 can generate when it comprises white LEDs of four different color temperatures C1 (2665K), C2 (3540K), C3 (5070K), and C4 (7165K). The CIE chromaticity diagram shows the chromaticity (color point: chromaticity coordinates CIE 0.4650, 0.4150-Δuv 0.0012) 432_{C1} for light of the first CCT (C1), the chromaticity (color point: chromaticity coordinates CIE 0.4050, 0.3950-Δuv 0.0019) 432_{C2} for light of the second CCT (C2), the chromaticity (color point: chromaticity coordinates CIE 0.3440, 0.3600-Δuv 0.0019) 432_{C3} for light of the third CCT (C3), and the chromaticity (color point: chromaticity coordinates 0.3030, 0.3200-Δuv 0.0036) 432_{C4} for light of the fourth CCT (C4). It is to be noted that the color points 432_{C1}, 432_{C2}, 432_{C3}, and 432_{C4} lie off of the black body locus (i.e., the points do not lie on the black body locus) and, as illustrated, all four points lie above the black body locus (dashed line). Straight lines 434_{C1C2}, 434_{C2C3}, 434_{C3C4}, and 434_{C1C4} connecting the points 432_{C1}, 432_{C2}, 432_{C3}, and 432_{C4} define a quadrilateral that represents the gamut of color temperatures (chromaticity) of light that light emitting device 426 can generate—i.e., the device can generate any color temperature of light lying within the quadrilateral or lying on the boundary of the quadrilateral. As will be noted from the chromaticity diagram, the device 426 can generate color temperatures of light from 2700K to 6500K lying on the black body locus (dashed line) or corresponding to the ANSI CCT center points.

[0209] FIG. 5 is a schematic top view of a color-temperature-tunable linear light emitting device 536 in accordance with an embodiment of the invention. The linear light emitting device 536 comprises a linear (elongate) substrate 538, such as for example a strip of Metal Core Printed Circuit Board (MCPCB) or a strip of flexible circuit board, and a plurality of color-temperature-tunable lighting devices 526 mounted on and electrically connected to the substrate. For the purposes of illustration only, the lighting devices 526 are shown as comprising the multi-LED package 310 of FIGS. 3E and 3F and comprise LEDs that generate light of two, three, or four different chromaticity/color temperatures. As illustrated, the lighting devices 526 can be arranged as a linear array extending in a direction of elongation of the substrate. The linear light emitting device 536 may, at opposite ends of the substrate 538, comprise an electrical connector 540 comprising respective electrical connections to the anode and cathode of the first, second, third, and fourth LEDs 528a-528d of the lighting devices 526. In such an arrangement the first, second, third and fourth LEDs 528a-528d of each device 526 are electrically connected in

in the third and fourth cavities, comprise a single-layer PC white LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, (Y,Lu)₃(Al,Ga)₅O₁₂ and red nitride phosphor CaAlSiN₃:Eu (CASN)). The green to red phosphors are incorporated in a light transmissive encapsulant and the mixture dispensed into the third and fourth cavities of the package to cover the violet to blue InGaN LED chip. Color-temperature-tunable lighting device Dev.1 is configured to generate light with a CCT from 2700K to 5000K and a general Color Rendering Index CRI Ra of 80.

[0212] TABLE 1 tabulates the optical characteristics of the PC Warm White (WW) LED (2700K) and Cool White (CW) LED of lighting device Dev.1. As can be seen from TABLE 1, the WW LEDs have a luminous efficacy of 127.5 lm/W and generates light with a CCT of 2700K with CRI Ra of 82.4 (about 80) and a CRI R9 of 6.0, while the CW LEDs have a luminous efficacy of 139.0 lm/W and generates light with a CCT of 5000K with CRI Ra of 80.7 (about 80) and a CRI R9 of 2.5.

TABLE 1

Dev. 1: Measured characteristics of PC Warm White LED (WW) and PC Cool White LED (CW)									
CCT (K)	Flux (lm)	Power (W)	LE (lm/W)	CIE		CCT (K)	CRI		
				x	Y		Ra	R9	Δuv
2700 (WW)	120.5	0.95	127.5	0.4585	0.4114	2725	82.4	6.0	0.0004
5000 (CW)	131.4	0.95	139.0	0.3487	0.3571	4886	80.7	2.5	0.0013

series. In other embodiments utilizing light emitting devices having a common cathode arrangement for the four LEDs, such as for example the multi-LED package 210 of FIGS. 2A-2D, the first, second, third and fourth LEDs of each device are electrically connected in parallel.

Experimental Test Data

[0210] In this specification, Dev. # is used to denote a color-temperature-tunable multi-LED packaged lighting device in accordance with the invention. The test method involves measuring light emission of the color-temperature-tunable lighting devices in an integrating sphere.

Color-Temperature-Tunable Lighting Device—Dev.1

[0211] A color-temperature-tunable lighting device, denoted Dev.1, comprises LEDs that generate light of two different chromaticity/CCTs: C1 and C2. Device, Dev.1 comprises the lighting device of FIGS. 3A-3D and comprises a 3838 four cavity package containing two cool white (CW) LEDs and two warm white (WW) LEDs. The CW LEDs, in the first and second cavities, comprise a PC white LED comprising a mixture of green to yellow photoluminescence materials (e.g., yttrium or lutetium aluminum garnet phosphors: GaYAG (Y₃(Al,Ga)₅O₁₂:Ce), GaLuAG (Lu₃(Al,Ga)₅O₁₂:Ce), (Y,Lu)₃(Al,Ga)₅O₁₂, YAG (Y₃Al₅O₁₂:Ce), and/or LuAG (Lu₃Al₅O₁₂:Ce)). The green to yellow phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the first and second cavities of the package to cover the violet to blue InGaN LED chip. The WW LEDs,

[0213] FIG. 6 is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light generated by lighting device (Dev.1) comprising LEDs of two different Color Temperatures: Warm White (WW) and Cool White (CW) for CCTs of 2700K, 3000K, 3500K, 4000K, and 5000K, light emission locus (solid line) for CCTs from 2700K to 5000K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses). Referring to FIG. 6, it is to be noted that light emission locus (solid line 642)—the chromaticity locus of light that lighting device Dev.1 is capable of generating—is a straight line 642 connecting the color points for the WW LED (2700K) and CW LED (5000K) and that color of light over the temperature range is within 3 SDCM.

Color-Temperature-Tunable Lighting Device—Dev.2

[0214] A color-temperature-tunable lighting device, denoted Dev.2, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.2 comprises the device of FIGS. 3A-3C and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED-C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3. The two LED-C1 LEDs, in the first and second cavities, comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or (Y,Lu)₃(Al,Ga)₅O₁₂ and CASN) and a narrowband red phosphor (e.g., K₂SiF₆:Eu (KSF)). The green to red and narrowband red phosphor are incorporated in a light trans-

missive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the first and second cavities to cover the violet to blue InGaN LED chip. LED-C2, in the third cavity, comprises a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or $(Y, Lu)_3(Al, Ga)_5O_{12}$ and CASN). The green to red phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the third cavity of the package to cover the violet to blue InGaN LED chip. LED-C3, in the fourth cavity, comprises a PC LED comprising a yellow photoluminescence material (e.g., Yttrium Aluminum Garnet (YAG) phosphor). The yellow phosphor is incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the fourth cavity of the package to cover the violet to blue InGaN LED chip. Color-temperature-tunable Lighting device Dev.2 is configured to generate light with a CCT from 2700K to 6500K and a general Color Rendering Index CRI Ra of 80.

[0215] TABLE 2A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LED-C3 of lighting device Dev.2 and TABLE 2B tabulates CIE region, CIE center point for light C1, C2, and C3 and CCT and Δuv for light C1 and C2. FIG. 7A-7D are measured characteristics for lighting device Dev.2 in which: FIG. 7A is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—solid square) of light C3, 1 SDCM, 2 SDCM, 3 SDCM and 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses—dotted and dashed lines) and CIE region (solid line—square) for light C3, FIG. 7B shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 7C is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that Dev.2 can generate, and black body locus (dotted line), and FIG. 7D is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by Dev.2 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Standard Deviation Color Matching—Mac Adam ellipses—thin solid line).

[0216] As can be seen from TABLE 2A: LED-C1 has a luminous efficacy (LE) of 119 lm/W and generates light of chromaticity (color point: CIE 0.5200, 0.4213, $\Delta uv=0.0021$) corresponding to a CCT of 2107K (about 2100K) with a CRI Ra of 78.1 (about 80); LED-C2 has a luminous efficacy of 170 lm/W and generates light of chromaticity (color point: CIE 0.3756, 0.4316, $\Delta uv=0.0251$) corresponding to a CCT of 4436K (about 4400K) with a CRI Ra of 61.7 (about 62); and LED-C3 has a luminous efficacy of 120 lm/W and generates light with a chromaticity (color point: CIE 0.2121, 0.2383) which is greenish blue to blue in color.

[0217] As is known, the black body locus represents the chromaticity locus of light generated by a black body (Planckian) radiator for temperatures from 1000K to infinity and goes from deep red at low temperatures through orange,

yellowish white, white to bluish white. Since the chromaticity (color point) of light in the greenish blue to blue region of the chromaticity diagram cannot be generated by a Planckian black body radiator, the chromaticity (color point) of light C3 cannot be ascribed a CCT. Since light C3 cannot be described in terms of color temperature, it will be described by a chromaticity center point and CIE chromaticity region (C3 CIE region). In this specification “a chromaticity region” is defined as an area on the CIE 1931 chromaticity diagram and may be defined as a quadrilateral composed of four chromaticity points CIE-1, CIE-2, CIE-3, and CIE-4 that are connected by straight lines. The C3 CIE region is defined by four CIE color points CIE-1, CIE-2, CIE-3, and CIE-4: CIE-1 (0.200, 0.244), CIE-2 (0.205, 0.212), CIE-3 (0.225, 0.232), and CIE-4 (0.219, 0.265)—TABLE 2B. The CIE region corresponds to 6 SDCM about the CIE center point and calculation of the CIE region is illustrated in FIG. 7A which is a CIE 1931 chromaticity diagram illustrating the chromaticity 732_{C1} (center point—solid square) and C3 CIE region 744_{C3} (solid line/square) for light C3. As can be seen from the figure, the CIE region 744_{C3} is a quadrilateral approximating to 6 SDCM (Mac Adam ellipses).

[0218] Similarly, CIE regions defined by four CIE color points CIE-1, CIE-2, CIE-3, and CIE-4 for light C1 and C2 were calculated and are given in TABLE 2B together with CCT and Δuv for each color point. C1 comprises a quadrilateral CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.502, 0.419), CIE-2 (0.510, 0.398), CIE-3 (0.538, 0.424), and CIE-4 (0.530, 0.445) connected by straight lines which in terms of CCT is from 1974K (about 1970K) to 2261K (about 2270K). C2 comprises a quadrilateral CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.364, 0.439), CIE-2 (0.365, 0.407), CIE-3 (0.388, 0.424), and CIE-4 (0.386, 0.456) connected by straight lines which in terms of CCT is from 4116K (about 4100K) to 4734K (about 4750K). The CIE chromaticity diagram of FIG. 7C shows each of the CIE regions: C1 CIE region 744_{C1} (diamond/solid line), C2 CIE region 744_{C2} (triangle/solid line), and C3 CIE Region 744_{C3} (square/solid line).

[0219] Referring to FIG. 7B the emission spectrum for LED-C1 (dotted line) primarily comprises a broadband peak (at about 610 nm) in the red region of the visible spectrum with multiple higher intensity narrowband peaks (FWHM $\approx$ 5 nm) from about 610-650 nm (maximum intensity peak at $\approx$ 632 nm) in the orange to red region of the visible spectrum. The emission spectrum for the LED-C2 (solid line) comprises a narrowband peak (FWHM $\approx$ 20 nm) with a peak emission wavelength at about 450 nm in the blue region of the visible spectrum and a broadband peak (FWHM $\approx$ 125 nm) with a peak emission wavelength of about 530 nm in the green region of the visible spectrum. The emission spectrum for the LED-C3 (dashed line) comprises a narrowband peak (FWHM $\approx$ 20 nm) with a peak emission wavelength of about 450 nm in the blue region of the visible spectrum and a lower intensity (about 25%) tail at wavelengths in the green to yellow region of the spectrum (460-580 nm).

TABLE 2A

Dev. 2: Measured characteristics of LED-C1, LED-C2, and LED-C3								
LED	Flux	Power	LE	CIE center point		CCT	CRI Ra	Δ_{uv}
	(lm)	(W)	(lm/W)	x	y	(K)		
LED-C1	44.6	0.376	119	0.5200	0.4213	2107	78.1	0.0021
LED-C2	63.8	0.376	170	0.3756	0.4316	4436	61.7	0.0251
LED-C3	45.0	0.376	120	0.2121	0.2383	n/a	n/a	n/a

TABLE 2B

Dev. 2: CIE region for light C1, C2, and C3										
CIE point	C1 CIE region				C2 CIE region				C3 CIE region	
	CIE		CCT		CIE		CCT		CIE	
	x	y	(K)	Δ_{uv}	x	y	(K)	Δ_{uv}	x	y
CIE-1	0.502	0.419	2261	0.0012	0.364	0.439	4734	0.0317	0.200	0.244
CIE-2	0.510	0.398	2045	-0.0053	0.365	0.407	4591	0.0184	0.205	0.212
CIE-3	0.538	0.424	1974	0.0036	0.388	0.424	4116	0.0185	0.225	0.232
CIE-4	0.530	0.445	2167	0.0092	0.386	0.456	4316	0.0314	0.219	0.265
Center	0.5200	0.4213	2107	0.0021	0.3756	0.4316	4436	0.0251	0.2121	0.2383

[0220] The CIE chromaticity diagram of FIG. 7C shows the chromaticity (center color point) 732_{C1} of light of color C1 generated by the LED-C1, the chromaticity (center color point) 732_{C2} of light of color C2 generated by LED-C2, and the chromaticity (center color point) 732_{C3} of light of color C3 generated by the LED-C3. Straight lines 734_{C1C2} , 734_{C2C3} , and 734_{C1C3} connecting the points 732_{C1} , 732_{C2} , and 732_{C3} define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.2 can generate—i.e., the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.2 can generate that lies on the black body locus (dotted line) is 2157K (about 2150K) which corresponds to the point of intersection **746** (CIE 0.510 0.415) of line 734_{C1C3} connecting color points 732_{C1} to 732_{C3} and the black body locus. It is to be noted that highest CCT of light that Dev.2 can generate that lies on the black body locus (dotted line) is 14012K (14000K) which corresponds to the point of intersection **748** (CIE 0.266, 0.270) of line 734_{C1C3} connecting color point 732_{C1} to color point 732_{C3} and the black body locus. It will be appreciated that Dev.2 is, therefore, capable of generating any CCT of light from about 2150K to about 14000K lying on the black body locus.

[0221] TABLE 3 tabulates forward drive current (I_F) for first and second LED-C1, LED-C2, and LED-C3 of Dev.2 for generating light with a nominal general color rendering index CRI Ra of 80 for nominal color temperatures (CCT) from 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K. TABLE 4 tabulates the measured optical and electrical characteristics for Dev.2 when operated to generate light with a nominal CRI Ra of 80 for nominal color temperatures (CCT) from 2700K to 6500K.

[0222] As can be seen from TABLE 4, the CCT of light generated by Dev.2 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of color C1. TABLE 4 demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and

LED-C3, lighting device Dev.2 can generate white light with a CCT from 2700 K to 6500K with a general color rendering index CRI Ra of about 80 (80.2 to 82.0) and CRI R9 of at least 9 (9.2 to 14.6) with a luminous efficacy from about 134 lm/W (133.6 lm/W) to about 143 lm/W (143 lm/W). TABLE 4 also includes the measured CCT of light generated by Dev.2.

[0223] Referring to FIG. 7D it is to be noted that Dev.2 can generate light **742** (solid line) of CCTs from 2700K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 4, the chromaticity of light generated by Dev.2 are within 0.0029 Δ_{uv} (about 0.003) or less of the black body locus for CCTs from 2700K to 6500K.

TABLE 3

Dev. 2: Forward drive current I_F for the 1 st LED-C1, 2 nd LED-C1, LED-C2, and LED-C3 for generating light with a nominal CRI Ra = 80 for nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Forward drive current I_F (mA)			
	1 st LED-C1	2 nd LED-C1	LED-C2	LED-C3
2700	117	117	72	13
3000	101	101	94	24
3500	83	83	114	38
4000	68	68	124	59
5000	49	49	128	94
5700	45	45	115	114
6500	37	37	106	139

TABLE 4

Dev. 2: Measured optical and Electrical characteristics for nominal CRI Ra = 80 for nominal color temperatures (CCT) from 2700K to 6500K									
Nom. CCT	Flux	Power	LE	CIE		CCT	CRI		
(K)	(lm)	(W)	(lm/W)	x	y	(K)	Ra	R9	Δuv
2700	130.1	0.97	133.6	0.4592	0.4121	2721	80.3	9.8	0.0006
3000	133.6	0.97	137.2	0.4345	0.4040	3041	80.6	12.0	0.0003
3500	136.9	0.97	140.7	0.4083	0.3937	3460	80.4	12.2	0.0007
4000	138.6	0.97	142.3	0.3819	0.3792	3979	80.5	12.7	0.0007
5000	139.5	0.97	143.2	0.3457	0.3565	4994	80.2	9.2	0.0022
5700	137.4	0.97	141.2	0.3310	0.3429	5563	81.7	14.6	0.0016
6500	136.0	0.97	139.7	0.3121	0.3278	6546	82.0	14.5	0.0029

Color-Temperature-Tunable Lighting Device—Dev.3

[0224] A color-temperature-tunable lighting device, denoted Dev.3, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.3 comprises the device of FIGS. 3A-3C and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED-C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3. The two LEDs LED-C1, in the first and second cavities, comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or (Y,Lu)₃(Al,Ga)₅O₁₂ and CASN). The green to red phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the first and second cavities to cover the violet to blue InGaN LED chip. LED-2, in the third cavity, comprises a PC LED comprising a mixture of green to yellow photoluminescence materials (e.g., GaYAG, GaLuAG, (Y,Lu)₃(Al,Ga)₅O₁₂), YAG, and/or LuAG). The green to yellow phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the third cavity of the package to cover the violet to blue InGaN LED chip. LED-C3, in the fourth cavity, comprises a PC LED comprising green to yellow photoluminescence material (e.g., GaYAG, GaLuAG, (Y,Lu)₃(Al,Ga)₅O₁₂), YAG, and/or LuAG). The green to yellow phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the fourth cavity of the package to cover the violet to blue InGaN LED chip. Color-temperature-tunable lighting device Dev.3 is configured to generate light with a CCT from 2700K to 6500K and a general Color Rendering Index CRI Ra of 80.

[0225] TABLE 5A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LED-C3 of Dev.3 and TABLE 5B tabulates CIE region, CIE center point for light C1, C2, and C3 and CCT and Δuv for light C1 and C2. FIG. 8A-8C are measured characteristics lighting device Dev.3 in which FIG. 8A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 8B is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that Dev.3 can generate, and black body locus (dotted line), and FIG. 8C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by Dev.3 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus

(solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line);

[0226] As can be seen from TABLE 5A: LED-C1 has a luminous efficacy (LE) of 119 lm/W and generates light of chromaticity (color point: CIE 0.5280, 0.4385, Δuv=0.0073) corresponding to a CCT of 2144K (about 2100K) with a CRI Ra of 74.4 (about 75); LED-C2 has a luminous efficacy of 164 lm/W and generates light of chromaticity (color point: CIE 0.3661, 0.4219, Δuv=0.0242) corresponding to a CCT of 4623K (about 4600K) with a CRI Ra of 69.5 (about 70); and LED-C3 has a luminous efficacy of 120 lm/W and generates light with a chromaticity (color point: CIE 0.2121, 0.2383). C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.511, 0.435), CIE-2 (0.518, 0.415), CIE-3 (0.545, 0.441), and CIE-4 (0.537, 0.461) which in terms of CCT is from 2203K (about 2200K) to 2279K (about 2300K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.353, 0.428), CIE-2 (0.356, 0.396), CIE-3 (0.379, 0.415), and CIE-4 (0.376, 0.448) which in terms of CCT is from about 4276K (about 4250K) to 4973K (about 4980K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.200, 0.244), CIE-2 (0.205, 0.212), CIE-3 (0.225, 0.232), and CIE-4 (0.219, 0.265)—TABLE 5B. The CIE chromaticity diagram of FIG. 8B shows each of the CIE regions: C1 CIE region **844_{C1}** (diamond/solid line), C2 CIE region **844_{C2}** (triangle/solid line), and C3 CIE Region **844_{C3}** (square/solid line).

[0227] As can be seen from FIG. 8A, the emission spectrum for the LED-C1 (solid line) is composed primarily of a broad emission peak in the orange to red region of the spectrum and the emission spectrum for LED-C3 (dashed line) is composed primarily of blue light with a low intensity tail at wavelengths in the green to yellow region of the spectrum.

[0228] Referring to FIG. 8A the emission spectrum for LED-C1 (dotted line) primarily comprises a broadband peak (FWHM≈90 nm) with a peak emission wavelength of about 530 nm at about 610 nm in the red region of the visible spectrum. The emission spectrum for the LED-C2 (solid line) comprises a narrowband peak (FWHM≈20 nm) with a peak emission wavelength at about 450 nm in the blue region if the visible spectrum and a broadband peak (FWHM Z130 nm) with a peak emission wavelength of about 570 nm in the yellow region of the visible spectrum. The emission spectrum for the LED-C3 (dashed line) comprises a narrowband peak (FWHM≈20 nm) with a peak emission wavelength of about 450 nm in the blue region of the visible spectrum and a lower intensity (about 25%) tail at wavelengths in the green to yellow region of the spectrum (460-580 nm).

TABLE 5A

Dev. 3: Measured characteristics of LED-C1, LED-C2, and LED-C3								
LED	Flux	Power	LE	CIE center point		CCT	CRI Ra	Δuv
	(lm)	(W)	(lm/W)	x	y	(K)		
LED-C1	44.9	0.376	119	0.5280	0.4385	2144	74.4	0.0073
LED-C2	61.6	0.376	164	0.3661	0.4219	4623	69.5	0.0242
LED-C3	45.0	0.376	120	0.2121	0.2383	n/a	n/a	n/a

TABLE 5B

Dev. 3: CIE region for light C1, C2, and C3										
CIE point	C1 CIE region				C2 CIE region				C3 CIE region	
	CIE		CCT		CIE		CCT		CIE	
	x	y	(K)	Δuv	x	Y	(K)	Δuv	x	y
CIE-1	0.511	0.435	2279	0.0061	0.353	0.428	4973	0.0310	0.200	0.244
CIE-2	0.518	0.415	2084	0.0002	0.356	0.396	4797	0.0167	0.205	0.212
CIE-3	0.545	0.441	2019	0.0085	0.379	0.415	4276	0.0174	0.225	0.232
CIE-4	0.537	0.461	2203	0.0137	0.376	0.448	4494	0.0314	0.219	0.265
Center	0.5280	0.4385	2144	0.0073	0.3661	0.4219	4623	0.0242	0.2121	0.2383

[0229] The CIE chromaticity diagram of FIG. 8B shows the chromaticity (center color point) $\mathbf{832}_{C1}$ of light of color C1 generated by the LED-C1, the chromaticity (center color point) $\mathbf{832}_{C2}$ of light of color C2 generated by LED-C2, and the chromaticity (center color point) $\mathbf{832}_{C3}$ of light of color C3 generated by the LED-C3. Straight lines $\mathbf{834}_{C1C2}$, $\mathbf{834}_{C2C3}$, and $\mathbf{832}_{C1C3}$ connecting the points $\mathbf{832}_{C1}$, $\mathbf{832}_{C2}$, and $\mathbf{832}_{C3}$ define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.3 can generate—i.e., the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.3 can generate that lies on the black body locus (dashed line) is 2349K (2350K) which corresponds to the point of intersection $\mathbf{846}$ (CIE 0.491, 0.415) of line $\mathbf{834}_{C1C3}$ connecting color points $\mathbf{832}_{C1}$ to $\mathbf{832}_{C3}$ and the black body locus. It is to be noted that highest CCT of light that Dev.3 can generate that lies on the black body locus (dashed line) is about 12404K (about 12000K) which corresponds to the point of intersection $\mathbf{848}$ (CIE 0.271, 0.275) of line $\mathbf{834}_{C1C3}$ connecting color point $\mathbf{832}_{C1}$ to color point $\mathbf{832}_{C3}$ and the black body locus. It will be appreciated that Dev.3 is therefore capable of generating any CCT of light from 2350K to about 12000K lying on the black body locus.

[0230] TABLE 6 tabulates forward drive current (I_F) for first and second LED-C1, LED-C2, and LED-C3 of Dev.3 for generating light with a nominal general color rendering index CRI Ra of 80 for nominal color temperatures (CCT) from 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K. TABLE 7 tabulates the measured optical and electrical characteristics for Dev.3 when operated to generate light with a nominal CRI Ra of 80 for nominal color temperatures (CCT) from 2700K to 6500K.

[0231] As can be seen from TABLE 6, the CCT of light generated by Dev.3 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of color C1. TABLE 7 demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and

LED-C3, Dev.3 can generate white light with a CCT from 2700 K to 6500K with a general color rendering index CRI Ra of about 80 (81.1 to 84.4) and CRI R9 of up to about 9 (−2.5 to 10.4) with a luminous efficacy from about 130 lm/W (130.1 lm/W) to about 140 lm/W (139.9 lm/W). The reduction of the value of CRI R9 compared with that of Dev.2 is due to the absence of the narrowband red phosphor. TABLE 7 also includes the measured CCT of light generated by Dev.3.

[0232] Referring to FIG. 8C it is to be noted that Dev.3 can generate light $\mathbf{842}$ (solid line) of CCTs from 2700K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 7, the chromaticity of light generated by Dev.3 are within 0.0034 Δuv (about 0.003) or less of the black body locus for CCTs from 2700K to 6500K.

TABLE 6

Dev. 3: Forward drive current I_F for the 1 st LED-C1, 2 nd LED-C1, LED-C2, and LED-C3 for generating light with a nominal CRI Ra = 80 for nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Forward drive current I_F (mA)			
	1 st LED-C1	2 nd LED-C1	LED-C2	LED-C3
2700	123	123	52	22
3000	106	106	78	30
3500	88	88	101	42
4000	71	71	120	58
5000	53	53	120	94
5700	46	46	116	112
6500	39	39	108	133

TABLE 7

Dev. 3: Measured optical and Electrical characteristics for nominal CRI Ra = 80 for nominal color temperatures (CCT) from 2700K to 6500K									
Nom.	Flux	Power	LE	CIE		CCT	CRI		
CCT (K)	(lm)	(W)	(lm/W)	x	y	(K)	Ra	R9	Δuv
2700	126.7	0.97	130.1	0.4599	0.4129	2717	81.1	-2.5	0.0008
3000	130.4	0.97	133.9	0.4341	0.4043	3050	82.5	2.1	0.0005
3500	133.5	0.97	137.2	0.4080	0.3935	3464	83.2	5.1	0.0006
4000	136.2	0.97	139.9	0.3818	0.3809	3993	83.3	5.6	0.0015
5000	136.2	0.97	139.9	0.3450	0.3552	5015	83.8	7.0	0.0018
5700	135.8	0.97	139.4	0.3296	0.3434	5625	84.3	9.3	0.0025
6500	134.6	0.97	138.3	0.3134	0.3301	6459	84.4	10.4	0.0034

Color-Temperature-Tunable Lighting Device—Dev.4

[0233] A color-temperature-tunable lighting device, denoted Dev.4, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.4 comprises the device of FIGS. 3A-3C and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED-C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3. The two C1 LEDs, in the first and second cavities, comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or (Y,Lu)₃(Al,Ga)₅O₁₂ and CASN) and a narrowband red phosphor (e.g., KSF). The green to red and narrowband red phosphor are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the first and second cavities to cover the violet to blue InGaN LED chip. LED-C2, in the third cavity, comprises a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or (Y,Lu)₃(Al,Ga)₅O₁₂ and CASN) and a narrowband red phosphor (e.g., KSF). The green to red phosphors and narrowband red phosphor are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the third cavity of the package to cover the violet to blue InGaN LED chip. LED-C3, in the fourth cavity, comprises a PC LED comprising green to yellow photoluminescence materials (e.g., YAG, LuAG, GaYAG, GaLuAG, and/or (Y,Lu)₃(Al,Ga)₅O₁₂). The green to yellow phosphor is incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the fourth cavity of the package to cover the violet to blue InGaN LED chip. Lighting device Dev.4 is configured to generate light with a general color rendering index CRI Ra of 90.

[0234] TABLE 8A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LED-C3 of lighting device Dev.4 and TABLE 8B tabulates CIE region, CIE center point for light C1, C2, and C3 and CCT and Δuv for light C1 and C2. FIG. 9A-9C are measured characteristics for Dev.4 in which FIG. 9A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 9B is a CIE 1931 chromaticity diagram illustrating the chromaticity (center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that Dev.4 can generate, and black body locus (dotted line), and FIG. 9C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by Dev.4 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light

emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line).

[0235] As can be seen from TABLE 8A: LED-C1 has a luminous efficacy (LE) of 107 lm/W and generates light of chromaticity (color point: CIE 0.5418, 0.4060, Δuv=−0.0012) corresponding to a CCT of 1840K (about 1800K) with a CRI Ra of 77.7 (about 78); LED-C2 has a luminous efficacy of 156 lm/W and generates light of chromaticity (color point: CIE 0.3481, 0.4405, Δuv=0.0375) corresponding to a CCT of 5122K (about 5100K) with a CRI Ra of 73.4 (about 70); and LED-C3 has a luminous efficacy of 120 lm/W and generates light with a chromaticity (color point: CIE 0.2121, 0.2383). C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.525, 0.405), CIE-2 (0.531, 0.383), CIE-3 (0.559, 0.407), and CIE-4 (0.553, 0.429) which in terms of CCT is from 1728K (about 1700K) to 1962K (about 1970K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.335, 0.445), CIE-2 (0.339, 0.415), CIE-3 (0.361, 0.436), and CIE-4 (0.357, 0.467) which in terms of CCT is from about 4796K (about 4800K) to 5455K (about 5460K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.200, 0.244), CIE-2 (0.205, 0.212), CIE-3 (0.225, 0.232), and CIE-4 (0.219, 0.265)—TABLE 8B. The CIE chromaticity diagram of FIG. 9B shows each of the CIE regions: C1 CIE region **944_{C1}** (diamond/solid line), C2 CIE region **944_{C2}** (triangle/solid line), and C3 CIE Region **944_{C3}** (square/solid line).

[0236] Referring to FIG. 9A the emission spectrum for LED-C1 (dotted line) primarily comprises a broadband peak (at about 610 nm) in the red region of the visible spectrum with multiple higher intensity narrowband peaks (FWHM≈5 nm) from about 610-650 nm (maximum intensity peak at ≈632 nm). The emission spectrum for the LED-C2 (solid line) comprises a narrowband peak (FWHM≈20 nm) with a peak emission wavelength at about 450 nm in the blue region if the visible spectrum and a higher intensity broadband peak (FWHM≈110 nm) with a peak emission wavelength of about 530 nm in the green region of the visible spectrum with multiple higher intensity narrowband peaks (FWHM≈5 nm) from about 610-650 nm (maximum intensity peak at 632 nm) in the orange to red region of the visible spectrum. The emission spectrum for the LED-C3 (dashed line) comprises a narrowband peak (FWHM≈20 nm) with a peak emission wavelength of about 450 nm in the blue region of the visible spectrum and a lower intensity (about 25%) tail at wavelengths in the green to yellow region of the spectrum (460-580 nm).

TABLE 8A

Dev. 4: Measured characteristics of LED-C1, LED-C2, and LED-C3								
LED	Flux	Power	LE	CIE center point		CCT	CRI Ra	Δuv
	(lm)	(W)	(lm/W)	x	y	(K)		
LED-C1	40.1	0.376	107	0.5418	0.4060	1840	77.7	-0.0012
LED-C2	58.7	0.376	156	0.3481	0.4405	5122	73.4	0.0375
LED-C3	45.0	0.376	120	0.2121	0.2383	n/a	n/a	n/a

TABLE 8B

Dev. 4: CIE region for light C1, C2, and C3										
CIE point	C1 CIE region				C2 CIE region				C3 CIE region	
	CIE		CCT		CIE		CCT		CIE	
	x	y	(K)	Δuv	x	y	(K)	Δuv	x	y
CIE-1	0.525	0.405	1962	-0.0025	0.335	0.445	5455	0.0438	0.200	0.244
CIE-2	0.531	0.383	1783	-0.0086	0.339	0.415	5328	0.0309	0.205	0.212
CIE-3	0.559	0.407	1728	0.0005	0.361	0.436	4796	0.0315	0.225	0.232
CIE-4	0.553	0.429	1891	0.0058	0.357	0.467	4966	0.0442	0.219	0.265
Center	0.5418	0.4060	1840	-0.0012	0.3481	0.4405	5122	0.0375	0.2121	0.2383

[0237] The CIE chromaticity diagram of FIG. 9B shows the chromaticity (center color point) $\mathbf{932}_{C1}$ of light of color C1 generated by the LED-C1, the chromaticity (center color point) $\mathbf{932}_{C2}$ of light of color C2 generated by LED-C2, and the chromaticity (center color point) $\mathbf{932}_{C3}$ of light of color C3 generated by the LED-C3. Straight lines $\mathbf{934}_{C1C2}$, $\mathbf{934}_{C2C3}$, and $\mathbf{932}_{C1C3}$ connecting the points $\mathbf{932}_{C1}$, $\mathbf{932}_{C2}$, and $\mathbf{932}_{C3}$ define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.4 can generate—i.e., the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.4 can generate that lies on the black body locus (dashed line) is about 2300K (2338K) which corresponds to the point of intersection $\mathbf{946}$ (CIE 0.492, 0.415) of line $\mathbf{934}_{C1C3}$ connecting color points $\mathbf{932}_{C1}$ to $\mathbf{932}_{C3}$ and the black body locus. It is to be noted that highest CCT of light that Dev.4 can generate that lies on the black body locus (dotted line) is about 16137K (about 16000K) which corresponds to the point of intersection $\mathbf{948}$ (CIE 0.262, 0.264) of line $\mathbf{934}_{C1C3}$ connecting color point $\mathbf{932}_{C1}$ to color point $\mathbf{932}_{C3}$ and the black body locus. It will be appreciated that Dev.4 is therefore capable of generating any CCT of light from 2300K to about 16000K lying on the black body locus.

[0238] TABLE 9 tabulates forward drive current (I_F) for first and second LED-C1, LED-C2, and LED-C3 of Dev.4 for generating light with a nominal general color rendering index CRI Ra of 90 for nominal color temperatures (CCT) from 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K. TABLE 10 tabulates the measured optical and electrical characteristics for Dev.4 when operated to generate light with a nominal CRI Ra of 90 for nominal color temperatures (CCT) from 2700K to 6500K.

[0239] As can be seen from TABLE 9, the CCT of light generated by Dev.4 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of

color C1. TABLE 10 demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and LED-C3, Dev.4 can generate white light with a CCT from 2700 K to 6500K with a general color rendering index CRI Ra of about 90 (91.3 to 92.6) and CRI R9 of at least 45 (46.3 to 64.8) with a luminous efficacy from about 126 lm/W (126.3 lm/W) to about 133 lm/W (133.1 lm/W). TABLE 10 also includes the measured CCT of light generated by Dev.4. Referring to FIG. 9C it is to be noted that Dev.4 can generate light $\mathbf{942}$ (solid line) of CCTs from 2700K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 10, the chromaticity of light generated by Dev.4 are within 0.0023 Δuv (about 0.002) or less of the black body locus for CCTs from 2700K to 6500K.

TABLE 9

Dev. 4: Forward drive current I_F for the 1 st LED-C1, 2 nd LED-C1, LED-C2, and LED-C3 for generating light with a nominal CRI Ra = 90 for nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Forward drive current I_F (mA)			
	1 st LED-C1	2 nd LED-C1	LED-C2	LED-C3
2700	105	105	103	6
3000	94	94	115	17
3500	81	81	125	32
4000	69	69	132	49
5000	54	54	125	86
5700	49	49	115	107
6500	43	43	103	130

TABLE 10

Dev. 4: Measured optical and Electrical characteristics for nominal CRI Ra = 80 for nominal color temperatures (CCT) from 2700 K to 6500 K									
Nom.	Flux	Power	LE	CIE		CCT	CRI		
CCT (K)	(lm)	(W)	(lm/W)	x	y	(K)	Ra	R9	Δ_{uv}
2700	123.0	0.97	126.3	0.4572	0.4107	2738	91.3	46.3	0.0003
3000	125.3	0.97	128.7	0.4340	0.4032	3043	92.2	53.3	0.0001
3500	127.4	0.97	130.8	0.4074	0.3921	3466	92.6	59.2	0.0002
4000	129.2	0.97	132.7	0.3828	0.3807	3966	92.3	61.6	0.0012
5000	129.6	0.97	133.1	0.3453	0.3552	5005	91.4	62.0	0.0017
5700	128.9	0.97	132.4	0.3292	0.3417	5644	91.6	64.8	0.0018
6500	128.0	0.97	131.5	0.3129	0.3274	6506	91.3	64.8	0.0023

Color-Temperature-Tunable Lighting Device—Dev.5

[0240] A color-temperature-tunable lighting device, denoted Dev.5, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.5 comprises the device of FIGS. 3A-3C and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3. In this embodiment, each of the LEDs, LED-C1, LED-C2, and LED-C3 comprise a broadband violet to blue InGaN LED chip with a FWHM from 25 nm to 50 nm. The broadband LED chips may comprise a multi-Quantum Well device.

[0241] The two LEDs LED-C1, in the first and second cavities, comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or $(Y, Lu)_3(Al, Ga)_5O_{12}$) and CASN). The green to red phosphor is incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the first and second cavities to cover the broadband violet to blue InGaN LED chip. LED-C2, in the third cavity, comprises a PC LED comprising a mixture of green to orange photoluminescence materials. The green to orange phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the third cavity of the package to cover the broadband violet to blue InGaN LED chip. LED-C3, in the fourth cavity, comprises a PC LED comprising green to yellow photoluminescence materials (e.g., GaYAG, GaLuAG, $(Y, Lu)_3(Al, Ga)_5O_{12}$, YAG and/or LuAG). The green to yellow phosphors are incorporated in a light transmissive encapsulant (e.g., phenyl silicone) and the mixture dispensed into the fourth cavity of the package to cover the broadband violet to blue InGaN LED chip. Color-temperature-tunable Lighting device Dev.5 is configured to generate light with a CCT from 2700K to 6500K and a general Color Rendering Index CRI Ra of 97.

[0242] TABLE 11A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LED-C3 of Dev.5 and TABLE 11B tabulates CIE region, CIE center point for light C1, C2, and C3 and CCT and Δ_{uv} for light C1 and C2. FIG. 10A-10C are measured characteristics for Dev.5 in which FIG. 10A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 11B is a CIE 1931 chromaticity diagram illustrating the chromaticity

(center point—cross) of light C1, C2, C3, the CIE region for light C1 (diamond/solid line), C2 (triangle/solid line), C3 (square/solid line), gamut of light (solid line) that Dev.5 can generate, and black body locus (dotted line), and FIG. 10C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by Dev.5 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 4500K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dashed line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line). FIGS. 10D-10I are measured emission spectra, normalized intensity (a.u.) versus wavelength (nm), for Dev.5 in which: FIG. 10D is the emission spectrum for operation for a nominal CCT of 2700K, FIG. 10E is the emission spectrum for operation for a nominal CCT of 3000K, FIG. 10F is the emission spectrum for operation for a nominal CCT of 3500K, FIG. 10G is the emission spectrum for operation for a nominal CCT of 4000K, FIG. 10H is the emission spectrum for operation for a nominal CCT of 5700K, and FIG. 10I is the emission spectrum for operation for a nominal CCT of 6500K.

[0243] As can be seen from TABLE 11A: LED-C1 has a luminous efficacy (LE) of 74 lm/W and generates light of chromaticity (color point: CIE 0.5323, 0.4015, $\Delta_{uv}=-0.0031$) corresponding to a CCT of 1883K (about 1900K) with a CRI Ra of 94.0; LED-C2 has a luminous efficacy of 160 lm/W and generates light of chromaticity (color point: CIE 0.3680, 0.4895, $\Delta_{uv}=0.0484$) corresponding to a CCT of 4789K (about 4800K) with a CRI Ra of 60.6 (about 61); and LED-C3 has a luminous efficacy of 148 lm/W and generates light with a chromaticity (color point: CIE 0.2228, 0.2782).

[0244] As can be seen from TABLE 11B C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.515, 0.399), CIE-2 (0.522, 0.378), CIE-3 (0.549, 0.404), and CIE-4 (0.542, 0.425) which in terms of CCT is a CIE region from 1778K (about 1750K) to 2008K (about 2020K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.356, 0.497), CIE-2 (0.358, 0.465), CIE-3 (0.380, 0.483), and CIE-4 (0.378, 0.514) which in terms of CCT is from 4,536K (about 4500K) to 5,046K (about 5050K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.217, 0.281), CIE-2 (0.220, 0.267), CIE-3 (0.228, 0.276), and CIE-4 (0.226, 0.290).

[0245] The CIE chromaticity diagram of FIG. 10B shows each of the CIE regions: C1 CIE region **1044_{C1}** (diamond/solid line), C2 CIE region **1044_{C2}** (triangle/solid line), and C3 CIE region **1044_{C3}** (square/solid line). FIG. 10B shows the chromaticity (center color point) **1032_{C1}** of light of color

C1 generated by the LED-C1, the chromaticity (center color point) 1032_{C2} of light of color C2 generated by LED-C2, and the chromaticity (center color point) 1032_{C3} of light of color C3 generated by the LED-C3. Straight lines 1034_{C1C2} , 1034_{C2C3} , and 1032_{C1C3} connecting the points 1032_{C1} , 1032_{C2} , and 1032_{C3} define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.5 can generate—i.e., the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.5 can generate that lies on the black body locus (dotted line) is 2185K (about 2150K) which corresponds to the point of intersection 1046 (CIE 0.507, 0.415) of line 1034_{C1C3} connecting color points 1032_{C1} to 1032_{C3} and the black body locus. It is to be noted that highest CCT of light that Dev.5 can generate that lies on the black body locus (dotted line) is 7620K (about 7650K) which corresponds to the point of intersection 1048 (CIE 0.299, 0.309) of line 1034_{C1C3} connecting color point 1032_{C1} to color point 1032_{C3} and the black body locus. It will be appreciated that Dev.5 is therefore capable of generating any CCT of light from 2150K to about 7650K lying on the black body locus.

[0246] As can be seen from FIG. 10A, the emission spectrum for the LED-C1 (dotted line) is composed primarily of a broadband peak (FWHM $\approx$ 100 nm) with a peak emission wavelength of about 640 nm in the red region of the visible spectrum, the emission spectrum for the LED-C2 (solid line) is composed primarily of a broadband peak (FWHM $\approx$ 140 nm) with a peak emission wavelength of about 550 nm in the green region of the visible spectrum, and the emission spectrum for the LED-C3 (dashed line) is composed of a broadband peak (FWHM $\approx$ 36 nm) with a peak emission wavelength of about 440 nm in the blue region of the visible spectrum and a broadband region at wavelengths in the green to yellow region of the spectrum (460-580 nm).

[0247] TABLE 12 tabulates forward drive current (I_F) for the first and second LEDs LED-C1, LED-C2, and LED-C3 of Dev.5 for generating light with a nominal general color rendering index CRI Ra of 90 for nominal color temperatures (CCT) from 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K. TABLE 13A tabulates the measured optical and electrical characteristics for Dev.5 when operated to generate light with a nominal CRI Ra of 97 for nominal color temperatures (CCT) from 2700K to 6500K.

[0248] As can be seen from TABLE 12, the CCT of light generated by Dev.5 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of color C1. TABLE 13A demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and LED-C3, the color-tunable multi-LED packaged light emitting device (Dev.5) can generate white light with a CCT from 2700 K to 6500K with a general color rendering index CRI Ra of about 97 (94.0 to 98.0) and CRI R9 of at least 92 (92.5 to 98.7) with a luminous efficacy from 100 lm/W (99.9 lm/W) to about 116 lm/W (116.0 lm/W). TABLE 13A also includes the measured CCT of light generated by Dev.5.

[0249] Referring to FIG. 10C it is to be noted that Dev.5 can generate light 1042 (solid line) of CCTs from 2700K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 13A, the chromaticity of light generated by Dev.5 are within 0.0026 Δuv (about 0.003) or less of the black body locus for CCTs from 2700K to 6500K.

TABLE 11A

Dev. 5: Measured characteristics of LED-C1, LED-C2, and LED-C3								
	Flux	Power	LE	CIE center point		CCT	CRI	
LED	(lm)	(W)	(lm/W)	x	y	(K)	Ra	Δuv
LED-C1	27.2	0.366	74	0.5323	0.4015	1883	94.0	-0.0031
LED-C2	58.7	0.366	160	0.3680	0.4895	4789	60.6	0.0484
LED-C3	54.2	0.366	148	0.2228	0.2782	n/a	n/a	n/a

TABLE 11B

Dev. 5: CIE region for light C1, C2, and C3										
C1 CIE region				C2 CIE region				C3 CIE region		
CIE		CCT		CIE		CCT		CIE		
CIE point	x	y	(K)	Δuv	x	y	(K)	Δuv	x	y
CIE-1	0.515	0.399	2008	-0.0048	0.356	0.497	5046	0.0546	0.217	0.281
CIE-2	0.522	0.378	1820	-0.0108	0.358	0.465	4948	0.0433	0.220	0.267
CIE-3	0.549	0.404	1778	-0.0012	0.380	0.483	4536	0.0426	0.228	0.276
CIE-4	0.542	0.425	1949	0.0041	0.378	0.514	4660	0.0534	0.226	0.290
Center	0.5323	0.4015	1883	-0.0031	0.3680	0.4895	4789	0.0484	0.2228	0.2782

TABLE 12

Dev. 5: Forward drive current I_F for the 1 st LED-C1, 2 nd LED-C1, LED-C2, and LED-C3 for generating light with a nominal CRI Ra = 97 for nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Forward drive current I_F (mA)			
	1 st LED-C1	2 nd LED-C1	LED-C2	LED-C3
2700	129	129	57	13
3000	118	118	64	26
3500	106	106	68	47
4000	95	95	70	66
5000	81	81	52	113
5700	75	75	40	137
6500	68	68	30	162

TABLE 13A

Dev. 5: Measured optical and Electrical characteristics for nominal CRI Ra = 97 for nominal color temperatures (CCT) from 2700 K to 6500 K									
Nom. CCT (K)	Flux (lm)	Power (W)	LE (lm/W)	CIE		CCT (K)	CRI		Δuv
				x	y		Ra	R9	
2700	96.9	0.97	99.9	0.4605	0.4133	2712	97.0	98.7	0.0009
3000	102.0	0.97	105.2	0.4359	0.4057	3031	97.0	97.2	0.0008
3500	106.9	0.97	110.2	0.4071	0.3930	3480	97.0	94.9	0.0006
4000	111.0	0.97	114.4	0.3841	0.3287	3946	98.0	97.2	0.0017
5000	112.5	0.97	116.0	0.3463	0.3549	4966	97.0	94.3	0.0012
5700	111.6	0.97	115.1	0.3307	0.3422	5577	95.0	92.5	0.0014
6500	111.1	0.97	114.5	0.3149	0.3299	6380	94.0	94.5	0.0026

[0250] FIGS. 10D-10I are measured emission spectra, normalized intensity (normalized to a CIE 1931 XYZ luminance $Y=100$) versus wavelength (nm), for lighting device Dev.5 in which: FIG. 10D is the emission spectrum for operation for a nominal CCT of 2700K and black body spectrum of 2700K, FIG. 10E is the emission spectrum for operation for a nominal CCT of 3000K and a black body spectrum of 3000K, FIG. 10F is the emission spectrum for operation for a nominal CCT of 3500K and a black body spectrum of 3500K, FIG. 10G is the emission spectrum for operation for a nominal CCT of 4000K and a black body spectrum of 4000K, FIG. 10H is the emission spectrum for operation for a nominal CCT of 5700K and a CIE Standard Illuminant D57 (5700K), and FIG. 10I is the emission spectrum for operation for a nominal CCT of 6500K and a CIE Standard Illuminant D65 (6500K). As can be seen from the figures, light generated by Dev.5 has a spectrum that closely resembles the black body spectrum/Standard illuminant for wavelengths from about 450 nm to 650 nm for CCT from 2700K to 6500K.

[0251] A metric for quantifying how closely the spectrum resembles the black body spectrum/Standard illuminant is a maximum (largest) intensity deviation ($I_{max\Delta}$) from the intensity of light of a black body radiator of the same Correlated Color Temperature. That is, over a selected wavelength range, $I_{max\Delta}$ is the maximum (largest) intensity difference between the intensity of the spectrum and the intensity of the spectrum of a black body radiator. The maximum deviation can be positive where the spectrum intensity is greater than the black body spectrum intensity

(e.g., a peak in the emission spectrum) or negative where the spectrum intensity is less than the black body spectrum intensity (e.g., a trough in the emission spectrum). To compare the spectra, each spectrum is normalized to have the same CIE 1931 XYZ luminance Y (e.g., $Y=100$), that is, each spectrum has the same lumen brightness. Each spectrum is normalized using the photopic luminosity function $y(\lambda)$ —sometimes referred to as the photopic or visual luminous efficiency function $v(\lambda)$ —of a standard observer which takes account of the photopic (visual) response of an observer and are for the same correlated color temperature. $I_{max\Delta}$ is thus the maximum (greatest) intensity difference between the normalized intensity of the spectrum and the normalized intensity of the black body spectrum over a selected wavelength range. $I_{max\Delta}$ is defined as:

$$I_{max\Delta} = \left[\frac{\text{Spectrum intensity @ } \lambda_{max\Delta} \times 100}{\text{Black body (Standard illuminant) spectrum intensity @ } \lambda_{max\Delta}} \right] - 100$$

where $\lambda_{max\Delta}$ is the wavelength at which the intensity difference (deviation) is a maximum.

[0252] Each of FIGS. 10D-10I indicate the maximum intensity deviation ($I_{max\Delta}$) and wavelength at which the intensity difference (deviation) is a maximum ($\lambda_{max\Delta}$) for a wavelength range 440 nm to 650 nm.

[0253] For example, when Dev.5 is operable to generate light with a color temperature of 2700K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of a black body of the same color temperature (i.e., 2700K) is about 12% (0.12) corresponding to peak 1050 at a wavelength $\lambda_{max\Delta}$ of 635 nm (FIG. 10D). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted 1052 and the black body spectrum intensity at $\lambda_{max\Delta}$ is denoted 1054. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 2700K has a maximum percentage intensity deviation $I_{max\Delta}$ of 12%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 112% of the normalized black body spectrum intensity at the same wavelength.

[0254] When Dev.5 is operable to generate light with a color temperature of 3000K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of the black body of the same color temperature (i.e., 3000K) is about 23% (0.23) corre-

sponding to peak **1050** at a wavelength $\lambda_{max\Delta}$ of 450 nm (FIG. 10E). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted **1052** and the black body spectrum intensity at $\lambda_{max\Delta}$ is denoted **1054**. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 3000K has a maximum percentage intensity deviation $I_{max\Delta}$ of 23%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 123% of the normalized black body spectrum intensity at the same wavelength.

[0255] When Dev.5 is operable to generate light with a color temperature of 3500K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of the black body of the same color temperature (i.e., 3500K) is about 23% (0.23) corresponding to peak **1050** at a wavelength $\lambda_{max\Delta}$ of 450 nm (FIG. 10F). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted **1052** and the black body spectrum intensity at $\lambda_{max\Delta}$ is denoted **1054**. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 3500K has a maximum percentage intensity deviation $I_{max\Delta}$ of 23%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 123% of the normalized black body spectrum intensity at the same wavelength.

[0256] When Dev.5 is operable to generate light with a color temperature of 4000K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of the black body of the same color temperature (i.e., 4000K) is about 28% (0.28) corresponding to peak **1050** at a wavelength $\lambda_{max\Delta}$ of 440 nm (FIG. 10G). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted **1052** and the black body spectrum intensity at $\lambda_{max\Delta}$ is denoted **1054**. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 4000K has a maximum percentage intensity deviation $I_{max\Delta}$ of 28%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 128% of the normalized black body spectrum intensity at the same wavelength.

[0257] When Dev.5 is operable to generate light with a color temperature of 5700K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of the Standard Illuminant D57 of the same color temperature (i.e., 5700K) is about 38% (0.38) corresponding to peak **1050** at a wavelength $\lambda_{max\Delta}$ of 440 nm (FIG. 10H). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted **1052** and the Standard Illuminant spectrum intensity at $\lambda_{max\Delta}$ is denoted **1054**. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 5700K has a maximum percentage intensity deviation $I_{max\Delta}$ of 38%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 138% of the Standard Illuminant spectrum intensity at the same wavelength.

[0258] When Dev.5 is operable to generate light with a color temperature of 6500K, over a wavelength range from 440 nm to 650 nm, the maximum deviation of the spectrum of Dev.5 from the spectrum of the Standard Illuminant D65 of the same color temperature (i.e., 6500K) is about 33% (0.33) close to peak **1050** at a wavelength $\lambda_{max\Delta}$ of 440 nm (FIG. 10I). In the figure, the spectrum intensity at $\lambda_{max\Delta}$ is denoted **1052** and the Standard Illuminant spectrum inten-

sity at $\lambda_{max\Delta}$ is denoted **1054**. Therefore, over the wavelength range from 440 nm to 650 nm, light generated by Dev.5 when operated to generate light of color temperature 6500K has a maximum percentage intensity deviation $I_{max\Delta}$ of 38%, that is at the maximum intensity deviation the normalized spectrum intensity at wavelength $\lambda_{max\Delta}$ is 138% of the Standard Illuminant spectrum intensity at the same wavelength.

[0259] TABLE 13B tabulates maximum intensity deviation ($I_{max\Delta}$) and wavelength of maximum deviation ($\lambda_{max\Delta}$) for wavelength ranges 440 nm to 650 nm and 460 nm to 650 nm for Dev.5 operable to generate light of nominal CCTs from 2700K to 6500K. As can be seen from the table Dev.5 is capable of generating light of color temperatures from 2700K to 6500K having an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, has a maximum deviation between the normalized intensity of light generated by the lighting device compared with the normalized intensity of light of the spectrum of a black body radiator or standard illuminant which is less than 0.38 (38%). As can be seen from the table Dev.5 is capable of generating light of color temperatures from 2700K to 6500K having an intensity versus wavelength spectrum, which over a wavelength range 460 nm to 650 nm, has a maximum deviation between the normalized intensity of light generated by the lighting device compared with the normalized intensity of light of the spectrum of a black body radiator or standard illuminant which is less than 0.25 (25%). Test data indicates that through further optimization, devices in accordance with the invention can generate light with color temperatures from 2700K to 6500K with the maximum deviation of less than 0.3, 0.2, and/or less than 0.1 over a wavelength range 440 nm to 650 nm.

TABLE 13B

Maximum intensity deviation ($I_{max\Delta}$) and wavelength of maximum deviation ($\lambda_{max\Delta}$) for wavelength ranges 440 nm to 650 nm and 460 nm to 650 nm for Dev. 5 operable to generate light of nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Wavelength range 440 nm to 650 nm		Wavelength range 460 nm to 650 nm	
	$I_{max\Delta}$	$\lambda_{max\Delta}$	$I_{max\Delta}$	$\lambda_{max\Delta}$ (nm)
2700	12% (0.12)	635 nm	12% (0.12)	635 nm
3000	23% (0.23)	450 nm	23% (0.23)	450 nm
3500	23% (0.23)	450 nm	12% (0.12)	630 nm
4000	28% (0.28)	440 nm	-17% (-0.17)	460 nm
5700	38% (0.38)	440 nm	-25% (-0.25)	460 nm
6500	38% (0.36)	440 nm	-25% (-0.25)	460 nm

CIE Region and Center Points for C1, C2, C3

[0260] TABLE 14 tabulates CIE regions, CIE center points for light C1, C2, and C3 and CCT and Auv for light C1 and C2. The CIE regions for each color C1, C2, and C3 encompass each of the CIE regions for Devs.2-5, that is for lighting devices with a CRI Ra from 80 to 97. As can be seen from TABLE 14, C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.498, 0.422), CIE-2 (0.522, 0.370), CIE-3 (0.562, 0.410), and CIE-4 (0.537, 0.461) which in terms of CCT is a CIE region from 1725K (about 1720K) to 2323K (about 2350K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.330, 0.480), CIE-2 (0.342,

0.385), CIE-3 (0.392, 0.420), and CIE-4 (0.378, 0.514) which in terms of CCT is from 4004K (about 4000K) to 5580K. C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.196, 0.261), CIE-2 (0.206, 0.212), CIE-3 (0.234, 0.243), and CIE-4 (0.226, 0.290).

TABLE 14

CIE regions and center points for light C1, C2, and C3										
	C1				C2				C3	
	CIE		CCT		CIE		CCT		CIE	
CIE point	x	y	(K)	Δuv	x	y	(K)	Δuv	x	y
CIE-1	0.498	0.422	2323	0.0022	0.330	0.480	5580	0.0575	0.196	0.261
CIE-2	0.522	0.370	1771	-0.0132	0.342	0.385	5195	0.0171	0.206	0.212
CIE-3	0.562	0.410	1725	0.0016	0.392	0.420	4004	0.0157	0.234	0.243
CIE-4	0.537	0.461	2203	0.0137	0.378	0.514	4665	0.0534	0.226	0.290
Center	0.530	0.415	1983	0.0006	0.360	0.450	4859	0.0371	0.216	0.251

[0261] FIGS. 11A-11D are CIE 1931 chromaticity diagrams illustrating various CIE regions and center points for light C1, C2, and C3.

[0262] FIG. 11A shows C1 CIE regions **1144** and C1 center points **1132** for lighting devices Dev.2 (**1144**_{Dev.2}—solid square/solid line and **1132**_{Dev.2}—solid square), Dev.3 (**1144**_{Dev.3}—dot/dotted line and **1132**_{Dev.3}—dot), Dev.4 (**1144**_{Dev.4}—star/dashed line and **1132**_{Dev.4}—star), and Dev.5 (**1144**_{Dev.5}—solid triangle/dashed line and **1132**_{Dev.5}—solid triangle) and C1 CIE region (**1144**_{C1}—diamond/dashed dot line) and C1 center point (**1132**_{C1}—diamond) that encompass lighting devices Devs.2-5. As can be seen from FIG. 11A, the C1 CIE region **1144**_{C1} encompasses (fully encloses) the C1 CIE regions **1144**_{Dev.2}, **1144**_{Dev.3}, **1144**_{Dev.4}, **1144**_{Dev.5}.

[0263] FIG. 11B shows C2 CIE regions **1144** and C2 center points **1132** for lighting devices Dev.2 (**1144**_{Dev.2}—solid square/solid line and **1132**_{Dev.2}—solid square), Dev.3 (**1144**_{Dev.3}—dot/dotted line and **1132**_{Dev.3}—dot), Dev.4 (**1144**_{Dev.4}—star/dashed line and **1132**_{Dev.4}—star), and Dev.5 (**1144**_{Dev.5}—solid triangle/dashed line and **1132**_{Dev.5}—solid triangle) and C2 CIE region (**1144**_{C2}—diamond/dashed dot line) and C2 center point (**1132**_{C2}—diamond) that encompass lighting devices Devs.2-5. As can be seen from FIG. 11B, the C2 CIE region **1144**_{C2} encompasses (fully encloses) the C2 CIE regions **1144**_{Dev.2}, **1144**_{Dev.3}, **1144**_{Dev.4}, **1144**_{Dev.5}.

[0264] FIG. 11C shows C3 CIE regions **1144** and C3 center points **1132** for lighting devices Devs.2-4 (**1144**_{Devs.2-4}—solid square/solid line and **1132**_{Devs.2-4}—solid square) and Dev.5 (**1144**_{Dev.5}—solid triangle/dotted line and **1132**_{Dev.5}—solid triangle) and C3 CIE region (**1144**_{C3}—diamond/dashed dot line) and C3 center point (**1132**_{C3}—diamond) that encompass lighting devices Devs.2-5. As can be seen from FIG. 11C, the C3 CIE region **1144**_{C3} encompasses (fully encloses) the C3 CIE regions **1144**_{Dev.2}, **1144**_{Dev.3}, **1144**_{Dev.4}, **1144**_{Dev.5}.

[0265] FIG. 11D shows CIE regions **1144** and center points **1132** for C1 (**1144**_{C1}—solid diamond/solid line and **1132**_{C1}—solid diamond), C2 (**1144**_{C2}—solid triangle/solid line and **1132**_{C2}—solid triangle), and C3 (**1144**_{C3}—solid square/solid line and **1132**_{C3}—solid square) that encompass lighting devices Dev.2-5.

Lighting Devices Comprising Light of Four Different Colors/CCTs: C1, C2, C3 and C4

[0266] As described herein, lighting devices in accordance with embodiments of the invention may comprise LEDs that generate light of four different colors/CCTs: C1, C2, C3 and C4.

[0267] In embodiments, a four color/CCT lighting device may comprise, for example, the lighting devices Dev.2-5 that further comprises a fourth LED that generates light of a fourth chromaticity in a fourth chromaticity region. In embodiments, the fourth center point and chromaticity region can be (i) between the first and second chromaticity regions or (ii) between the second and third chromaticity regions. TABLES 15A and 15B tabulate CIE regions, CIE center points for light C1, C2, C3, C4a, and C4b, CCT and Δuv for light C1, C2, C4a, C4b. The CIE regions for each color C1, C2, C3, C4a and C4b encompass lighting devices that can generate light with a CRI Ra from 80 to 97. Center point C4a and CIE region C4a are between the first and second chromaticity regions and center point C4b and CIE region C4b are between the second and third chromaticity regions. The center points and CIE regions for C1, C2, and C3 are tabulated in TABLE 15A. As can be seen from TABLE 15B, C4a comprises a CIE region (C4a CIE region) defined by CIE color coordinates CIE-1 (0.420, 0.468), CIE-2 (0.418, 0.383), CIE-3 (0.475, 0.430), and CIE-4 (0.479, 0.514) which in terms of CCT is a CIE region from 2644K (about 2620K) to 3736K (about 3750K). C4b comprises a CIE region (C4b CIE Region) defined by CIE color coordinates CIE-1 (0.249, 0.362), CIE-2 (0.265, 0.283), CIE-3 (0.309, 0.335), and CIE-4 (0.295, 0.414) which in terms of CCT is from 6654K (about 6650K) to 12482K (about 12500K).

[0268] FIG. 12 shows CIE regions **1244** and center points **1232** for C1 (**1244**_{C1}—solid diamond/solid line and **1232**_{C1}—solid diamond), C2 (**1244**_{C2}—solid triangle/solid line and **1232**_{C2}—solid triangle), C3 (**1244**_{C3}—solid square/solid line and **1232**_{C3}—solid square), C4a (**1244**_{C4a}—star/dashed line and **1232**_{C4a}—star), and C4b (**1244**_{C4b}—cross/dashed line and **1232**_{C4b}—cross).

TABLE 15A

CIE regions and center points for light C1, C2, and C3										
CIE point	C1				C2				C3	
	CIE		CCT	Δuv	CIE		CCT	Δuv	CIE	
	x	y	(K)		x	y	(K)		x	y
CIE-1	0.498	0.422	2323	0.0022	0.330	0.480	5580	0.0575	0.196	0.261
CIE-2	0.522	0.370	1771	-0.0132	0.342	0.385	5195	0.0171	0.206	0.212
CIE-3	0.562	0.410	1725	0.0016	0.392	0.420	4004	0.0157	0.234	0.243
CIE-4	0.537	0.461	2203	0.0137	0.378	0.514	4665	0.0534	0.226	0.290
Center	0.530	0.415	1983	0.0006	0.360	0.450	4859	0.0371	0.216	0.251

TABLE 15B

CIE regions and center points for light C4a and C4b								
CIE point	C4a				C4b			
	CIE		CCT	Δuv	CIE		CCT	Δuv
	x	y	(K)		x	y	(K)	
CIE-1	0.420	0.468	3736	0.0274	0.249	0.362	9582	0.0508
CIE-2	0.418	0.383	3167	-0.0059	0.265	0.283	12482	0.0073
CIE-3	0.475	0.430	2644	0.0058	0.309	0.335	6654	0.0081
CIE-4	0.479	0.514	3135	0.0324	0.295	0.414	6700	0.0483
Center	0.450	0.450	3135	0.0155	0.280	0.350	8029	0.0299

[0269] In embodiments, a four color/CCT lighting device may comprise for example a first LED that generates light of a first chromaticity C1, a second LED that generates light of a second chromaticity C2, a third LED that generates light of a third chromaticity C3, and a fourth LED that generates light of a fourth chromaticity C4. TABLES 16A and 16B tabulate CIE regions, CIE center points for light C1, C2, C3 and C4, CCT and Δuv for light C1, C2, and C3. The CIE regions for each color C1, C2, C3, and C4 encompass lighting devices that can generate light with a CRI Ra from 80 to 97. As can be seen from TABLES 16A and 16B, C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.498, 0.422), CIE-2 (0.522, 0.370), CIE-3 (0.562, 0.410), and CIE-4 (0.537, 0.461) which in terms of CCT is a CIE region from 1725K (about 1720K) to 2323K (about 2350K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.390, 0.488), CIE-2 (0.398, 0.403), CIE-3 (0.450, 0.450), and CIE-4 (0.442, 0.534) which in terms of CCT is from 3135K (about 3120K) to 4366K (about 4370K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.277, 0.418), CIE-2 (0.283, 0.333), CIE-3 (0.334, 0.380), and CIE-4 (0.327, 0.464) which in terms of CCT is from 5456K (about 5450K) to 8192K (about 8200K). C4 comprises a CIE region (C4 CIE Region) defined by CIE color coordinates CIE-1 (0.196, 0.261), CIE-2 (0.206, 0.212), CIE-3 (0.234, 0.243), and CIE-4 (0.226, 0.290).

[0270] FIG. 13 shows CIE regions 1344 and center points 1332 for C1 (1344_{C1}—solid diamond/solid line and 1332_{C1}—solid diamond), C2 (1344_{C2}—solid triangle/solid line and 1332_{C2}—solid triangle), C3 (1344_{C3}—star/solid line and 1332_{C3}—star), and C4 (1344_{C4}—solid square/solid line and 1332_{C4}—solid square).

TABLE 16A

CIE regions and center points for light C1 and C2								
CIE point	C1				C2			
	CIE		CCT	Δuv	CIE		CCT	Δuv
	x	y	(K)		x	y	(K)	
CIE-1	0.498	0.422	2323	0.0022	0.390	0.488	4366	0.0417
CIE-2	0.522	0.370	1771	-0.0132	0.398	0.403	3757	0.0071
CIE-3	0.562	0.410	1725	0.0016	0.450	0.450	3135	0.0154
CIE-4	0.537	0.461	2203	0.0137	0.442	0.534	3728	0.0445
Center	0.530	0.415	1983	0.0006	0.420	0.470	3747	0.0281

TABLE 16B

CIE regions and center points for light C3 and C4							
CIE point	C3				C4		
	CIE		CCT	Δuv	CIE		Y
	x	y	(K)		x	Y	
CIE-1	0.277	0.418	7302	0.0575	0.196	0.261	
CIE-2	0.283	0.333	8192	0.0207	0.206	0.212	
CIE-3	0.334	0.380	5456	0.0180	0.234	0.243	
CIE-4	0.327	0.464	5654	0.0533	0.226	0.290	
Center	0.305	0.400	6426	0.0387	0.216	0.251	

Color-Temperature-Tunable Lighting Device—Dev.6

[0271] A color-temperature-tunable lighting device, denoted Dev.6, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.6 comprises the device of FIGS. 3A-3C and comprises

a 3838 four cavity package containing an LED, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED C2, that generates of a second chromaticity/color temperature C2, and two LEDs, LED-C3, that generates light of a third chromaticity/color temperature C3. In this embodiment, each of the LEDs, LED-C1, LED-C2, and LED-C3 comprise a broadband violet to blue InGaN LED chip with a FWHM from 25 nm to 50 nm. The broadband LED chips may comprise a multi-Quantum Well device.

[0272] The two LEDs LED-C3 comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or $(Y, Lu)_3(Al, Ga)_5O_{12}$ and CASN). LED-C2, in the third cavity, comprises a PC LED comprising a mixture of green to orange photoluminescence materials. LED-C1 comprises a PC LED comprising green to yellow photoluminescence materials (e.g., GaYAG, GaLuAG, $(Y, Lu)_3(Al, Ga)_5O_{12}$, YAG and/or LuAG). Color-temperature-tunable Lighting device Dev.6 is configured to generate light with a CCT from 2700K to 6500K and a general Color Rendering Index CRI Ra of 97.

[0273] TABLE 17A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LEDs, LED-C3 of Dev.6 and TABLES 17B-17D tabulates CIE region, CIE center point, CCT, and Δuv for light C1, C2, and C3.

[0274] FIG. 14A-14C are measured characteristics for a color-temperature-tunable lighting device (Dev.6) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. 14A shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. 14B is a CIE 1931 chromaticity diagram illustrating the chromaticity (CIE center point—cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.6 can generate, and black body locus (dotted line), and FIG. 14C is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.6 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dashed line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line). FIGS. 14D-14J are measured emission spectra (solid line), normalized intensity versus wavelength (nm), for lighting device Dev.6 in which: FIG. 14D is the emission spectrum for operation for a nominal CCT of 2700K, FIG. 14E is the emission spectrum for operation for a nominal CCT of 3000K, FIG. 14F is the emission spectrum for operation for a nominal CCT of 3500K, FIG. 14G is the emission spectrum for operation for a nominal CCT of 4000K, FIG. 10H is the emission spectrum for operation for a nominal CCT of 5000K, FIG. 14I is the emission spectrum for operation for a nominal CCT of 5700K, and FIG. 14J is the emission spectrum for operation for a nominal CCT of 6500K.

[0275] As can be seen from TABLE 17A: LED-C1 has a luminous efficacy (LE) of 102 lm/W and generates light of chromaticity (color point: CIE 0.511, 0.386, $\Delta uv = -0.0090$) corresponding to a CCT of 1955K (about 1950K) with a CRI Ra of 91.8; LED-C2 has a luminous efficacy of 137 lm/W and generates light of chromaticity (color point: CIE 0.443, 0.453, $\Delta uv = 0.0175$) corresponding to a CCT of 3267K (about 3300K) with a CRI Ra of 88.7 (about 90); and LED-C3 has a luminous efficacy of 135 lm/W and generates

light with a chromaticity of chromaticity (color point: CIE 0.276, 0.309, $\Delta uv = 0.0136$) corresponding to a CCT of 9461K (about 9500K) with a CRI Ra of 89.9 (about 90).

[0276] As can be seen from TABLE 17B: C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.495, 0.384), CIE-2 (0.501, 0.363), CIE-3 (0.529, 0.388), and CIE-4 (0.522, 0.409) which in terms of CCT is a CIE region from 1831K (about 1800K) to 2094K (about 2100K). The C1 CIE region corresponds to about 6 SDCM centered on a CIE center (0.511, 0.386). As can be seen from TABLE 17C: C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.430, 0.457), CIE-2 (0.433, 0.426), CIE-3 (0.455, 0.449), and CIE-4 (0.452, 0.479) which in terms of CCT is from 3046K (about 3050K) to 3479K (about 3500K). The C2 CIE region corresponds to about 6 SDCM centered on a CIE center (0.443, 0.453). As can be seen from TABLE 17D: C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.270, 0.310), CIE-2 (0.274, 0.297), CIE-3 (0.281, 0.308), and CIE-4 (0.278, 0.321) which in terms of CCT is from 8873K (about 8500K) to 10234K (about 10500K).

[0277] The CIE chromaticity diagram of FIG. 14B shows each of the CIE regions: C1 CIE region 1444_{C1} (diamond/solid line), C2 CIE region 1044_{C2} (triangle/solid line), and C3 CIE region 1044_{C3} (square/solid line). FIG. 14B shows the chromaticity (center color point) 1432_{C1} of light of color C1 generated by the LED-C1, the chromaticity (center color point) 1432_{C2} of light of color C2 generated by LED-C2, and the chromaticity (center color point) 1432_{C3} of light of color C3 generated by the LED-C3. Straight lines 1434_{C1C2}, 1434_{C2C3}, and 1432_{C1C3} connecting the points 1432_{C1}, 1432_{C2}, and 1432_{C3} define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.6 can generate—i.e., the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.6 can generate that lies on the black body locus (dotted line) is 1695K (about 1700K) which corresponds to the point of intersection 1446 (CIE 0.4825, 0.4140) of line 1434_{C1C3} connecting color points 1432_{C1} to 1432_{C3} and the black body locus. It is to be noted that highest CCT of light that Dev.6 can generate that lies on the black body locus (dotted line) is 6668K (about 6600K) which corresponds to the point of intersection 1448 (CIE 0.3110, 0.3210) of line 1434_{C1C3} connecting color point 1432_{C1} to color point 1432_{C3} and the black body locus. It will be appreciated that Dev.6 is therefore capable of generating any CCT of light from 1700K to about 6600K lying on the black body locus.

[0278] As can be seen from FIG. 14A, the emission spectrum for the LED-C1 (dotted line) is composed primarily of a broadband peak (FWHM≈100 nm) with a peak emission wavelength of about 640 nm in the red region of the visible spectrum, the emission spectrum for the LED-C2 (solid line) is composed primarily of a broadband peak (FWHM≈200 nm) centered at a wavelength of about 590 nm in the yellow region of the visible spectrum, and the emission spectrum for the LED-C3 (dashed line) is composed of a broadband peak (FWHM≈36 nm) with a peak emission wavelength of about 440 nm in the blue region of the visible spectrum and a broadband tail region at wavelengths in the green to red region of the spectrum (460-680 nm).

TABLE 17A

Dev. 6: Measured characteristics of LED-C1, LED-C2, and LED-C3								
LED	Flux	Power	LE	CIE center point		CCT	CRI	Δ_{uv}
	(lm)	(W)	(lm/W)	x	y	(K)	Ra	
LED-C1	17.2	0.17	102	0.5114	0.3860	1955	91.8	-0.0090
LED-C2	23.8	0.17	137	0.4426	0.4526	3267	88.7	0.0175
LED-C3	23.8	0.18	135	0.2757	0.3092	9461	89.9	0.0136

TABLE 17B

Dev. 6: CIE region for light C1				
CIE point	C1 CIE region		CCT (K)	Δ_{uv}
	x	y		
CIE-1	0.495	0.384	2094	-0.0103
CIE-2	0.501	0.363	1892	-0.0170
CIE-3	0.529	0.388	1831	-0.0007
CIE-4	0.522	0.409	2016	-0.0015
Center	0.511	0.386	1955	-0.0090

TABLE 17C

Dev. 6: CIE region for light C2				
CIE point	C2 CIE region		CCT (K)	Δ_{uv}
	x	y		
CIE-1	0.430	0.457	3497	0.0212
CIE-2	0.433	0.426	3236	0.0093
CIE-3	0.455	0.449	3046	0.0142
CIE-4	0.452	0.479	3294	0.0220
Center	0.443	0.453	3267	0.0175

TABLE 17D

Dev. 6: CIE region for light C3				
CIE point	C3 CIE region		CCT (K)	Δ_{uv}
	x	Y		
CIE-1	0.270	0.310	9927	0.0174
CIE-2	0.274	0.297	10234	0.0090
CIE-3	0.281	0.308	9013	0.0099
CIE-4	0.278	0.321	8873	0.0181
Center	0.276	0.309	9465	0.0136

[0279] TABLE 18 tabulates forward drive current (I_F) for the first and second LEDs LED-C3, LED-C2, and LED-C1 of Dev.6 for generating light with a nominal general color rendering index CRI Ra of 97 for nominal color temperatures (CCT) of 2700K, 3000K, 3500K, 4000K, 5000K,

5700K, and 6500K. TABLE 19 tabulates the measured optical and electrical characteristics for Dev.6 when operated to generate light with a nominal CRI Ra of 97 for nominal color temperatures (CCT) from 2700K to 6500K.

[0280] As can be seen from TABLE 18, the CCT of light generated by Dev.6 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of color C1 and C2. TABLE 19 demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and LED-C3, the color-tunable multi-LED packaged light emitting device (Dev.6) can generate white light with a CCT from 2700 K to 6500K with a general color rendering index CRI Ra of at least 97 (97.1 to 99.0) and CRI R9 of at least 90 (90.4 to 99.0) with a luminous efficacy from 102 lm/W (102.1 lm/W) to about 122 lm/W (121.6 lm/W). TABLE 19 also includes the measured CCT of light generated by Dev.6.

[0281] Referring to FIG. 14C it is to be noted that Dev.6 can generate light 1442 (solid line) of CCTs from 2700K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 19, the chromaticity of light generated by Dev.6 are within 0.0032 Δ_{uv} (about 0.003) or less of the black body locus for CCTs from 2700K to 6500K.

TABLE 18

Dev. 6: Forward drive current I_F for the 1 st LED-C3, 2 nd LED-C3, LED-C2, and LED-C1 for generating light with a nominal CRI Ra = 97 for nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Forward drive current I_F (mA)			
	LED-C1	LED-C2	1 st LED-C3	2 nd LED-C3
2700	149.9	134.5	7.8	7.8
3000	119.8	145.7	17.3	17.3
3500	89.7	148.3	31.0	31.0
4000	59.2	146.9	46.9	46.9
5000	44.6	97.0	79.2	79.2
5700	44.4	63.4	96.1	96.1
6500	42.9	27.6	114.8	114.8

TABLE 19A

Dev. 6: Measured optical and Electrical characteristics for nominal CRI Ra = 97 for nominal color temperatures (CCT) from 2700 K to 6500 K									
Nom.	Flux	Power	LE	CIE		CCT	CRI		
CCT (K)	(lm)	(W)	(lm/W)	x	y	(K)	Ra	R9	Δuv
2700	91.4	0.90	102.1	0.4571	0.4102	2736	97.7	90.4	0.0001
3000	95.3	0.90	106.8	0.4340	0.4025	3038	97.1	95.2	-0.0002
3500	99.3	0.89	112.1	0.4068	0.3913	3471	98.4	98.5	0.0001
4000	102.8	0.89	116.0	0.3813	0.3805	4005	99.0	97.0	0.0015
5000	106.5	0.87	121.6	0.3445	0.3551	5033	97.9	96.7	0.0020
5700	106.1	0.87	121.5	0.3290	0.3424	5654	97.6	99.0	0.0022
6500	106.2	0.88	120.3	0.3130	0.3293	6488	97.2	97.8	0.0032

[0282] FIGS. 14D-14J are measured emission spectra, normalized intensity (normalized to a CIE 1931 XYZ luminance $Y=100$) versus wavelength (nm), for lighting device Dev.6 in which: FIG. 14D is the emission spectrum for operation for a nominal CCT of 2700K and black body spectrum of 2700K, FIG. 14E is the emission spectrum for operation for a nominal CCT of 3000K and a black body spectrum of 3000K, FIG. 14F is the emission spectrum for operation for a nominal CCT of 3500K and a black body spectrum of 3500K, FIG. 14G is the emission spectrum for operation for a nominal CCT of 4000K and a black body spectrum of 4000K, FIG. 14H is the emission spectrum for operation for a nominal CCT of 5000K and a black body spectrum of 5000K, FIG. 14I is the emission spectrum for operation for a nominal CCT of 5700K and a CIE Standard Illuminant D57 (5700K), and FIG. 14J is the emission spectrum for operation for a nominal CCT of 6500K and a CIE Standard Illuminant D65 (6500K). As can be seen from the figures, light generated by Dev.6 has a spectrum that closely resembles the black body spectrum/Standard illuminant for wavelengths from about 450 nm to 650 nm. Each of FIGS. 14D-10J show the maximum intensity deviation ($I_{max\Delta}$) and wavelength at which the intensity difference (deviation) is a maximum ($\lambda_{max\Delta}$) for a wavelength range 440 nm to 650 nm.

[0283] TABLE 19B tabulates maximum intensity deviation ($I_{max\Delta}$) and wavelength of maximum deviation ($\lambda_{max\Delta}$) for wavelength ranges 440 nm to 650 nm and 460 nm to 650 nm for Dev.6 operable to generate light of nominal CCTs from 2700K to 6500K. As can be seen from the table Dev.6 is capable of generating light of color temperatures from 2700K to 6500K having an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, has a maximum deviation between the normalized intensity of light generated by the lighting device compared with the normalized intensity of light of the spectrum of a black body radiator or standard illuminant which is less than 0.38 (38%). As can be seen from the table Dev.6 is capable of generating light of color temperatures from 2700K to 6500K having an intensity versus wavelength spectrum, which over a wavelength range 460 nm to 650 nm, has a maximum deviation between the normalized intensity of light generated by the lighting device compared with the normalized intensity of light of the spectrum of a black body radiator or standard illuminant which is less than 0.22 (22%). Test data indicates that through further optimization, devices in accordance with the invention can generate light with color temperatures from 2700K to 6500K with the maximum

deviation of less than 0.3, 0.2, and/or less than 0.1 over a wavelength range 440 nm to 650 nm.

TABLE 19B

Maximum intensity deviation ($I_{max\Delta}$) and wavelength of maximum deviation ($\lambda_{max\Delta}$) for wavelength ranges 440 nm to 650 nm and 460 nm to 650 nm for Dev. 6 operable to generate light of nominal CCTs from 2700K to 6500K				
Nom. CCT (K)	Wavelength range 440 nm to 650 nm		Wavelength range 460 nm to 650 nm	
	$I_{max\Delta}$	$\lambda_{max\Delta}$	$I_{max\Delta}$	$\lambda_{max\Delta}$ (nm)
2700	38% (0.40)	450 nm	16% (0.16)	475 nm
3000	36% (0.35)	455 nm	12% (0.12)	475 nm
3500	29% (0.29)	455 nm	11% (0.11)	630 nm
4000	19% (0.19)	450 nm	14% (0.14)	460 nm
5000	19% (0.19)	460 nm	19% (0.19)	460 nm
5700	23% (0.23)	440 nm	22% (0.22)	462 nm
6500	23% (0.23)	440 nm	22% (0.22)	462 nm

Color-Temperature-Tunable Lighting Device—Dev.7

[0284] A color-temperature-tunable lighting device, denoted Dev.7, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.7 comprises the device of FIGS. 3A-3C and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED-C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3. The two C1 LEDs, in the first and second cavities, comprise a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or $(Y,Lu)_3(Al,Ga)_5O_{12}$ and CASN) and a narrowband red phosphor (e.g., KSF). LED-C2, in the third cavity, comprises a PC LED comprising a mixture of green to red photoluminescence materials (e.g., GaYAG, GaLuAG, and/or $(Y,Lu)_3(Al,Ga)_5O_{12}$ and CASN) and a narrowband red phosphor (e.g., KSF). LED-C3, in the fourth cavity, comprises a PC LED comprising green to yellow photoluminescence materials (e.g., YAG, LuAG, GaYAG, GaLuAG, and/or $(Y,Lu)_3(Al,Ga)_5O_{12}$). The Lighting device, Dev.7, is configured to generate light with a general color rendering index CRI Ra of 90.

[0285] TABLE 20A tabulates measured optical/electrical characteristics of LED-C1, LED-C2, and LED-C3 of lighting device Dev.7 and TABLE 20B tabulates CIE region, CIE center point, CCT and Δuv for light C1, C2, and C3. FIG.

15A-15C are measured characteristics for a color-temperature-tunable lighting device (Dev.7) comprising LEDs of three chromaticity/color temperatures C1, C2, C3 in which FIG. **15A** shows spectra, normalized intensity (a.u.) versus wavelength (nm), for light C1 (dotted line), light C2 (solid line), and light C3 (dashed line), FIG. **15B** is a CIE 1931 chromaticity diagram illustrating the chromaticity (CIE center point-cross) of light C1, C2, C3, the CIE region for light C1 (solid line/diamond), C2 (solid line/triangle), C3 (solid line/square), gamut of light (solid line) that lighting device Dev.7 can generate, and black body locus (dotted line), and FIG. **15C** is a CIE 1931 chromaticity diagram illustrating the chromaticity (color) of light (solid square) generated by lighting device Dev.7 for nominal CCTs of 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K, light emission locus (solid line) for CCTs from 2700K to 6500K, black body locus (dotted line), and 3 SDCM and 6 SDCM (Mac Adam ellipses—thin solid line)

[0286] As can be seen from TABLE 20A: LED-C1 has a luminous efficacy (LE) of 121.7 lm/W (122 lm/W) and generates light of chromaticity (color point: CIE 0.5557, 0.3974, $\Delta uv = -0.0023$) corresponding to a CCT of 1700K with a CRI Ra of 81.5 (about 82); LED-C2 has a luminous efficacy of 176.9 lm/W (177 lm/W) and generates light of chromaticity (color point: CIE 0.4575, 0.4991, $\Delta uv = 0.0309$) corresponding to a CCT of 3333K (about 3300K) with a CRI Ra of 81.2 (about 81); and LED-C3 has a luminous efficacy of 143.9 lm/W (144 lm/W) and generates light with a chromaticity (color point: CIE 0.218, 0.264). C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.539, 0.396), CIE-2 (0.545, 0.374), CIE-3 (0.572, 0.399), and CIE-4 (0.567, 0.421) which in terms of

CCT is from 1605K (about 1600K) to 1800K. C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.445, 0.503), CIE-2 (0.448, 0.473), CIE-3 (0.470, 0.496), and CIE-4 (0.467, 0.526) which in terms of CCT is from about 3139K (about 3100K) to 3536K (about 3550K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.207, 0.275), CIE-2 (0.210, 0.240), CIE-3 (0.230, 0.254), and CIE-4 (0.227, 0.289)—TABLE 20B. The CIE chromaticity diagram of FIG. **15B** shows each of the CIE regions: C1 CIE region **1544**_{C1} (diamond/solid line), C2 CIE region **1544**_{C2} (triangle/solid line), and C3 CIE Region **1544**_{C3} (square/solid line).

[0287] Referring to FIG. **15A** the emission spectrum for LED-C1 (dotted line) primarily comprises a broadband peak (at about 610 nm) in the red region of the visible spectrum with multiple higher intensity narrowband peaks (FWHM $\approx$ 5 nm) from about 610-650 nm (maximum intensity peak at $\approx$ 632 nm). The emission spectrum for the LED-C2 (solid line) comprises a narrowband peak (FWHM $\approx$ 20 nm) with a peak emission wavelength at about 450 nm in the blue region of the visible spectrum and a higher intensity broadband peak (FWHM $\approx$ 110 nm) with a peak emission wavelength of about 550 nm in the yellow to green region of the visible spectrum with multiple higher intensity narrowband peaks (FWHM $\approx$ 5 nm) from about 610-650 nm (maximum intensity peak at 632 nm) in the orange to red region of the visible spectrum. The emission spectrum for the LED-C3 (dashed line) comprises a narrowband peak (FWHM $\approx$ 20 nm) with a peak emission wavelength of about 450 nm in the blue region of the visible spectrum and a lower intensity (about 30%) tail at wavelengths in the green to yellow region of the spectrum (460-580 nm).

TABLE 20A

Dev. 7: Measured characteristics of LED-C1, LED-C2, and LED-C3								
LED	Flux	Power	LE	CIE center point		CCT	CRI	
	(lm)	(W)	(lm/W)	x	y	(K)	Ra	Δuv
LED-C1	43.8	0.36	121.7	0.5557	0.3974	1700	81.5	-0.0023
LED-C2	63.7	0.36	176.9	0.4575	0.4991	3333	81.2	0.0309
LED-C3	51.8	0.36	143.9	0.2183	0.2644	n/a	n/a	n/a

TABLE 20B

Dev. 7: CIE region for light C1, C2, and C3										
CIE point	C1 CIE region				C2 CIE region				C3 CIE region	
	CIE		CCT	Δuv	CIE		CCT	Δuv	CIE	
	x	y	(K)		x	y	(K)		x	y
CIE-1	0.539	0.396	1800	-0.0042	0.445	0.503	3536	0.0342	0.207	0.275
CIE-2	0.545	0.374	1636	-0.0099	0.448	0.473	3308	0.0237	0.210	0.240
CIE-3	0.572	0.399	1605	-0.0002	0.470	0.496	3139	0.0279	0.230	0.254
CIE-4	0.567	0.421	1754	0.0047	0.467	0.526	3356	0.0377	0.227	0.289
Center	0.556	0.397	1693	-0.0024	0.458	0.499	3332	0.031	0.218	0.264

[0288] The CIE chromaticity diagram of FIG. 15B shows the chromaticity (CIE center point) 1532_{C1} of light of color C1 generated by the LED-C1, the chromaticity (CIE center point) 1532_{C2} of light of color C2 generated by LED-C2, and the chromaticity (CIE center point) 932_{C3} of light of color C3 generated by the LED-C3. Straight lines 1534_{C1C2} , 1534_{C2C3} , and 1532_{C1C3} connecting the points 1532_{C1} , 1532_{C2} , and 1532_{C3} define a triangle that represents the gamut of chromaticity (colors)/color temperatures of light that Dev.7 can generate—i.e. the device can generate any chromaticity (color)/color temperature of light lying on the boundary or within the triangle. It is to be noted that lowest CCT of light that Dev.7 can generate that lies on the black body locus (dashed line) is about 1800K (1821K) which corresponds to the point of intersection 1546 (CIE 0.546, 0.408) of line 1534_{C1C3} connecting color points 1532_{C1} to 1532_{C3} and the black body locus. It is to be noted that highest CCT of light that Dev.7 can generate light that lies on the black body locus (dotted line) is about 9800K (9809K) which corresponds to the point of intersection 1548 (CIE 0.282, 0.289) of line 1534_{C1C3} connecting color point 1532_{C1} to color point 1532_{C3} and the black body locus. It will be appreciated that Dev.7 is therefore capable of generating any CCT of light from about 1800K to about 9800K lying on the black body locus.

[0289] TABLE 21 tabulates forward drive current (I_F) for first and second LED-C1, LED-C2, and LED-C3 of Dev.7 for generating light with a nominal general color rendering index CRI Ra of 90 for nominal color temperatures (CCT)

[0291] Referring to FIG. 15C it is to be noted that Dev.7 can generate light 1542 (solid line) of CCTs from 2200K to 6500K with a chromaticity that closely matches the black body locus (dotted line). As can be seen from TABLE 22, the chromaticity of light generated by Dev.7 are within 0.0051 Δuv (about 0.005) or less of the black body locus for CCTs from 2200K to 6500K.

TABLE 21

Dev. 7: Forward drive current I_F for the 1 st LED-C3, 2 nd LED-C3, LED-C2, and LED-C1 for generating light with a nominal CRI Ra = 90 for nominal CCTs from 2200K to 6500K				
Nom. CCT	Forward drive current I_F (mA)			
(K)	1 st LED-C1	2 nd LED-C1	LED-C2	LED-C3
2200	104.9	104.9	89.1	1.1
2700	77.3	77.3	127.4	18.0
3000	64.3	64.3	139.8	31.6
3500	51.9	51.9	147.7	48.5
4000	40.4	40.4	149.4	69.7
5000	30.5	30.5	129.7	109.4
5700	28.8	28.8	110.5	132.0
6500	25.9	25.9	92.5	155.6

TABLE 22

Dev. 7: Measured optical and Electrical characteristics for nominal CRI Ra = 90 for nominal color temperatures (CCT) from 2200 K to 6500 K									
Nom.	Flux	Power	LE	CIE		CCT	CRI		
CCT (K)	(lm)	(W)	(lm/W)	x	y	(K)	Ra	R9	Δuv
2200	126.8	0.89	143.1	0.5019	0.4193	2264	91.9	44.0	0.0013
2700	134.8	0.88	153.5	0.4585	0.4145	2749	94.0	59.5	0.0016
3000	155.5	0.88	155.5	0.4336	0.4071	3082	94.3	65.2	0.0017
3500	157.5	0.88	157.5	0.4082	0.3975	3492	94.0	69.0	0.0023
4000	158.7	0.88	158.7	0.3817	0.3854	4029	93.2	69.7	0.0036
5000	157.8	0.88	157.8	0.3460	0.3614	4999	92.4	68.6	0.0045
5700	155.7	0.89	155.7	0.3304	0.3478	5589	92.4	70.3	0.0043
6500	151.3	0.90	151.3	0.3144	0.3344	6377	92.1	69.3	0.0051

of 2200K, 2700K, 3000K, 3500K, 4000K, 5000K, 5700K, and 6500K. TABLE 22 tabulates the measured optical and electrical characteristics for Dev.7 when operated to generate light with a nominal CRI Ra of 90 for nominal color temperatures (CCT) from 2200K to 6500K.

[0290] As can be seen from TABLE 21, the CCT of light generated by Dev.7 is increased by increasing the proportion of light of color C3 while reducing the proportion of light of color C1. TABLE 22 demonstrates that by selection of the drive currents to the first and second LED-C1, LED-C2, and LED-C3, Dev.7 can generate white light with a CCT from 2200 K to 6500K with a general color rendering index CRI Ra of at least 90 (91.9 to 94.3) and CRI R9 of at least 45 (44.0 to 70.3) with a luminous efficacy from about 143 lm/W (143.1 lm/W) to about 159 lm/W (158.7 lm/W). TABLE 22 also includes the measured CCT of light generated by Dev.7.

CIE Region and Center Points for C1, C2, C3: Dev.6 and Dev.7

[0292] TABLES 23A, 23B, and 23C tabulate CIE regions, CIE center points, CCT and Δuv for light C1, C2, and C3 respectively. The CIE regions for each color C1, C2, and C3 encompass the CIE regions for Dev.6 and Dev.7, that is for lighting devices with a CRI Ra from 80 to 97.

[0293] As can be seen from the tables, C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.495, 0.384), CIE-2 (0.509, 0.340), CIE-3 (0.576, 0.402), and CIE-4 (0.557, 0.442) which in terms of CCT is a CIE region from 1601K (about 1600K) to 2089K (about 2100K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.427, 0.484), CIE-2 (0.433, 0.427), CIE-3 (0.474, 0.468), and CIE-4 (0.467, 0.526) which in terms of CCT is from 2920K (about

2900K) to 3713K (about 3750K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.207, 0.275), CIE-2 (0.210, 0.240), CIE-3 (0.284, 0.294), and CIE-4 (0.279, 0.326).

TABLE 23A

Devs. 6-7: CIE region and CIE center point for light C1				
CIE point	C1 CIE region			
	CIE		CCT (K)	Δuv
	x	y		
CIE-1	0.495	0.384	2089	-0.0103
CIE-2	0.509	0.340	1685	-0.0240
CIE-3	0.576	0.402	1601	0.0009
CIE-4	0.557	0.442	1801	0.0108
Center	0.536	0.394	1813	-0.0052

TABLE 23B

Devs. 6-7: CIE region and CIE center point for light C2				
CIE point	C2 CIE region			
	CIE		CCT (K)	Δuv
	x	y		
CIE-1	0.427	0.484	3713	0.0315
CIE-2	0.433	0.427	3267	0.0094
CIE-3	0.474	0.468	2920	0.0185
CIE-4	0.467	0.526	3356	0.0377
Center	0.450	0.476	3298	0.0244

TABLE 23C

Devs. 6-7: CIE region and CIE center point for light C3				
	C3 CIE region			
CIE	CIE		CCT	
point	x	y	(K)	Δuv
CIE-1	0.207	0.275	n/a	n/a
CIE-2	0.210	0.240	n/a	n/a
CIE-3	0.284	0.294	9351	0.0009
CIE-4	0.279	0.326	8634	0.0197
Center	0.249	0.286	15053	0.0192

[0294] FIGS. 16A-16D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light C1, C2, and C3 for lighting devices Dev.6 and Dev.7.

[0295] FIG. 16A shows C1 CIE regions **1644** and C1 CIE center points **1632** for lighting devices Dev.6 (**1144**_{Dev.6}—solid square/solid line and **1632**_{Dev.6}—solid square), Dev.7 (**1644**_{Dev.7}—dot/dotted line and **1632**_{Dev.7}—dot), and C1 CIE region (**1644**_{C1}—diamond/dashed dot line) and C1 CIE center point (**1632**_{C1}—diamond) that encompass lighting devices Dev.6 and Dev.7. As can be seen from FIG. 16A the C1 CIE region **1644**_{C1} encompasses (fully encloses) the C1 CIE regions **1644**_{Dev.6} and **1644**_{Dev.7}.

[0296] FIG. 16B shows C2 CIE regions **1644** and C2 CIE center points **1632** for lighting devices Dev.6 (**1644**_{Dev.6}—solid square/solid line and **1632**_{Dev.6}—solid square), Dev.7

(**1644**_{Dev.7}—dot/dotted line and **1632**_{Dev.7}—dot), and C2 CIE region (**1644**_{C2}—diamond/dashed dot line) and C2 CIE center point (**1632**_{C2}—diamond) that encompass lighting devices Dev.6 and Dev.7. As can be seen from FIG. 16B the C2 CIE region **1644**_{C2} encompasses (fully encloses) the C2 CIE regions **1644**_{Dev.6} and **1644**_{Dev.7}.

[0297] FIG. 16C shows C3 CIE regions **1644** and C3 CIE center points **1632** for lighting devices Dev.6 (**1144**_{Dev.6}—solid square/solid line and **1632**_{Dev.6}—solid square), Dev.7 (**1644**_{Dev.7}—dot/dotted line and **1632**_{Dev.7}—dot), and C3 CIE region (**1644**_{C3}—diamond/dashed dot line) and C3 CIE center point (**1632**_{C3}—diamond) that encompass lighting devices Dev.6 and Dev.7. As can be seen from FIG. 16B the C3 CIE region **1644**_{C3} encompasses (fully encloses) the C3 CIE regions **1644**_{Dev.6} and **1644**_{Dev.7}.

[0298] FIG. 16D shows CIE regions **1644** and CIE center points **1632** for C1 (**1644**_{C1}—solid diamond/solid line and **1632**_{C1}—solid diamond), C2 (**1644**_{C2}—solid triangle/solid line and **1632**_{C2}—solid triangle), and C3 (**1644**_{C3}—solid square/solid line and **1632**_{C3}—solid square) that encompass lighting devices Dev.6 and Dev.7.

CIE Region and Center Points for C1, C2, C3—Devs.2-7

[0299] TABLES 24A, 24B, and 24C tabulate CIE regions, CIE center points, CCT and Δuv for light C1, C2, and C3 respectively. The CIE regions for each color C1, C2, and C3 encompass the CIE regions for lighting devices Devs.2-7.

[0300] As can be seen from the tables, C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.486, 0.410), CIE-2 (0.509, 0.340), CIE-3 (0.578, 0.405), and CIE-4 (0.537, 0.461) which in terms of CCT is a CIE region from 1602K (about 1600K) to 2368K (about 2400K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.328, 0.495), CIE-2 (0.342, 0.390), CIE-3 (0.478, 0.444), and CIE-4 (0.464, 0.545) which in terms of CCT is from 2706K (about 2700K) to 5621K (about 5700K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.195, 0.266), CIE-2 (0.205, 0.212), CIE-3 (0.286, 0.293), and CIE-4 (0.276, 0.328).

TABLE 24A

Devs. 2-7: CIE region and CIE center point for light C1				
CIE point	C1 CIE region			
	CIE		CCT	Δuv
	x	y	(K)	
CIE-1	0.486	0.410	2368	-0.0016
CIE-2	0.509	0.340	1681	-0.0240
CIE-3	0.578	0.405	1602	0.0019
CIE-4	0.537	0.461	2203	0.0137
Center	0.525	0.408	1981	-0.0016

TABLE 24B

Devs. 2-7: CIE region and CIE center point for light C2				
CIE point	C2 CIE region		CCT	
	CIE		(K)	Δuv
	x	y		
CIE-1	0.328	0.495	5621	0.0630
CIE-2	0.342	0.390	5204	0.0193
CIE-3	0.478	0.444	2706	0.0103
CIE-4	0.464	0.545	3486	0.0438
Center	0.405	0.469	4008	0.0312

TABLE 24C

Devs. 2-7: CIE region and CIE center point for light C3				
CIE point	C3 CIE region		CCT	
	CIE		(K)	Δuv
	x	y		
CIE-1	0.195	0.266	n/a	n/a
CIE-2	0.205	0.212	n/a	n/a
CIE-3	0.286	0.293	9198	-0.0008
CIE-4	0.276	0.328	8789	0.0223
Center	0.247	0.282	16002	0.0186

[0301] FIGS. 17A-16D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light of chromaticity C1, C2, and C3 for lighting devices Dev.2-7.

[0302] FIG. 17A shows C1 CIE regions **1744** and C1 CIE center points **1732** for lighting devices Dev.2 (**1744**_{Dev.2}—solid square/solid line and **1744**_{Dev.2}—solid square), Dev.3 (**1744**_{Dev.3}—dot/dotted line and **1732**_{Dev.3}—dot), Dev.4 (**1744**_{Dev.4}—star/dashed line and **1732**_{Dev.4}—dot), Dev.5 (**1744**_{Dev.5}—triangle/dashed line and **1732**_{Dev.5}—triangle), Dev.6 (**1744**_{Dev.6}—dot/solid line and **1732**_{Dev.6}—dot), Dev.7 (**1744**_{Dev.7}—square/dotted line and **1732**_{Dev.7}—square), and C1 CIE region (**1744**_{C1}—diamond/dashed dot line) and C1 CIE center point (**1732**_{C1}—diamond) that encompass lighting devices Devs.2-7. As can be seen from FIG. 17A the C1 CIE region **1744**_{C1} encompasses (fully encloses) the C1 CIE regions **1744**_{Dev.2}, **1744**_{Dev.3}, **1744**_{Dev.4}, **1744**_{Dev.5}, **1744**_{Dev.6}, and **1744**_{Dev.7} for lighting devices Devs.2-7.

[0303] FIG. 17B shows C2 CIE regions **1744** and C2 CIE center points **1732** for lighting devices Dev.2 (**1744**_{Dev.2}—solid square/solid line and **1744**_{Dev.2}—solid square), Dev.3 (**1744**_{Dev.3}—dot/dotted line and **1732**_{Dev.3}—dot), Dev.4 (**1744**_{Dev.4}—star/dashed line and **1732**_{Dev.4}—dot), Dev.5 (**1744**_{Dev.5}—triangle/dashed line and **1732**_{Dev.5}—triangle), Dev.6 (**1744**_{Dev.6}—dot/solid line and **1732**_{Dev.6}—dot), Dev.7 (**1744**_{Dev.7}—square/dotted line and **1732**_{Dev.7}—square), and C2 CIE region (**1744**_{C2}—diamond/dashed dot line) and C2 CIE center point (**1732**_{C2}—diamond) that encompass lighting devices Devs.2-7. As can be seen from FIG. 17B the C2 CIE region **1744**_{C2} encompasses (fully encloses) the C2 CIE regions **1744**_{Dev.2}, **1744**_{Dev.3}, **1744**_{Dev.4}, **1744**_{Dev.5}, **1744**_{Dev.6}, and **1744**_{Dev.7} for lighting devices Devs.2-7.

[0304] FIG. 17C shows C3 CIE regions **1744** and C3 CIE center points **1732** for lighting devices Dev.2-4 (**1744**_{Devs.}

₂₋₄—diamond/solid line and **1744**_{Devs.2-4}—diamond), Dev.5 (**1744**_{Dev.5}—triangle/dashed line and **1732**_{Dev.5}—triangle), Dev.6 (**1744**_{Dev.6}—dot/solid line and **1732**_{Dev.6}—dot), Dev.7 (**1744**_{Dev.7}—square/dotted line and **1732**_{Dev.7}—square), and C3 CIE region (**1744**_{C3}—diamond/dashed dot line) and C3 CIE center point (**1732**_{C3}—diamond) that encompass lighting devices Devs.2-7. As can be seen from FIG. 17C the C3 CIE region **1744**_{C3} encompasses (fully encloses) the C3 CIE regions **1744**_{Devs.2-4}, **1744**_{Dev.5}, **1744**_{Dev.6}, and **1744**_{Dev.7} for lighting devices Devs.2-7.

[0305] FIG. 17D shows CIE regions **1744** and CIE center points **1732** for C1 (**1744**_{C1}—solid diamond/solid line and **1732**_{C1}—solid diamond), C2 (**1744**_{C2}—solid triangle/solid line and **1732**_{C2}—solid triangle), and C3 (**1744**_{C3}—solid square/solid line and **1732**_{C3}—solid square) that encompass lighting devices Devs.2-7.

CIE Region and Center Points for C1, C2, C3—Devs.2,3,6,7

[0306] TABLES 25A, 25B, and 25C tabulate CIE regions, CIE center points, CCT and Δuv for light C1, C2, and C3 respectively. The CIE regions for each color C1, C2, and C3 encompass the CIE regions for lighting devices Dev.2, Dev.3, Dev.6 and Dev.7.

[0307] As can be seen from TABLES 25A, 25B, and 25C, C1 comprises a CIE region (C1 CIE region) defined by CIE color coordinates CIE-1 (0.486, 0.410), CIE-2 (0.509, 0.340), CIE-3 (0.578, 0.405), and CIE-4 (0.537, 0.461) which in terms of CCT is a CIE region from 1602K (about 1600K) to 2368K (about 2400K). C2 comprises a CIE region (C2 CIE Region) defined by CIE color coordinates CIE-1 (0.348, 0.478), CIE-2 (0.356, 0.396), CIE-3 (0.477, 0.444), and CIE-4 (0.467, 0.526) which in terms of CCT is from 2719K (about 2700K) to 5180K (about 5200K). C3 comprises a CIE region (C3 CIE Region) defined by CIE color coordinates CIE-1 (0.195, 0.266), CIE-2 (0.205, 0.212), CIE-3 (0.286, 0.293), and CIE-4 (0.276, 0.328).

TABLE 25A

Devs. 2-3, 6-7: CIE region and CIE center point for light C1				
CIE point	C1 CIE region		CCT	
	CIE		(K)	Δuv
	x	y		
CIE-1	0.486	0.410	2368	-0.0016
CIE-2	0.509	0.340	1681	-0.0240
CIE-3	0.578	0.405	1602	0.0019
CIE-4	0.537	0.461	2203	0.0137
Center	0.525	0.408	1981	-0.0016

TABLE 25B

Devs. 2-3, 6-7: CIE region and CIE center point for light C2				
CIE point	C2 CIE region		CCT	
	CIE		(K)	Δuv
	x	y		
CIE-1	0.348	0.478	5180	0.0508
CIE-2	0.356	0.396	4797	0.0167
CIE-3	0.477	0.444	2719	0.0104

TABLE 25B-continued

Devs. 2-3, 6-7: CIE region and CIE center point for light C2				
CIE point	C2 CIE region			
	CIE		CCT (K)	Δuv
	x	y		
CIE-4	0.467	0.526	3352	0.0377
Center	0.412	0.462	3844	0.0270

TABLE 25C

Devs. 2-3, 6-7: CIE region and CIE center point for light C3				
	C3 CIE region			
CIE	CIE		CCT	
point	x	y	(K)	Δuv
CIE-1	0.195	0.266	n/a	n/a
CIE-2	0.205	0.212	n/a	n/a
CIE-3	0.286	0.293	9198	-0.0008
CIE-4	0.276	0.328	8789	0.0223
Center	0.247	0.282	16002	0.0186

[0308] FIGS. 18A-18D are CIE 1931 chromaticity diagrams illustrating various CIE regions and CIE center points for light of chromaticity C1, C2, and C3 for lighting devices Dev.2, Dev.3, Dev.6, and Dev.7.

[0309] FIG. 18A shows C1 CIE regions **1844** and C1 CIE center points **1832** for lighting devices Dev.2 (**1844**_{Dev.2}—solid triangle/solid line and **1844**_{Dev.2}—solid triangle), Dev.3 (**1844**_{Dev.3}—solid diamond/dashed line and **1832**_{Dev.3}—solid triangle), Dev.6 (**1844**_{Dev.6}—dot/solid line and **1832**_{Dev.6}—dot), Dev.7 (**1844**_{Dev.7}—square/dotted line and **1832**_{Dev.7}—square), and C1 CIE region (**1844**_{C1}—diamond/dashed dot line) and C1 CIE center point (**1832**_{C1}—diamond) that encompass lighting devices Devs.2, Dev.3, Dev.6, and Dev.7. As can be seen from FIG. 18A the C1 CIE region **1844**_{C1} encompasses (fully encloses) the C1 CIE regions **1844**_{Dev.2}, **1844**_{Dev.3}, **1844**_{Dev.6}, and **1844**_{Dev.7} for lighting devices Devs.2-3,6-7.

[0310] FIG. 18B shows C2 CIE regions **1844** and C2 CIE center points **1832** for lighting devices Dev.2 (**1844**_{Dev.2}—solid triangle/solid line and **1844**_{Dev.2}—solid triangle), Dev.6 (**1844**_{Dev.6}—dot/solid line and **1832**_{Dev.6}—dot), Dev.7 (**1844**_{Dev.7}—solid square/dotted line and **1832**_{Dev.7}—solid triangle), and C2 CIE region (**1844**_{C2}—diamond/dashed dot line) and C2 CIE center point (**1832**_{C2}—diamond) that encompass lighting devices Devs.2, Dev.3, Dev.6, and Dev.7. As can be seen from FIG. 18B the C2 CIE region **1844**_{C2} encompasses (fully encloses) the C2 CIE regions **1844**_{Dev.2}, **1844**_{Dev.3}, **1844**_{Dev.6}, and **1844**_{Dev.7} for lighting devices Devs.2-3,6-7.

[0311] FIG. 18C shows C3 CIE regions **1844** and C3 CIE center points **1832** for lighting devices Dev.2-3 (**1844**_{Devs.2-3}—triangle/solid line and **1844**_{Devs.2-3}—triangle), Dev.6 (**1844**_{Dev.6}—dot/solid line and **1832**_{Dev.6}—dot), Dev.7 (**1844**_{Dev.7}—solid square/dotted line and **1832**_{Dev.7}—solid square), and C3 CIE region (**1844**_{C3}—diamond/dashed dot line) and C3 CIE center point (**1832**_{C3}—diamond) that encompass lighting devices Devs.2, Dev.3, Dev.6, and Dev.7. As can be seen from FIG. 18C the C3 CIE region **1844**_{C3}

encompasses (fully encloses) the C3 CIE regions **1844**_{Devs.2-3}, **1844**_{Dev.6}, and **1844**_{Dev.7} for lighting devices Devs.2-3, 6-7.

[0312] FIG. 18D shows CIE regions **1844** and CIE center points **1832** for C1 (**1844**_{C1}—solid diamond/solid line and **1832**_{C1}—solid diamond), C2 (**1844**_{C2}—solid triangle/solid line and **1832**_{C2}—solid triangle), and C3 (**1844**_{C3}—solid square/solid line and **1832**_{C3}—solid square) that encompass the CIE regions **1844**_{Dev.2}, **1844**_{Dev.3}, **1844**_{Dev.6}, and **1844**_{Dev.7} for lighting devices Devs.2-3,6-7.

Dominant Wavelength

[0313] In color space, the dominant wavelength is a way of characterizing a color's hue. A color's dominant wavelength is the wavelength of monochromatic light (single wavelength) that to an observer evokes an identical perception of hue. FIG. 19 is a 1931 CIE chromaticity diagram illustrating calculation of dominant wavelength λ_d of light of a given chromaticity (color point). As shown in FIG. 19, the chromaticity (color point), CIE 0.356, 0.500, is indicated on the chromaticity diagram by a cross **1958**, and a "white standard illuminant", CIE (1/3, 1/3), is indicated by a circle **1960**. A straight line **1962** (dashed line) connecting the color point **1958** to the "white standard illuminant" **1960** is extended so that it intersects the outer curved gamut boundary (perimeter) of the CIE color space. The point of intersection **1964** (large solid dot) nearest to the color point **1958** corresponds to the dominant wavelength λ_d (560 nm in the figure) of the color as the wavelength of the pure spectral (monochromatic) color at that intersection point **1964**. It will be appreciated that in the example illustrated all chromaticity (color points) lying on the line **1962** have a dominant wavelength λ_d , 560 nm.

Color-Temperature-Tunable Lighting Device—Dev.8

[0314] A color-temperature-tunable lighting device, denoted Dev.8, comprises LEDs that generate light of three different colors/CCTs: C1, C2, and C3. Lighting device Dev.8 comprises the device of FIGS. 3E-3F and comprises a 3838 four cavity package containing two LEDs, LED-C1, that generate light with a first chromaticity/color temperature C1, an LED, LED-C2, that generates of a second chromaticity/color temperature C2, and an LED, LED-C3, that generates light of a third chromaticity/color temperature C3.

[0315] The two LEDs, LED-C1, are for generating "warm white" light of a color temperature from 1400K to 3000K with a chromaticity that lies on the black body locus or near to the black body locus. In this specification, near to the black body locus means within 5 SDCM, preferably within 3SDCM or within 0.003 Δuv of the black body locus. The LED, LED-3, is for generating "cool white" light of a color temperature from 5000K to 15000K with a chromaticity that lies on the black body locus or near to the black body locus (i.e., within 5 SDCM or 0.003 Δuv). The LED, LED C2, is for generating light with a dominant wavelength λ_d from 535 nm to 575 nm. Use of an LED with a dominant wavelength from 535 nm to 575 nm has the benefit of increasing luminous efficacy of the light emitting device since such a wavelength is close to or corresponds to the maxima of the photopic response curve at 555 nm.

[0316] Dev.8 can generate light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a

chromaticity that is within 0.003 Δuv of the black body locus for CCTs from 2200K to 6500K with a luminous efficacy of at least 150 lm/W.

TABLE 26

Devs.8: CIE region for light C2					
C2 CIE region					
CIE	CIE		λ_d	CCT	
point	x	y	(nm)	(K)	Δuv
CIE-1	0.256	0.580	535	6926	0.1087
CIE-2	0.334	0.400	555	5466	0.0267
CIE-3	0.436	0.466	575	3458	0.0234
CIE-4	0.336	0.560	555	5468	0.0788

[0317] TABLE 26 tabulates the CIE region, dominant wavelength λ_d , CCT, and Δuv for light of chromaticity C2. As can be seen from TABLE 26, C2 comprises a quadrilateral CIE region (C2 CIE region) defined by straight lines connecting CIE color coordinates CIE-1 (0.256, 0.580), CIE-2 (0.334, 0.400), CIE-3 (0.436, 0.466), and CIE-4 (0.336, 0.560) which in terms of dominant wavelength λ_d is from 535 nm to 575 nm. In terms of CCT, the C2 CIE region is from 3458K (about 3500K) to 6926K (about 6900K) with a Δuv of 0.0234 to 0.1087, that is, at least 0.020 Δuv from the black body locus.

[0318] FIG. 20 is a CIE 1931 chromaticity diagram illustrating CIE regions for light of chromaticity C1, C2, and C3 for lighting device Dev.8. The CIE chromaticity diagram shows the chromaticity region (C2 CIE region) 2044_{C2} of light of color C2 generated by the LED-C2. The chromaticity region is a quadrilateral with a solid line connecting the points (solid triangle) CIE-1, CIE-2, CIE-3, and CIE-4. The chromaticity of light for the chromaticity region (CIE region) can comprise any chromaticity value (CIE x, y) within the quadrilateral or lying on the straight line sides connecting of the quadrilateral. As described herein, for light of chromaticity C1 and C3 generated by LED-C1 and LED-C3 respectively, the chromaticity region may be defined in terms of a color temperature range $C1_{CCT}$ and $C3_{CCT}$ respectively. As shown in FIG. 20, chromaticity region $C1_{CCT}$ is indicated by a curved doubled headed arrow $C1_{CCT}$ and corresponds to a chromaticity lying on or near the black body locus for a color temperature from 1400K to 3000K (solid diamond). The chromaticity region $C3_{CCT}$ is indicated in the figure by a near-straight doubled headed arrow $C3_{CCT}$ and corresponds to a chromaticity lying on or near the black body locus for a color temperature from 5000K to 15000K (cross).

[0319] In embodiments, the first LED may be for generating light with a CCT from 1400K to 2500K, optionally 1400K to 2000K; the second LED may be for generating light with a dominant wavelength from 545 nm to 565 nm, optionally from 550 nm to 560 nm, optionally about 555 nm; and the third LED may be for generating light with a CCT from 10000K to 15000K.

LIST OF REFERENCE NUMERALS

FIG. 1:

- [0320] 1 Multi-LED package
- [0321] 2 Lead frame

- [0322] 3 Direct-Emitting LED chip
- [0323] 3R Direct-Emitting Red LED chip
- [0324] 3G Direct-Emitting Green LED chip
- [0325] 3B Direct-Emitting Blue LED chip
- [0326] 4 Housing
- [0327] 5 Cavity (recess)
- [0328] 6 Light-transmissive encapsulant
- [0329] 7 Anode electrical terminal
- [0330] 7R Anode electrical terminal Red LED
- [0331] 7G Anode electrical terminal Green LED
- [0332] 7B Anode electrical terminal Blue LED
- [0333] 8 Cathode electrical terminal
- [0334] 8R Cathode electrical terminal Red LED
- [0335] 8G Cathode electrical terminal Green LED
- [0336] 8B Cathode electrical terminal Blue LED

FIGS. 2 to 9 (#=Figure Number):

- [0337] #10 Multi-LED package (package)
- [0338] #12 Anode lead frame
- [0339] #12a Anode lead frame—first cavity
- [0340] #12b Anode lead frame—second cavity
- [0341] #12c Anode lead frame—third cavity
- [0342] #12d Anode lead frame—fourth cavity
- [0343] #12e Common cathode lead frame—cavities 1-4
- [0344] #14 Cathode lead frame
- [0345] #14a-d Cathode lead frame—common to cavities 1-4
- [0346] #14a Cathode lead frame—first cavity
- [0347] #14b Cathode lead frame—second cavity
- [0348] #14c Cathode lead frame—third cavity
- [0349] #14d Cathode lead frame—fourth cavity
- [0350] #16 Housing
- [0351] #18 Cavity (recess or cup)
- [0352] #18a First cavity
- [0353] #18b Second cavity
- [0354] #18c Third cavity
- [0355] #18d Fourth cavity
- [0356] #20 LED chip
- [0357] #20a LED chip—first cavity
- [0358] #20b LED chip—second cavity
- [0359] #20c LED chip—third cavity
- [0360] #20d LED chip—fourth cavity
- [0361] #22 Anode electrical terminal
- [0362] #22a Anode electrical terminal first cavity (recess)
- [0363] #22b Anode electrical terminal second cavity (recess)
- [0364] #22c Anode electrical terminal third cavity (recess)
- [0365] #22d Anode electrical terminal fourth cavity (recess)
- [0366] #24 Cathode electrical terminal
- [0367] #24a-d Cathode electrical terminal—common to cavities 1-4
- [0368] #24a Cathode electrical terminal first cavity (recess)
- [0369] #24b Cathode electrical terminal second cavity (recess)
- [0370] #24c Cathode electrical terminal third cavity (recess)
- [0371] #24d Cathode electrical terminal fourth cavity (recess)

[0372] #26 Color-temperature-tunable lighting device (lighting device)
 [0373] #28 LED (Light Emitting Diode)
 [0374] #28a First LED
 [0375] #28b Second LED
 [0376] #28c Third LED
 [0377] #28d Fourth LED
 [0378] #30 Photoluminescence layer
 [0379] #30a Photoluminescence layer—first cavity
 [0380] #30b Photoluminescence layer—second cavity
 [0381] #30c Photoluminescence layer—third cavity
 [0382] #30d Photoluminescence layer—fourth cavity
 [0383] #32 Chromaticity (color point)
 [0384] #32_{C1} Chromaticity (color point) of light C1
 [0385] #32_{C2} Chromaticity (color point) of light C2
 [0386] #32_{C3} Chromaticity (color point) of light C3
 [0387] #32_{C4} Chromaticity (color point) of light C4
 [0388] #34 Straight line connecting chromaticity (color points)
 [0389] #34_{C1C2} Straight line connecting color points C1 and C2
 [0390] #34_{C1C3} Straight line connecting color points C1 and C3
 [0391] #34_{C1C4} Straight line connecting color points C1 and C4
 [0392] #34_{C2C3} Straight line connecting color points C2 and C3
 [0393] #34_{C3C4} Straight line connecting color points C3 and C4
 [0394] #36 Color-tunable linear light emitting device (linear lighting device)
 [0395] #38 Substrate
 [0396] #40 Electrical connector
 [0397] #42 Light emission locus
 [0398] #44 LED CIE region
 [0399] #44_{C1} C1 CIE Region
 [0400] #44_{C2} C2 CIE Region
 [0401] #44_{C3} C3 CIE Region
 [0402] #46 Lowest CCT
 [0403] #48 Highest CCT
 [0404] #50 Peak
 [0405] #52 Spectrum intensity at $\lambda_{max\Delta}$
 [0406] #54 Black body spectrum intensity at $\lambda_{max\Delta}$
 [0407] #56 Trough
 [0408] #58 Chromaticity (color point)
 [0409] #60 “White standard illuminant”—CIE (1/3, 1/3)
 [0410] #62 Straight line connecting CIE (1/3,1/3) to color point #58
 [0411] #64 Dominant wavelength λ_d
 1. A light emitting device comprising:
 a first LED for generating light of a first chromaticity in a first chromaticity region;
 a second LED for generating light of a second chromaticity in a second chromaticity region; and
 a third LED for generating light of a third chromaticity in a third chromaticity region,
 wherein the device is for generating light with a CCT that is tunable within a range of CCTs from 1800K to 6500K and has a chromaticity that is within 5 SDCM of the black body locus;
 wherein the device has a luminous efficacy of at least 95 lm/W; and

wherein the light generated by the device comprises a combination of light generated by the first, second, and third LEDs and wherein the CCT is tunable by controlling power to the first, second and third LEDs.

2. The light emitting device of claim 1, wherein the device is for generating light with a chromaticity that is within 0.004 Auv of the black body locus, or is within 3 SDCM of the black body locus.

3. The light emitting device of claim 1, wherein the device has a luminous efficacy of at least 135 lm/W, or at least 150 lm/W.

4. The light emitting device of claim 1, wherein light generated by the first LED and third LED has a chromaticity that is within 5 SDCM of the black body locus, is within 3 SDCM of the black body locus, or is within 0.003 Auv of the black body locus.

5. The light emitting device of claim 1, wherein light generated by the second LED has a chromaticity that is at least 2 SDCM from the black body locus, is at least 7 SDCM from the black body locus, or is at least 0.020 Auv from the black body locus.

6. The light emitting device of claim 1, wherein the first LED is for generating warm white light; the second LED is for generating light with a dominant wavelength from 535 nm to 575 nm; and the third LED is for generating cool white light.

7. The light emitting device of claim 1, comprising a fourth LED for generating light of the first chromaticity, the second chromaticity, or the third chromaticity.

8. The light emitting device of claim 1, wherein the LED that generates light with the lowest CCT comprises a narrowband red phosphor selected from the group consisting of: $\text{K}_2\text{SiF}_6:\text{Mn}^{4+}$, $\text{K}_2\text{GeF}_6:\text{Mn}^{4+}$, and/or $\text{K}_2\text{TiF}_6:\text{Mn}^{4+}$.

9. The light emitting device of claim 1, wherein at least one of the first, second, or third LED comprises an LED chip for generating blue light having a FWHM of at about 30 nm to about 80 nm.

10. The light emitting device of claim 1, comprising at least two first LEDs for generating light of a lowest CCT.

11. The light emitting device of claim 1, wherein the luminous efficacy of the first LED is lower than that of the second and third LEDs.

12. The light emitting device of claim 1, wherein the first chromaticity region is defined by chromaticity coordinates (0.486, 0.410), (0.509, 0.340), (0.578, 0.405), and (0.537, 0.461);

the second chromaticity region is defined by chromaticity coordinates (0.328, 0.495), (0.342, 0.390), (0.478, 0.444), and (0.464, 0.545); and

the third chromaticity region is defined by chromaticity coordinates (0.195, 0.266), (0.205, 0.212), (0.286, 0.293), and (0.276, 0.328).

13. The light emitting device of claim 12, wherein the second chromaticity region is defined by chromaticity coordinates (0.348, 0.478), (0.356, 0.396), (0.477, 0.444), and (0.467, 0.526).

14. The light emitting device of claim 12, wherein

the first chromaticity region is defined by chromaticity coordinates (0.495, 0.384), (0.509, 0.340), (0.576, 0.402), and (0.557, 0.442);

the second chromaticity region is defined by chromaticity coordinates (0.427, 0.484), (0.433, 0.427), (0.474, 0.468), and (0.467, 0.526); and

the third chromaticity region is defined by chromaticity coordinates (0.207, 0.275), (0.210, 0.240), (0.284, 0.294), and (0.279, 0.326).

15. The light emitting device of claim **12**, wherein the first chromaticity region is defined by chromaticity coordinates (0.495, 0.384), (0.501, 0.363), (0.529, 0.388), and (0.522, 0.409);

the second chromaticity region is defined by chromaticity coordinates (0.430, 0.457), (0.433, 0.426), (0.455, 0.449), and (0.452, 0.479); and

the third chromaticity region is defined by chromaticity coordinates (0.270, 0.310), (0.274, 0.297), (0.281, 0.308), and (0.278, 0.321).

16. The light emitting device of claim **15**, wherein the first chromaticity region is 6 SDCM centered on a chromaticity (0.511, 0.386);

the second chromaticity region is 6 SDCM centered on a chromaticity (0.443, 0.453); and

the third chromaticity region is 6 SDCM centered on a chromaticity (0.276, 0.309).

17. The light emitting device of claim **12**, wherein the first chromaticity region is defined by chromaticity coordinates (0.539, 0.396), (0.545, 0.374), (0.572, 0.399), and (0.567, 0.421);

the second chromaticity region is defined by chromaticity coordinates (0.445, 0.503), (0.448, 0.473), (0.470, 0.496), and (0.467, 0.526); and

the third chromaticity region is defined by chromaticity coordinates (0.207, 0.275), (0.210, 0.240), (0.230, 0.254), and (0.227, 0.289).

18. The light emitting device of claim **17**, wherein the first chromaticity region is 6 SDCM centered on a chromaticity (0.556, 0.397);

the second chromaticity region is 6 SDCM centered on a chromaticity (0.458, 0.499); and

the third chromaticity region is 6 SDCM centered on a chromaticity (0.218, 0.264).

19. The light emitting device of claim **1**, wherein the first chromaticity region is defined by chromaticity coordinates (0.495, 0.384), (0.501, 0.363), (0.529, 0.388), and (0.522, 0.409);

the second chromaticity region is defined by chromaticity coordinates (0.430, 0.457), (0.433, 0.426), (0.455, 0.449), and (0.452, 0.479); and

the third chromaticity region is defined by chromaticity coordinates (0.270, 0.310), (0.274, 0.297), (0.281, 0.308), and (0.278, 0.321).

20. The light emitting device of claim **1**, wherein the device is for generating light with an intensity versus wavelength spectrum, which over a wavelength range 440 nm to 650 nm, a maximum percentage deviation between the intensity of light generated by the device and the intensity of light of the spectrum of a black body radiator or standard illuminant is at least one of less than 30%, less than 20%, or less than 10%.

* * * * *